



# SMART COCKPIT.COM

## ***A319/A320/A321***

### ***Navigation***



**DESCRIPTION**

The Air Data and Inertial Reference System (ADIRS) supplies temperature, anemometric, barometric and inertial parameters to the EFIS system (PFD and ND) and to other user systems (FMGC, FADEC, ELAC, SEC, FAC, FWC, SFCC, ATC, GPWS, CFDIU, CPC).

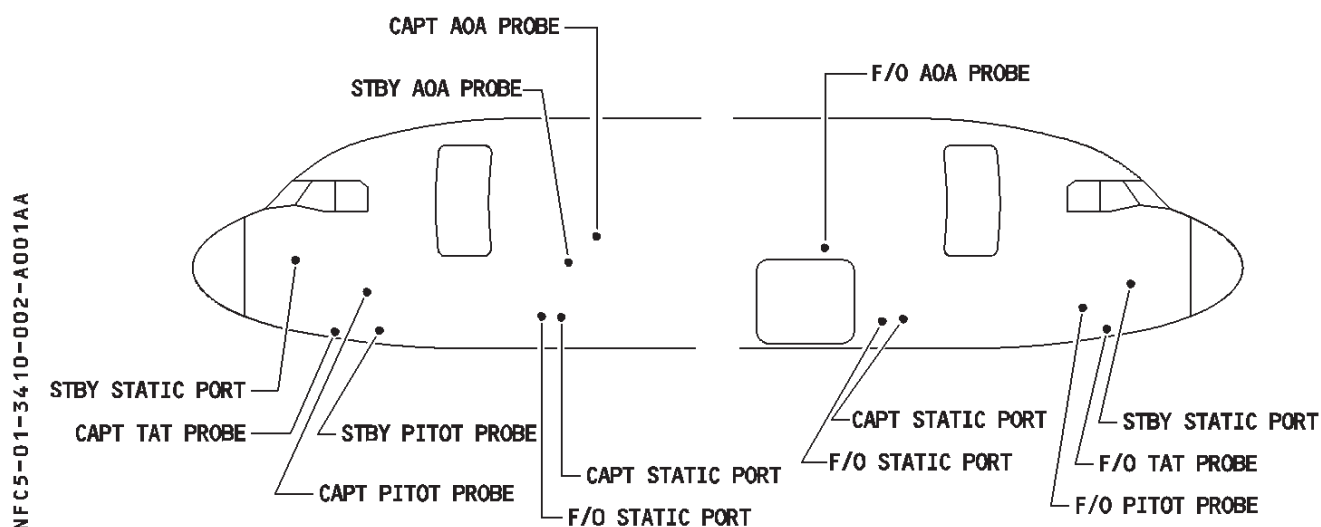
The system includes :

- three identical ADIRU's (Air Data and Inertial Reference Units).  
Each ADIRU is divided in two parts, either of which can work separately in case of failure in the other :
  - the ADR part (Air Data Reference) which supplies barometric altitude, airspeed, mach, angle of attack, temperature and overspeed warnings.
  - the IR part (Inertial Reference) which supplies attitude, flight path vector, track, heading, accelerations, angular rates, ground speed and aircraft position.

*Note : Above 73° North and above 60° South, the ADIRU gives true heading instead of magnetic heading.*

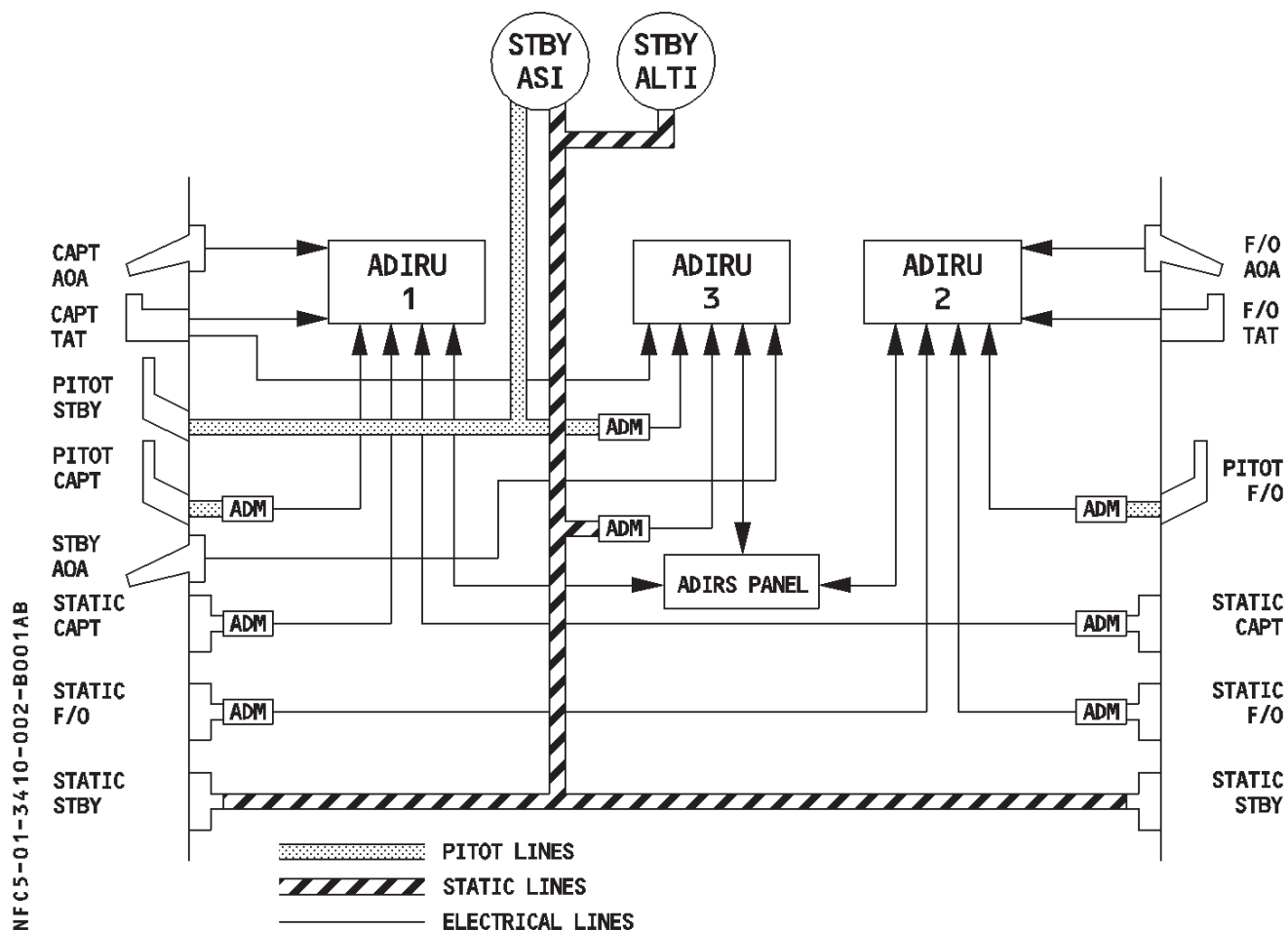
- one ADIRS control panel (ADIRS CDU) on the overhead panel for selection of modes (NAV, ATT, OFF) and indications of failures.  
The IR is normally initialized through the FMGS, but the ADIRS CDU may be used as a back up.
- four types of sensors :
  - pitot probes (3)
  - static pressure probes (STAT) (6)
  - angle of attack sensors (AOA) (3)
  - total air temperature probes (TAT) (2)These sensors are electrically heated to prevent from icing up.
- eight ADMs (Air Data Modules) which convert pneumatic data from PITOT and STAT probes into numerical data for the ADIRUs.
- a switching facility for selecting ADR3 or IR3 for instrument displays in case of ADIRU 1 or 2 failure.

## PROBES LOCATION



## PROBES SCHEMATIC

R



Note : ADIRU (1) is supplied by CAPT probes,  
 (2) is supplied by F/O probes,  
 (3) is supplied by STBY probes and CAPT TAT.

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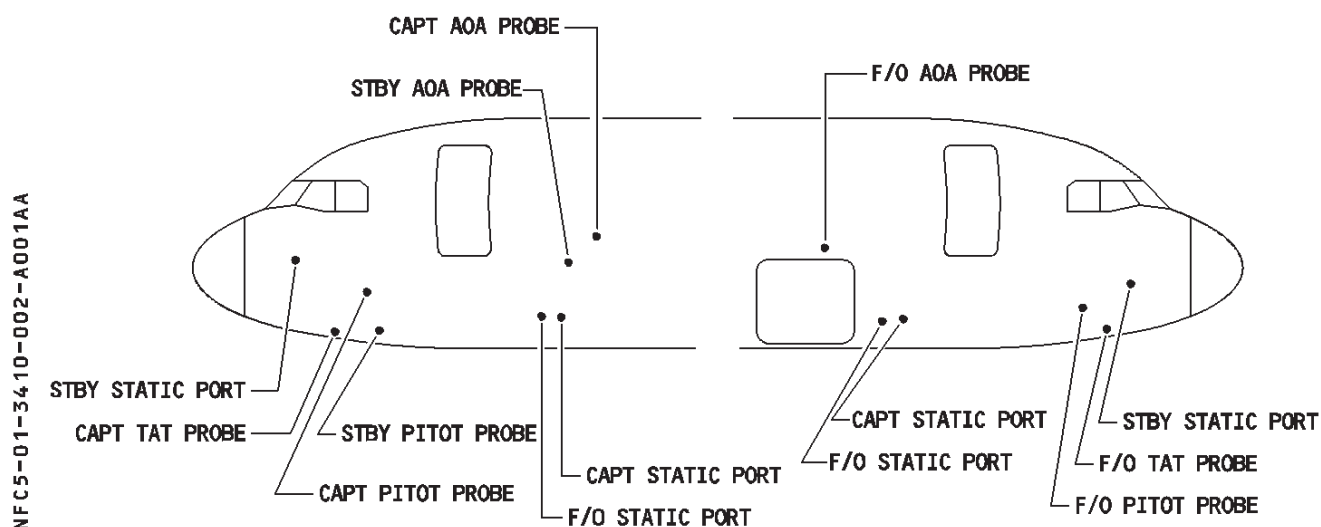
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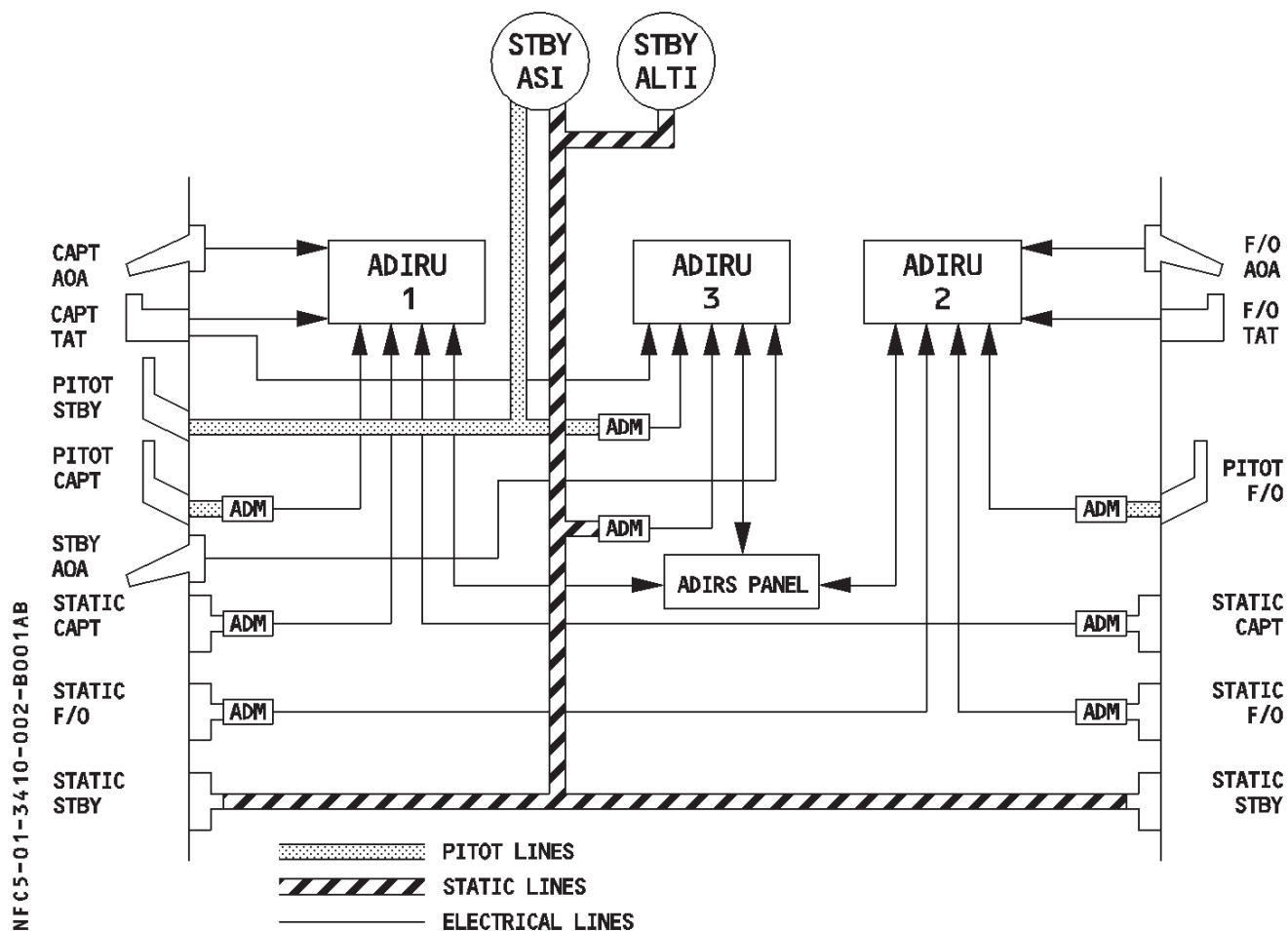
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- above 82° North
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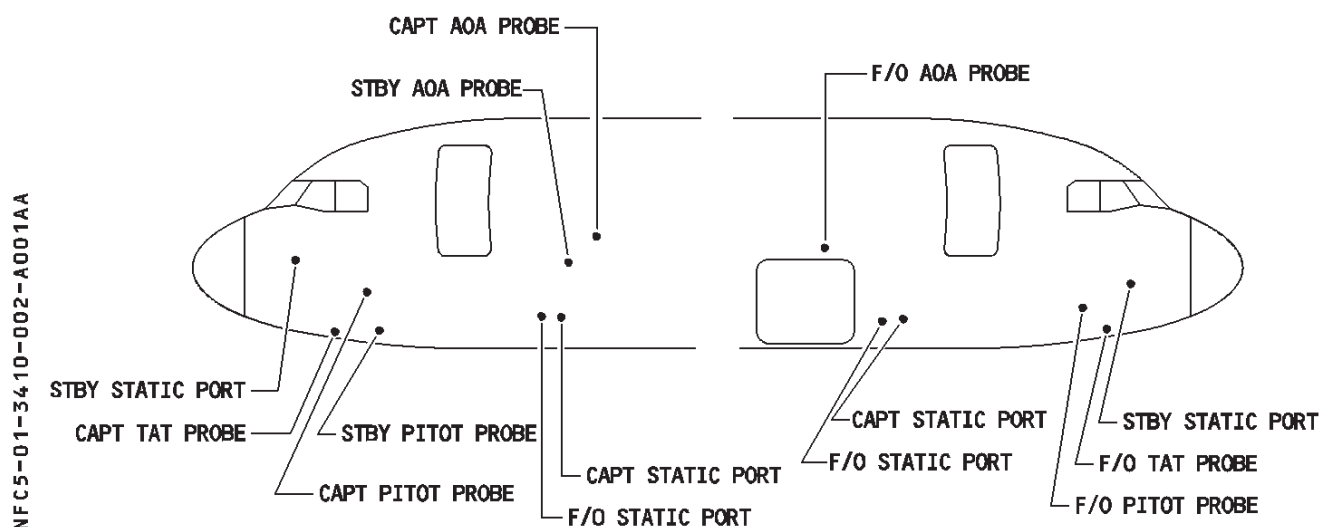
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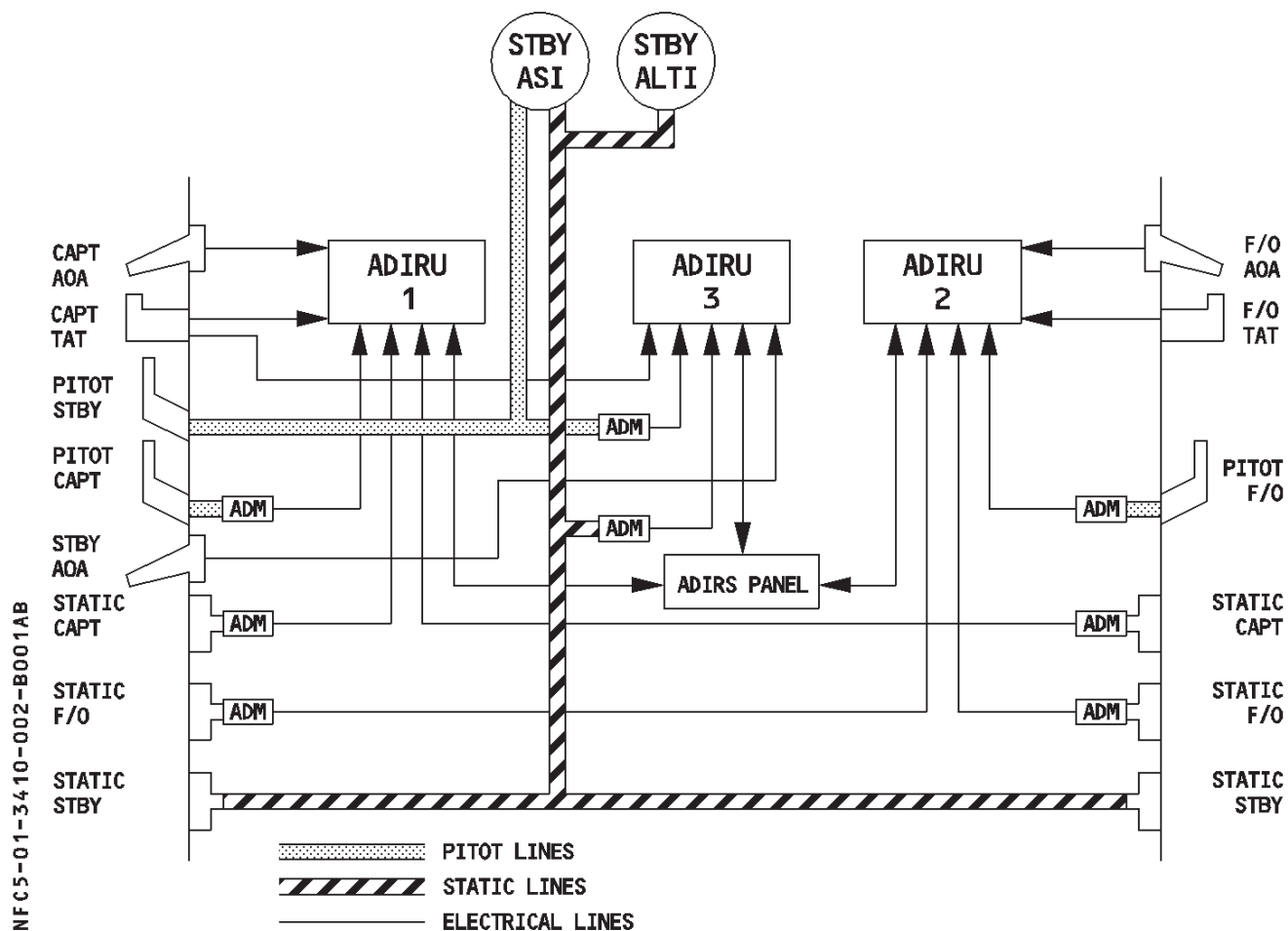
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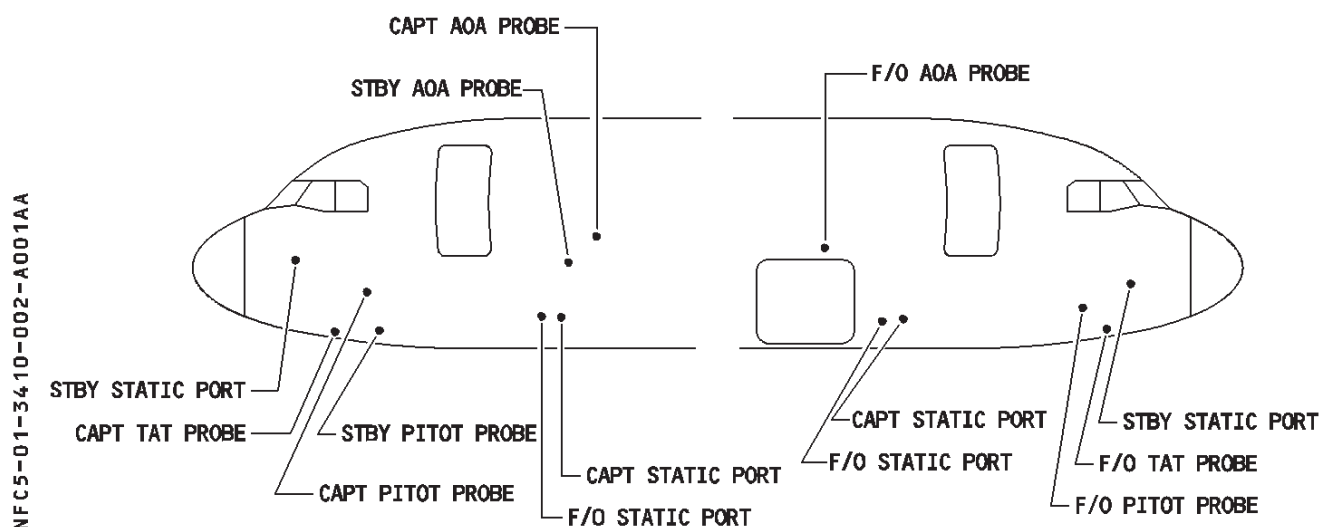
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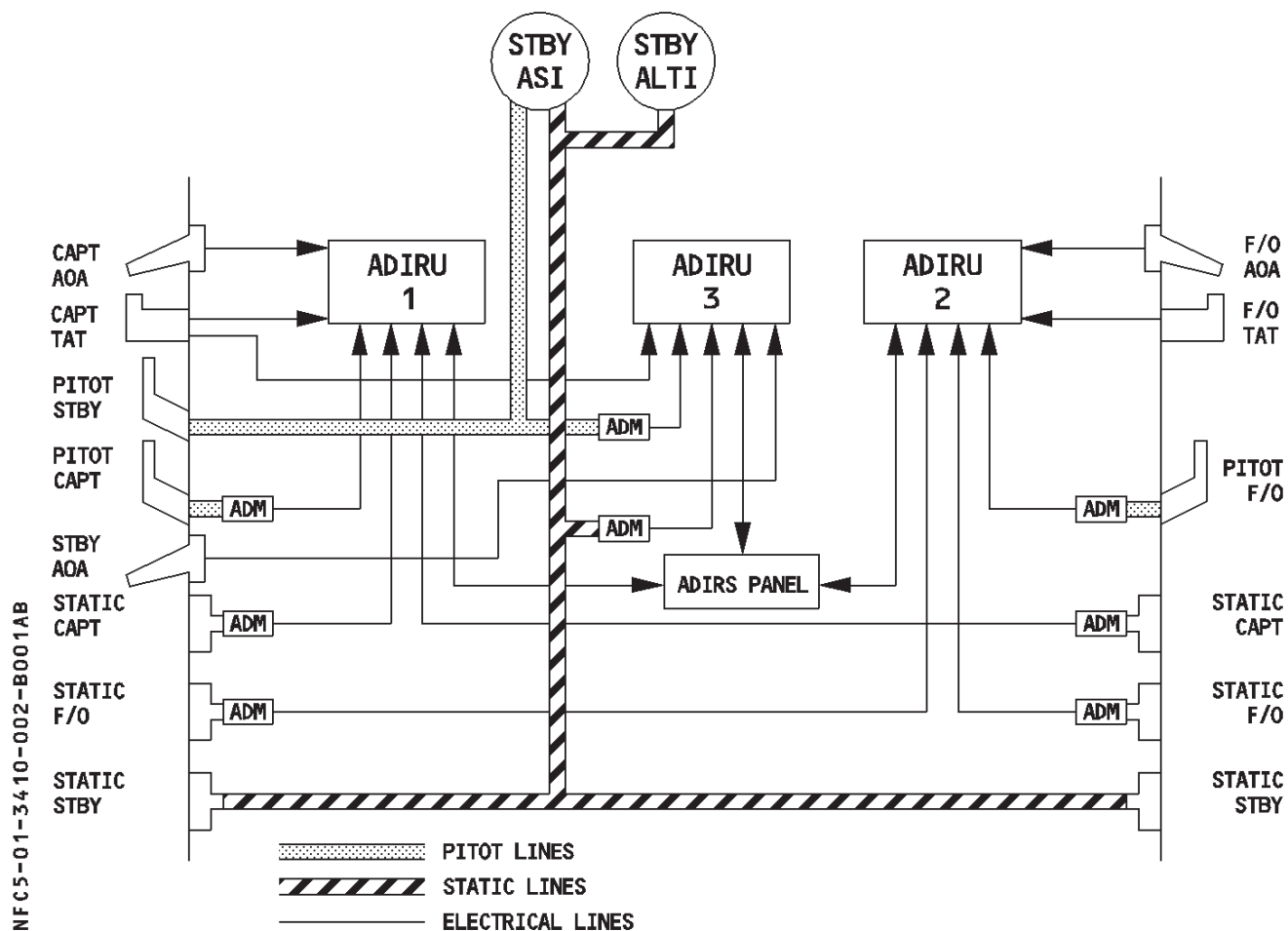
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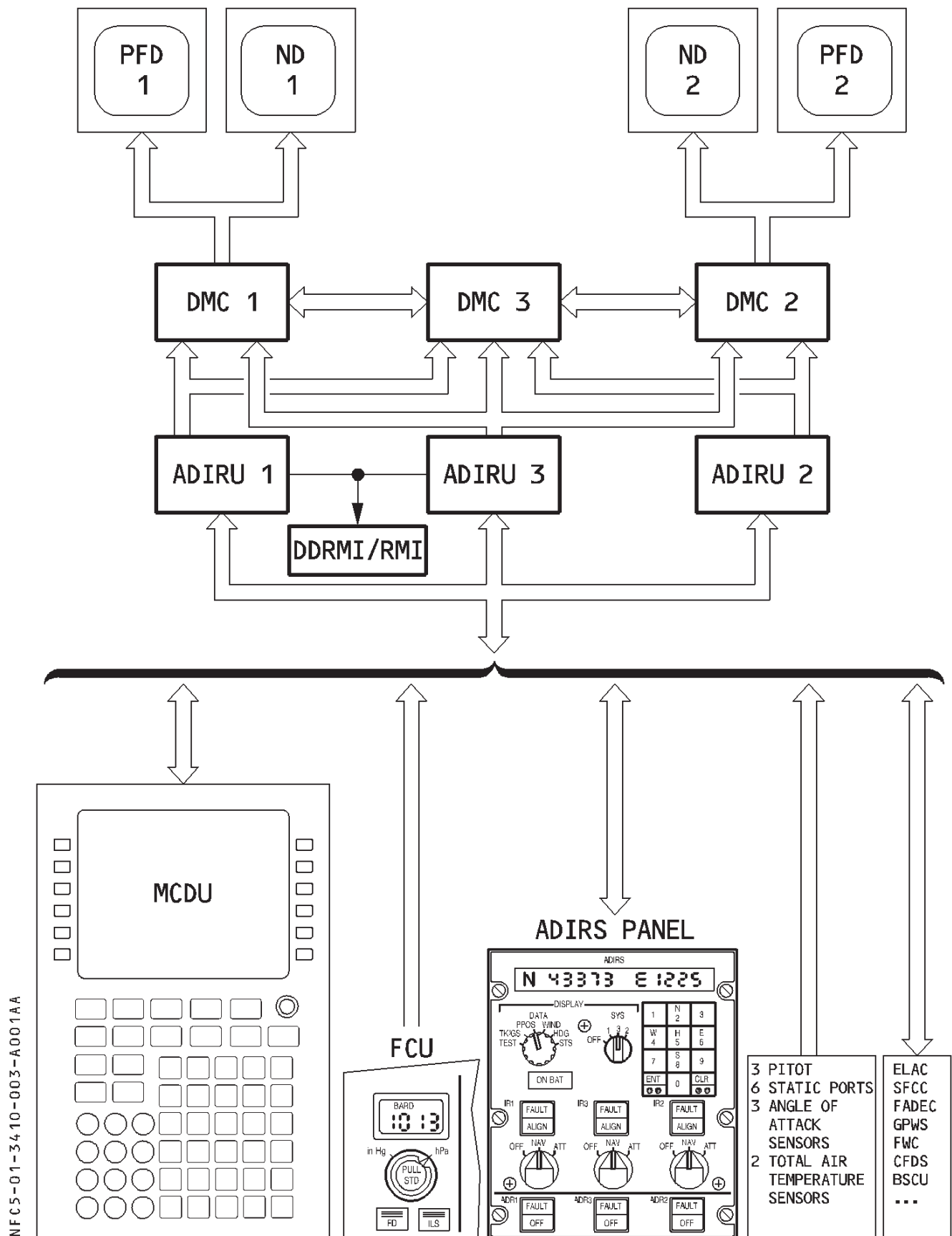
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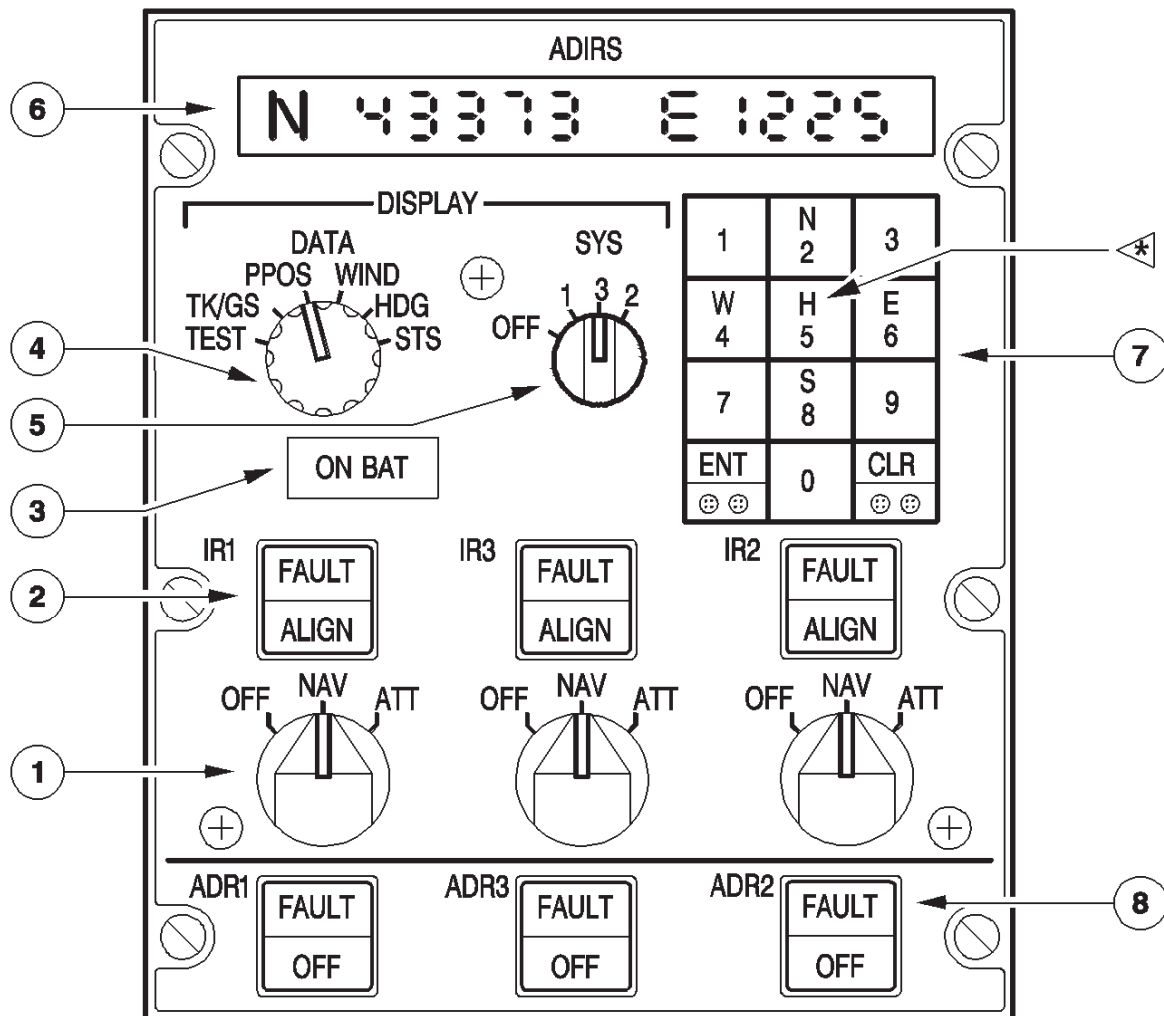
## ADIRS SCHEMATIC



## CONTROLS AND INDICATORS

### OVERHEAD PANEL

- R The ADIRS CDU on the overhead panel provides the controls and indicators to permit :
- selection of power supplies to the ADR and IR systems
  - selection and display of navigation data
  - manual initialization (normally performed through the FMGC)
  - status and fault indication of IRs or ADRs.

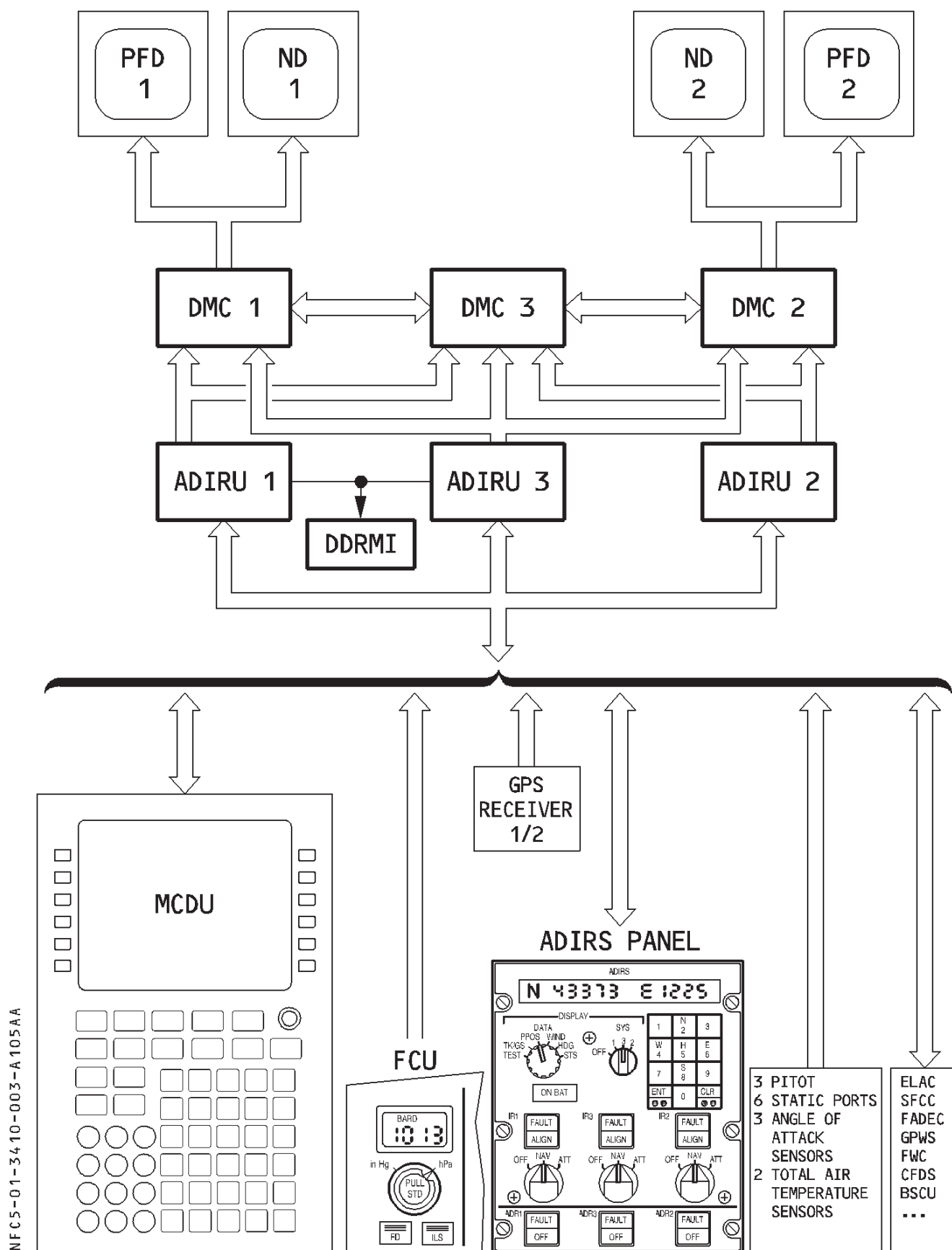


NFC5-01-3410-004-A001AA

#### ① IR 1(2)(3) Mode rotary sel :

- OFF : The ADIRU is not energized. ADR and IR data are not available.  
 NAV : Normal mode of operation. Supplies full inertial data to aircraft systems.  
 ATT : IR mode supplying only attitude and heading information if the system loses its ability to navigate.  
 The heading must be entered through the CDU keyboard and has to be reset frequently (about every 10 minutes)

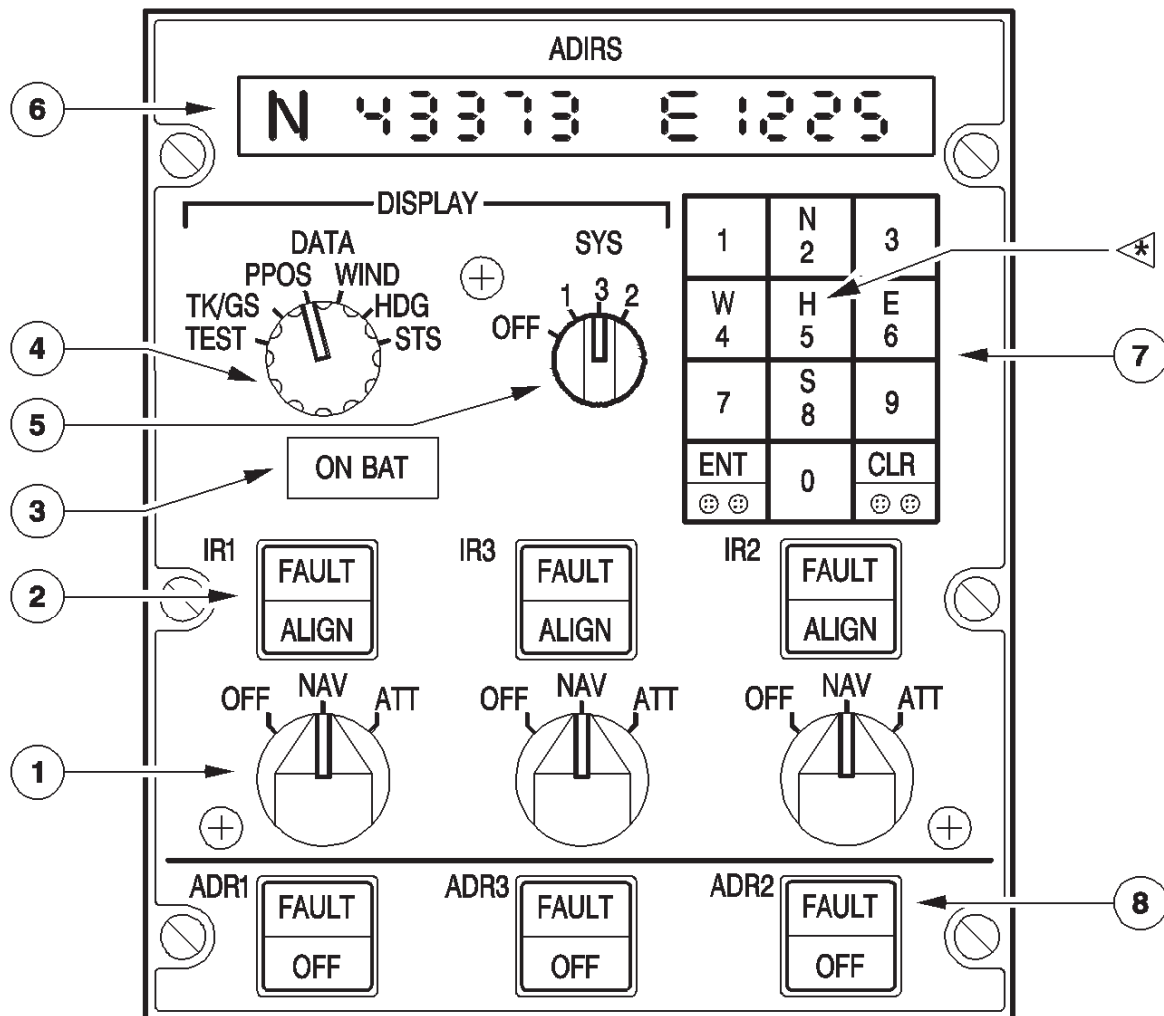
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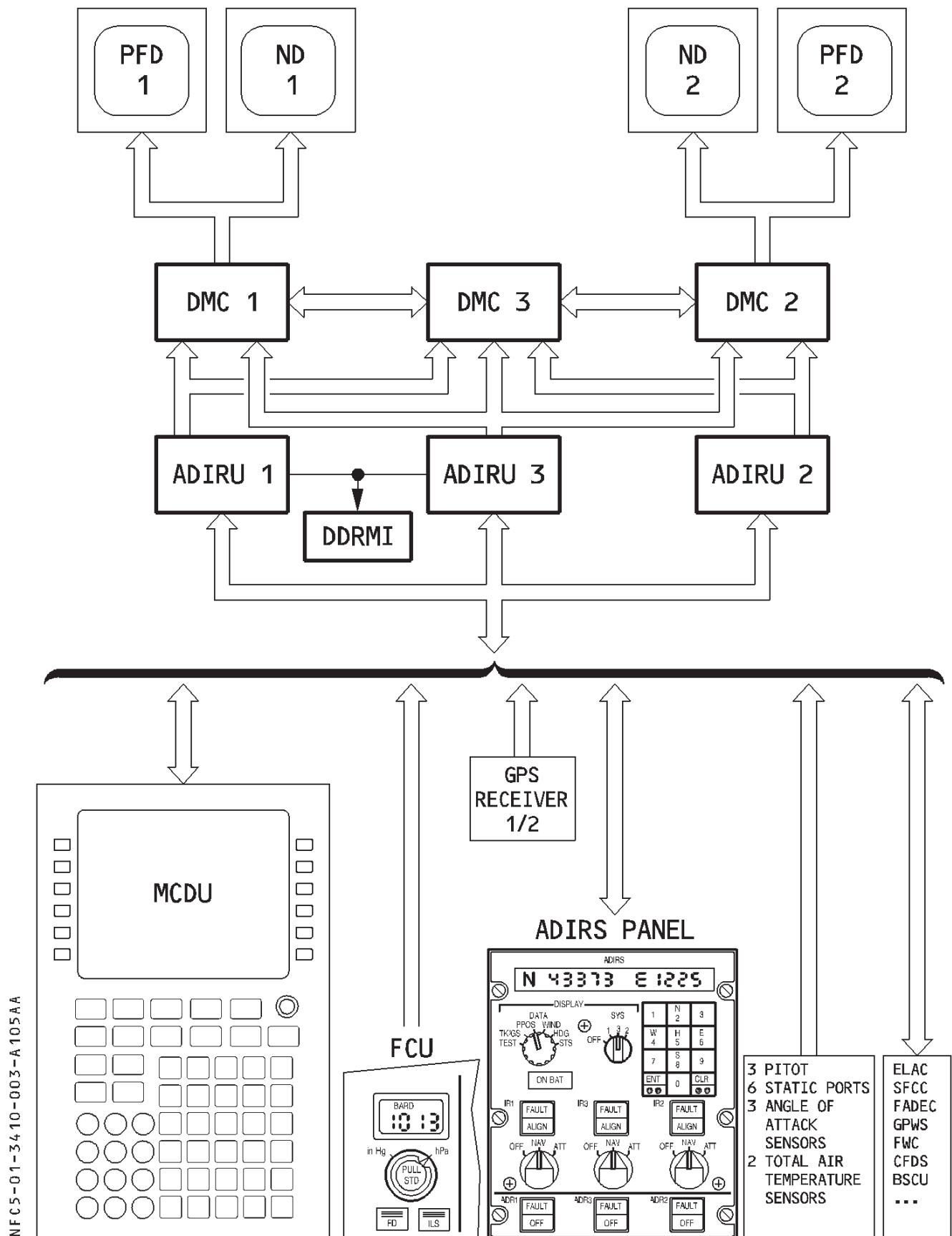


NFC5-01-3410-004-A001AA

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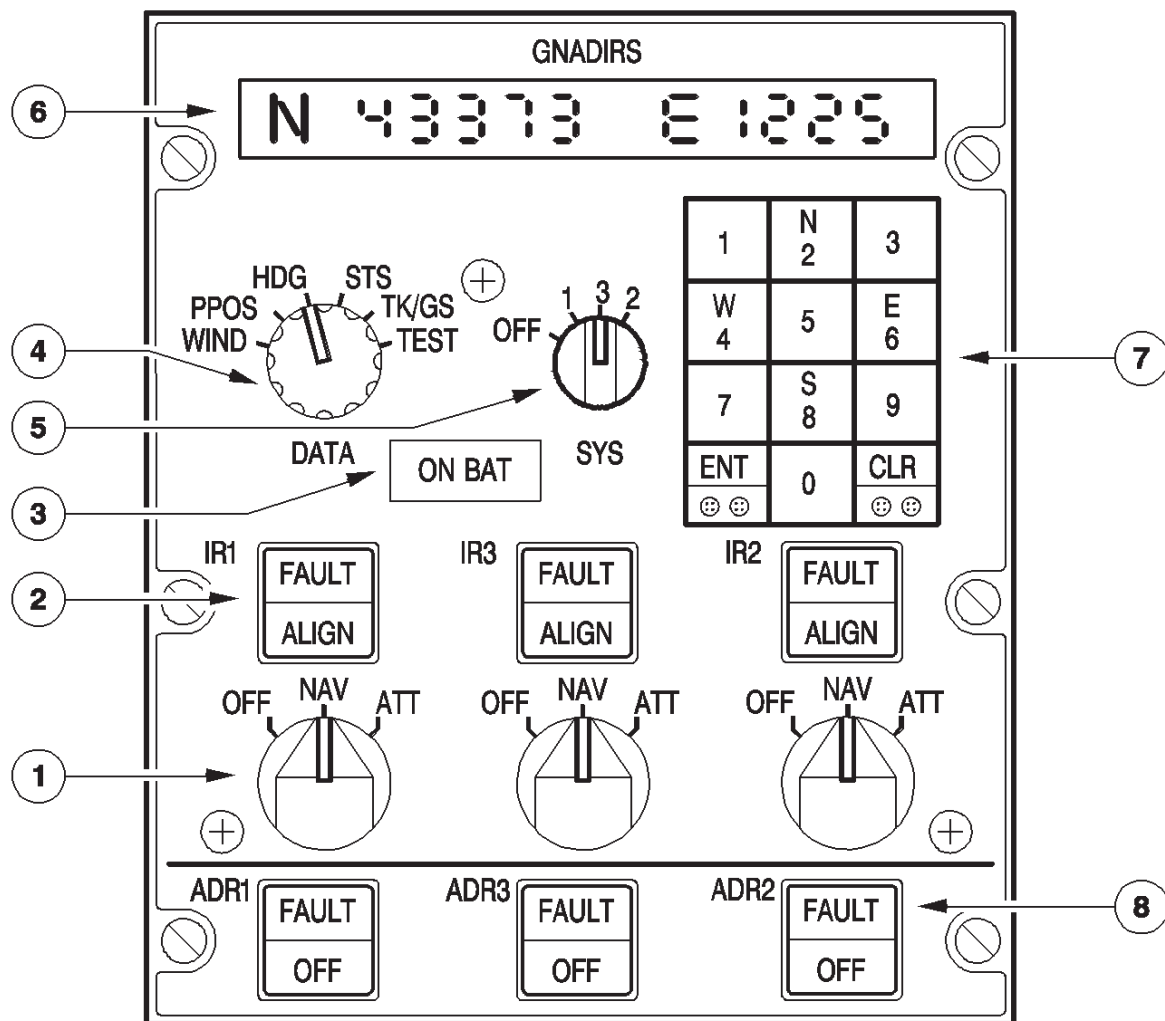
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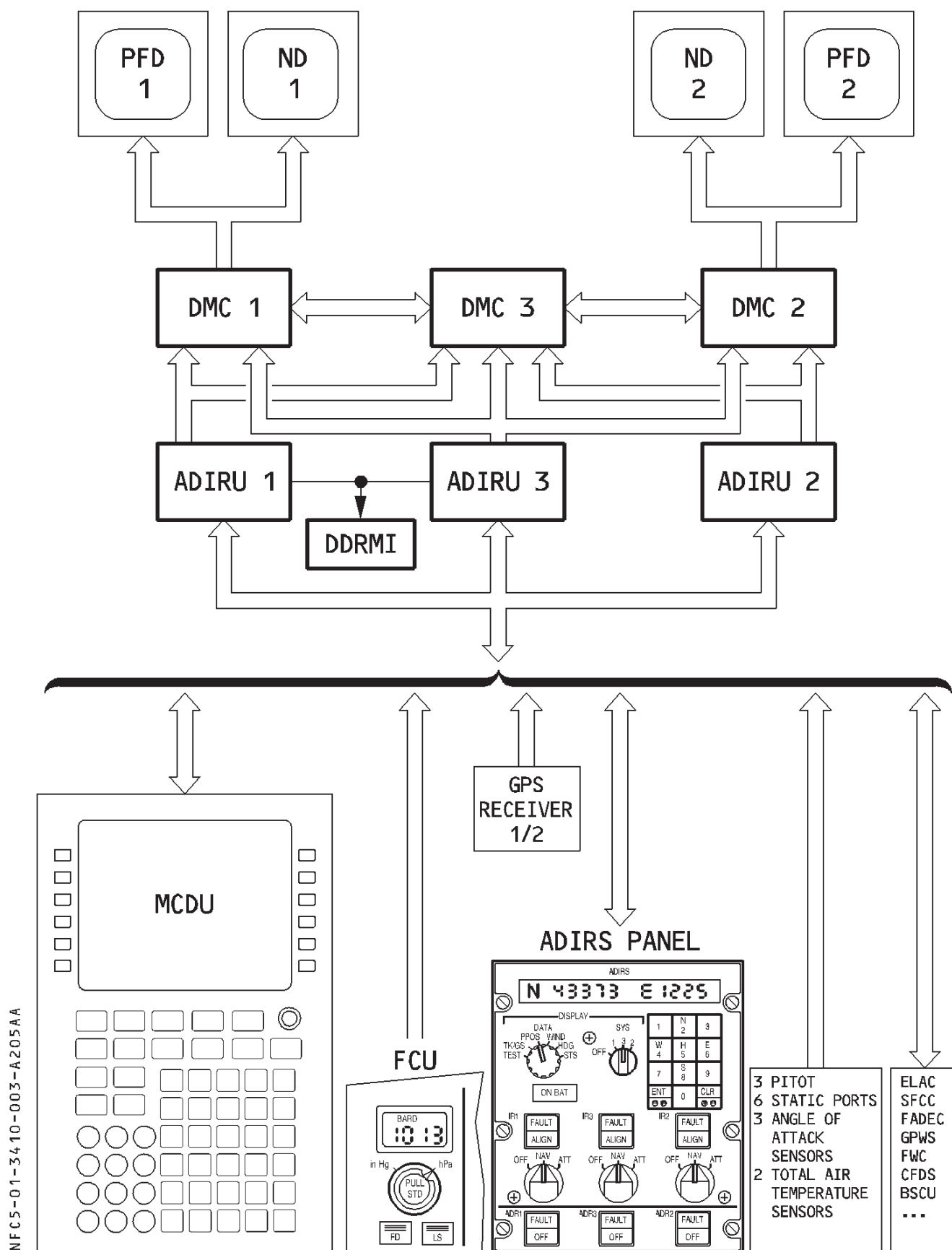
**NFC5-01-3410-004-A100A**

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ATT : IR mode supplies only attitude and heading information, if the system loses its ability to navigate.

The heading must be entered through the CDU keyboard and has to be reset frequently (about every 10 minutes)

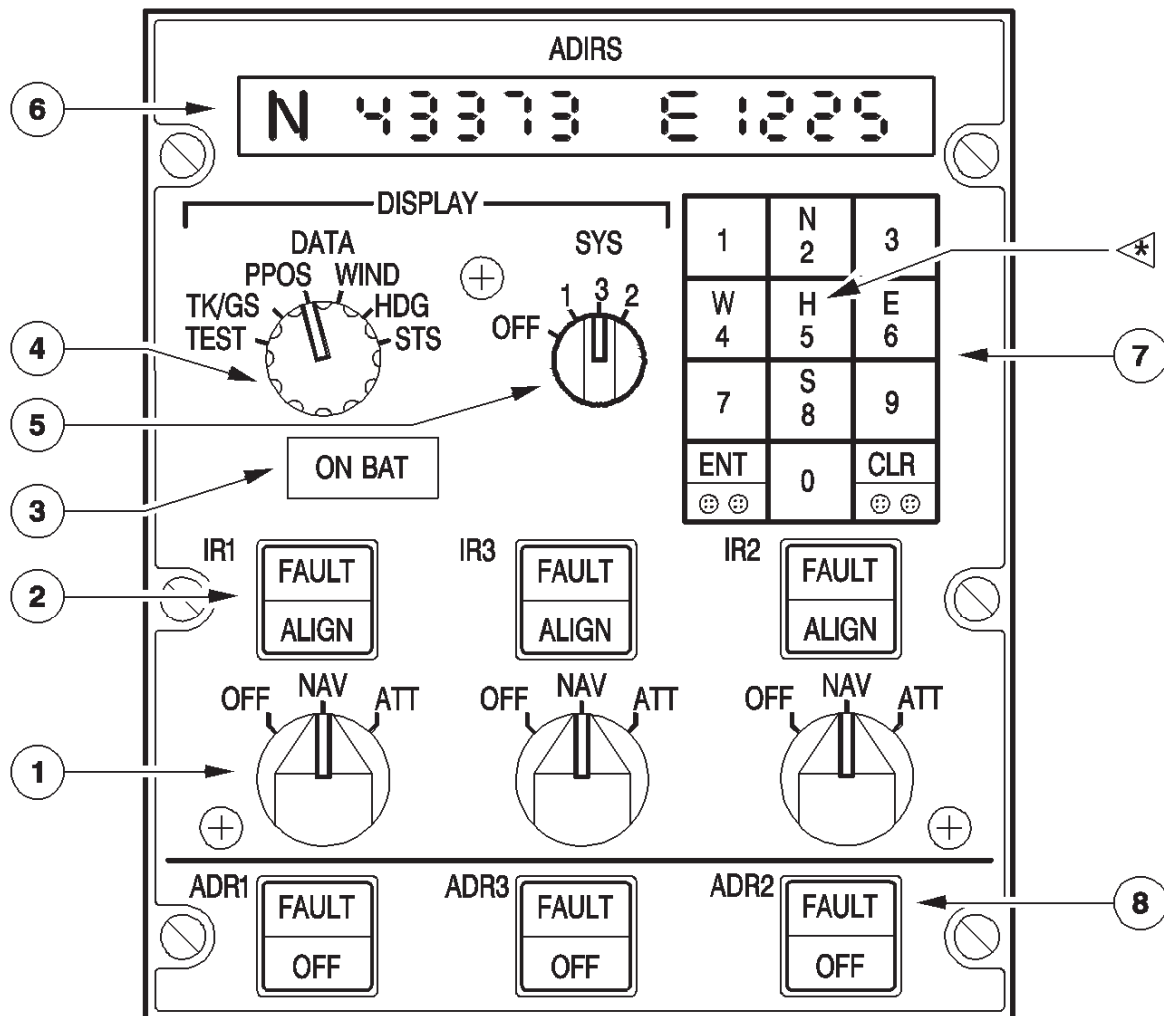
## ADIRS SCHEMATIC



## CONTROLS AND INDICATORS

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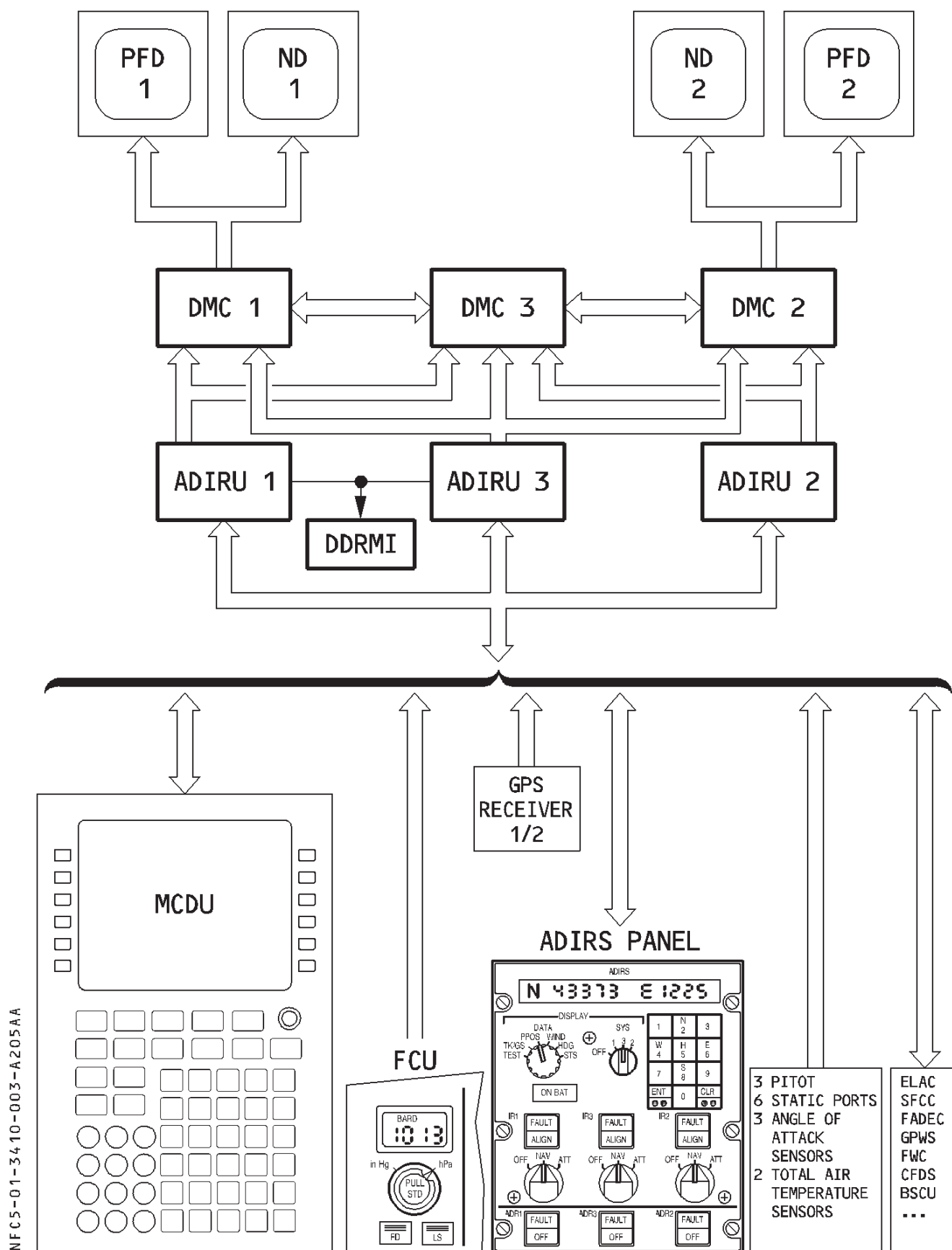


NFC5-01-3410-004-A001AA

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## ADIRS SCHEMATIC

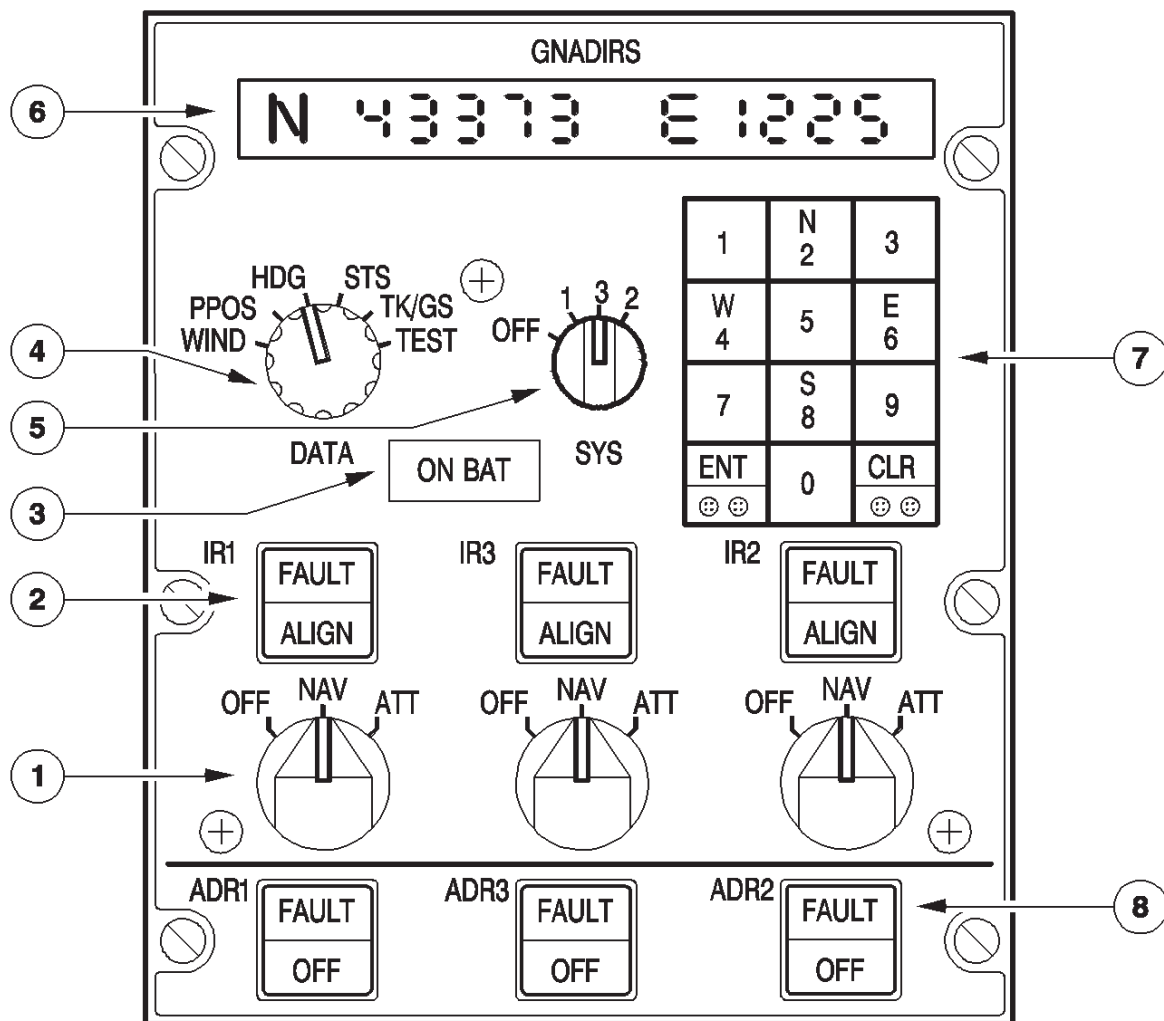


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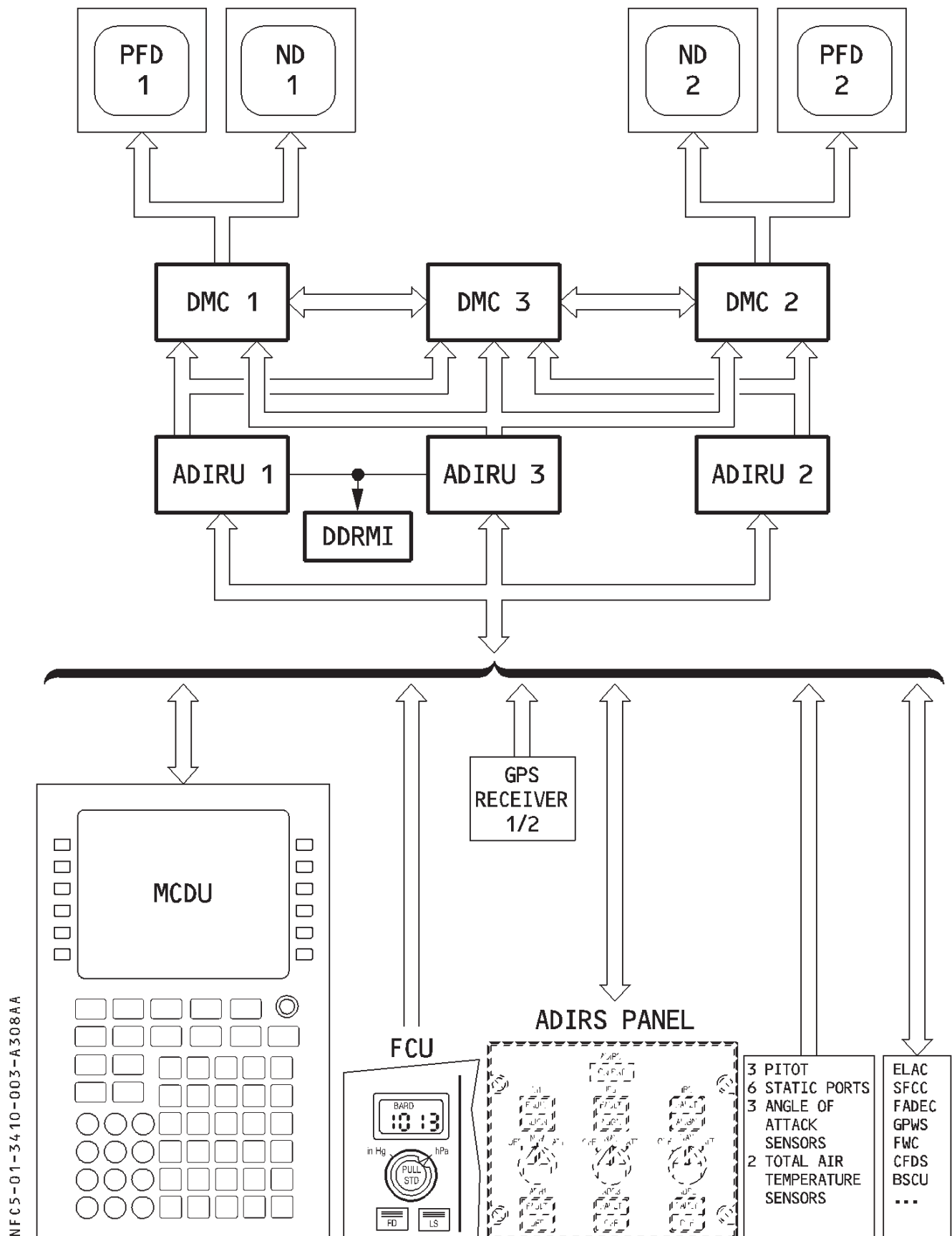


NFC5-01-3410-004-A100AA

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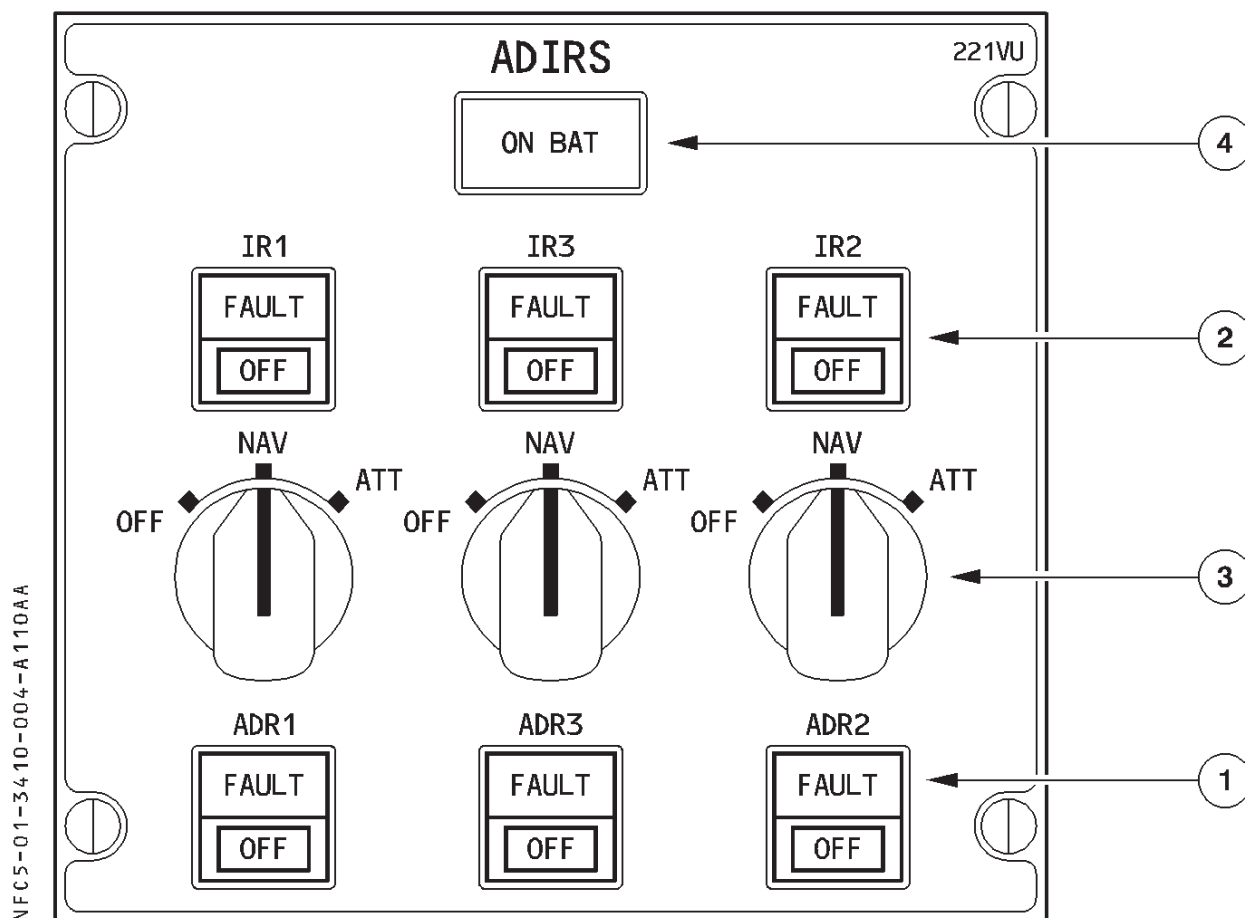
## ADIRS SCHEMATIC



NFC5-01-3410-003-A308AA

## CONTROLS AND INDICATORS

### OVERHEAD PANEL



#### ① ADR 1 (2) (3) pushbutton

OFF : Air data output disconnected.

FAULT light : This amber light comes on with an ECAM caution if a fault is detected in the air data reference part.

#### ② IR 1 (2) (3) pushbutton

OFF : Inertial data output disconnected.

FAULT light : This amber light comes on with an ECAM caution when a fault affects the respective IR.

Steady : the respective IR is lost.

Flashing : the attitude and heading information may be recovered in ATT mode.

### ② IR 1 (2) (3) It

- FAULT** : Comes on amber associated with an ECAM caution when a fault affects the respective IR.
- Steady : the respective IR is lost.
  - Flashing : the attitude and heading information may be recovered in ATT mode.
- ALIGN** : Steady : the respective IR is operating normally in align mode.
- Flashing if : IR alignment fault, or no present position entry after 10 min, or difference between position at shutdown and entered position exceeds 1° of latitude or longitude.
  - Extinguished : Alignment has been completed.

### ③ ON BAT It

Comes on amber when one or more IRs is supplied only by the a/c battery. It also comes on for a few sec at the beginning of the alignment, but not for a fast realignment.

Note : *If, when the aircraft is on the ground, at least one ADIRU is supplied by aircraft batteries :*

- an external horn sounds
- the ADIRU and AVNCS light comes on amber on the EXTERNAL POWER panel.

R

### ④ DATA selector knob

This knob selects the information to be displayed in the ADIRS display window.

- TEST** The ENT and CLR buttons on the keyboard come on, and the display shows all 8's.
- TK/GS** The display shows true track and ground speed.
- PPOS** The display shows present latitude and longitude
- WIND** The display shows true wind direction and speed.
- HDG** The display shows true heading and the minutes remaining until alignment is completed.
- STS** The display shows an action code.

### ⑤ SYS selector knob

- OFF** : The CDU display is not energized. ADIRS are still energized if the associated IR mode rotary selectors are not at OFF.
- 1.2.3** : System selected for data display.

## ⑥ Display

The display presents the data selected by the DATA selector.  
A keyboard entry overrides the selected display.

## ⑦ Keyboard

The flight crew can use the keyboard to enter the present position, or the heading in ATT mode, into the selected system.

Letter keys : Used to enter N, S, E, or W for position, or entering H (<A>) for heading (ATT mode).

Number keys : Used to enter the present position (or the present magnetic heading in ATT mode).

CLR key : The integral cue light comes on after an entry operation, if the data has an unreasonable value.  
Pressing this key clears the data display, that has been keyed in but not yet entered.

ENT Key : The integral cue light comes on when a crewmember has keyed in a number for N, S, W, E or H (<A>).  
Pressing the key enters data into the ADIRS.

## R ⑧ ADR 1 (2) (3) pb (momentary action)

OFF Air data output disconnected.

FAULT This amber light comes on with an ECAM caution, if a fault is detected in the air data reference part.

③ IR 1 (2) (3) mode rotary selector

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ADR and IR data are not available.

NAV : Normal mode of operation.

Supplies full inertial data to aircraft systems.

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The heading must be entered through the MCDU and has to be reset frequently (about every 10 minutes).

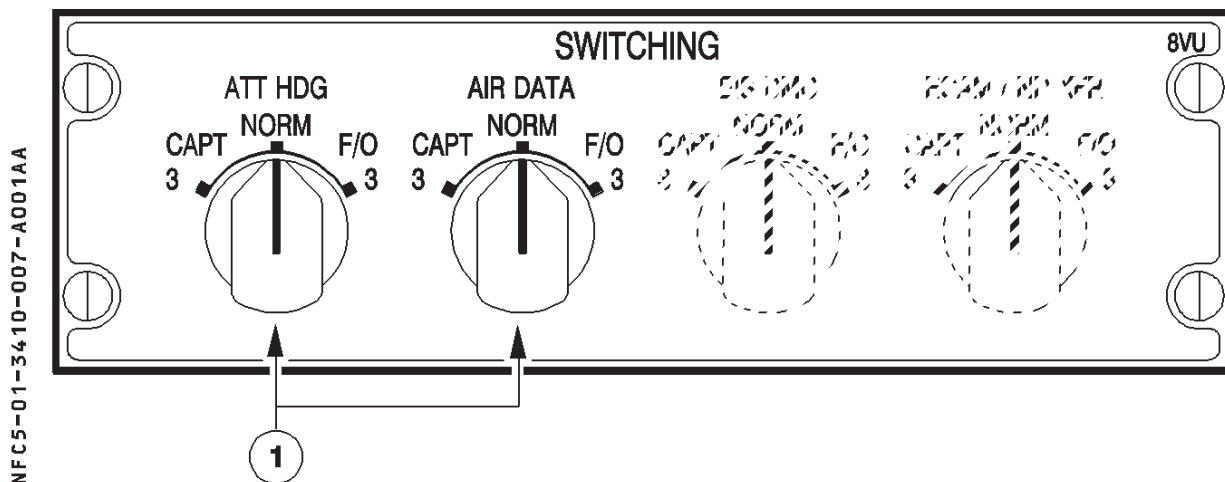
④ ON BAT light

The ON BAT light come on in amber when the aircraft battery supplies at least one IRS. The ON BAT light also comes on for a few seconds at the beginning of a complete IRS alignment.

Its does not come on in the case of a fast alignment.

**INTENTIONALLY LEFT BLANK**

## PEDESTAL



### ① ATT HDG and AIR DATA sel

**NORM** : ADIRU 1 supplies data to PFD 1, ND 1, RMI and VOR/DME.

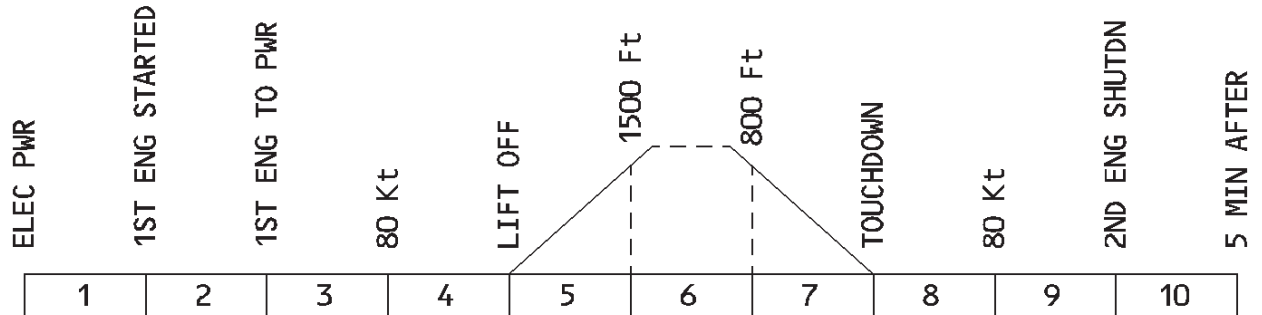
ADIRU 2 supplies data to PFD 2, and ND2.

**CAPT 3** : ADR 3 or IR 3 replaces ADR 1 or IR 1.

**F/O 3** : ADR 3 or IR 3 replaces ADR 2 or IR 2.

# WARNINGS AND CAUTIONS

NFC5-01-3410-008-A001AA



R

| E / WD: FAILURE TITLE<br>conditions                                                                                                                                                                                                                                          | AURAL<br>WARNING                     | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING | FLT<br>PHASE<br>INHIB |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------|----------------------|------------------|-----------------------|
| STALL WARNING (No ECAM message)<br>An aural stall warning is triggered when the AOA is greater than a predetermined angle<br>This angle depends on the :<br>– Slats/Flap position<br>– Speed/Mach<br>– F/CTL law (normal, alternate/direct)                                  | CRICKET + STALL<br>(SYNTHETIC VOICE) | MASTER<br>WARN  | NIL                  | NIL              | GND                   |
| OVERSPEED<br>– VMO/MMO<br>Aircraft speed/mach greater than<br>VMO + 4 kt/MMO + 0.006<br>– VLE<br>Aircraft speed greater than VLE + 4 kt, with L/G not uplocked, or L/G doors not closed<br>– VFE<br>Aircraft speed greater than VFE + 4 kt, with slats or/and flaps extended | CRC                                  |                 |                      |                  | 2, 3, 4<br>8, 9,10    |
| ADR 1 (2) FAULT                                                                                                                                                                                                                                                              | SINGLE<br>CHIME                      | MASTER<br>CAUT  |                      | ADR<br>FAULT It  | 1, 4,8,10             |
| ADR 3 FAULT                                                                                                                                                                                                                                                                  |                                      |                 |                      |                  | 1,4,5<br>7,8,10       |
| ADR (1+2) (1+3) (2+3) FAULT                                                                                                                                                                                                                                                  |                                      |                 |                      |                  | 1,4,8,10              |
| IR 1 (2) FAULT                                                                                                                                                                                                                                                               |                                      |                 |                      | IR<br>FAULT It   | 4,5,7,8               |
| IR 3 FAULT                                                                                                                                                                                                                                                                   |                                      |                 |                      |                  | 4,5<br>7,8            |
| IR (1 + 3) (2+3) FAULT                                                                                                                                                                                                                                                       |                                      |                 |                      |                  | 4, 8                  |
| IR 1 + 2 FAULT                                                                                                                                                                                                                                                               |                                      |                 |                      |                  |                       |

| E / WD: FAILURE TITLE<br>conditions                                                                                                                                             | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING                   | FLT<br>PHASE<br>INHIB |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|------------------------------------|-----------------------|
| HDG DISCREPANCY<br>difference between heading on CAPT and F/O displays<br>greater than 5°                                                                                       | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | CHECK<br>HDG<br>(on PFD<br>and ND) | 4,8                   |
| ATT DISCREPANCY<br>difference between roll or pitch angle displayed on<br>CAPT and F/O PFD greater than 5°                                                                      |                  |                 |                      | CHECK<br>ATT<br>(on PFD)           | 3,4,8                 |
| ALTI DISCREPANCY<br>difference between altitude displayed on CAPT and F/O<br>PFD greater than :<br>– 500 ft if baro ref STD is selected<br>– 250 ft if QNH (or QFE) is selected |                  |                 |                      | CHECK<br>ALT<br>(on PFD)           |                       |

### MEMO DISPLAY

- R · This displays shows IRS IN ALIGN X MN during phase 1 or 2 if :
  - R – at least one active IRS is in ALIGN submode
  - R – the time remaining until NAV mode is obtained, is X minutes ( $1 < x < 10$ ).
- R · IRS IN ALIGN appears if one of the 3 IRS is still in alignment
- R These 2 messages are displayed :
  - R – in green if both engines are stopped
  - R – in amber if one engine is running.

| E / WD: FAILURE TITLE<br>conditions                                                                                                                                                        | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING                   | FLT<br>PHASE<br>INHIB |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|------------------------------------|-----------------------|
| HDG DISCREPANCY<br>Difference between the heading on CAPT and F/O displays is greater than 5°.                                                                                             | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | CHECK<br>HDG<br>(on PFD<br>and ND) | 4, 8                  |
| ATT DISCREPANCY<br>Difference between roll or pitch angle, displayed on the CAPT and F/O PFD is greater than 5°.                                                                           |                  |                 |                      | CHECK<br>ATT<br>(on PFD)           | 3, 4, 8               |
| ALTI DISCREPANCY<br>Difference between the altitude displayed on the CAPT and F/O PFD is greater than :<br>– 500 ft, if baro ref STD is selected.<br>– 250 ft if QNH (or QFE) is selected. |                  |                 |                      | CHECK<br>ALT<br>(on PFD)           |                       |
| ADR DISAGREE<br>One ADR is faulty or has been rejected by the ELAC and there is a speed or angle-of-attack disagree between the two remaining ADR.                                         |                  |                 |                      | FAULT Its<br>on ELAC<br>pbs        | 3, 4, 5, 7            |
| IR DISAGREE<br>Disagree between two IR, with the third one failed.                                                                                                                         |                  |                 |                      | PFD<br>message                     |                       |

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| HDG DISCREPANCY<br>Difference between the heading on CAPT and F/O displays is greater than 5°.                                                                                             | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | CHECK<br>HDG<br>(on PFD<br>and ND) | 4, 8                  |
| ATT DISCREPANCY<br>Difference between roll or pitch angle, displayed on the CAPT and F/O PFD is greater than 5°.                                                                           |                  |                 |                      | CHECK<br>ATT<br>(on PFD)           | 3, 4, 8               |
| ALTI DISCREPANCY<br>Difference between the altitude displayed on the CAPT and F/O PFD is greater than :<br>– 500 ft, if baro ref STD is selected.<br>– 250 ft if QNH (or QFE) is selected. |                  |                 |                      | CHECK<br>ALT<br>(on PFD)           |                       |
| ADR DISAGREE<br>One ADR is faulty or has been rejected by the ELAC and there is a speed or angle-of-attack disagree between the two remaining ADR.                                         |                  |                 |                      | FAULT Its<br>on ELAC<br>pbs        | 3, 4, 5, 7            |
| IR DISAGREE<br>Disagree between two IR, with the third one failed.                                                                                                                         |                  |                 |                      | PFD<br>message                     |                       |
| IR NOT ALIGNED<br>Problem detected during IR alignment                                                                                                                                     |                  |                 |                      | NIL                                | 1                     |

## MEMO DISPLAY

- “IRS IN ALIGN X MN” memo appears during phase 1 or 2, during IRS alignment. X MN indicates the number of minutes remaining ( $1 < x < 10$ ), until NAV mode is reached.
  - Before any engine is started, “IRS IN ALIGN X MN” memo :
    - \* Appears in green when at least one active IRS is being aligned,
    - \* Pulses in green if the alignment of one IRS is faulty.
  - When one engine is started, “IRS IN ALIGN X MN” memo appears in amber during IRS alignment. If the alignment of one IRS is faulty, “IRS IN ALIGN X MN” memo is replaced by the “IR NOT ALIGNED” ECAM caution.
- “IR IN ATT ALIGN” appears in green during the IR alignment in Attitude mode.

| E / WD: FAILURE TITLE<br>conditions                                                                                                                                                               | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING                   | FLT<br>PHASE<br>INHIB |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|------------------------------------|-----------------------|
| HDG DISCREPANCY<br>Difference between the heading on the CAPT and F/O displays is greater than 5°.                                                                                                | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | CHECK<br>HDG<br>(on PFD<br>and ND) | 4, 8                  |
| ATT DISCREPANCY<br>Difference between the roll or pitch angle, displayed on the CAPT and F/O PFD, is greater than 5°.                                                                             |                  |                 |                      | CHECK<br>ATT<br>(on PFD)           | 3, 4, 8               |
| ALTI DISCREPANCY<br>Difference between the altitude, displayed on the CAPT and F/O PFD, is greater than :<br>– 500 feet, if baro ref STD is selected.<br>– 250 feet, if QNH (or QFE) is selected. |                  |                 |                      | CHECK<br>ALT<br>(on PFD)           |                       |
| IAS DISCREPANCY<br>Caution activated when there is a discrepancy between the speeds displayed on the PFDs.                                                                                        |                  |                 |                      | NIL                                | 4, 5, 7, 8            |
| ADR DISAGREE<br>One ADR is faulty, or has been rejected by the ELAC, and there is a speed or angle-of-attack disagree between the two remaining ADRs.                                             |                  |                 |                      | FAULT Its<br>on ELAC<br>pbs        | 3, 4, 5, 7            |
| IR DISAGREE<br>Disagree between two IRs, with the third one failed.                                                                                                                               |                  |                 |                      | PFD<br>message                     |                       |

## MEMO DISPLAY

- This displays “IRS IN ALIGN X MN” during Phase 1 or 2, if :
  - At least one active IRS is in ALIGN submode.
  - The time remaining until NAV mode is obtained, is X minutes ( $1 < x < 10$ ).
- “IRS IN ALIGN” appears, if one of the 3 IRS is still in alignment.

These two messages are displayed :

- In green, if both engines are stopped.
- In amber, if one engine is running.

| E / WD: FAILURE TITLE<br>conditions                                                                                                                                                        | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING                   | FLT<br>PHASE<br>INHIB |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|------------------------------------|-----------------------|
| HDG DISCREPANCY<br>Difference between the heading on CAPT and F/O displays is greater than 5°.                                                                                             | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | CHECK<br>HDG<br>(on PFD<br>and ND) | 4, 8                  |
| ATT DISCREPANCY<br>Difference between roll or pitch angle, displayed on the CAPT and F/O PFD is greater than 5°.                                                                           |                  |                 |                      | CHECK<br>ATT<br>(on PFD)           | 3, 4, 8               |
| ALTI DISCREPANCY<br>Difference between the altitude displayed on the CAPT and F/O PFD is greater than :<br>– 500 ft, if baro ref STD is selected.<br>– 250 ft if QNH (or QFE) is selected. |                  |                 |                      | CHECK<br>ALT<br>(on PFD)           |                       |
| IAS DISCREPANCY<br>Caution activated when there is a discrepancy between the speeds displayed on the PFDs.                                                                                 |                  |                 |                      | NIL                                | 4, 5, 7, 8            |
| ADR DISAGREE<br>One ADR is faulty or has been rejected by the ELAC and there is a speed or angle-of-attack disagree between the two remaining ADR.                                         |                  |                 |                      | FAULT Its<br>on ELAC<br>pbs        | 3, 4, 5, 7            |
| IR DISAGREE<br>Disagree between two IR, with the third one failed.                                                                                                                         |                  |                 |                      | PFD<br>message                     |                       |
| IR NOT ALIGNED<br>Problem detected during IR alignment                                                                                                                                     |                  |                 |                      | NIL                                | 1                     |

## MEMO DISPLAY

- “IRS IN ALIGN X MN” memo appears during phase 1 or 2, during IRS alignment. X MN indicates the number of minutes remaining ( $1 < x < 10$ ), until NAV mode is reached.
  - Before any engine is started, “IRS IN ALIGN X MN” memo :
    - \* Appears in green when at least one active IRS is being aligned,
    - \* Pulses in green if the alignment of one IRS is faulty.
  - When one engine is started, “IRS IN ALIGN X MN” memo appears in amber during IRS alignment. If the alignment of one IRS is faulty, “IRS IN ALIGN X MN” memo is repalced by the “IR NOT ALIGNED” ECAM caution.
- “IR IN ATT ALIGN” appears in green during the IR alignment in Attitude mode.

## DESCRIPTION

The Global Positioning System (GPS) is a satellite based radio navigation aid.

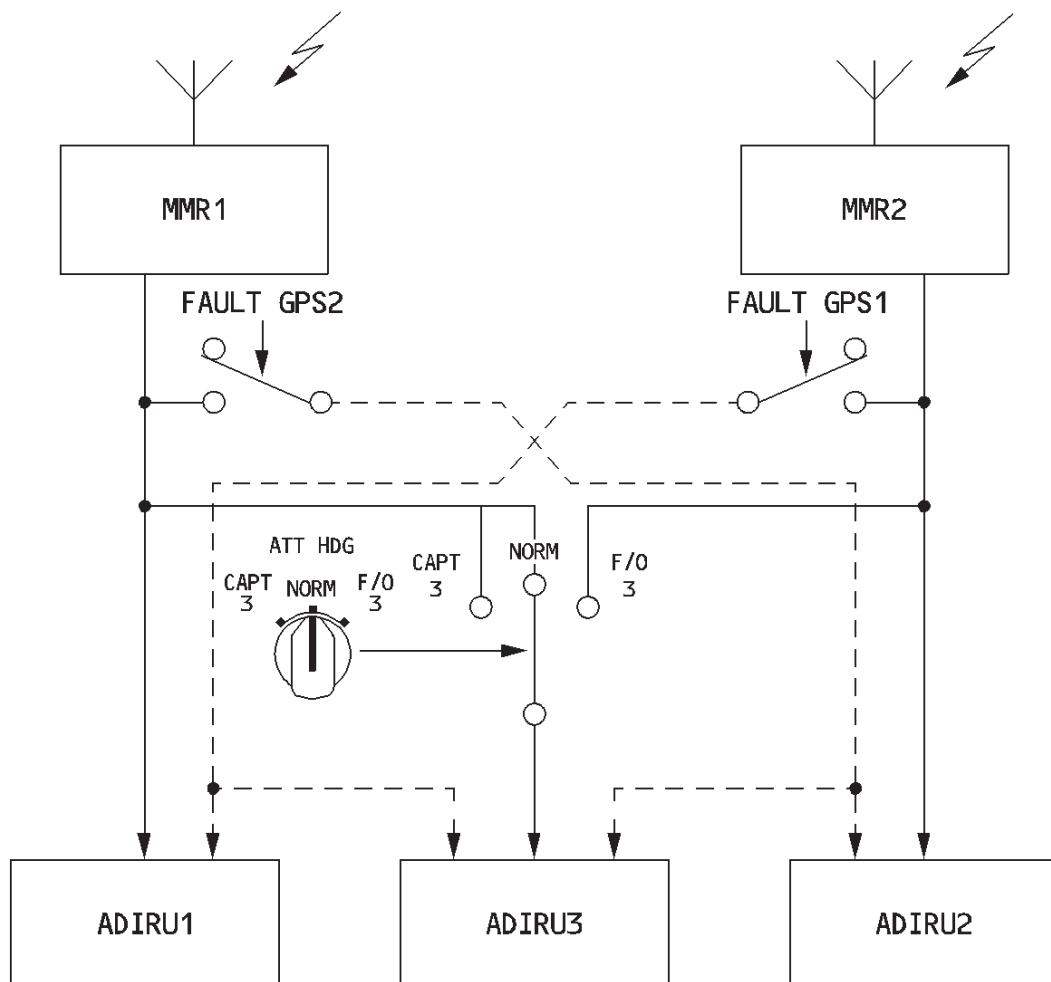
Worldwide 24 satellites broadcast accurate navigation data that aircraft can use for the precise determination of its position.

The aircraft has two independent GPS receivers. Each GPS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (GPS 1 receiver in MMR1, GPS2 receiver in MMR2).

The MMR processes the data received and transfers them to the ADIRUs, which then perform a GP-IRS hybrid position calculation. The FMGCs use the hybrid position. The GPS MONITOR page on MCDU1 or MCDU2 can display pure GPS position, true track, ground speed, estimated position, accuracy level, and mode of operation for the information and use of the flight crew.

*Note : Flight crew can use the MCDU NAVAID page to deselect the use of GPS data for calculating position (refer to FCOM 4.03).*

R



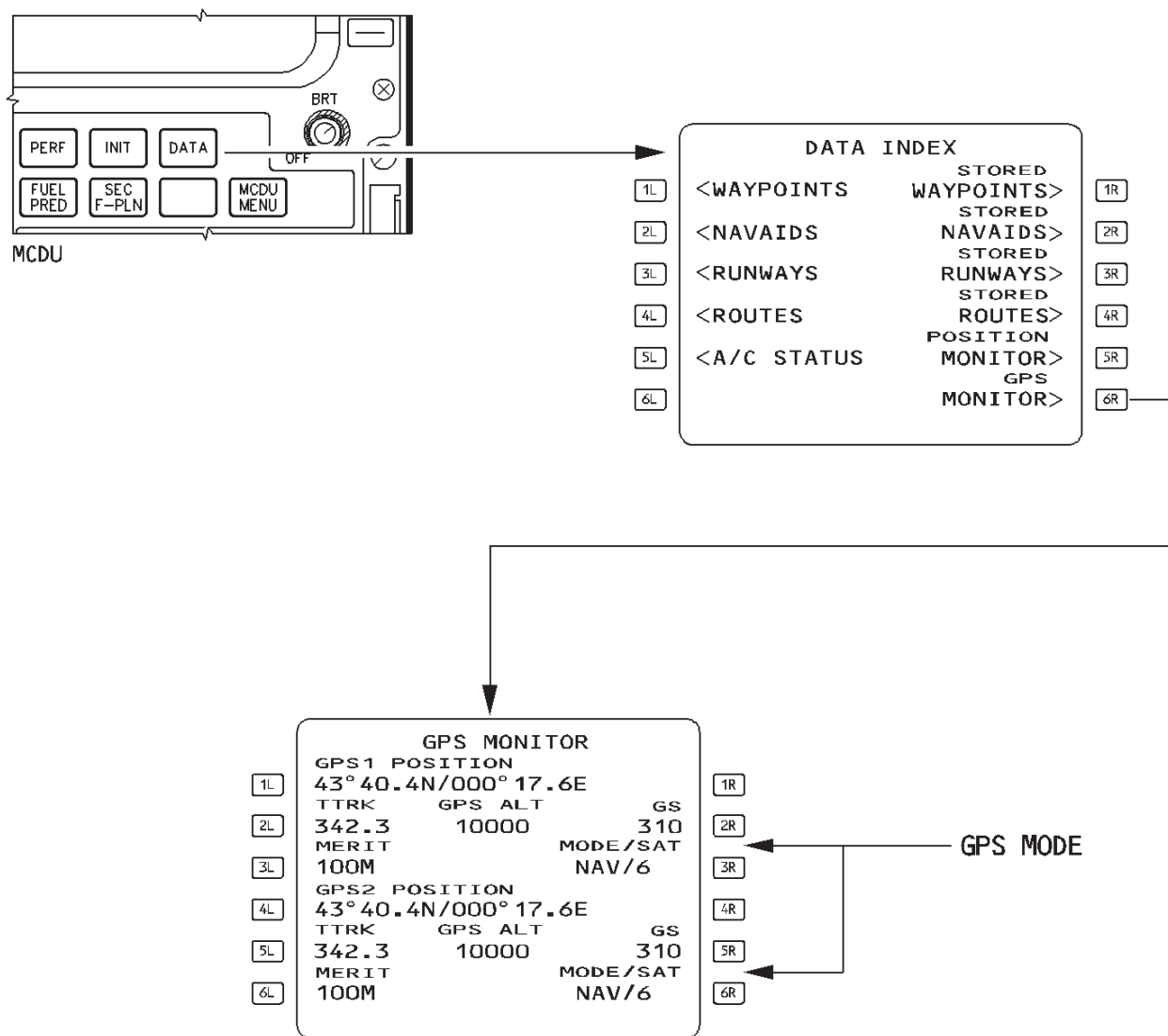
NFC5-01-3415-001-A110AA

## NORMAL OPERATION

In normal operation, the GPS receiver 1 supplies ADIRU 1 and ADIRU 3, the GPS receiver 2 supplies ADIRU 2.

The MMR operates in different modes which are indicated on the GPS MONITOR page :

R



NFC5-01-3415-002-A120AA

### R – Initialization mode (INIT)

R When this mode is entered, the GPS hardware and software are initialized.

### – Acquisition mode (ACQ)

The MMR enters in this mode after power-up or during long periods of lost satellite signal. It remains in this mode until it is able to track at least 4 satellites, then transfers to NAV mode. To enter navigation mode more quickly, MMR uses initial position, time and altitude from IRS.

– **Navigation Mode (NAV)**

When the MMR can track 4 or more satellites, it enters NAV mode and continuously supplies data to the ADIRUs.

– **Altitude Aiding (ALTAID)**

If the MMR can track at least 4 satellites, it uses the GPS altitude and the IR altitude to calculate an altitude bias.

If the number of satellites drops to three, the altitude bias is frozen and the MMR enters ALTAID mode, using the IR altitude (corrected with the altitude bias).

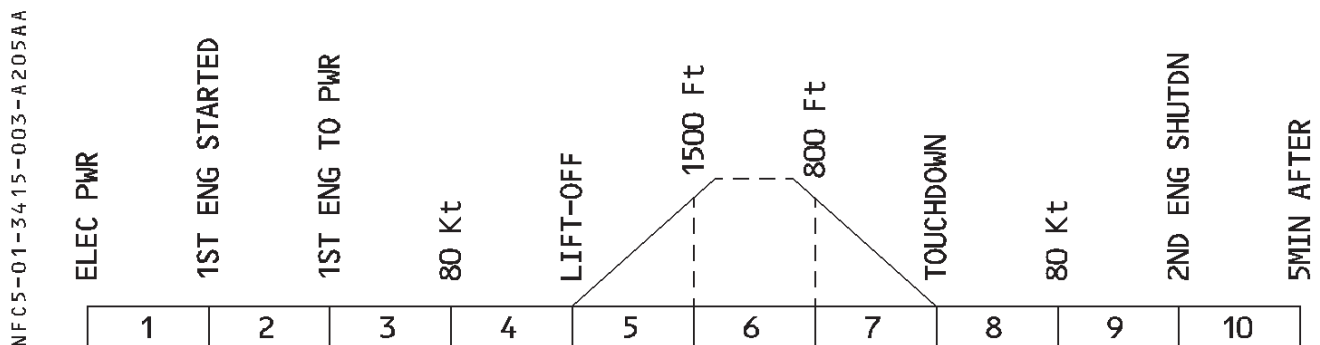
– **Fault Mode (FAULT)**

The fault mode is entered when a failure, which may prevent the MMR from transmitting valid data, has been detected.

### OPERATION IN CASE OF FAILURE

If one GPS receiver fails, the three ADIRUs automatically select the only operative GPS receiver. If ADIRU 1 fails, ADIRU 3 is supplied by MMR 1, and ADIRU 2 is supplied by MMR 2. To maintain Side 1 and Side 2 segregation, in case ADIRU 2 fails, the ATT HDG selector must be set to F/O 3, so that ADIRU 3 will be supplied with MMR 2 data. If two ADIRUs fail, the remaining ADIRU is supplied by its own side GPS receiver.

### WARNINGS AND CAUTIONS

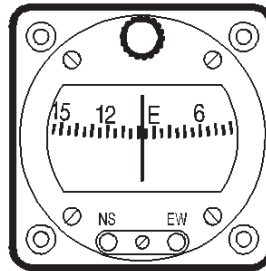


| E / WD: FAILURE TITLE<br>conditions | AURAL<br>WARNING                                         | MASTER<br>LIGHT   | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING | FLT<br>PHASE<br>INHIB |
|-------------------------------------|----------------------------------------------------------|-------------------|----------------------|------------------|-----------------------|
| GPS 1 (2) FAULT                     | SINGLE<br>CHIME                                          | MASTER<br>CAUTION | NIL                  | NIL              | 4, 5, 7,<br>8         |
| FM/GPS POS DISAGREE                 |                                                          |                   |                      |                  | 1, 3, 4,<br>10        |
| GPS PRIMARY LOST (No ECAM warning)  | TRIPLE<br>CLICK<br>During<br>non ILS<br>approach<br>only | NIL               |                      |                  | 2,3,4,5<br>8,9,10     |

## COMPASS

There is a compass located is on top of the windshield center post.  
The deviation card is located above the compass.

*Note : Because of the location of the APU power on contactor in the cockpit, the APU start sequence may disturb the compass reading.*

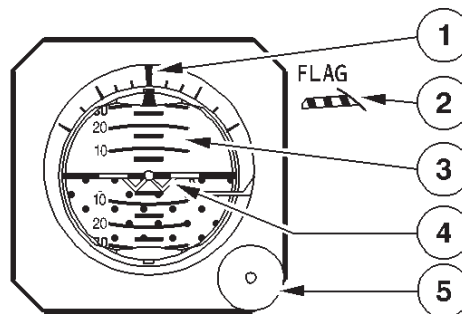


NFC5-01-3420-001-AD01AA

## HORIZON

The electric standby horizon normally obtains current from the DC ESS BUS. In the case of a total electrical failure, the horizon remains usable for 5 minutes.

*Note : When leveling the wings, after performing a small turn of a small bank angle, the displayed roll attitude may temporarily be incorrect by a few degrees.*



NFC5-01-3420-001-B001AA

### ① Roll scale

The roll scale indicates the bank angle. It has bank angle graduations up to 60°. There is no rotation limit.

### ② Flag

The flag appears if the instrument fails or if power supply fails.

### ③ Pitch scale

The pitch scale indicates the pitch attitude. It can show pitch angle up to  $\pm 85^\circ$ .

#### ④ Aircraft reference

Fixed symbol.

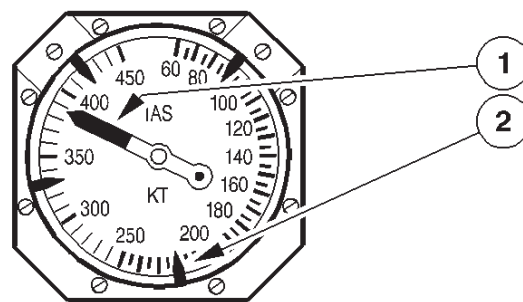
#### ⑤ Caging knob

The flight crew pulls it out to reinitialize the gyro, and to level and center the horizon. (The airplane should be level during this procedure).

**R** Note : After low-rate turns, the standby horizon may not give accurate indications. To  
**R** correct this behavior, use the caging knob when the aircraft is level.

### AIRSPEED INDICATOR

NFC5-01-3420-002-A001AA



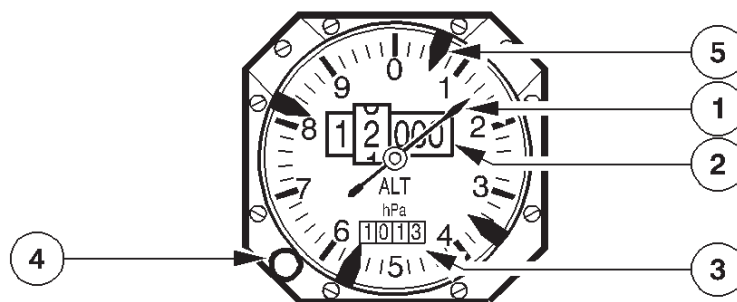
#### ① Airspeed pointer

#### ② Airspeed bugs (4)

For marking airspeed references.

### ALTIMETER

NFC5-01-3420-002-B001AA



#### ① Altitude pointer

#### ② Altitude counter (feet)

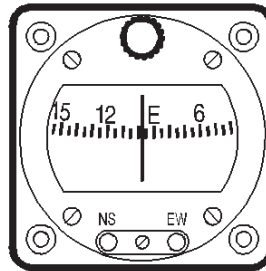
#### ③ Altimeter setting

Displays the pressure setting in hPa.

## COMPASS

There is a compass located on top of the windshield center post.  
The deviation card is located above the compass.

*Note : Because of the location of the APU power on contactor in the cockpit, the APU start sequence may disturb the compass reading.*

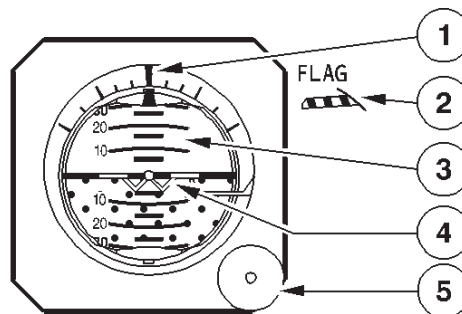


NFC5-01-3420-001-A100AA

## HORIZON

The electric standby horizon is powered by the DC ESS BUS. Backup power is supplied by the HOT BUS 1.

*Note : When leveling the wings, after performing a small turn of a small bank angle, the displayed roll attitude may temporarily be incorrect by a few degrees.*



NFC5-01-3420-001-B100AA

### ① Roll scale

The roll scale indicates the bank angle. It has bank angle graduations up to 60°. There is no rotation limit.

### ② Flag

The flag appears if the instrument fails or if power supply fails.

### ③ Pitch scale

The pitch scale indicates the pitch attitude. It can show pitch angle up to  $\pm 85^\circ$ .

#### ④ Aircraft reference

Fixed symbol.

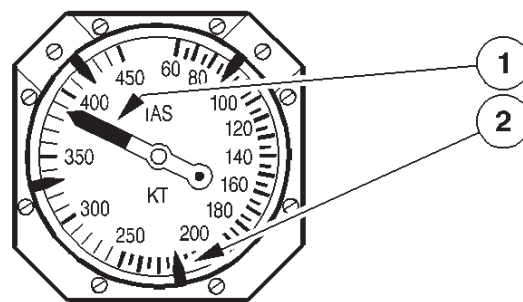
#### ⑤ Caging knob

The flight crew pulls it out to reinitialize the gyro, and to level and center the horizon. (The airplane should be level during this procedure).

**R** Note : After low-rate turns, the standby horizon may not give accurate indications. To  
**R** correct this behavior, use the caging knob when the aircraft is level.

### AIRSPEED INDICATOR

NFC5-01-3420-002-A001AA



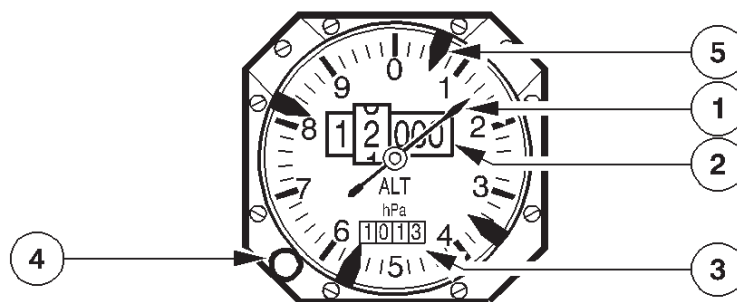
#### ① Airspeed pointer

#### ② Airspeed bugs (4)

For marking airspeed references.

### ALTIMETER

NFC5-01-3420-002-B001AA



#### ① Altitude pointer

#### ② Altitude counter (feet)

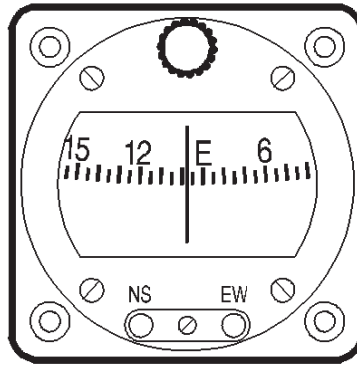
#### ③ Altimeter setting

Displays the pressure setting in hPa.

**COMPASS**

This instrument is on top of windshield center post.  
The deviation card is above it.

Note : *Because of the location of the APU power on contactor in the cockpit, the APU start sequence may disturb the compass reading.*



NFC5-01-3420-001-A104AA

**INTENTIONALLY LEFT BLANK**

④ Altimeter setting knob

⑤ Altitude bugs (4)

For marking of altitude references.

**ALTIMETER (in meter)**

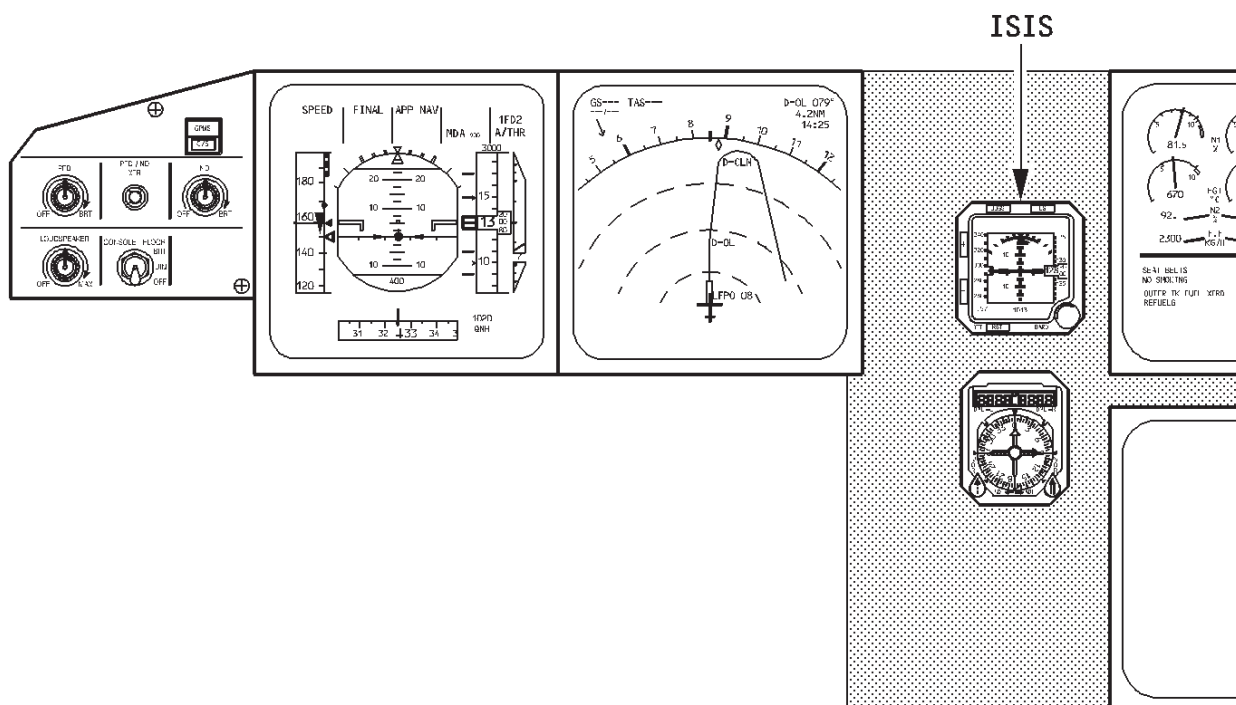
NOT APPLICABLE.

**ALTIMETER (in meter)**

**INTENTIONALLY LEFT BLANK**

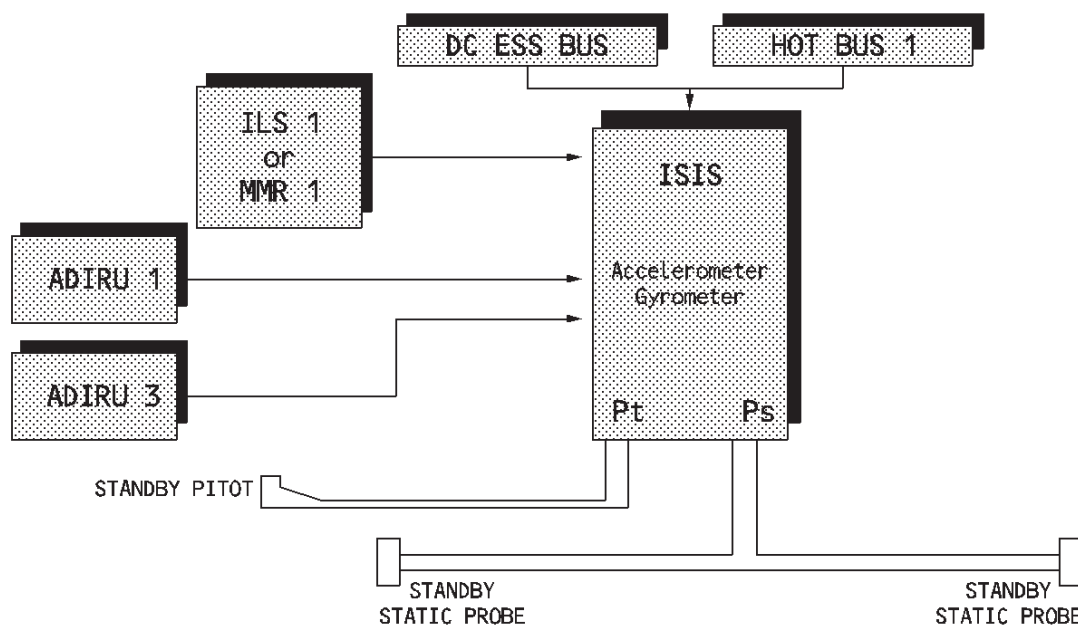
## INTRODUCTION

The Integrated Standby Instrument System (ISIS) provides a third source of information and display to the crew. It is mounted in the center of the instrument panel.



NFC5-01-3425-001-A100AA

## ARCHITECTURE



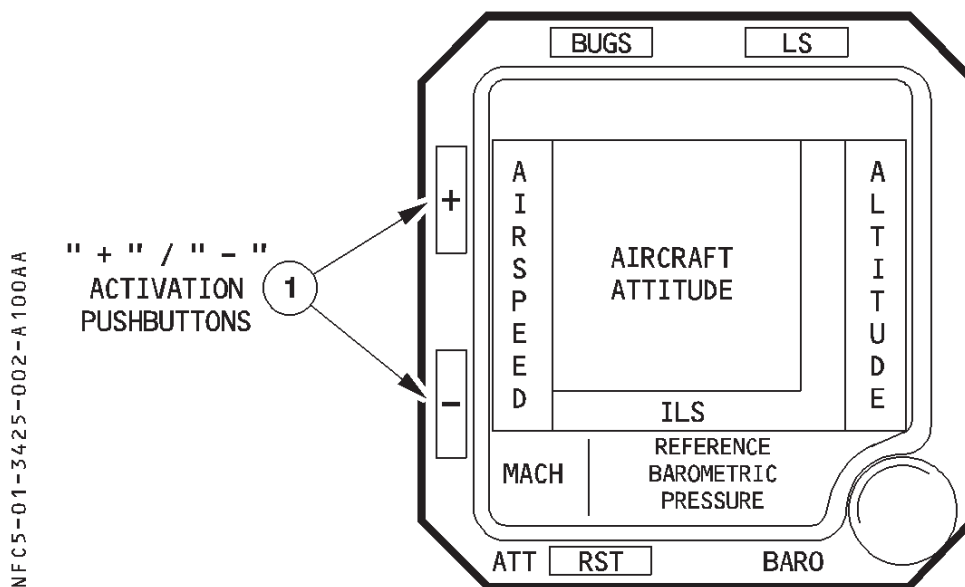
NFC5-01-3425-001-B100AA

## GENERAL

The ISIS system displays the following information :

- Attitude
- Airspeed and mach
- Altitude
- Barometric pressure
- LS function
- Bugs

R Note : When leveling the wings after performing a small turn of a small bank angle, the  
R displayed roll attitude may temporarily be incorrect by a few degrees.



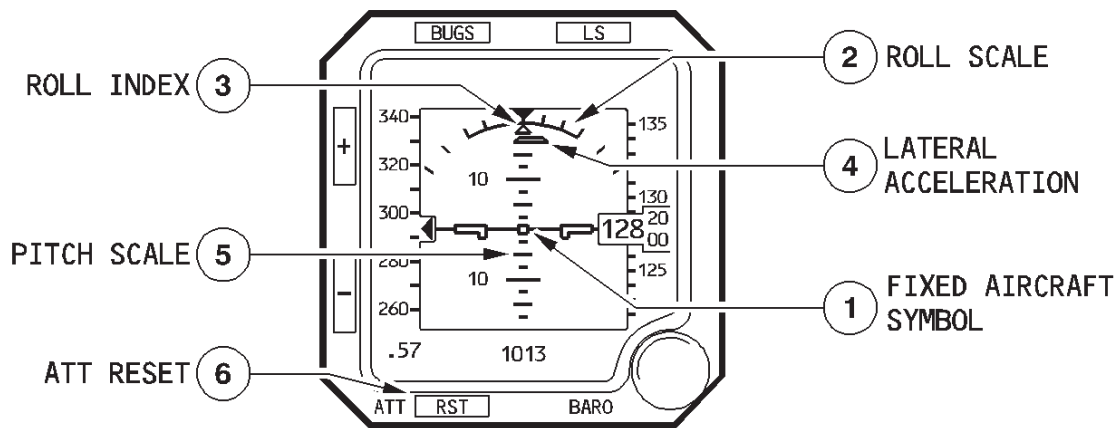
R ① "+" / "-" activation pushbuttons

R Two pushbuttons labelled "+" and "-" are used to adjust the level of brightness.  
R The brightness of the screen automatically adjusts after power-up tests.  
R This is because of the photosensitive cell located on the surface of the ISIS system  
R display.  
R The "+" and "-" pushbuttons then allow this initial brightness to be manually adjusted  
R and changed.

## ATTITUDE

R

NFC5-01-3425-003-A100AA



### ① Fixed aircraft symbol

The fixed aircraft symbol is in black, and outlined in yellow.

### ② Roll scale

R The roll scale is in white. The markers are at 0 (small yellow triangle), 10, 20, 30, 45, 60 degrees of bank.

### ③ Roll index

R The roll index is in black, and outlined in white. It indicates the bank angle.

### ④ Lateral acceleration index

The trapezoidal index moves beneath the roll index. It represents the aircraft's lateral acceleration.

### ⑤ Pitch scale

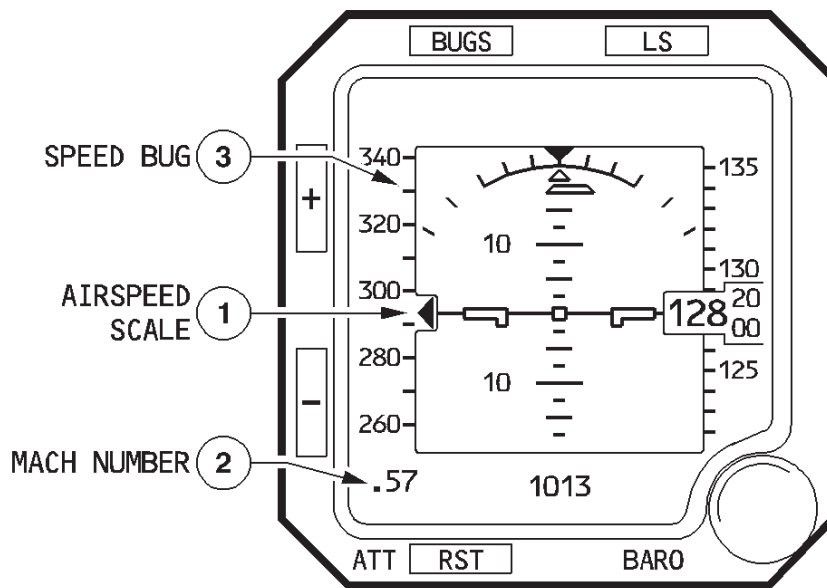
The pitch scale is in white. The scale shows markers every 2.5° between 30° nose up and 30° nose down. Beyond 30°, large red arrowheads (V-shaped) indicate that the attitude has become excessive, and show the direction to follow in order to reduce it. The minimum pitch scale displayed is  $-17.5^\circ + 15^\circ$  at 0° pitch.

### ⑥ ATT RST

The attitude indication can be reset by pressing this pushbutton for at least 2 seconds. The aircraft must be level during this procedure. During the reset time (approximately 10 seconds), the "ATT 10s" message is displayed on the screen. This pushbutton is also used to realign the system, if excessive aircraft movement is detected during the alignment phase.

## AIRSPEED

NFC5-01-3425-004-A100AA



### ① Airspeed scale

A white scale moves in front of a yellow triangle indicating the airspeed. The scale ranges from 5 to 250 knots, with a mark every 5 knots, and from 250 to 520 knots, with a mark every 10 knots.

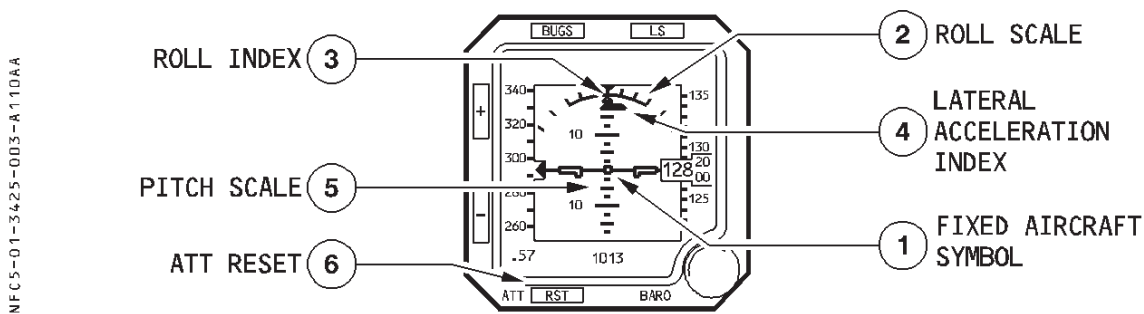
### ② Mach Number

R The Mach number is displayed in green, when greater than 0.5, but disappears only when it goes lower than 0.45.

### ③ Speed bug

When a speed bug is entered via the BUGS function, the corresponding speed mark is indicated by a cyan dash.

## ATTITUDE



### ① Fixed aircraft symbol

The fixed aircraft symbol is in black, and outlined in yellow.

### ② Roll scale

The roll scale is in white.

The markers are at 0 (small yellow triangle), 10, 20, 30, 45, 60 degrees of bank.

### ③ Roll index

The roll index indicates the bank angle. It is in black, and outlined in white.

### ④ Lateral acceleration index

Lateral acceleration index indicates the aircraft's lateral acceleration. It is represented by a trapezoidal index that moves beneath the roll index.

### ⑤ Pitch scale

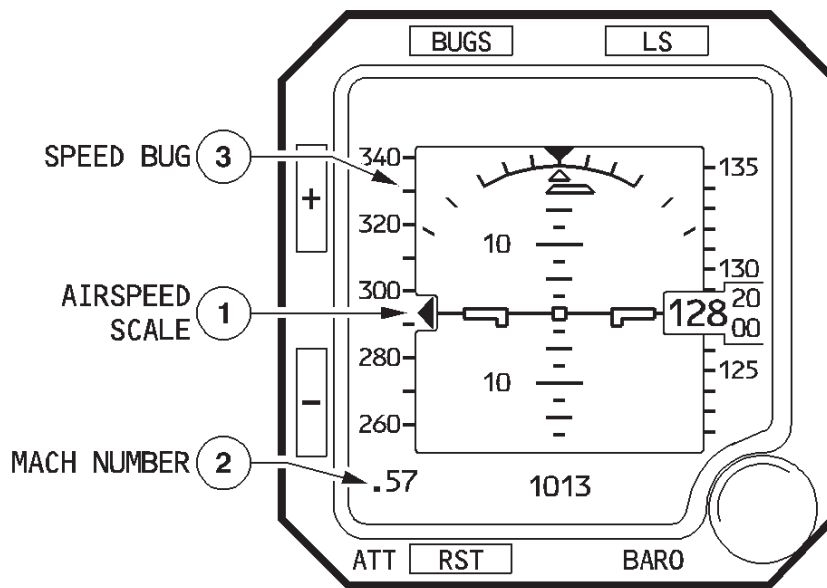
The pitch scale indicates the aircraft attitude, and it is in white. The scale has markers every 2.5 degrees, between 30 degrees nose up and 30 degrees nose down. If the pitch exceeds 30 degrees, large (V-shaped) red arrowheads indicate that the attitude is excessive. They also indicate the direction to follow, to resume normal attitude. The minimum displayed pitch scale is  $-17.5$  degrees  $+15$  degrees at 0 degrees pitch.

### ⑥ ATT RST pushbutton

Attitude indication is reset, by pressing "ATT RST" for at least 2 seconds. The aircraft must be level during this procedure. During the reset time (approx. 10 seconds), the "ATT 10s" message is displayed on the screen. "ATT RST" is also used to realign the system, if excessive aircraft movement is detected during the alignment phase, or after 350 hours of continuous electrical supply.

## AIRSPEED

NFC5-01-3425-004-A100AA



### ① Airspeed scale

A white scale moves in front of a yellow triangle indicating the airspeed. The scale ranges from 5 to 250 knots, with a mark every 5 knots, and from 250 to 520 knots, with a mark every 10 knots.

### ② Mach Number

R The Mach number is displayed in green, when greater than 0.5, but disappears only when it goes lower than 0.45.

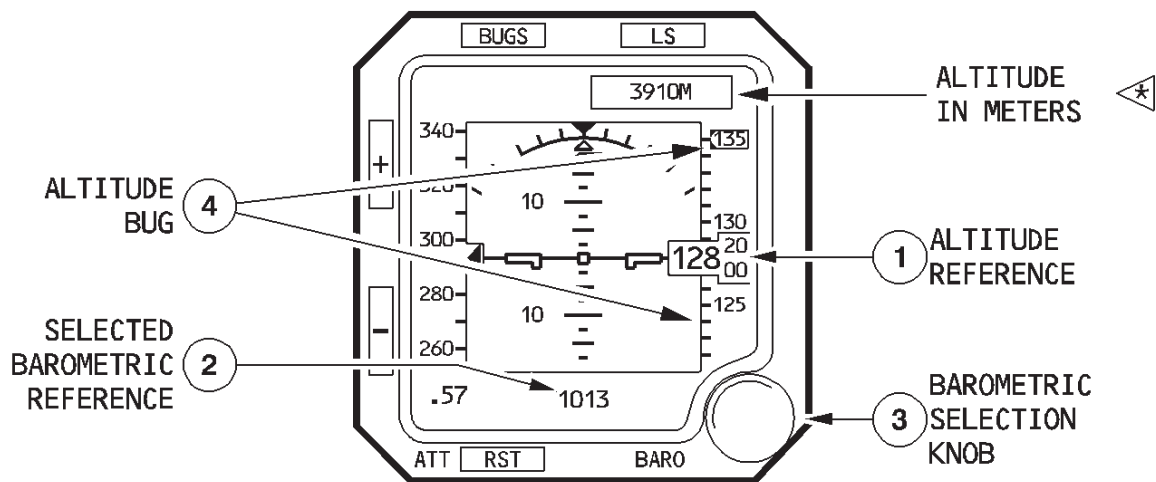
### ③ Speed bug

When a speed bug is entered via the BUGS function, the corresponding speed mark is indicated by a cyan dash.

## ALTITUDE

R

NFC5-01-3425-005-A100AA



### 1 Altitude indication

The altitude indication is given as a white moving scale, and a green digital readout on a gray background.

The altitude scale ranges from – 2000 feet to 50 000 feet every 100 feet, with altitude digital indications every 500 feet. “NEG” appears in white in the negative altitudes window.

If the option is installed, the altitude is also given in meters.

The altitude box changes to cyan, when it also corresponds to a bug value.

### 2 Barometric reference

The barometric reference pressure is displayed in cyan, in hectoPascal (hPa). It corresponds to either the selected barometric pressure, or standard pressure. The barometric pressure ranges from 745 hPa to 1100 hPa.

### 3 Barometric selection knob

It enables the selection of a barometric pressure, setting a variation of 10 hPa per rotation of the knob.

The standard barometric pressure can be selected by pressing the barometric knob. “STD” is then displayed in place of the pressure value.

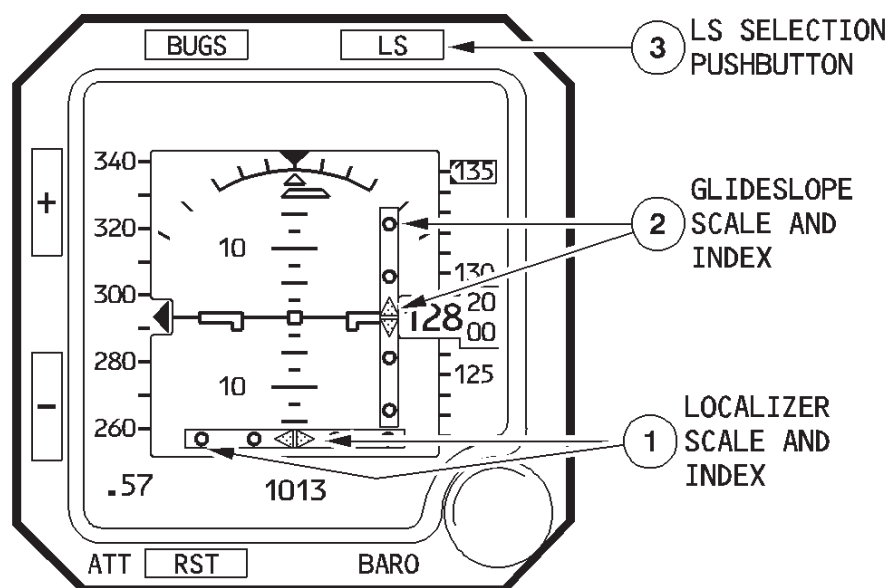
Pressing the knob again will display the selected barometric pressure.

### 4 Altitude bug

When an altitude bug is entered through the bugs function, the corresponding altitude mark is indicated by a cyan dash, or a cyan box, when the dash covers the digital indication on the scale.

## LANDING SYSTEM FUNCTION

NFC5-01-3425-006-A100AA



- ① Localizer scale and index
- ② Glideslope scale and index

The deviation scales appear when the LS pushbutton is pressed. The indexes appear when the Glideslope and Localizer signals are valid and deviation scales are displayed.

- ③ LS selection pushbutton

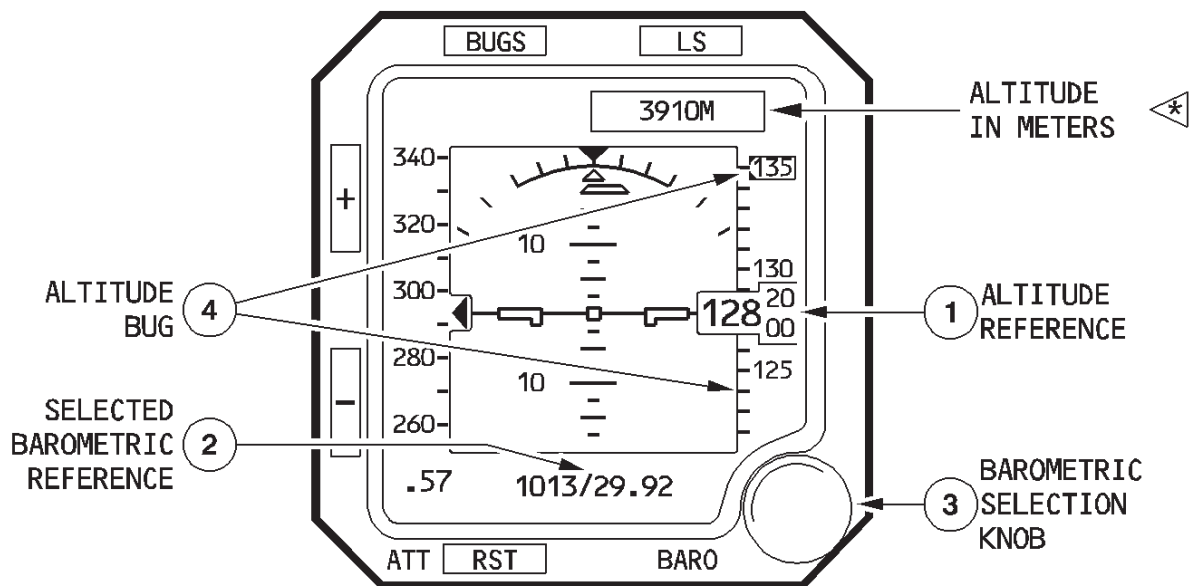
Pressing the LS pushbutton will display the ILS scales. Pressing it again will de-select the ILS scales.

### CAUTION

Do not use the ISIS LS for takeoff using the localizer of the opposite runway, or for a back-course localizer approach. The LOC deviations are given in the wrong sense.

R  
R

NFC5-01-3425-005-A210AA



### ① Altitude indication

The altitude scale ranges from – 2000 feet to 50 000 feet every 100 feet, with altitude digital indications every 500 feet. “NEG” appears in white, in the negative altitudes window.

The altitude box changes to cyan, when it also corresponds to a bug value.

② Barometric reference

The barometric reference pressure is displayed in cyan. It is indicated in both hecto Pascal (hPa) and inches of mercury (Hg).

③ Barometric selection knob

The standard barometric pressure can be selected by pressing the barometric knob. "STD" is then displayed in place of the pressure value.

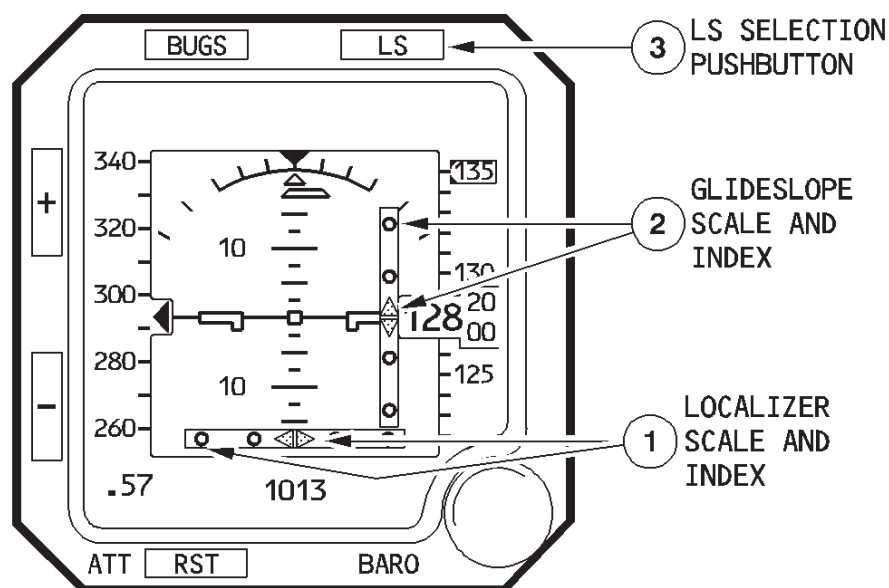
Pressing the knob again will display the selected barometric pressure.

④ Altitude bug

When an altitude bug is entered through the bugs function, the corresponding altitude mark is indicated by a cyan dash, or a cyan box, when the dash covers the digital indication on the scale.

## LANDING SYSTEM FUNCTION

NFC5-01-3425-006-A100AA



- ① Localizer scale and index
- ② Glideslope scale and index

The deviation scales appear when the LS pushbutton is pressed. The indexes appear when the Glideslope and Localizer signals are valid and deviation scales are displayed.

- ③ LS selection pushbutton

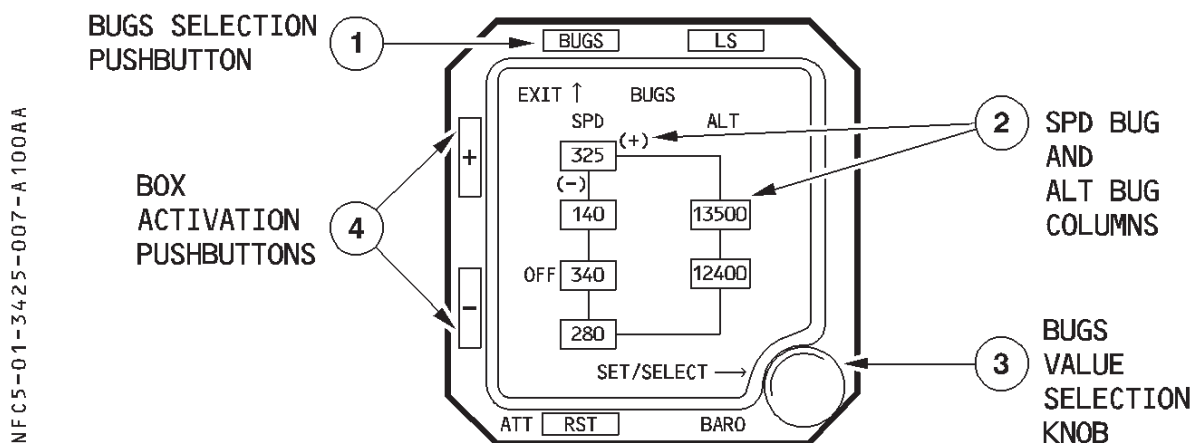
Pressing the LS pushbutton will display the ILS scales. Pressing it again will de-select the ILS scales.

### CAUTION

Do not use the ISIS LS for takeoff using the localizer of the opposite runway, or for a back-course localizer approach. The LOC deviations are given in the wrong sense.

R  
R

## BUGS FUNCTION



### ① BUGS function selection pushbutton

Pressing the BUGS pushbutton will activate the BUGS function and display the bug values to be selected.

### ② SPD BUG and ALT BUG columns

The SPD BUG column gives four speed values (in knots) that can be selected by the crew.

The ALT BUG column gives two altitude values (feet) to be selected by the crew.

### ③ BUGS value selection knob

It allows the bug value to be set by rotating the BARO knob. This value cannot be lower than 30 knots for a speed bug, or a negative value for an altitude bug.

Pressing the BARO setting knob, once a bug value box is activated, will deselect the bug value. The "OFF" label comes on close to the activated box.

The entered values are memorized by the system, when exiting the screen, by pressing the BUGS pushbutton (1), or after 15 seconds without any pilot action.

### ④ "+"/"-" box activation buttons

Access from one box to another is obtained by pressing the "+" or "-" pushbutton.

When a bug value is entered, access to the next box is obtained by pressing the "-" pushbutton. The box becomes active and flashes.

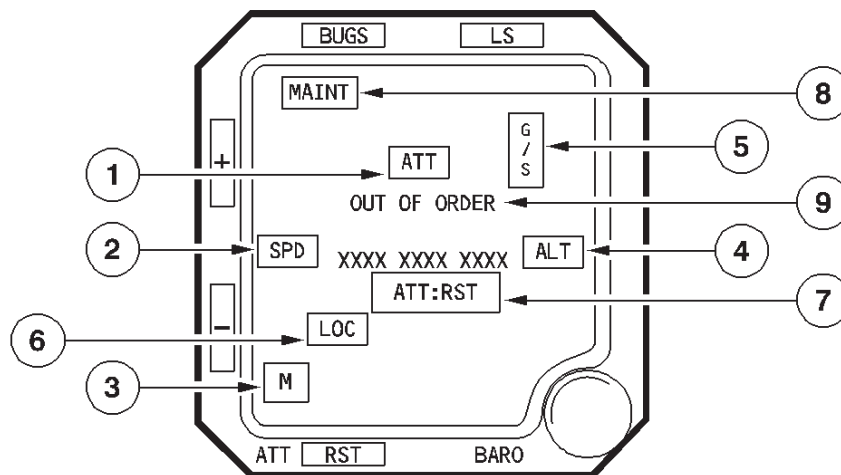
The "+" pushbutton can be used to return to a previous box.

R *Note : Use of the ISIS bugs function is not recommended because, in the event that both*  
 R *PFDs are lost in flight, when the ISIS bugs were previously set for takeoff, then*  
 R *for the approach, the bugs would remain at the takeoff characteristic speed*  
 R *settings.*

## FLAGS

R

NFC5-01-3425-008-A100AA



### ① ATT flag (red)

When attitude data is lost, the red ATT flag appears.

### ② SPD flag (red)

When airspeed data is lost, the red SPD flag appears.

### ③ M flag (red)

When mach number is lost, the red M flag appears.

### ④ ALT flag (red)

When altitude data is lost, the red ALT flag appears.

### ⑤ G/S flag (red)

When glideslope information is lost, the red G/S flag appears.

### ⑥ LOC flag (red)

When localizer data is lost, the red LOC flag appears.

⑦ ATT : RST (yellow)

This message appears when excessive aircraft movement is detected during the alignment phase, or after 350 hours of continuous electrical supply. In this case, press the ATT reset pushbutton to realign and recover the attitude indication.

⑧ MAINT flag (white)

Failure not affecting ISIS operation. Service ISIS, when convenient.

R ⑨ OUT OF ORDER (white)

R When an internal failure of the ISIS indicator occurs, the "OUT OF ORDER" message  
R appears, accompanied by a fault code.

## TUNING

The FMGC is the basic means for tuning nav aids.  
Three modes of tuning are available.

### AUTOMATIC TUNING

In normal operation, the FMGC tunes nav aids automatically, with each FMGC controlling its own receivers.

If one FMGC fails, the remaining one controls both sides receivers.

### MANUAL TUNING

The crew can use the MCDU to override the FMGC's automatic selection and tuning of nav aids and select a specific nav aid for visual display.

This does not affect the automatic function of the FMGC. Any entry on one MCDU is sent to both FMGC in dual mode, or the remaining FMGC in single.

### BACK UP TUNING

If both FMGCs fail, the flight crew can use the RMPs (Radio Management Panels 1 and 2) on the pedestal for back up tuning.

R The CAPT RMP controls VOR 1 and ADF 1 .

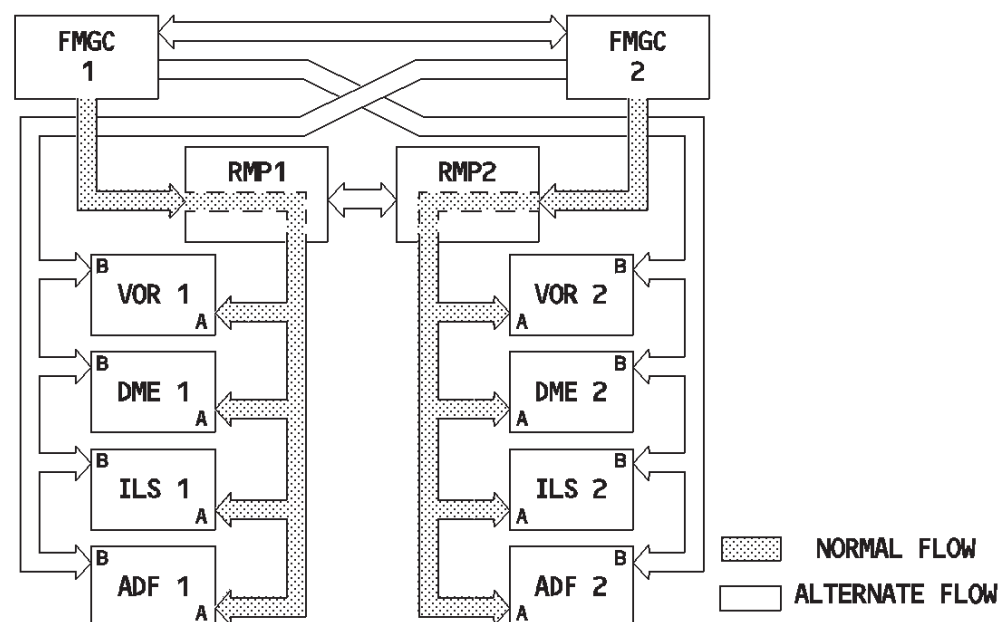
R The F/O RMP controls VOR 2 and ADF 2 .

Either RMP controls both ILSs (provided NAV back up is selected on RMP 1 and RMP 2).

RMP 3 (if installed) is not used for nav aids tuning.

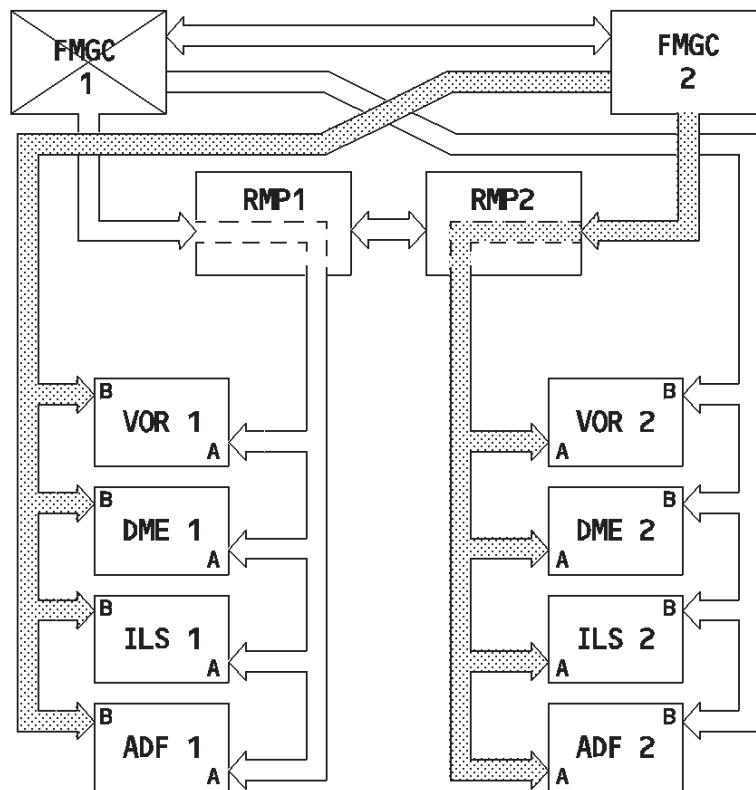
### ARCHITECTURE

#### Normal operation



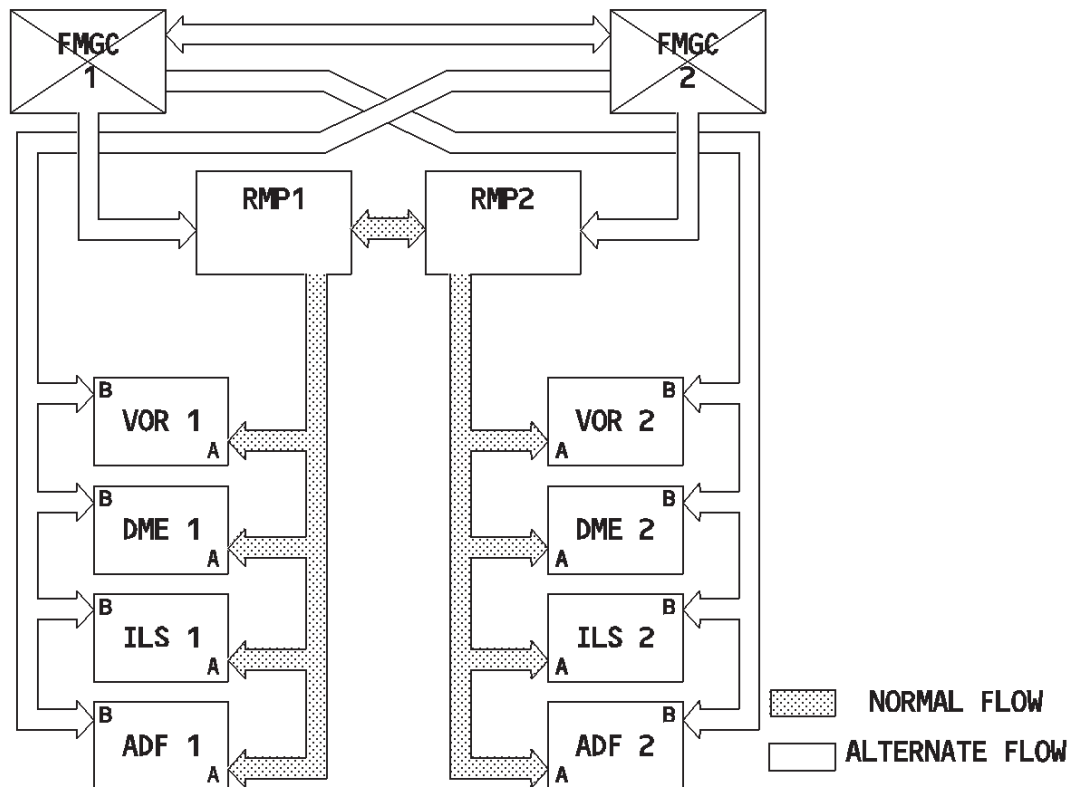
NFC5-01-3430-001-A001AA

## FMGC 1 failure



NFC5-01-3430-002-A001AA

## Back up tuning



NFC5-01-3430-002-B001AA

## NAVAIDS

### VOR

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

### ILS

The aircraft has two ILS receivers.

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can display the ILS information on each PFD by pressing the ILS pushbutton on the onside EFIS control panel (deviation scales and deviation indexes come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

### ADF

The aircraft has two ADF systems.

(For tuning instructions, refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

R The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

### DME

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

### MARKER BEACON

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

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## NAVAIDS

### VOR

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

### ILS

The aircraft has two ILS receivers.

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can display the ILS information on each PFD by pressing the ILS pushbutton on the onside EFIS control panel (deviation scales and deviation indexes come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

### ADF

The aircraft has two ADF systems.

(For tuning instructions, refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

R The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

### DME

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

### MARKER BEACON

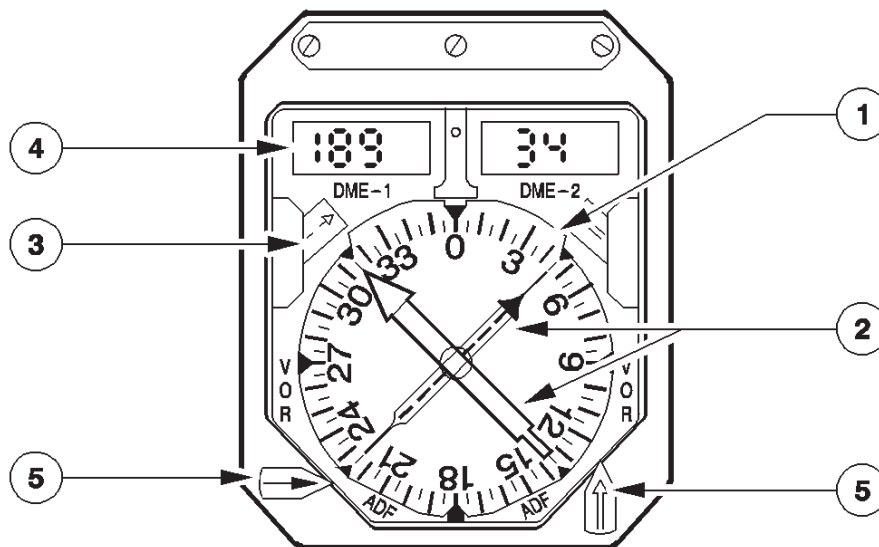
One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### R DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)

NFC5-01-3430-004-A103AB



#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 or ADF 1 (dashed pointer) and VOR 2 or ADF 2 ◀ (double pointer).

*Note : Depending on the quality of the VOR beacon signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR/ADF flags

The indicators display these flags if :

- the VOR or ADF receiver fails (VOR/ADF selector position indicates the failed receiver)
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer moves into horizontal position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

## NAVAIDS

### VOR

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

### ILS

The aircraft has two ILS receivers. Each ILS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (ILS1 receiver in MMR1, ILS2 receiver in MMR2).

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the ILS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

### ADF

The aircraft has two ADF systems.

(For tuning instructions refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

R The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

### DME

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

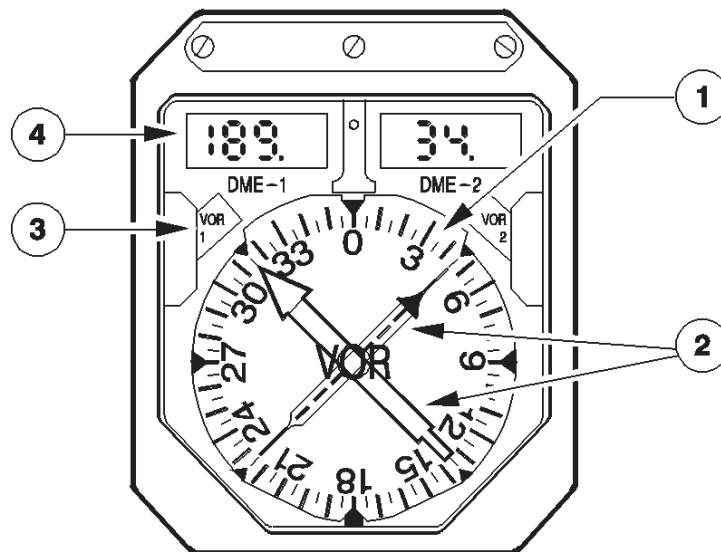
### MARKER BEACON

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A001AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 (dashed pointer) and VOR 2 (double pointer).

*Note : Depending on the quality of the VOR beacon's signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR 1(2) flags

The indicators display these flags if :

- the VOR receiver fails
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

## NAVAIDS

### VOR

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

### ILS

The aircraft has two ILS receivers. Each ILS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (ILS1 receiver in MMR1, ILS2 receiver in MMR2).

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the ILS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

### ADF

The aircraft has two ADF systems.

(For tuning instructions refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

R The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

### DME

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

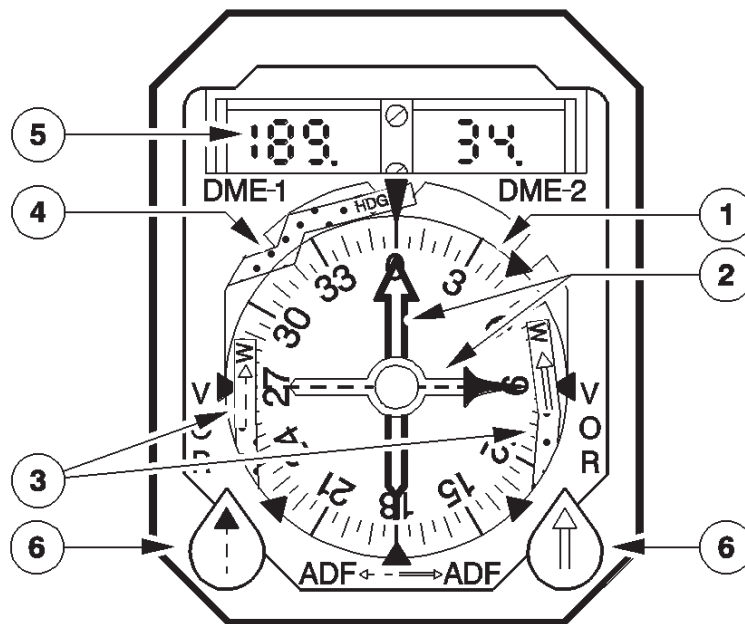
### MARKER BEACON

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A100AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 or ADF 1 (dashed pointer) and, VOR 2 or ADF 2 (double pointer).

*Note : Depending on the quality of the VOR beacon signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR/ADF flags

The indicators display these flags if :

- the VOR or ADF receiver fails (VOR/ADF selector position indicates the failed receiver).
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

**NAVAIDS****VOR**

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

**ILS**

The aircraft has two ILS receivers.

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

**ADF**

The aircraft has two ADF systems.

(For tuning instructions, refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

**DME**

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

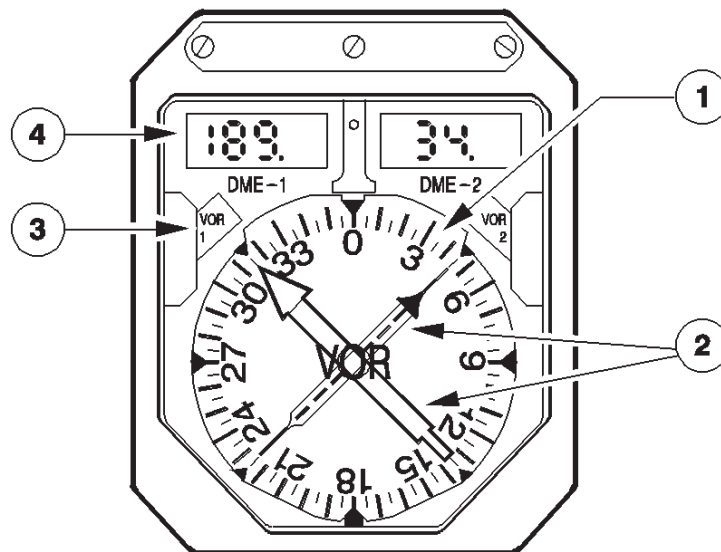
**MARKER BEACON**

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A001AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 (dashed pointer) and VOR 2 (double pointer).

*Note : Depending on the quality of the VOR beacon's signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR 1(2) flags

The indicators display these flags if :

- the VOR receiver fails
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

**NAVAIDS****VOR**

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

**ILS**

The aircraft has two ILS receivers.

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

**ADF**

The aircraft has two ADF systems.

(For tuning instructions, refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

**DME**

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

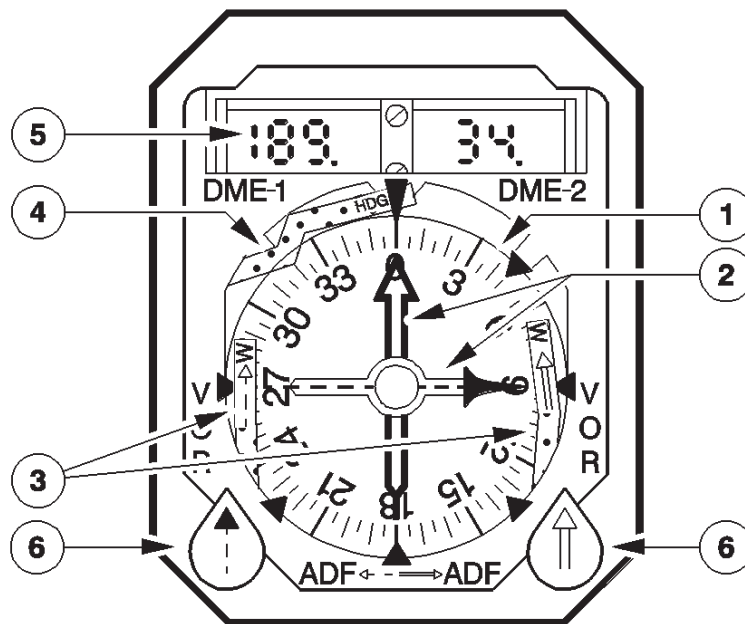
**MARKER BEACON**

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A100AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 or ADF 1 (dashed pointer) and, VOR 2 or ADF 2 (double pointer).

*Note : Depending on the quality of the VOR beacon signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR/ADF flags

The indicators display these flags if :

- the VOR or ADF receiver fails (VOR/ADF selector position indicates the failed receiver).
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

**NAVAIDS****VOR**

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

**ILS**

The aircraft has two ILS receivers.

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE ILS mode on the EFIS control panel (refer to 1.31).

**ADF**

The aircraft has two ADF systems.

(For tuning instructions, refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).

The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

**DME**

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

**MARKER BEACON**

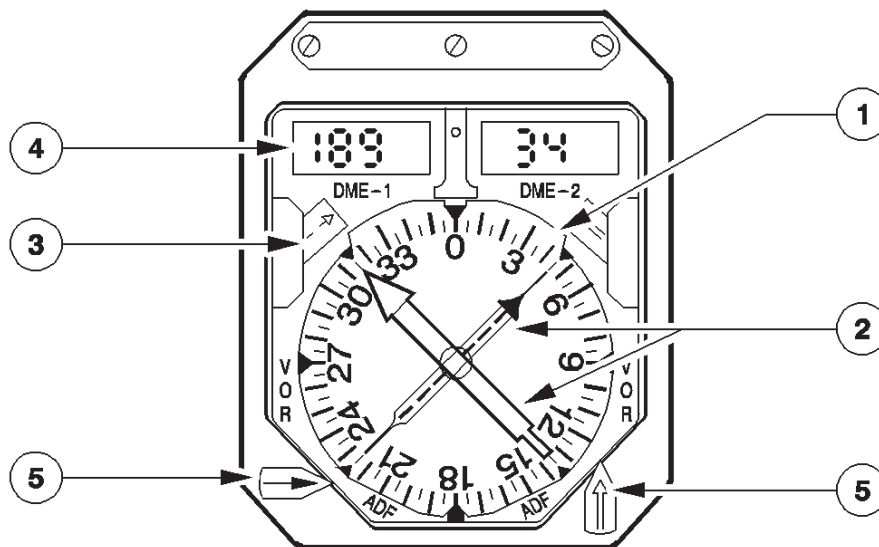
One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### R DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)

NFC5-01-3430-004-A103AB



#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 or ADF 1 (dashed pointer) and VOR 2 or ADF 2 ◀ (double pointer).

*Note :* Depending on the quality of the VOR beacon signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.

#### ③ VOR/ADF flags

The indicators display these flags if :

- the VOR or ADF receiver fails (VOR/ADF selector position indicates the failed receiver)
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer moves into horizontal position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

## NAVAIDS

### VOR

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

### ILS

The aircraft has two ILS receivers. Each ILS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (ILS1 receiver in MMR1, ILS2 receiver in MMR2).

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE LS mode on the EFIS control panel (refer to 1.31).

### R ADF

The aircraft has one ADF system.

(For tuning instructions, refer to the TUNING paragraph).

- R – The ND displays ADF information, depending on the position of the ADF/VOR selector on the EFIS control panel (refer to 1.31).  
The DDRMI ◀ also displays ADF bearing, depending on the position of the ADF/VOR selector (on the DDRMI).

### DME

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

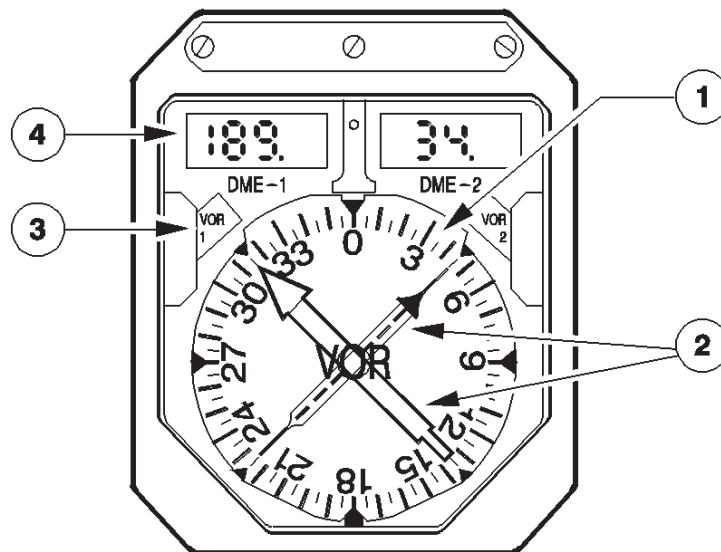
### MARKER BEACON

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A001AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 (dashed pointer) and VOR 2 (double pointer).

*Note : Depending on the quality of the VOR beacon's signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR 1(2) flags

The indicators display these flags if :

- the VOR receiver fails
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

**NAVAIDS****VOR**

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

**ILS**

The aircraft has two ILS receivers. Each ILS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (ILS1 receiver in MMR1, ILS2 receiver in MMR2).

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE LS mode on the EFIS control panel (refer to 1.31).

**ADF**

The aircraft has two ADF systems.

(For tuning instructions refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).  
The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

**DME**

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

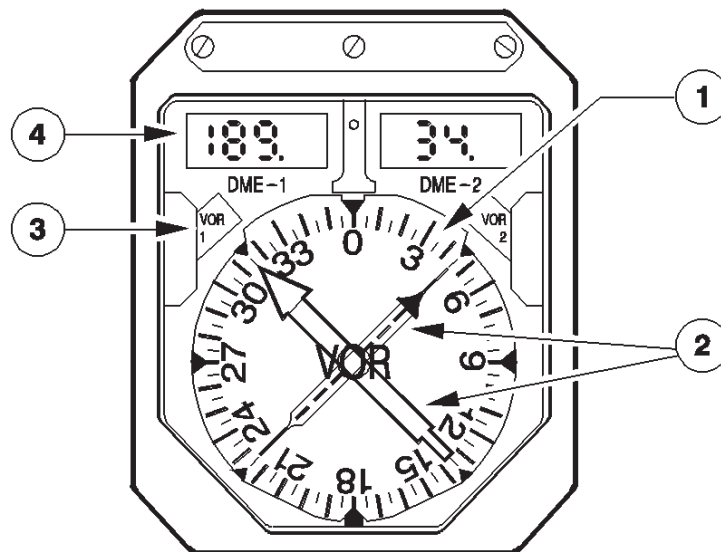
**MARKER BEACON**

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A001AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 (dashed pointer) and VOR 2 (double pointer).

*Note : Depending on the quality of the VOR beacon's signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR 1(2) flags

The indicators display these flags if :

- the VOR receiver fails
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

**NAVAIDS****VOR**

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

**ILS**

The aircraft has two ILS receivers. Each ILS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (ILS1 receiver in MMR1, ILS2 receiver in MMR2).

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE LS mode on the EFIS control panel (refer to 1.31).

**ADF**

The aircraft has two ADF systems.

(For tuning instructions refer to the TUNING paragraph).

- The NDs display ADF information, depending on the position of the ADF/VOR selectors on the EFIS control panel (refer to 1.31).  
The DDRMI  also displays ADF1 and ADF2 bearings, depending on the position of the ADF/VOR selector (on the DDRMI).

**DME**

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (refer to 1.31).

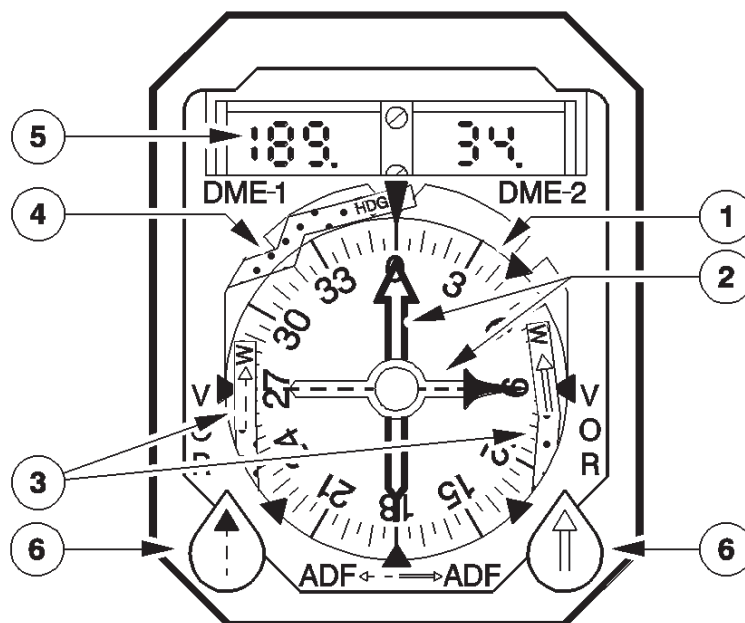
**MARKER BEACON**

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (refer to 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A100AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 or ADF 1 (dashed pointer) and, VOR 2 or ADF 2 ◀ (double pointer).

*Note : Depending on the quality of the VOR beacon signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR/ADF flags

The indicators display these flags if :

- the VOR or ADF receiver fails (VOR/ADF selector position indicates the failed receiver).
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

**NAVAIDS****VOR**

The aircraft has two VOR receivers.

(For tuning instructions, refer to the TUNING paragraph).

- The Navigation Displays (NDs) show VOR1 and VOR2 information, in accordance with the position of the VOR selectors on the EFIS control panel (Ref. FCOM 1.31).
- The DDRMI on the center panel also displays VOR1 and VOR2 bearings, if the heading signal is valid.

**ILS**

The aircraft has two ILS receivers. Each ILS receiver is integrated in a modular avionics unit called MMR (Multi Mode Receiver) (ILS1 receiver in MMR1, ILS2 receiver in MMR2).

(For tuning instructions, refer to the TUNING paragraph).

- PFD1 and ND2 display ILS1 information.
- PFD2 and ND1 display ILS2 information.
- The flight crew can put the same ILS information on each PFD by pressing the LS button on the EFIS control panel (the green bars come on).
- The NDs display ILS information, if the flight crew selects the ROSE LS mode on the EFIS control panel (Ref. FCOM 1.31).

**DME**

The aircraft has two DMEs.

The frequency that is automatically set on the DME corresponds to the one that is set on the VOR or ILS.

The NDs and the DDRMI can display VOR DME information, and the PFDs can display ILS DME information (Ref. FCOM 1.31).

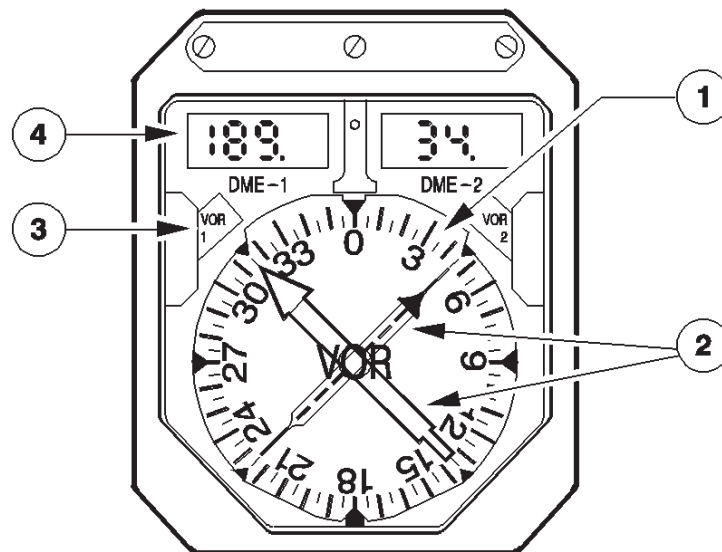
**MARKER BEACON**

One marker beacon system is included in VOR receiver 1.

The PFD displays the outer, middle, and inner marker signals (Ref. FCOM 1.31).

## CONTROLS AND INDICATORS

### DIGITAL DISTANCE AND RADIO MAGNETIC INDICATOR (DDRMI)



NFC5-01-3430-004-A001AA

#### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

#### ② Bearings pointers

Indicate the magnetic bearing to the station received by VOR 1 (dashed pointer) and VOR 2 (double pointer).

*Note : Depending on the quality of the VOR beacon's signal, and mainly at distances greater than 25 NM from the station, the processing of the signal, on aircraft equipped with COLLINS or BENDIX VOR may lead to bearing pointer oscillations.*

#### ③ VOR 1(2) flags

The indicators display these flags if :

- the VOR receiver fails
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer remains at the last valid position.

#### ④ DME 1(2) counters

The counters indicate distances in NM and 1/10th at less than 20 NM.  
At less than 1 NM, 0 is shown.

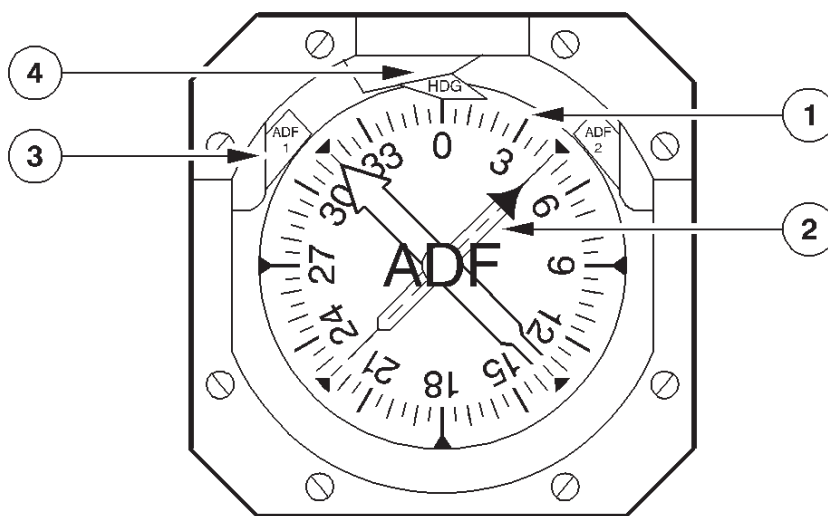
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**INTENTIONALLY LEFT BLANK**

**INTENTIONALLY LEFT BLANK**

## RADIO MAGNETIC INDICATOR (RMI)

NFC5-01-3430-006-A200AA



### ① Compass card

ADIRU 1 normally supplies the signal that positions the compass card. ADIRU3 supplies it when selected by the ATT HDG SWITCHING selector.

### ② Bearing pointers

Indicate the magnetic bearing to ADF 1 (dashed pointer) and ADF 2 (double pointer).

### ③ ADF flag

The indicator displays this flag if :

- the ADF receiver fails
- the RMI has an internal failure
- the heading signal from ADIRS is not valid
- the power supply fails.

As long as the flag shows, the relevant pointer goes to the 3 o'clock position (except in case of power supply failure).

### ④ HDG flag

This red flag is displayed when :

- the heading signal from ADIRS is not valid
- malfunction of heading indication
- the power supply fails.

The HDG flag appears associated with the ADF flag.

⑤ VOR/ADF sel

- R
- VOR 1 or ADF 1 on the dashed pointer.
  - VOR 2 or ADF 2 (◀ ; if not installed, then ADF 1) on the double pointer.

**INTENTIONALLY LEFT BLANK**

④ HDG flag

Appears, associated with VOR/ADF flags display, when :

- The heading signal from the supplying ADIRS is not valid ;
- The RMI has an internal failure ;
- The power supply fails.

⑤ DME 1(2) counters

- The counters indicate distances in NM and 1/10 th at less than 20 NM.  
At less than 1NM, 0 is displayed.

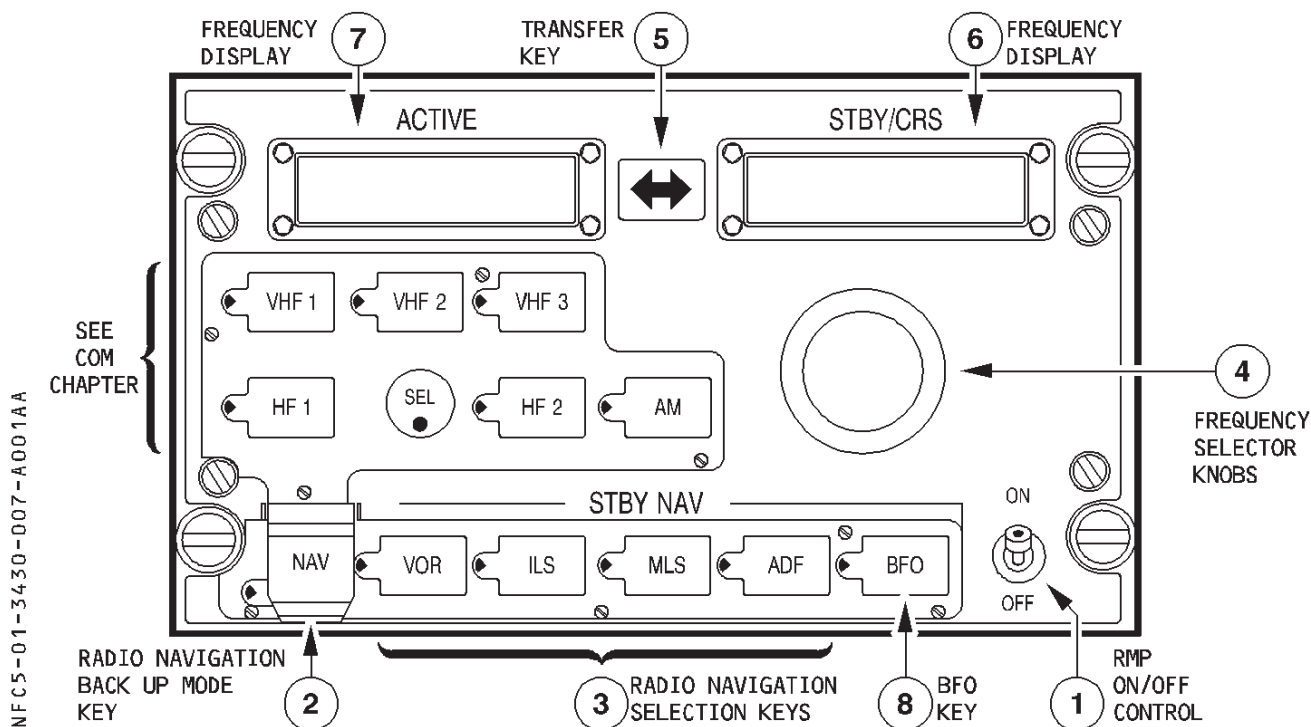
⑥ VOR/ADF sel

- R
- VOR 1 or ADF 1 on the dashed pointer.
  - VOR 2 or ADF 2  (or ADF 1 if ADF 2 is not installed).

**INTENTIONALLY LEFT BLANK**

## RADIO MANAGEMENT PANEL (RMP)

R



### 1 ON/OFF sw

This switch controls the power supply to the panel.

### 2 NAV key (transparent switchguard)

- Pressing this key engages the radio navigation backup mode. It takes control of the VOR, ILS, MLS, and ADF (<img alt="ADF symbol" data-bbox="385 585 405 605"/>) receivers away from the FMGC and gives it to the RMP.
- The green monitor light comes on.
- Pressing the NAV key a second time returns control of the navigation radios to the FMGC.

**Note :** – The flight crew must select this backup tuning mode on both RMP1 and RMP2 if both FMGCs or both MCDUs fail. In the emergency electrical configuration, only RMP1 receives power.

- Pressing the NAV key on RMP3 (if installed) has no effect.
  - In the NAV backup mode, the flight crew can select radio communication systems as it would in the normal mode.
- Setting one RMP to NAV backup mode removes nav aids tuning from both FMGCs.
- When the flight crew uses an RMP to turn an ILS/DME, the PFDs do not display the DME distance.

### ③ STBY NAV keys

When the NAV key is on and the flight crew presses one of these STBY NAV keys, the ACTIVE window displays the frequency to which that receiver is tuned. The green monitor light on the selected key comes on, and the one on the previously selected STBY NAV or COM key goes out.

### ④ Frequency selector knob

Two concentric knobs allow the flight crew to preselect frequencies for communication radios and stand-by navigation systems and select courses for VOR and ILS.

The desired frequency or course is set in the STBY/CRS window.

– setting frequency :

The outer knob controls the most significant digits, the inner knob controls the least significant digits. A rate multiplier speeds up the tuning when the knob is rotated rapidly.

– setting course :

Selected by inner knob only.

### ⑤ Transfer key

The flight crew presses this key to interchange ACTIVE and STBY frequencies. This action tunes the selected receiver to the new ACTIVE frequency.

### ⑥ STBY/CRS window

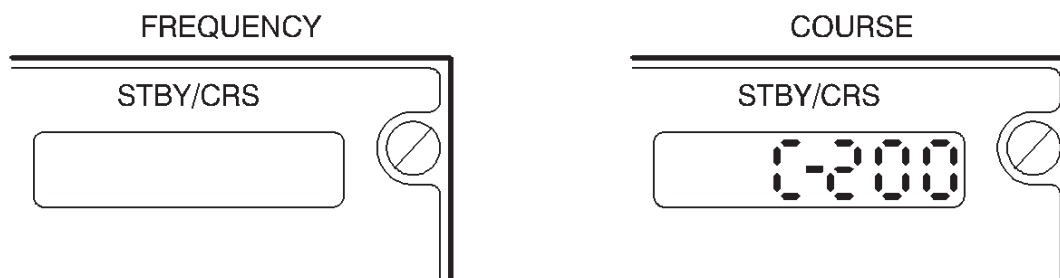
The flight crew can make the frequency displayed in this window become the active frequency by pressing the transfer key, or change it by rotating the tuning knob.

If this window displays a course, then the ACTIVE window displays the associated frequency.

*Note : If the STBY/CRS window is displaying a course, then pressing the transfer key displays the active frequency in both windows.*

R

NFC5-01-3430-008-A001AA



⑦ ACTIVE window

This window displays the frequency of the selected navaid, which is identified by a green monitor light on the selection key.

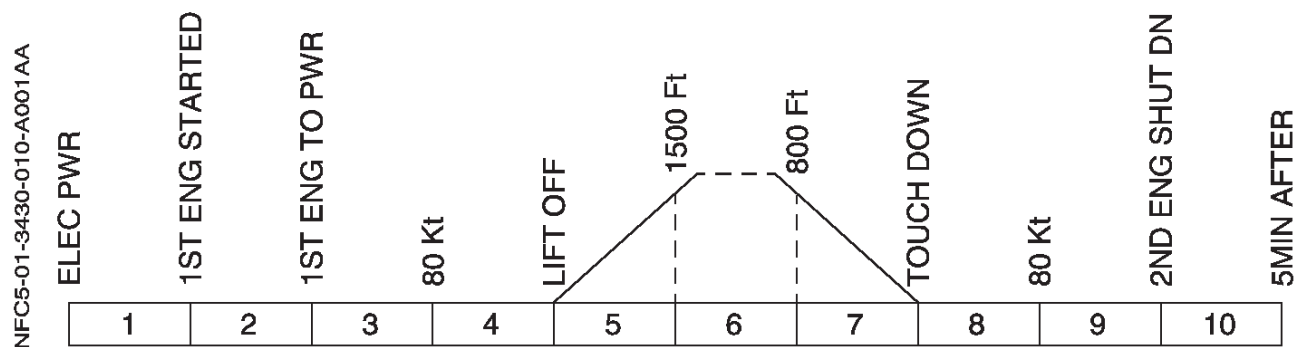
⑧ BFO key

Pressing this key activates the BFO (Beat Frequency Oscillator), if the ADF receiver is selected.

The green monitor light comes on.

R For most ADF, with BFO activated, the audio identification is heard. However there are  
R some ADF where the BFO must be deactivated in order to hear the audio identification.

WARNINGS AND CAUTIONS



| E / WD: FAILURE TITLE<br>conditions | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING               | FLT<br>PHASE<br>INHIB |
|-------------------------------------|------------------|-----------------|----------------------|--------------------------------|-----------------------|
| ILS1 (2) FAULT                      | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | Flag<br>on<br>PFD<br>and<br>ND | 3, 4, 5               |

⑦ ACTIVE window

This window displays the frequency of the selected navaid, which is identified by a green monitor light on the selection key.

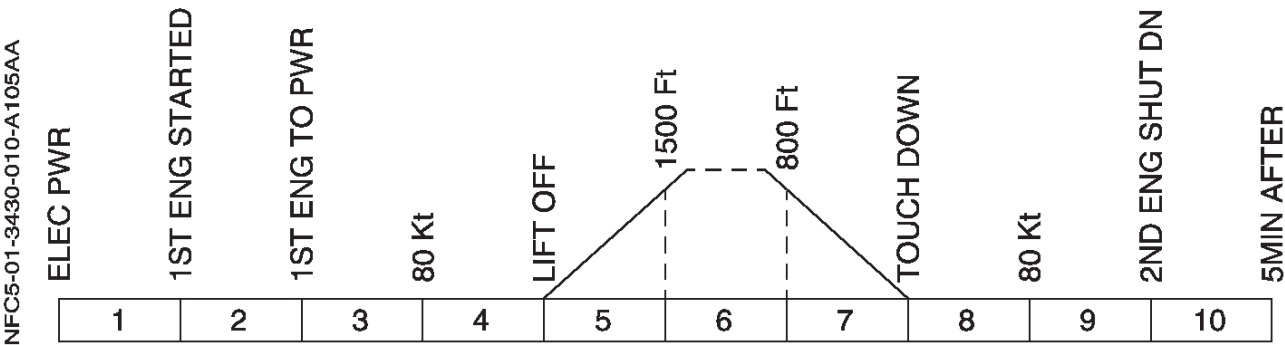
⑧ BFO key

Pressing this key activates the BFO (Beat Frequency Oscillator), if the ADF receiver is selected.

The green monitor light comes on.

R For most ADF, with BFO activated, the audio identification is heard. However there are  
R some ADF where the BFO must be deactivated in order to hear the audio identification.

WARNINGS AND CAUTIONS



| E / WD: FAILURE TITLE<br>conditions | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING               | FLT<br>PHASE<br>INHIB |
|-------------------------------------|------------------|-----------------|----------------------|--------------------------------|-----------------------|
| ILS 1(2) (1+2) FAULT                | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | Flag<br>on<br>PFD<br>and<br>ND | 3, 4, 5               |
| MLS1 (2) FAULT                      |                  |                 |                      |                                |                       |

## DESCRIPTION

The aircraft has two radio altimeters.

Normally the CAPT PFD displays the RA1 height and the F/O PFD displays the RA2 height. If either radio altimeter fails, both PFDs display the height from the remaining one.

## INDICATIONS ON PFD

(Refer to 1.31).

## AUTOMATIC CALL OUT

FWC generates synthetic voice for radio height announcement below 2500 feet. These announcements come through the cockpit loudspeakers even if the speakers are turned off.

### Predetermined call out

The altitude call out uses the following predetermined threshold :

| height (ft) | call out                            |
|-------------|-------------------------------------|
| 2500        | TWO THOUSAND FIVE HUNDRED OR TWENTY |
| 2000        | FIVE HUNDRED                        |
| 1000        | TWO THOUSAND                        |
| 500         | ONE THOUSAND                        |
| 400         | FIVE HUNDRED                        |
| 300         | FOUR HUNDRED                        |
| 200         | THREE HUNDRED                       |
| 100         | TWO HUNDRED                         |
| 50          | ONE HUNDRED                         |
| 40          | FIFTY                               |
| 30          | FORTY                               |
| 20          | THIRTY                              |
| 10          | TWENTY                              |
| 5           | TEN                                 |
| 5           | FIVE                                |
| DH + 100    | HUNDRED ABOVE                       |
| DH          | MINIMUM                             |

Pin programming enables Operators to select the required callouts. If the aircraft remains at a height that is in the detection zone for a height callout, the corresponding message is repeated at regular intervals.

### Intermediate callout

If time between two consecutive predetermined call outs exceeds a certain threshold, the present height is repeated at regular intervals.

The threshold is : 11 seconds above 50 feet

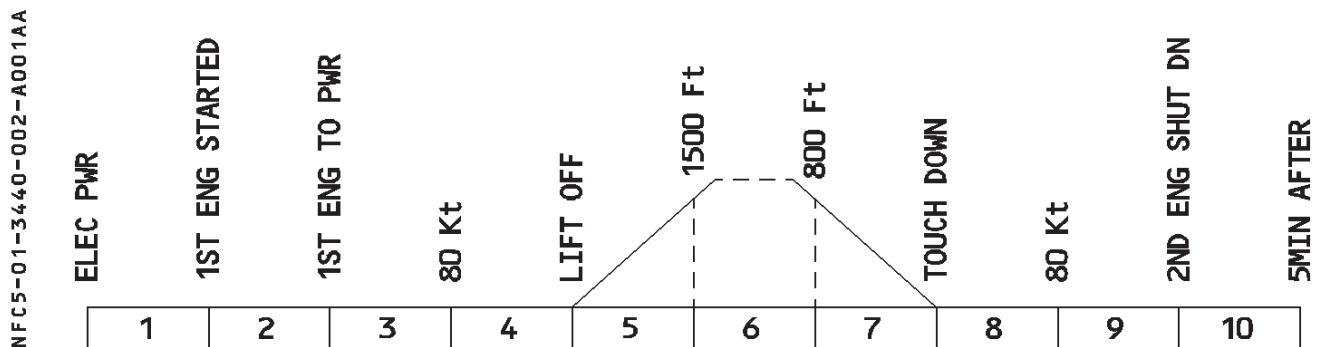
4 seconds below 50 feet

The repeating interval is 4 seconds.

### RETARD announcement

The loudspeaker announces RETARD at 20 feet or at 10 feet if autothrust is active and one autopilot is in LAND mode.

### WARNINGS AND CAUTIONS



| E / WD: FAILURE TITLE<br>conditions | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING | FLT<br>PHASE<br>INHIB |
|-------------------------------------|------------------|-----------------|----------------------|------------------|-----------------------|
| RA 1 (2) FAULT                      | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | NIL              | 3, 4<br>5, 8          |

## DESCRIPTION

The aircraft has two radio altimeters.

Normally, the CAPT's PFD displays the RA1 height, and the F/O's PFD displays the RA2 height. If either radio altimeter fails, both PFDs display the height from the remaining one.

## INDICATIONS ON PFD

(Refer to 1.31).

## AUTOMATIC CALLOUT

The FWC generates a synthetic voice for radio height announcement below 2500 feet. These announcements come through the cockpit loudspeakers, even if the speakers are turned off.

### Predetermined callout

The altitude callout uses the following predetermined threshold :

| Height (feet)         | Callout                                          |
|-----------------------|--------------------------------------------------|
| 2500                  | TWO THOUSAND FIVE HUNDRED OR TWENTY FIVE HUNDRED |
| 2000                  | TWO THOUSAND                                     |
| 1000                  | ONE THOUSAND                                     |
| 500                   | FIVE HUNDRED                                     |
| 400                   | FOUR HUNDRED                                     |
| 300                   | THREE HUNDRED                                    |
| 200                   | TWO HUNDRED                                      |
| 100                   | ONE HUNDRED                                      |
| 50                    | FIFTY                                            |
| 40                    | FORTY                                            |
| 30                    | THIRTY                                           |
| 20                    | TWENTY                                           |
| 10                    | TEN                                              |
| 5                     | FIVE                                             |
| DH (or MDA/MDH) + 100 | HUNDRED ABOVE                                    |
| DH (or MDA/MDH)       | MINIMUM                                          |

Note : The reference altitude for callouts is the radio altitude for precision approaches (DH), and baro altitude (MDA/MDH) for non precision approaches.

Pin programming enables Operators to select the required callouts. If the aircraft remains at a height that is in the detection zone for a height callout, the corresponding message is repeated at regular intervals.

### Intermediate callout

If time between two consecutive predetermined call outs exceeds a certain threshold, the present height is repeated at regular intervals.

The threshold is : 11 seconds above 50 feet

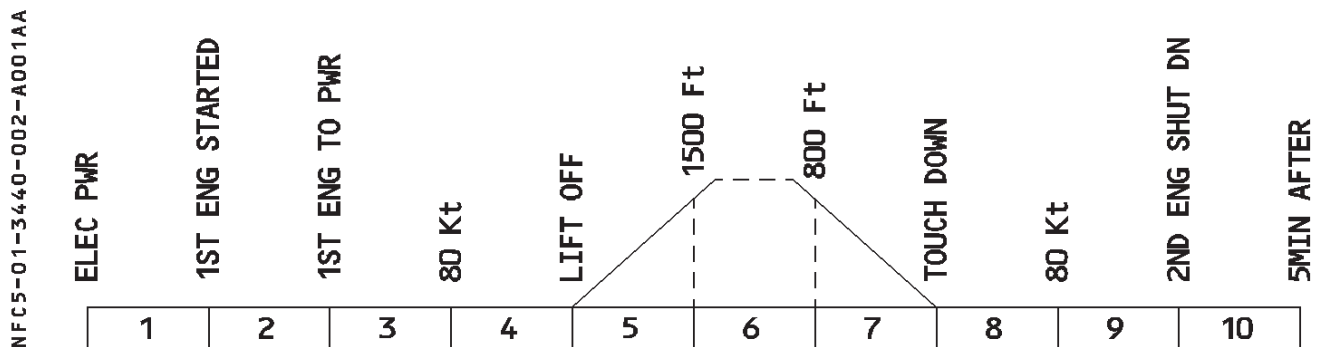
4 seconds below 50 feet

The repeating interval is 4 seconds.

### RETARD announcement

The loudspeaker announces RETARD at 20 feet or at 10 feet if autothrust is active and one autopilot is in LAND mode.

### WARNINGS AND CAUTIONS



| E / WD: FAILURE TITLE<br>conditions | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING | FLT<br>PHASE<br>INHIB |
|-------------------------------------|------------------|-----------------|----------------------|------------------|-----------------------|
| RA 1 (2) FAULT                      | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | NIL              | 3, 4<br>5, 8          |

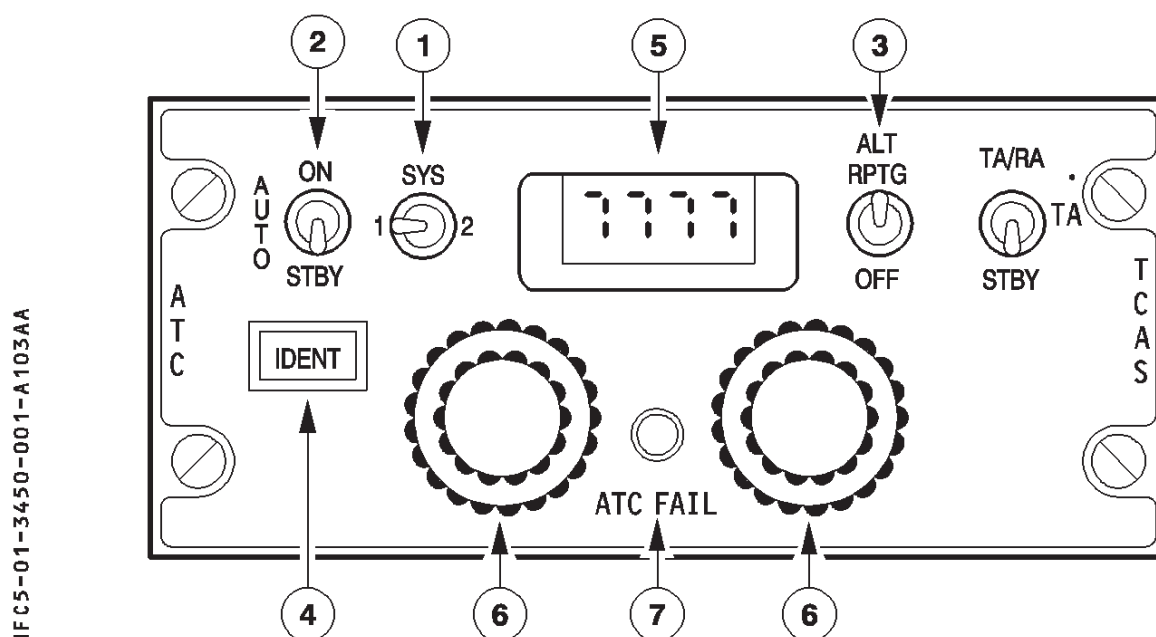
## DESCRIPTION

The aircraft has two ATC transponders which are controlled by a dual control box on the center pedestal.

Only the selected transponder operates.

The associated ADIRS (1 for transponder 1, etc ...) supplies the altitude for altitude reporting. In case of a failure, ADIRS 3 can do this when selected by the AIR DATA SWITCHING selector.

## CONTROL PANEL



### ① SYS sel

This switch selects transponder 1 or 2.

### ② Mode sel

STBY : Both ATC transponders are electrically supplied but do not operate.

ON : Selected transponder operates.

AUTO : In flight : Selected transponder operates.

AUTO : On ground : Selected transponder operates only in mode S (Selective A/C interrogation mode).

R  
R  
R

③ ALT RPTG sw

ALT RPTG : The transponder sends barometric standard altitude data.

OFF : No altitude data transmission. If the TCAS is installed, the upper ECAM displays "TCAS STBY" in green.

④ IDENT sw

This flight crew presses this button to send the aircraft identification signal.

⑤ Display

The window displays the code selected.

⑥ Knobs

The flight crew uses these knobs to set the code assigned by ATC.

⑦ ATC FAIL light

The light comes on if the selected transponder fails.

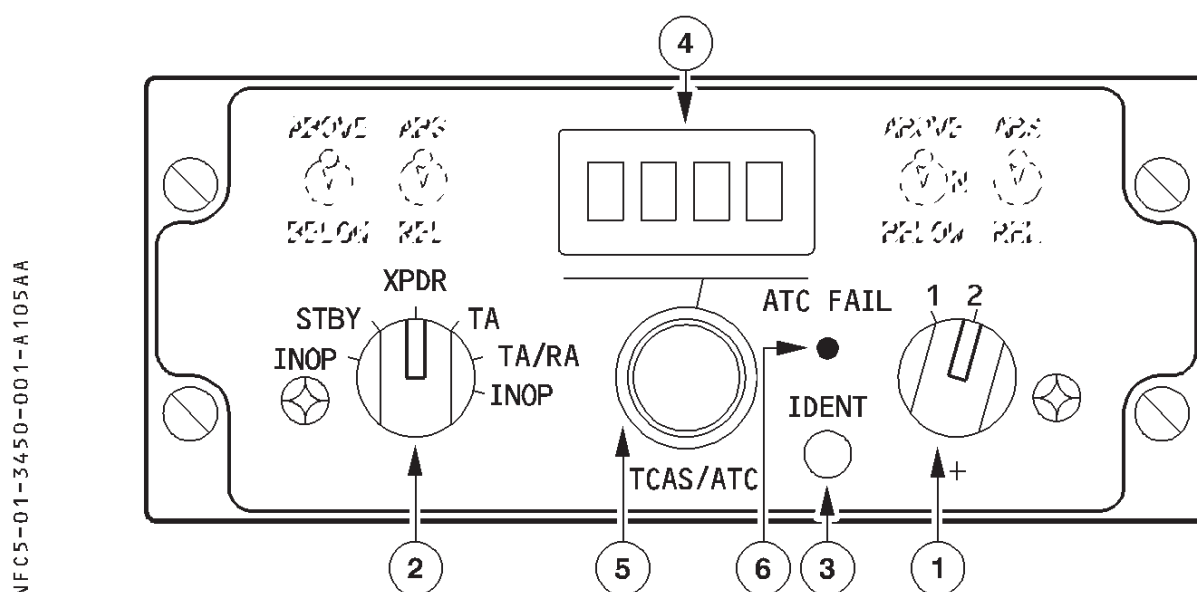
## DESCRIPTION

The aircraft has two ATC transponders. They are controlled from a dual control box that is located on the center pedestal.

Only the selected Transponder operates.

The associated ADIRS (1 for Transponder 1, etc ...) provides altitude information for altitude reporting. In case of a failure, ADIRS 3 can provide altitude information, when selected via the AIR DATA SWITCHING selector.

## CONTROL PANEL



### ① ATC selector

Selects Transponder 1 or 2.

### ② Mode selector

**STBY** : The two ATC Transponders and the TCAS are electrically-supplied but are not operating.

**XPDR** : – The TCAS is on standby.  
 – On ground: The selected ATC Transponder only operates in the selective aircraft interrogation mode of Mode S.  
 – In flight : The selected ATC Transponder operates.

**TA-TA/RA** : Refer to 1.34.80.

R  
R  
R  
R

③ IDENT sw

The flight crew presses this button to send the aircraft identification signal.

④ Code display

The window displays the selected code.

⑤ Code selector knobs

Flight crew uses these knobs to set the code assigned by ATC.

⑥ ATC FAIL lt

This light comes on if the selected transponder fails.

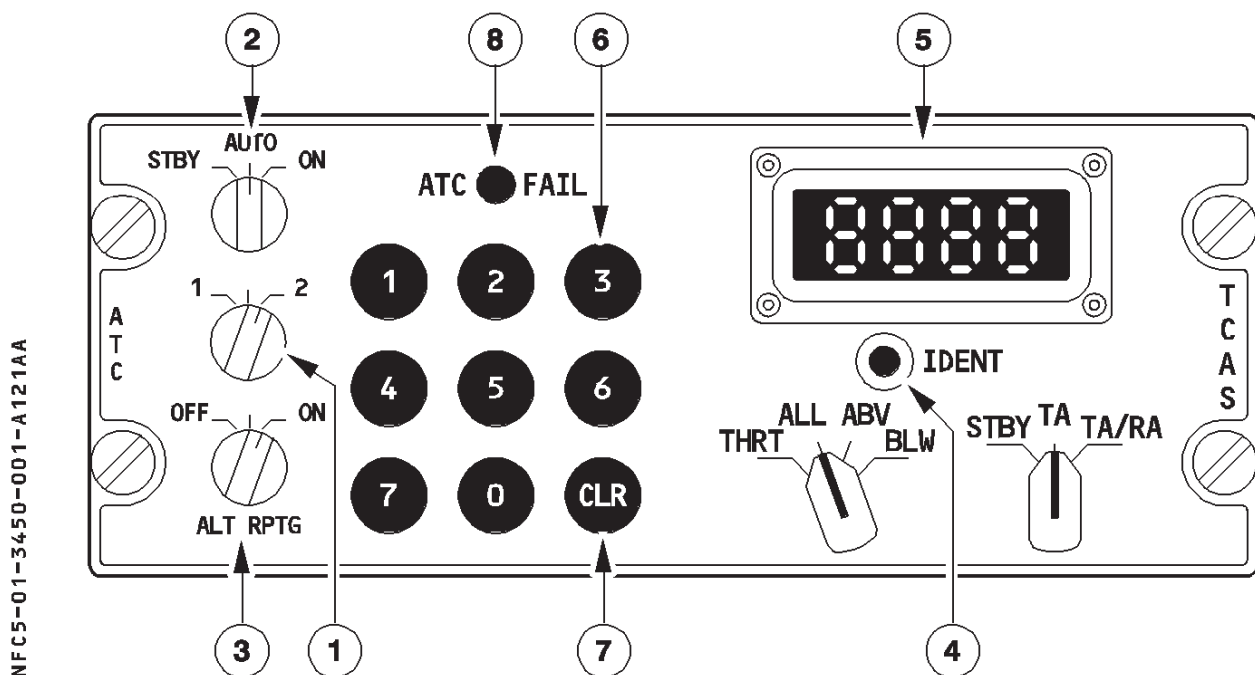
## DESCRIPTION

The aircraft has two ATC transponders which are controlled by a dual control box on the center pedestal.

Only the selected transponder operates.

The associated ADIRS (1 for transponder 1, etc ...) supplies the altitude for altitude reporting. In case of a failure, ADIRS 3 can do this when selected by the AIR DATA SWITCHING selector.

## CONTROL PANEL



### ① ATC sel

This switch selects transponder 1 or 2.

### ② Mode sel

STBY : Both ATC transponders are electrically supplied but do not operate.

ON : Selected transponder operates.

AUTO : In flight : Selected transponder operates.

AUTO : On ground : Selected transponder operates only in mode S (Selective aircraft interrogation mode).

R  
R  
R

③ ALT RPTG sw

- ON : The transponder sends barometric standard altitude data.  
OFF : No altitude data transmission. If the TCAS is installed, the upper ECAM displays "TCAS STBY" in green.

④ IDENT sw

The flight crew presses this button to send the aircraft identification signal.

⑤ Code display

The window displays the selected code.

⑥ Keyboard

The flight crew uses these pushbuttons to set the code assigned by ATC.

⑦ CLR pb

The flight crew uses this pushbutton to clear the code display.

*Note : As long as the four figures of the new code are not entirely written, the previous code remains.*

⑧ ATC FAIL

This light comes on if the selected transponder fails.

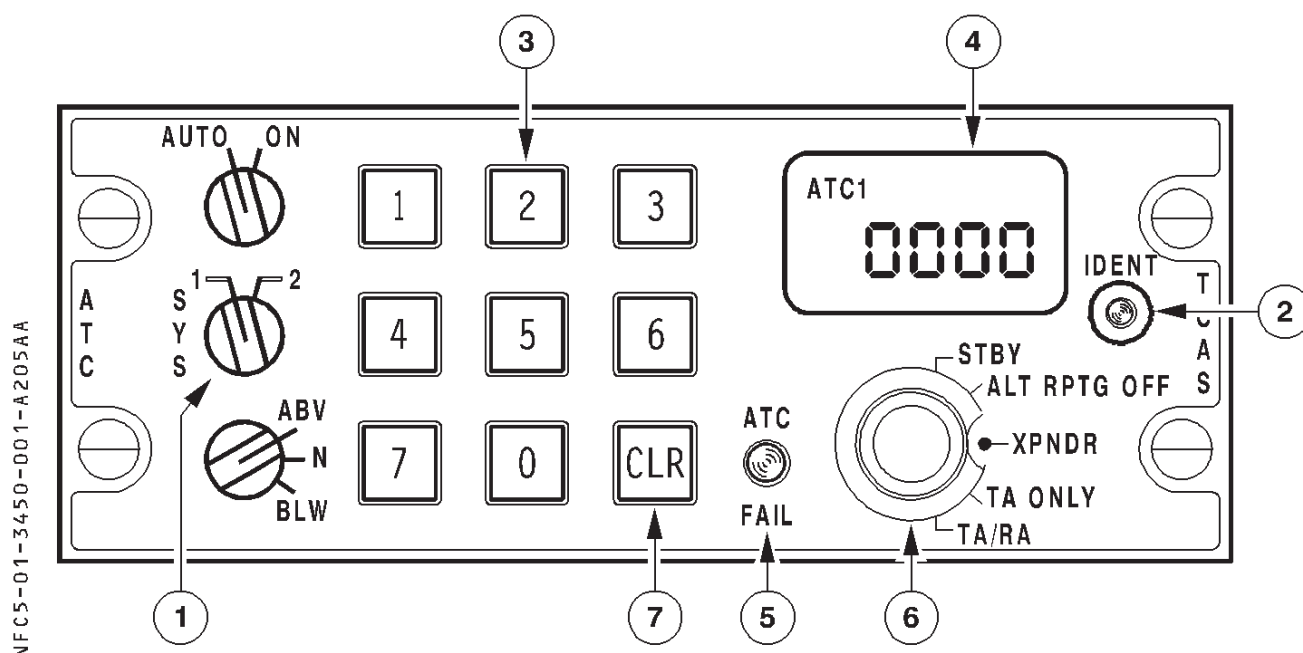
## DESCRIPTION

Two ATC transponders are installed and are controlled by a dual control box located on the center pedestal.

Only the selected transponder operates.

The associated ADIRS (1 for transponder 1, etc ...) supplies the altitude for altitude reporting. In case of a failure, ADIRS 3 can do this when selected by the AIR DATA SWITCHING selector.

## CONTROL PANEL



### ① XPNDR sel

This switch selects transponder 1 or 2.

### ② Mode sel

IDENT sw The flight crew presses this button to send the aircraft identification signal.

### ③ Pushbuttons

Set(s) the code that is assigned by the ATC.

### ④ Code display

This window displays the selected code.

### ⑤ ATC FAIL light

This light comes on, if the selected Transponder fails.

### ⑥ Mode selector

STBY : The two ATC Transponders and the TCAS are electrically-supplied, but are not operating.

ALT RPTG OFF : No altitude data is transmitted.

R  
R  
R  
XPNDR : – On ground: The selected ATC Transponder only operates in the selective aircraft interrogation mode of Mode S.  
– In flight : The selected ATC Transponder operates.  
Baro altitude data is transmitted.

ATC 1 uses ADR 1 or ADR 3. ATC 2 uses ADR 2 or ADR 3.  
The TCAS is on standby.

TA ONLY/TA/RA : Refer to 1.34.80.

### ⑦ CLR pushbutton

This pushbutton clears the code display.

Note : The previous code remains, until a new four-digit code is entered.

## DESCRIPTION

The aircraft has two ATC transponders which are controlled by a dual control box on the center pedestal.

Only the selected ATC transponder operates.

The associated ADIRS (1 for transponder 1, etc ...) supplies the altitude for altitude reporting.

The ATC transponder is capable of enhanced surveillance : It transmits the following parameters upon ground request :

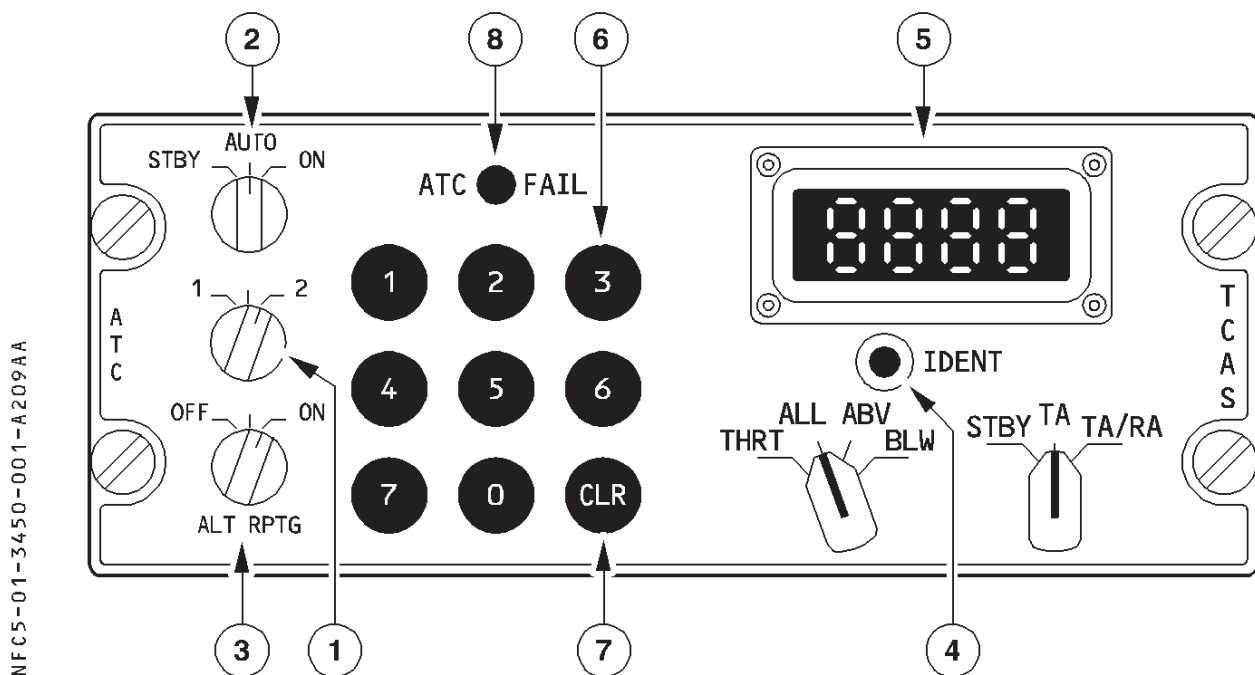
- Indicated airspeed, Mach number, and baro vertical speed, that are all supplied by the ADRs.
- R – Magnetic heading, roll angle, ground speed, track angle, track angle rate, and inertial vertical speed, that are all supplied by the IRs.
- Selected altitude and barometric reference settings supplied by the FCUs.

In the case of an ADR (1 or 2) failure, ADR 3 replaces the faulty ADR, when the AIR DATA SWITCHING Selector is set to CAP on 3 or F/O on 3.

IR/FCU parameters are only transmitted to :

- ATC 1 by IR 1/FCU 1
- ATC 2 by IR 2/FCU 2

## CONTROL PANEL



① ATC sel

Selects ATC transponder 1 or 2.

② Mode sel

STBY : Both ATC transponders are electrically-supplied, but are on standby.

ON : Selected ATC transponder operates in all modes.

AUTO : In flight : Selected ATC transponder operates in all modes.

On ground : Selected ATC transponder only operates in mode S (Selective aircraft interrogation mode).

③ ALT RPTG sw

ON : The transponder sends barometric standard altitude data.

OFF : No altitude data transmission. If the TCAS is installed, the upper ECAM displays "TCAS STBY" in green.

④ IDENT sw

The flight crew presses this button to send the aircraft identification signal.

⑤ Code display

The window displays the selected code.

⑥ Keyboard

The flight crew uses these pushbuttons to set the code assigned by ATC.

⑦ CLR pb

The flight crew uses this pushbutton to clear the code display.

Note : *As long as the four figures of the new code are not entirely written, the previous code remains.*

⑧ ATC FAIL

This light comes on if the selected transponder fails.

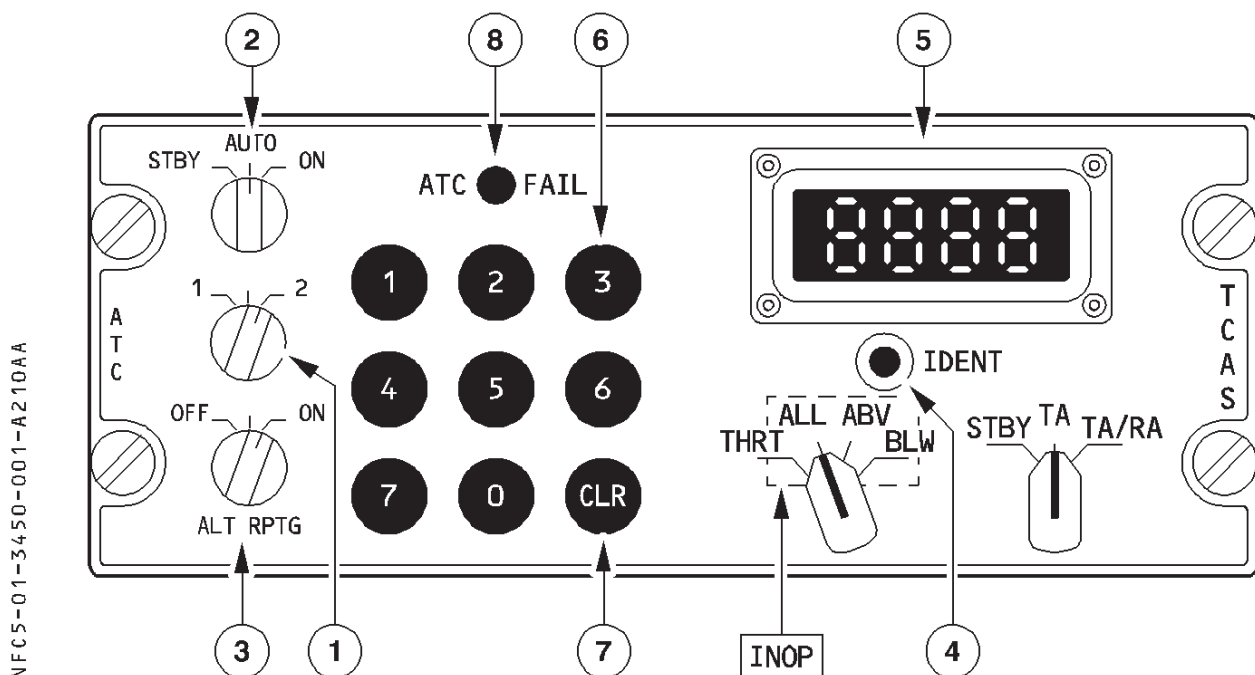
## DESCRIPTION

The aircraft has two ATC transponders which are controlled by a dual control box on the center pedestal.

Only the selected transponder operates.

The associated ADIRS (1 for transponder 1, etc ...) supplies the altitude for altitude reporting. In case of a failure, ADIRS 3 can do this when selected by the AIR DATA SWITCHING selector.

## CONTROL PANEL



### ① ATC sel

This switch selects transponder 1 or 2.

### ② Mode sel

STBY : Selected ATC transponder is electrically supplied but does not operate.

ON : Selected ATC operates.

AUTO : In flight : Selected transponder operates.

AUTO : On ground : Selected transponder operates only in mode S (Selective aircraft interrogation mode).

R  
R  
R

③ ALT RPTG sw

- ON : The transponder sends barometric standard altitude data.  
OFF : No altitude data transmission. If the TCAS is installed, the upper ECAM displays "TCAS STBY" in green.

④ IDENT sw

The flight crew presses this button to send the aircraft identification signal.

⑤ Code display

The window displays the selected code.

⑥ Keyboard

The flight crew uses these pushbuttons to set the code assigned by ATC.

⑦ CLR pb

The flight crew uses this pushbutton to clear the code display.

*Note : As long as the four figures of the new code are not entirely written, the previous code remains.*

⑧ ATC FAIL

This light comes on if the selected transponder fails.

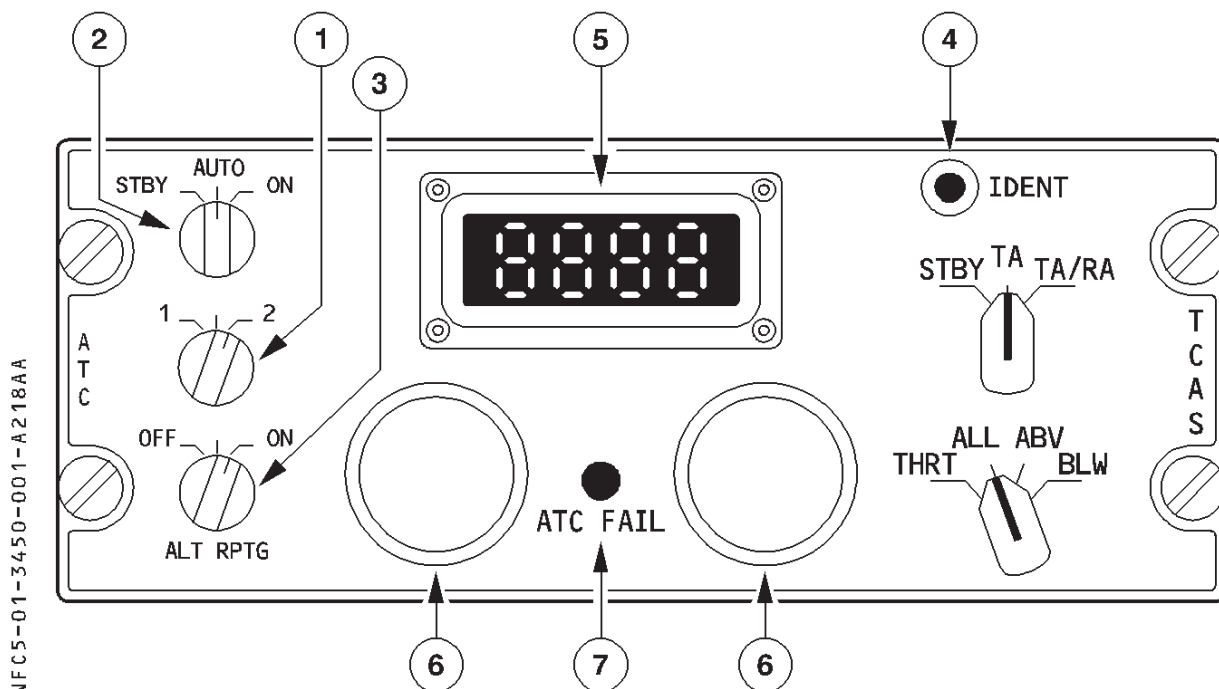
## DESCRIPTION

The aircraft has two ATC transponders which are controlled by a dual control box on the center pedestal.

Only the selected transponder operates.

The associated ADIRS (1 for transponder 1, etc ...) supplies the altitude for altitude reporting. In case of a failure, ADIRS 3 can do this when selected by the AIR DATA SWITCHING selector.

## CONTROL PANEL



### ① ATC sel

This switch selects transponder 1 or 2.

### ② Mode sel

STBY : Both ATC transponders are electrically supplied but do not operate.

ON : Selected transponder operates.

AUTO : In flight : Selected transponder operates.

AUTO : On ground : Selected transponder operates only in mode S (Selective A/C interrogation mode).

R  
R  
R

③ ALT RPTG selector

- ON : The transponder sends barometric standard altitude data.  
OFF : No altitude data transmission. If the TCAS is installed, the upper ECAM displays "TCAS STBY" in green.

④ IDENT sw

The flight crew presses this button to send the aircraft identification signal.

⑤ Display

The window displays the code selected.

⑥ Knobs

The flight crew uses these knobs to set the code assigned by ATC.

⑦ ATC FAIL lt

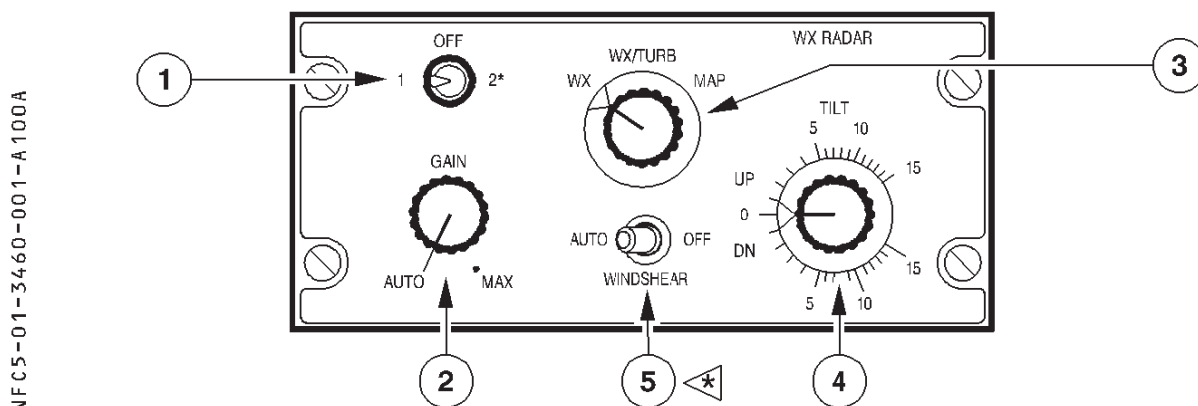
This light comes on if the selected transponder fails.

## DESCRIPTION

The aircraft has two weather radar systems. Only one transceiver is working at a time. It can display the weather image on the ND, in any ND mode except PLAN. Each pilot may remove the weather image from his ND by setting the associated brightness control to the minimum (refer to 1.31).

R Note : Some aircraft may be fitted with one weather radar system only.

## CONTROL PANEL



\* If only one radar system is installed, "2" is replaced by an "INOP" sticker.

### ① Transceiver 1-2 sel

This switch allow to select one radar or to turn both radars to off.

R Note : When one radar only is fitted on aircraft, no weather image will be displayed  
R when switching on system 2.

### ② GAIN knob

This knob adjusts the sensitivity of the receiver in all modes.  
AUTO automatically adjusts the gain to the optimum setting.

### ③ Mode sel

- WX : Weather mode : colors indicate the intensity of precipitation (black for the lowest intensity, green, amber and red indicating progressively higher intensities).
- WX / TURB : The screen shows turbulence areas (in precipitation areas) in magenta (within 40 NM).
- MAP : Radar operates in ground mapping mode : black indicates water, green ground and amber cities and mountains.

④ TILT knob

This knob controls antenna tilt.

- R Radar 1 depends on ADIRS 1, radar 2  on ADIRS 2. ADIRS 3 replaces either ADIRS if  
R ATT HDG selector is used.  
Zero represents the horizon as seen by ADIRS.

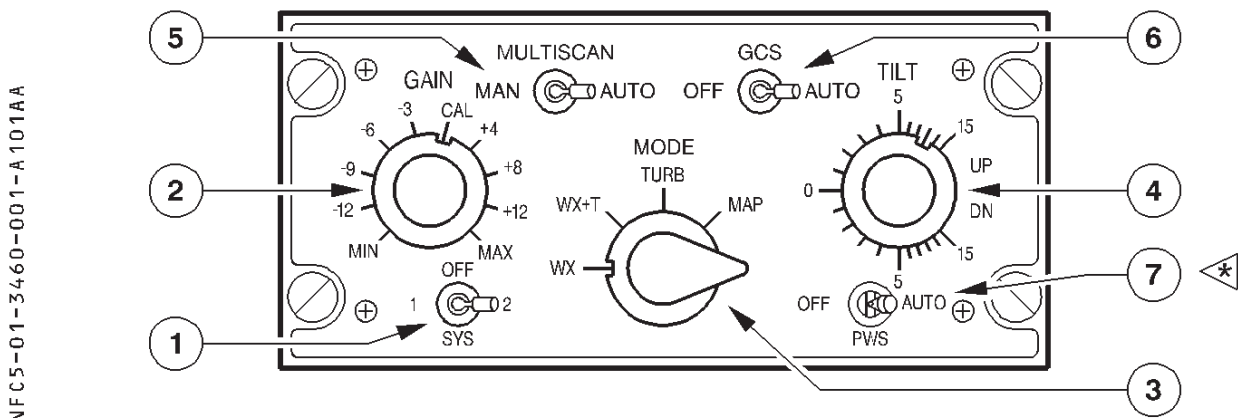
R ⑤ WINDSHEAR sw  (operative only if the predictive windshear function is embodied)

- R AUTO : Predictive windshear function is activated : windshear areas are detected by  
R the antenna scanning below 2300 feet RA, even if the transceiver 1-2  
R selector is set to OFF, and displayed on the ND if below 1500 feet.  
R OFF : No predictive windshear function.

## DESCRIPTION

The aircraft has two weather radar systems. Only one transceiver works at a time. It can display the weather image on the ND, in any ND mode except PLAN. Each pilot may remove the weather image from their ND by setting the associated brightness control to minimum (Refer to 1.31).

## CONTROL PANEL



### ① SYS toggle switch

This switch selects one radar, or turns OFF both radars.

### ② GAIN knob

It adjusts the receiver's sensitivity in weather mode (WX), or in ground mapping mode (MAP).

CAL is the normal position of the knob. It adjusts the gain to a calibrated setting.

### ③ Mode selector

- WX : Weather mode : Colors indicate the intensity of precipitation (black indicates the lowest intensity. Green, amber and red indicate progressively higher intensities).
- WX + T : The screen only displays turbulence areas (in precipitation areas) in magenta (within 40 NM).
- TURB : The screen only shows turbulence areas.
- MAP : The radar operates in ground mapping mode. Black indicates water. Green indicates the ground. Amber indicates cities and mountains.

④ TILT knob

This knob controls the antenna tilt.

Radar 1 depends on ADIRS 1. Radar 2 depends on ADIRS 2. ADIRS 3 replaces either one of the ADIRS, if ATT HDG selector is set to CAPT 3 or F/O 3.

Zero represents the horizon, as seen by the ADIRS.

⑤ MULTISCAN toggle switch

AUTO : Automatically activates tilt to optimize weather detection and minimize ground clutter.

MAN : When set to MAN, the crew can then manually adjust the tilt by using the TILT knob.

⑥ GCS toggle switch

R The Ground Clutter Suppression (GCS) toggle switch is spring-loaded to the AUTO position.

R AUTO : – If MULTISCAN toggle switch is set to AUTO, the radar is in normal use. Ground clutter is not displayed on the screen.

R – If MULTISCAN toggle switch is set to MAN, the GCS switch has no utility. Ground clutter is displayed on the screen.

R OFF : Ground clutter is displayed on the screen.

⑦ PWS toggle switch  (only operative, if the predictive windshear function is installed)

AUTO : Activates the predictive windshear function. Windshear areas are detected by antenna scanning below 2300 feet RA, even if the SYS switch is set to OFF. They are displayed on the ND, if below 1500 feet.

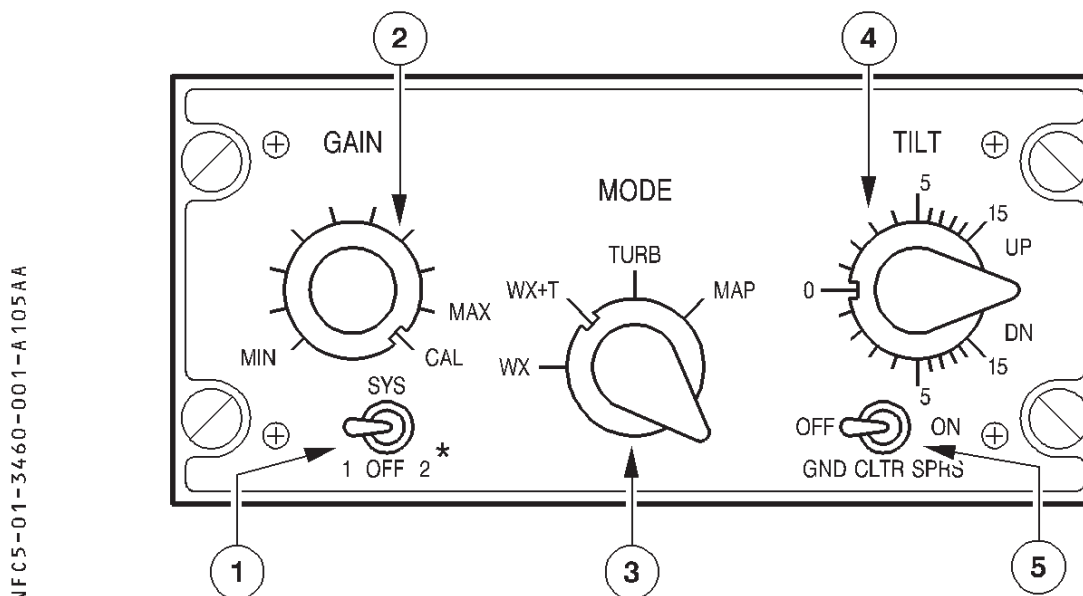
OFF : The predictive windshear function is off.

## DESCRIPTION

The aircraft has two weather radar systems. Only one transceiver works at a time. It can display the weather image on the ND in any ND mode except PLAN. Each pilot may remove the weather image from their ND by setting the associated brightness control to minimum (Refer to 1.31).

## CONTROL PANEL

R



R \* If only one radar system is installed, "2" is replaced by an "INOP" sticker.

### ① SYS sel

This switch is used to select the radar, or to turn both radars to OFF.

### ② GAIN knob

This knob adjusts the sensitivity of the transceiver in all modes. CAL is the normal position, it adjusts the gain to a calibrated setting.

### ③ Mode sel

- WX : Weather mode : Colors indicate the intensity of precipitation (black indicates the lowest intensity, green, amber and red indicate progressively higher intensities).
- WX + T : The screen shows turbulence areas (in precipitation areas) in magenta (within 40 NM).
- TURB : The screen only shows turbulence areas.
- MAP : Radar operates in ground mapping mode : Black indicates water, green indicates the ground, and amber indicates cities and mountains.

④ TILT knob

This knob controls antenna tilt.

R Radar 1 depends on ADIRS 1, radar 2  on ADIRS 2. ADIRS 3 replaces either ADIRS if ATT HDG selector is used.

Zero represents the horizon as seen by ADIRS.

⑤ GND CLTR SPRS sw

ON : Suppresses the ground echo on the screen.

OFF : Normal use of the radar.

## DESCRIPTION

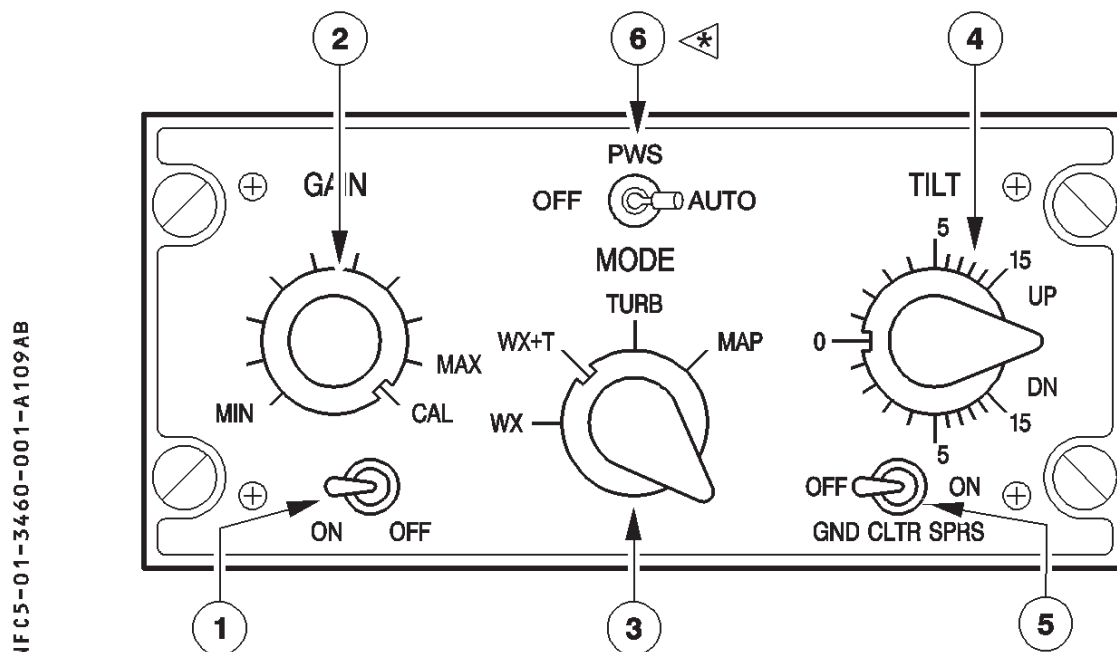
The aircraft has one weather radar system.

It can display the weather image on the ND in any ND mode except PLAN.

Each pilot may remove the weather image from his ND by setting the associated brightness control to the minimum (refer to 1.31).

## CONTROL PANEL

R



### ① SYS sw

This switch turns the radar on or off.

### ② GAIN knob

This knob is used to adjust the sensitivity of the receiver in all modes. CAL is the normal position. It adjusts the gain to a calibrated setting.

### ③ Mode sel

- WX : Weather mode : colors indicate the intensity of precipitation (black for the lowest intensity, green, amber and red indicating progressively higher intensities).
- WX + T : The screen shows turbulence areas (in precipitation areas) in magenta (within 40 NM).
- TURB : The screen shows only turbulence areas.
- MAP : Radar operates in ground mapping mode : black indicates water, green, ground and amber, cities and mountains.

### ④ TILT knob

This knob controls antenna tilt. Zero represents the horizon as ADIRS 1 sees it (or ADIRS 3 if ATT HDG selector is at CAPT 3).

### ⑤ GND CLTR SPRS sw

- ON : Suppresses the ground echo on the screen.
- OFF : Normal use of the radar.

### R ⑥ PWS sw ◀ (operative only if the predictive windshear function is embodied)

- R AUTO : Predictive windshear function is activated : windshear areas are detected by  
 R the antenna scanning below 2300 feet RA, even if the SYS switch is set to  
 R OFF, and displayed on the ND if below 1500 feet.
- R OFF : No predictive windshear function.

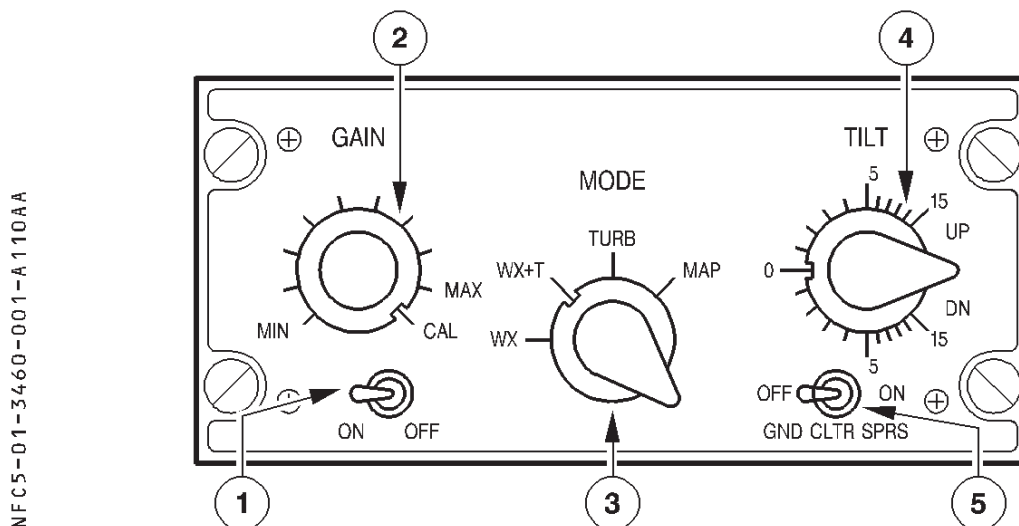
## DESCRIPTION

The aircraft has one weather radar system.

It can display the weather image on the ND, in any ND mode except PLAN.

Each pilot may remove the weather image from their ND by setting the associated brightness control to minimum (refer to 1.31).

## CONTROL PANEL



### ① SYS sw

This switch turns the radar ON or OFF.

### ② GAIN knob

This knob adjusts the sensitivity of the receiver in all modes.

“CAL” is the normal position ; it adjusts the gain to a calibrated setting.

### ③ Mode sel

- WX : Weather mode : Colors indicate the intensity of precipitation (black indicates the lowest intensity, green, amber and red indicate progressively higher intensities, magenta indicates saturated areas.
- WX + T : Weather and turbulence mode : Precipitation and turbulence areas are indicated.
- TURB : The screen shows turbulence areas only. (Turbulence is shown in magenta).
- MAP : Radar operates in ground mapping mode : Black indicates water, green indicates the ground, and amber indicates cities and mountains.

④ TILT knob

This knob controls antenna tilt. Zero represents the horizon as ADIRS 1 sees it (or ADIRS 3 if ATT HDG selector is at CAPT 3).

⑤ GND CLTR SPRS sw

ON : Suppresses the ground echo on the screen

OFF : Normal use of the radar

## DESCRIPTION

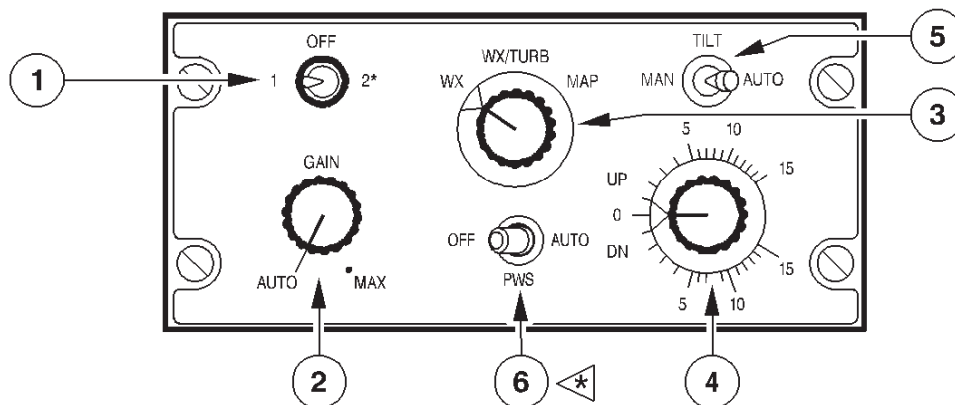
The aircraft has two weather radar systems. Only one transceiver works at a time. It can display the weather image on the ND, in any ND mode except PLAN. Each pilot may remove the weather image from their ND, by setting the associated brightness control to the minimum (Refer to 1.31).

*Note : Some aircraft may be fitted with only one weather radar system only.*

## CONTROL PANEL

R

NFC5-01-3460-001-A115AA



### ① Transceiver 1-2 switch

This switch selects one radar or turns off both radars.

*Note : If there is only one radar on the aircraft, no weather image is displayed when system 2 is switched on.*

### ② GAIN knob

Adjusts the sensitivity of the receiver in all modes.

AUTO : Automatically adjusts the gain to the optimum setting.

### ③ Mode selector

- WX : Weather mode : colors indicate the intensity of precipitation (Black indicates the lowest intensity. Green, amber and red indicate progressively higher intensities).
- WX / TURB : The screen shows turbulence areas (in precipitation areas) in magenta (within 40 NM).
- MAP : The radar operates in ground mapping mode. Black indicates water. Green indicates the ground and Amber indicates cities and mountains.

④ TILT knob

This knob controls the antenna tilt.

Radar 1 depends on ADIRS 1. Radar 2 ◀ depends on ADIRS 2. ADIRS 3 replaces either one of the ADIRS, if ATT HDG selector is set to CAPT 3 or F/O 3.

Zero represents the horizon as seen by the ADIRS.

⑤ TILT toggle switch

AUTO : Automatically selects tilt to optimize weather detection and minimize ground clutter.

MAN : When set to MAN, the crew can then manually adjust the tilt by using the TILT knob.

⑥ PWS sw ◀ (operative only if the predictive windshear function is embodied)

AUTO : Predictive windshear function is activated : windshear areas are detected by the antenna scanning below 2300 feet RA, even if the SYS switch is set to OFF, and displayed on the ND if below 1500 feet.

OFF : No predictive windshear function.

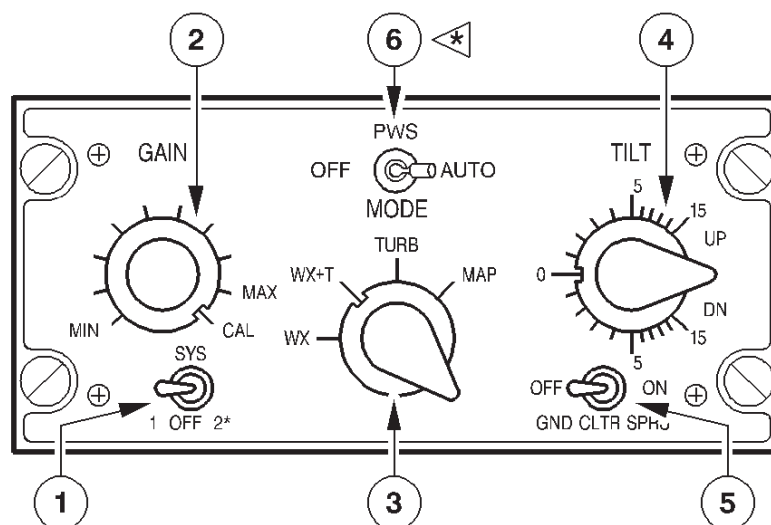
## DESCRIPTION

The aircraft has two weather radar systems. Only one transceiver works at a time. It can display the weather image on the ND, in any ND mode except PLAN. Each pilot may remove the weather image from their ND by setting the associated brightness control to minimum (refer to 1.31).

*Note : Some aircraft may be fitted with only one weather radar.*

## CONTROL PANEL

R



\* IF ONLY ONE RADAR SYSTEM IS INSTALLED, "2" IS REPLACED BY AN "INOP" STICKER.

### ① SYS selector

This switch is used to select the radar, or to turn OFF both radars.

### ② GAIN knob

It adjusts the receiver's sensitivity in weather mode (WX), or in ground mapping mode (MAP). "CAL" is the normal position ; it adjusts the gain to a calibrated setting.

### ③ Mode selector

- WX : Weather mode : Colors indicate the intensity of precipitation (black indicates the lowest intensity, green, amber and red indicate progressively higher intensities).
- WX + T : The screen only shows turbulence areas (in precipitation areas) in magenta (within 40 NM).
- TURB : The screen only shows turbulence areas.
- MAP : Radar operates in ground mapping mode : Black indicates water, green indicates the ground, and amber indicates cities and mountains.

④ TILT knob

This knob controls antenna tilt.

- R Radar 1 depends on ADIRS 1, radar 2  on ADIRS 2. ADIRS 3 replaces either ADIRS if ATT HDG selector is used.  
Zero represents the horizon as seen by ADIRS.

⑤ GND CLTR SPRS sw

ON : Suppresses the ground echo on the screen.

OFF : Normal use of the radar.

R ⑥ PWS sw  (operative only if the predictive windshear function is embodied)

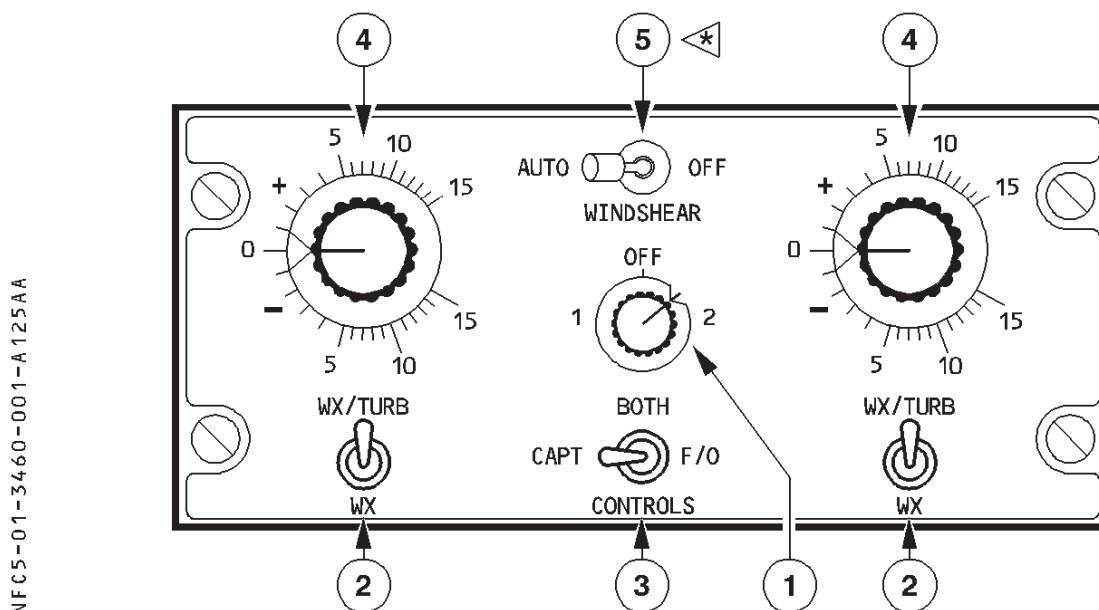
- R AUTO : Predictive windshear function is activated : windshear areas are detected by  
R the antenna scanning below 2300 feet RA, even if the SYS switch is set to  
R OFF, and displayed on the ND if below 1500 feet.  
R OFF : No predictive windshear function.

## DESCRIPTION

The aircraft has two weather radar systems. Only one receiver is working at a time. This unit can display on each ND one weather image different from the other. It is possible for the Captain and the First Officer to select separately different radar modes (WX, WX/TURB) and different TILT.

R Note : Some aircraft may be fitted with one weather radar only.

## CONTROL PANEL



### ① Transceiver 1-2 sel

This switch enables the crew to select one radar, or to turn both radars off.

R Note : When one radar only is fitted on aircraft, no weather image will be displayed  
R when switching on system 2.

### ② Mode sel

WX : Weather mode : the colors indicate the density of precipitation (black for the lowest intensity, green, amber and red indicating progressively for higher intensities).

WX + TURB : The screen displays the turbulence areas (in precipitation areas) in magenta (within 40 NM).

### ③ BOTH sel

CAPT : controls the settings (tilt and radar mode) from the Captain's side for both NDs.

F/O : controls the settings (tilt and radar modes) from the First Officer's side for both NDs.

BOTH : controls the image settings on each ND separately.

### ④ TILT knob

This knob controls the antenna tilt.

R Radar 1 depends on ADIRS 1, radar 2 ◀ on ADIRS 2. ADIRS 3 replaces either ADIRS  
R if ATT HDG selector is used.

Zero represents the horizon as seen by ADIRS.

### R ⑤ WINDSHEAR sw ◀ (operative only if the predictive windshear function is embodied)

R AUTO : The predictive windshear function is activated : the windshear areas are  
R detected by the antenna scanning below 2300 feet RA, even if the  
R transceiver selector (1) is set to OFF, and displayed on the ND if below  
R 1500 feet.

R OFF : No predictive windshear function.

## WINDSHEAR PREDICTION FUNCTION

The weather radars have a Predictive Windshear System (PWS) that operates when :

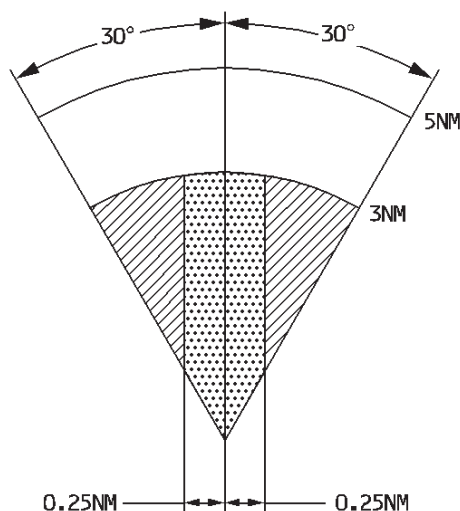
- R — The PWS switch is in the AUTO position (Even if the weather radar is OFF), and
- R — The aircraft is below 2300 feet AGL, and
- R — The ATC is switched to the ON, or AUTO, or XPDR, or XPNDR, position (depending on the ATC panel), and
- R — Either engine is running.

R *Note : When two weather radars are installed, if the selected weather radar fails, the PWS*  
 R *function is recovered by selecting the non-failed weather radar on the control panel.*

The system scans the airspace, within a range of 5 NM ahead of the aircraft, for windshears. Below 1500 feet, when the system detects windshear, depending on the range selected on the ND, a warning, caution, or advisory message appears on the ND. Predictive windshear warnings and cautions are associated with an aural warning.

## R WINDSHEAR ALERTS during takeoff roll, up to 100 knots

R

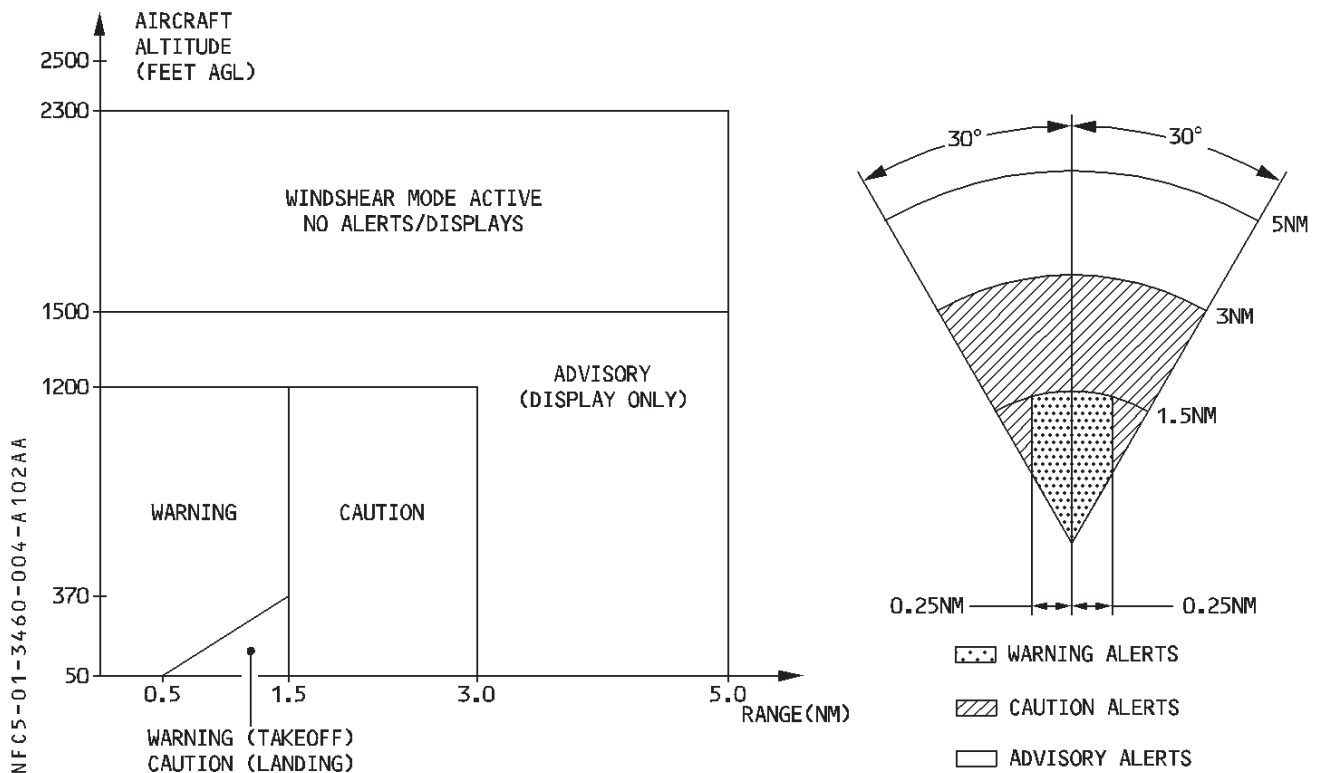


-  WARNING ALERTS
-  CAUTION ALERTS
-  ADVISORY ALERTS

NFC5-01-3460-003-A102AA

R During the takeoff roll, up to 100 knots, both warnings and cautions are available within a  
 R range of 3 NM.

## WINDSHEAR ALERTS above 50 feet



During final approach, the visual and aural warning alerts are downgraded to caution alerts between 370 feet AGL and 50 feet AGL, and range between 1.5 NM and 0.5 NM.

## WINDSHEAR ALERTS inhibition

At takeoff, alerts are inhibited above 100 knots and up to 50 feet. During landing, alerts are inhibited below 50 feet.

## WINDSHEAR PREDICTION FUNCTION

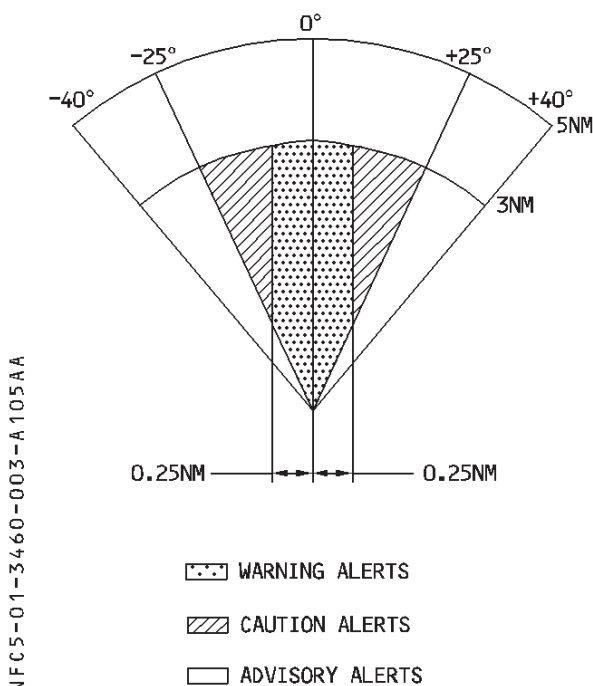
The weather radars have a Predictive Windshear System (PWS) that operates when :

- The PWS switch is in the AUTO position (Even if the weather radar is turned OFF), and
- The aircraft is below 2300 feet AGL, and
- The ATC is switched to the ON, or AUTO, or XPDR, or XPNDR, position (depending on the ATC panel), and
- Either engine is running.

*Note : When two weather radars are installed, if the selected weather radar fails, the PWS function is recovered by selecting the non-failed weather radar on the control panel.*

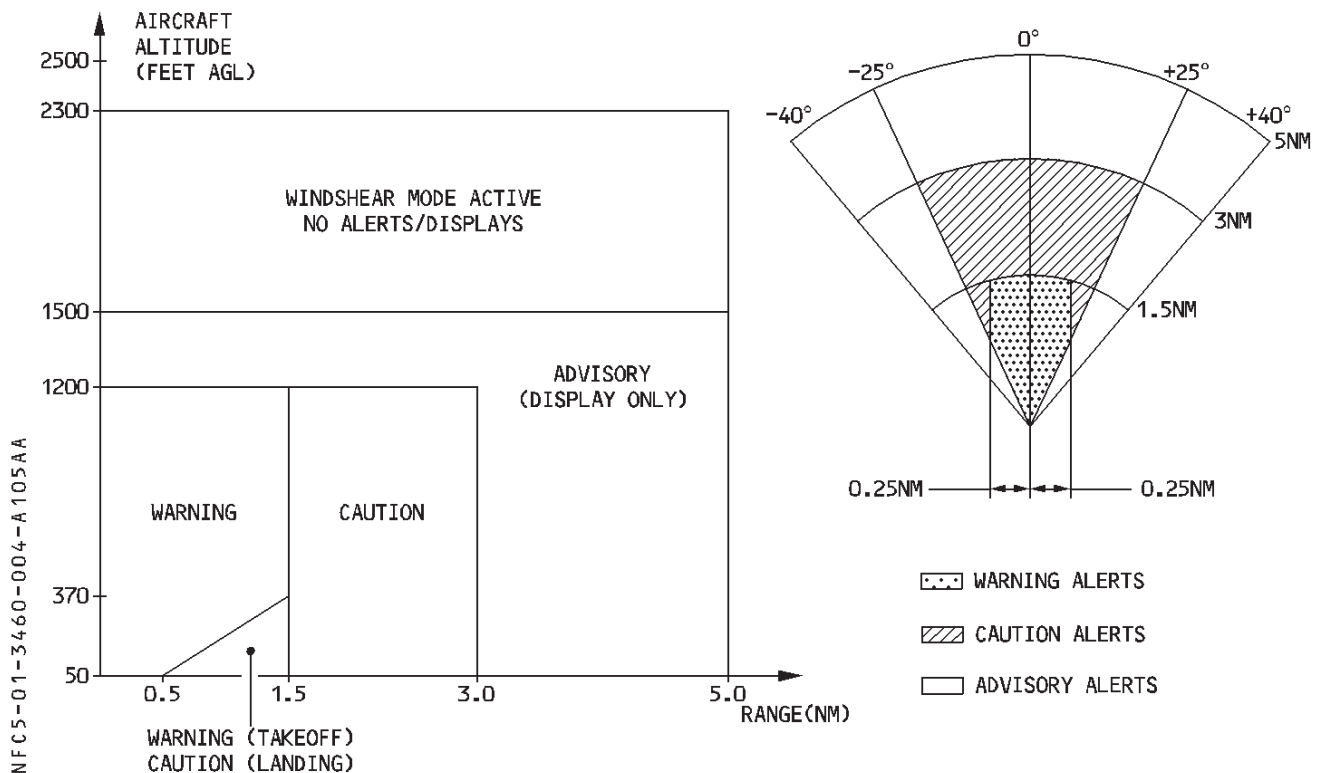
The system scans the airspace, within a range of 5 NM ahead of the aircraft, for windshears. Below 1500 feet, when the system detects windshear, depending on the range selected on the ND, a warning, caution, or advisory message appears on the ND. Predictive windshear warnings and cautions are associated with an aural warning.

### WINDSHEAR ALERTS during takeoff roll, up to 100 knots



During the takeoff roll, up to 100 knots, both warnings and cautions are available within a range of 3 NM.

## WINDSHEAR ALERTS above 50 feet



During final approach, the visual and aural warning alerts are downgraded to caution alerts between 370 feet AGL and 50 feet AGL, and range between 1.5 NM and 0.5 NM.

## WINDSHEAR ALERTS inhibition

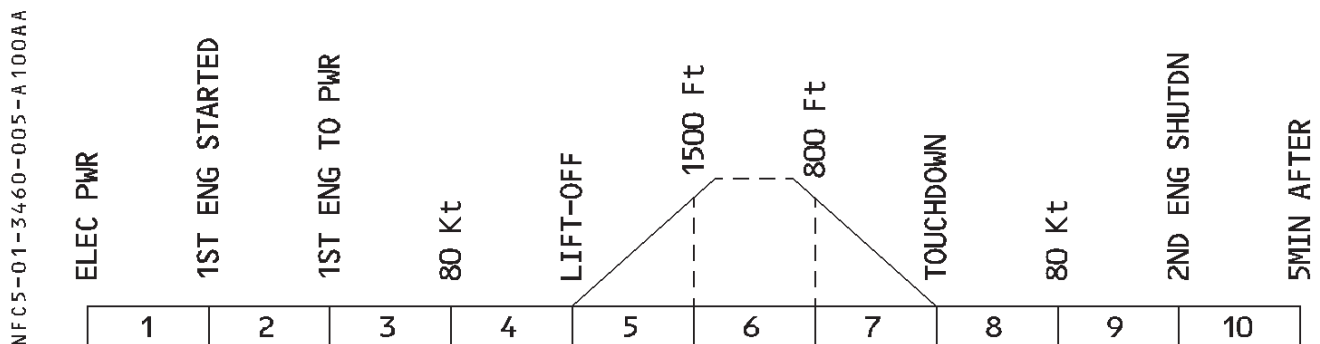
At takeoff, alerts are inhibited above 100 knots and up to 50 feet. During landing, alerts are inhibited below 50 feet.

| Alert Level        | Aural Warning               | PFD               | ND (refer to 1.31.45) |
|--------------------|-----------------------------|-------------------|-----------------------|
| Warning (Approach) | «GO AROUND WINDSHEAR AHEAD» | W/S AHEAD (red)   | Windshear icon        |
| Warning (Takeoff)  | «WINDSHEAR AHEAD» (twice)   | W/S AHEAD (red)   | Windshear icon        |
| Caution            | «MONITOR RADAR DISPLAY»     | W/S AHEAD (amber) | Windshear icon        |
| Advisory           | Nil                         | Nil               | Windshear icon        |

The aural alerts of the Predictive Windshear System (PWS) :

- Have priority over TCAS, GPWS, and other FWC aural warnings
- Are inhibited by windshear detection, via the FAC, stall warnings and aural messages.

## WARNINGS AND CAUTIONS



| E/WD : FAILURE TITLE<br>Conditions | AURAL<br>WARNING | MASTER<br>LIGHT   | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING | FLT<br>PHASE<br>INHIB |
|------------------------------------|------------------|-------------------|----------------------|------------------|-----------------------|
| PRED. W/S DET FAULT                | SINGLE<br>CHIME  | MASTER<br>CAUTION | NIL                  | NIL              | 3, 4, 5, 8            |

## MEMO DISPLAY

The “PRED W/S OFF” message appears, when windshear is set to OFF on the weather radar panel.

The “PRED W/S OFF”, message appears in green during flight phases 1, 2, 6 and 10.

It appears in amber :

- In flight phases 3, 4, 5, 7, 8, and 9
- When the T.O. CONFIG pushbutton is pressed during phase 2.

**DESCRIPTION**

The Ground Proximity Warning System (GPWS) generates aural and visual warnings, when one of the following conditions occurs between radio altitudes 30 and 2450 feet.

- Mode 1 : Excessive rate of descent.
- Mode 2 : Excessive terrain closure rate.
- Mode 3 : Altitude loss after takeoff, or go-around.
- Mode 4 : Unsafe terrain clearance when not in landing configuration.
- Mode 5 : Too far below glideslope.

In addition to the basic GPWS functions, the GPWS has an enhanced function (EGPWS) which provides, based on a worldwide terrain database :

- A Terrain Awareness Display (TAD), which predicts the terrain conflict, and displays the terrain on the ND.
- R – A Terrain Clearance Floor (TCF), which improves the low terrain warning during landing. The cockpit loudspeakers broadcast, even if turned off, the aural warning or caution messages associated with each mode. The audio volume of these messages is not controlled by the loudspeaker volume knobs. (These knobs only allow volume adjustment for radio communication).
- R GPWS lights come on to give a visual warning for modes 1 to 4, TAD, and TCF. For mode 5, the glideslope lights, on the Captain and First Officer instrument panels, come on.

- Note : A number of airports throughout the world have approaches or departures that are not entirely compatible with standard GPWS operation. These airports are identified in the envelope modulation database, in such a way that, when the GPWS recognizes such an airport, it modifies the profile to avoid nuisance warnings.
- R



**DESCRIPTION**

The Ground Proximity Warning System (GPWS) generates aural and visual warnings, when one of the following conditions occurs between radio altitudes 30 and 2450 feet.

- Mode 1 : Excessive rate of descent.
- Mode 2 : Excessive terrain closure rate.
- Mode 3 : Altitude loss after takeoff, or go-around.
- Mode 4 : Unsafe terrain clearance when not in landing configuration.
- Mode 5 : Too far below glideslope.

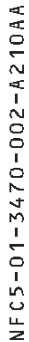
In addition to the basic GPWS functions, the GPWS has an enhanced function (EGPWS) which provides, based on a worldwide terrain database :

- A Terrain Awareness Display (TAD), which predicts the terrain conflict, and displays the terrain on the ND.

- R – A Terrain Clearance Floor (TCF), which improves the low terrain warning during landing. The cockpit loudspeakers broadcast, even if turned off, the aural warning or caution messages associated with each mode. The audio volume of these messages is not controlled by the loudspeaker volume knobs. (These knobs only allow volume adjustment for radio communication).

PULL UP or GPWS lights, on the Captain and First Officer instrument panels, come on to give a visual warning depending on the engaged GPWS mode.

- R Note : A number of airports throughout the world have approaches or departures that are not entirely compatible with standard GPWS operation. These airports are identified in the envelope modulation database, in such a way that, when the GPWS recognizes such an airport, it modifies the profile to avoid nuisance warnings.



**DESCRIPTION**

The Ground Proximity Warning System (GPWS) generates aural and visual warnings, when one of the following conditions occurs between radio altitudes 30 and 2450 feet.

- Mode 1 : Excessive rate of descent.
- Mode 2 : Excessive terrain closure rate.
- Mode 3 : Altitude loss after takeoff, or go-around.
- Mode 4 : Unsafe terrain clearance when not in landing configuration.
- Mode 5 : Too far below glideslope.

In addition to the basic GPWS functions, the GPWS has an enhanced function (EGPWS) which provides, based on a worldwide terrain database :

- A Terrain Awareness Display (TAD), which predicts the terrain conflict, and displays the terrain on the ND.
- A Terrain Clearance Floor (TCF), which improves the low terrain warning during landing.

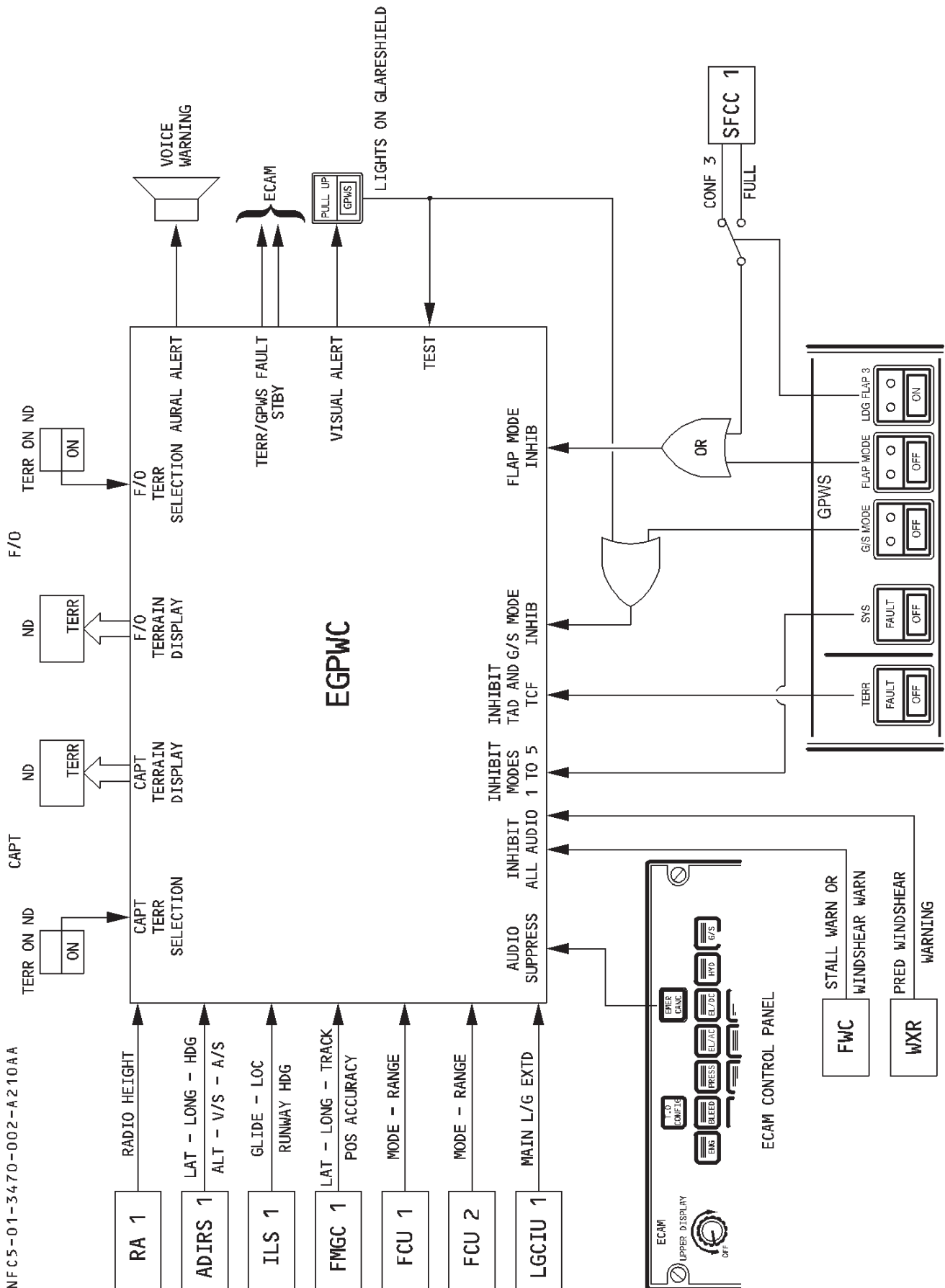
The EGPWS uses the geometric altitude. The geometric altitude is calculated by means of a specific algorithm that uses the following as inputs : The pressure altitude, GPS altitude, radio altitude, and data from the terrain database.

The cockpit loudspeakers broadcast, even if turned off, the aural warning or caution messages associated with each mode. The audio volume of these messages is not controlled by the loudspeaker volume knobs. (These knobs only allow volume adjustment for radio communication).

PULL UP or GPWS lights, on the Captain and First Officer instrument panels, come on to give a visual warning depending on the engaged GPWS mode.

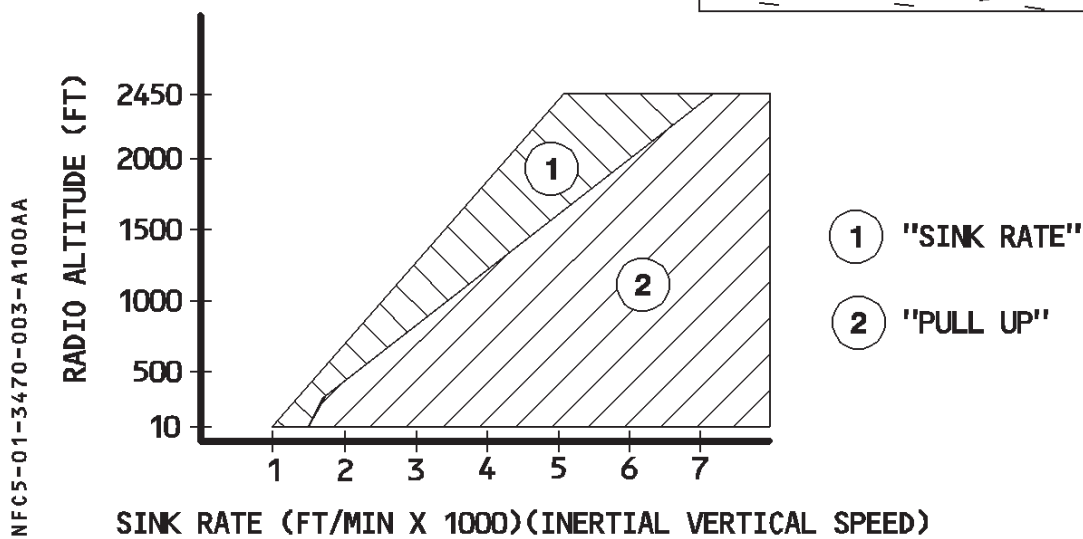
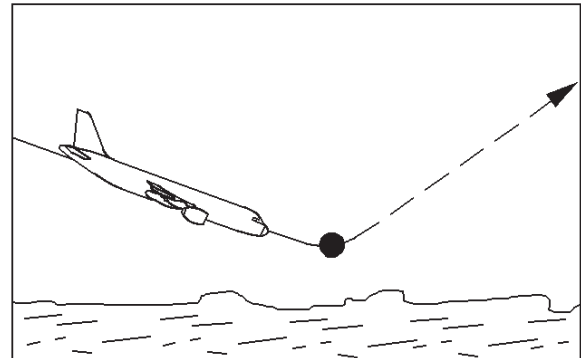
*Note : A number of airports throughout the world have approaches or departures that are not entirely compatible with standard GPWS operation. These airports are identified in the envelope modulation database, in such a way that, when the GPWS recognizes such an airport, it modifies the profile to avoid nuisance warnings.*

FOR INFO



## R GPWS BASIC MODES

### MODE 1: EXCESSIVE RATE OF DESCENT

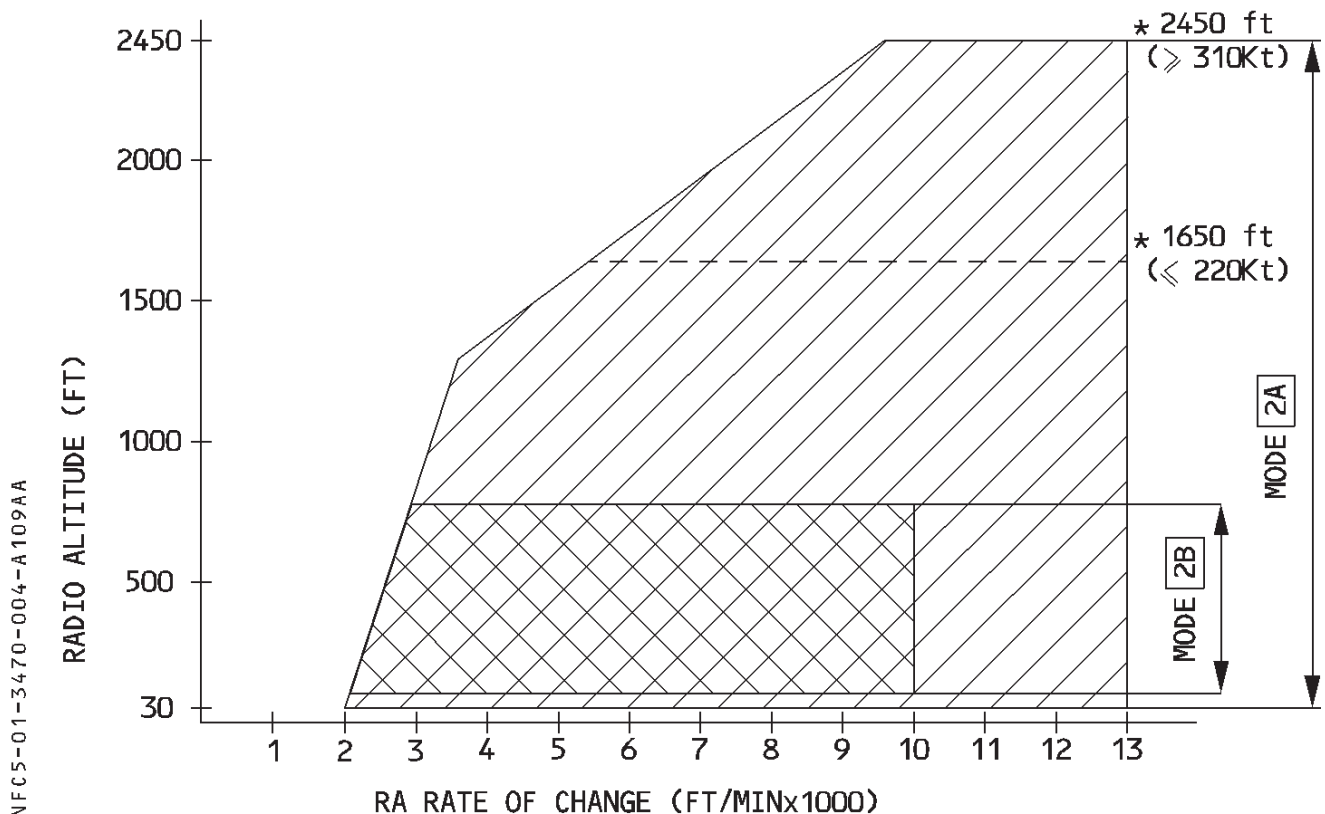
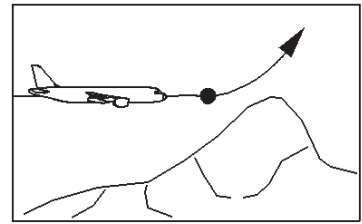


Mode 1 has two boundaries. Penetration of the first boundary generates the illumination of the GPWS light and a repeated aural alert "SINK RATE". Penetration of the second boundary generates a repetitive "PULL UP".

The upper cut-off limit is 2450 feet radio altitude.

The lower cut-off limit is 10 feet radio altitude.

## MODE 2 : EXCESSIVE TERRAIN CLOSURE RATE



2A — Flaps not in landing configuration and aircraft not on the glide slope beam.

Penetration of the boundary lights up the GPWS lights and sounds the repeated aural alert: "TERRAIN".

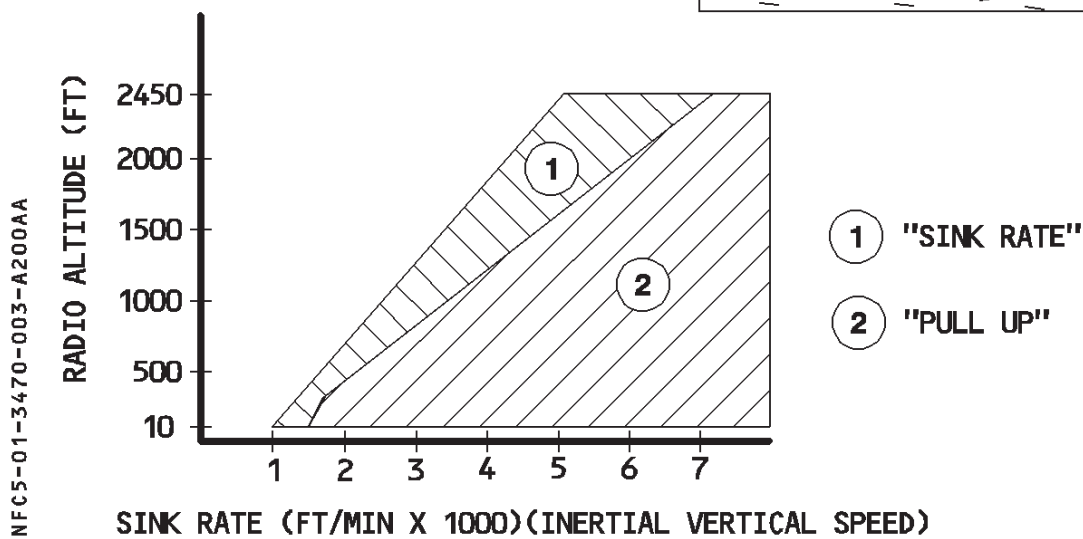
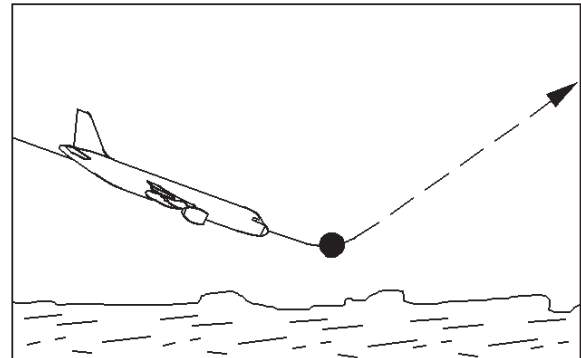
After "TERRAIN" has sounded twice, the warning switches to "PULL UP", repeated continually until the aircraft leaves the warning envelope.

After the aircraft leaves the boundary, the GPWS lights stay on and the voice message "TERRAIN" persists. These alerts cease when the aircraft increases either the barometric or inertial altitude by 300 feet. If it enters another alert region during this altitude-gain time, then the whole process begins again with a new reference altitude for the 300 feet altitude gain.

\* Upper cut-off limit varies from 1650 feet to 2450 feet radio altitude, depending on speed (between 220 knots to 310 knots). At certain airports, the upper boundary may be lowered down to 1250 feet to reduce the warning sensitivity and minimize the nuisance warnings.

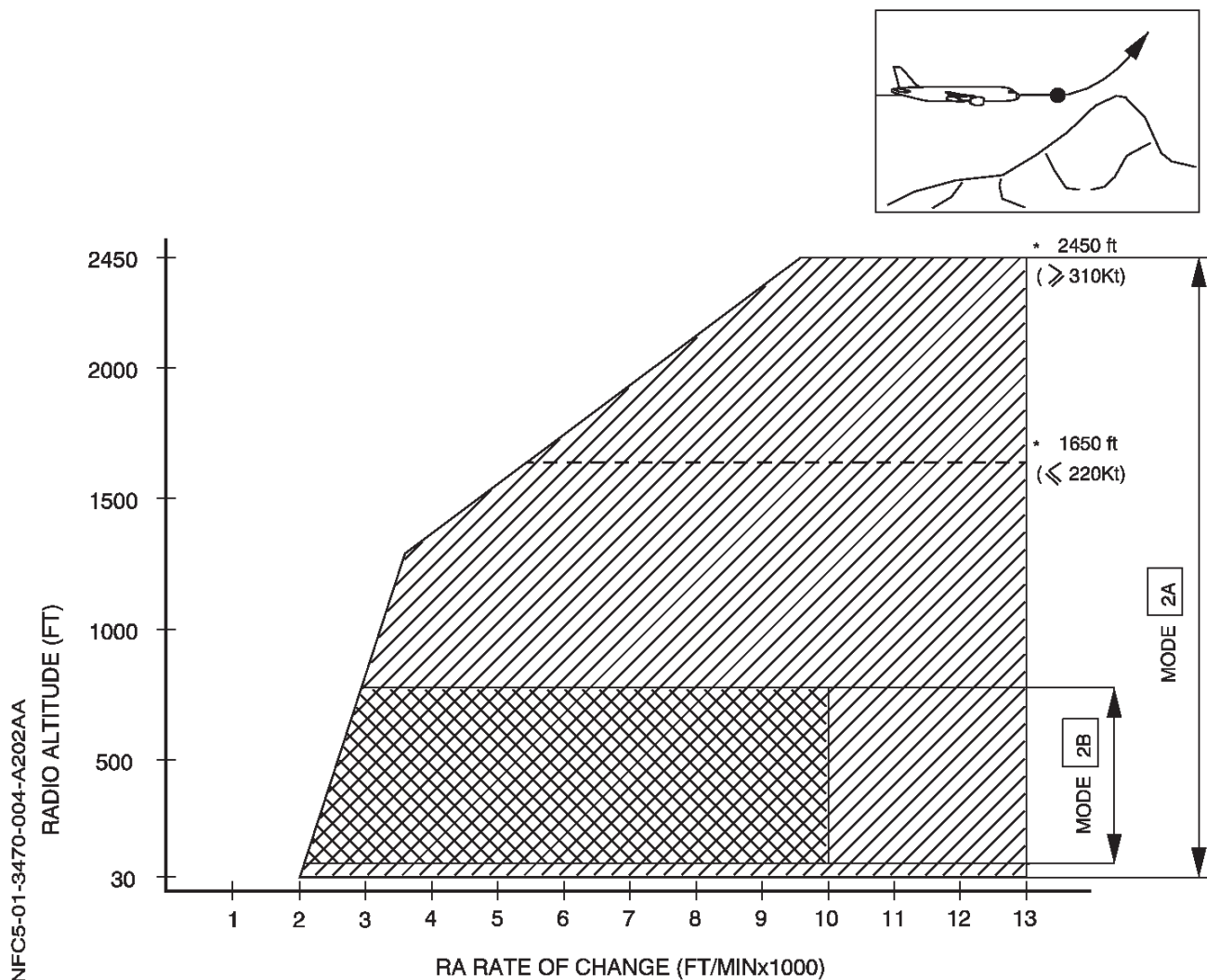
## GPWS BASIC MODES

### MODE 1: EXCESSIVE RATE OF DESCENT



- R Mode 1 has two boundaries. Penetration of the first boundary generates a repeated aural alert "SINK RATE" and causes the GPWS lights to come on. Penetration of the second boundary generates a repetitive "PULL UP" and illuminates both PULL UP lights. The upper cut-off limit is 2450 feet radio altitude. The lower cut-off limit is 10 feet radio altitude.

## MODE 2 : EXCESSIVE TERRAIN CLOSURE RATE



2A — Flaps not in landing configuration and aircraft not on the glide slope beam.

Penetration of the boundary lights up the GPWS lights and sounds the repeated aural alert: "TERRAIN".

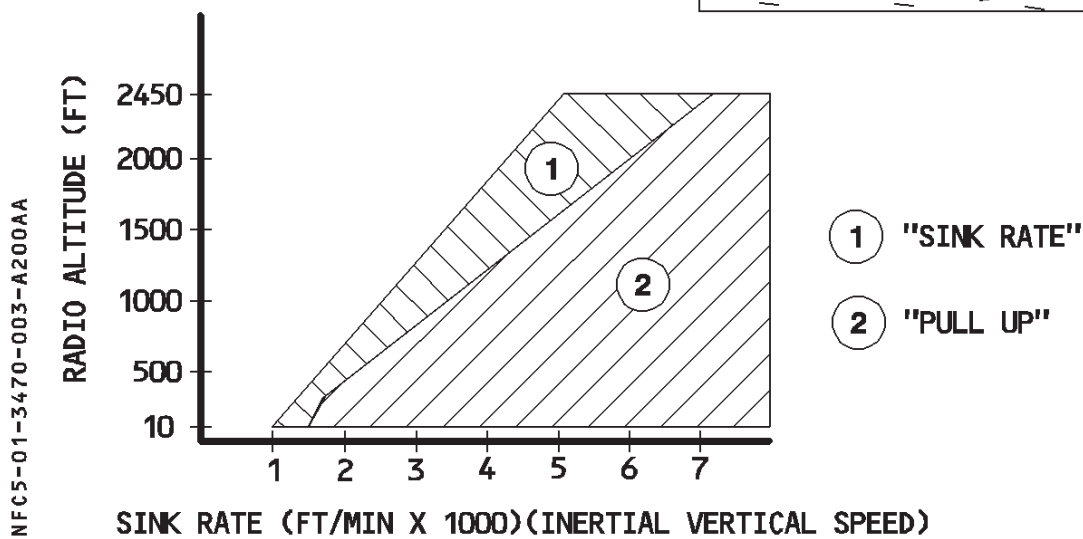
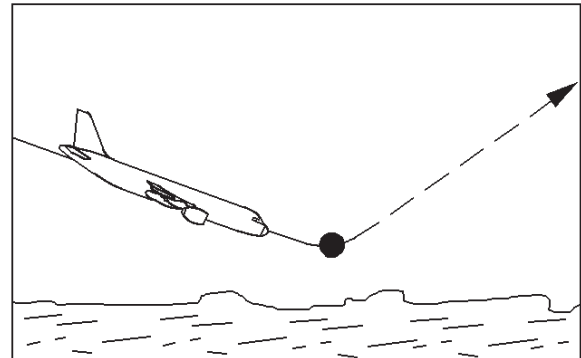
After "TERRAIN" has sounded twice, the warning switches to "PULL UP", repeated continually until the aircraft leaves the warning envelope. In addition, the PULL UP lights come on.

After the aircraft leaves the boundary, the PULL UP lights stay on and the voice message "TERRAIN" persists. These alerts cease when the aircraft increases either the barometric or inertial altitude by 300 feet. If it enters another alert region during this altitude-gain time, then the whole process begins again with a new reference altitude for the 300 feet altitude gain.

\* Upper cut-off limit varies from 1650 feet to 2450 feet radio altitude, depending on speed (between 220 knots to 310 knots). At certain airports, the upper boundary may be lowered down to 1250 feet to reduce the warning sensitivity and minimize the nuisance warnings.

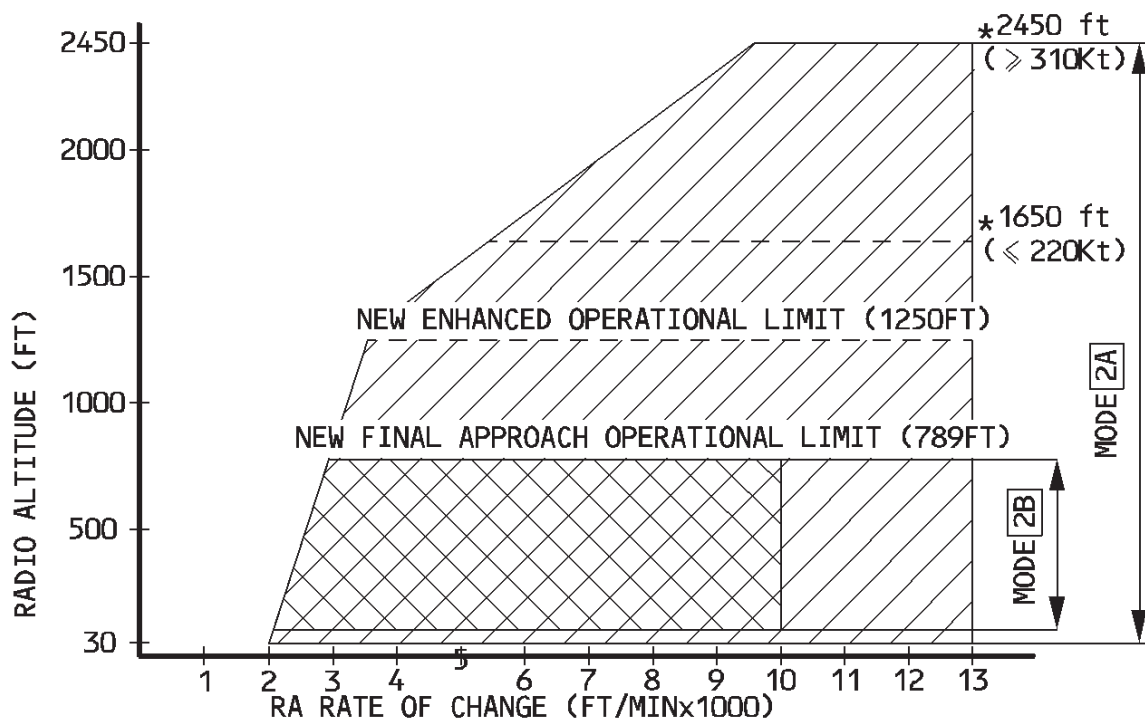
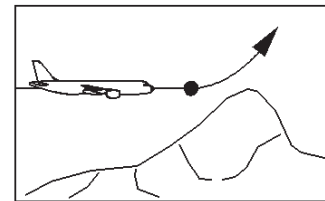
## GPWS BASIC MODES

### MODE 1: EXCESSIVE RATE OF DESCENT



- R Mode 1 has two boundaries. Penetration of the first boundary generates a repeated aural alert "SINK RATE" and causes the GPWS lights to come on. Penetration of the second boundary generates a repetitive "PULL UP" and illuminates both PULL UP lights. The upper cut-off limit is 2450 feet radio altitude. The lower cut-off limit is 10 feet radio altitude.

## MODE 2 : EXCESSIVE TERRAIN CLOSURE RATE



2A — Flaps not in landing configuration, and aircraft not on the glideslope beam.

R Penetration of the boundary causes the GPWS lights to come on, and generates the repeated aural alert: "TERRAIN".

R After "TERRAIN" has sounded twice, the warning switches to "PULL UP", repeated continually until the aircraft leaves the warning envelope. In addition, the PULL UP lights come on.

R After the aircraft leaves the boundary, the PULL UP lights stay on and the "TERRAIN" aural message persists. These alerts stop when the aircraft increases either the barometric or inertial altitude by 300 feet. If it enters another alert region during this altitude-gain time, then the whole process begins again with a new reference altitude for the 300 feet altitude gain.

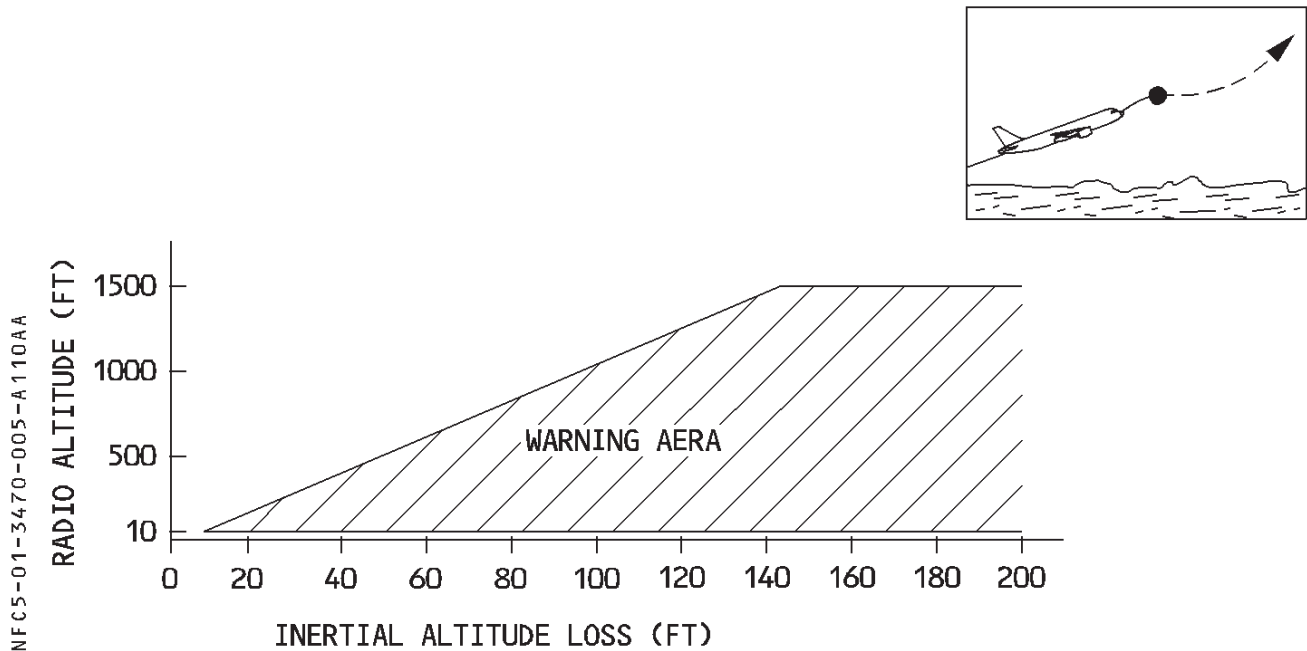
\* The upper cut-off limit varies from 1650 to 2450 feet radio altitude, depending on speed (between 220 to 310 knots). At certain airports, the upper boundary may be lowered down to 1250 feet to reduce the warning sensitivity and minimize the nuisance warnings.

The new enhanced operational upper limit is reduced to 1250 feet and to 789 feet in final approach, when the enhanced functions and the Geometric Altitude are of high integrity.

## 2B — Flaps in landing configuration

Lowering the flaps to the landing position, automatically switches GPWS to Mode 2B. In this case, lower boundary varies between 200 feet and 600 feet depending on altitude rate. In ILS approach (glide slope deviation  $< \pm 2$  dots) the lower boundary is fixed at 30 feet. When the aircraft enters the envelope, the alert is the same as for mode 2A. When gear and flaps are in the landing configuration, the aural message is "TERRAIN" only, and is not followed by "PULL UP" if the aircraft remains in the envelope.

## MODE 3 : ALTITUDE LOSS AFTER TAKEOFF



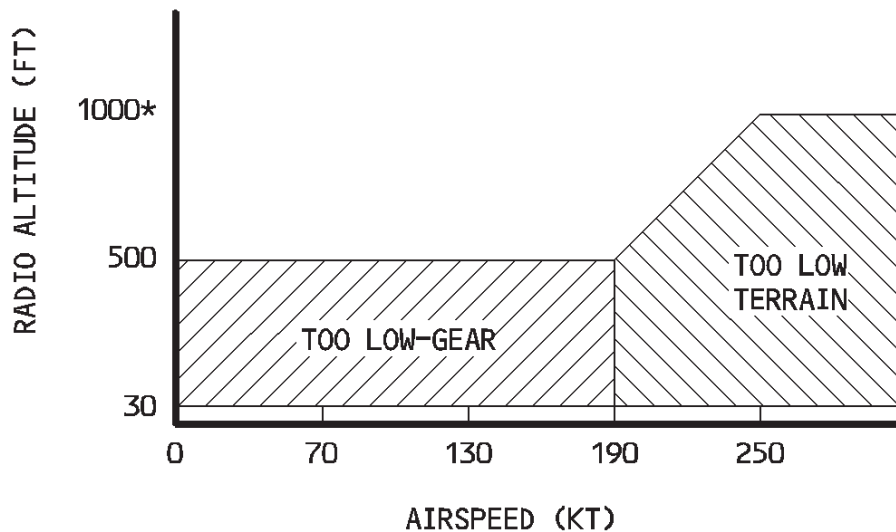
If the aircraft descends during the initial takeoff climb or during a go around, GPWS lights come on and the aural alert "DON'T SINK" sounds repeatedly.

The lower cut-off limit is 10 feet radio altitude.

Mode 3 is desensitized according to the time accumulated after departure and the radio altitude.

## MODE 4 : UNSAFE TERRAIN CLEARANCE WHEN NOT IN LANDING CONFIGURATION

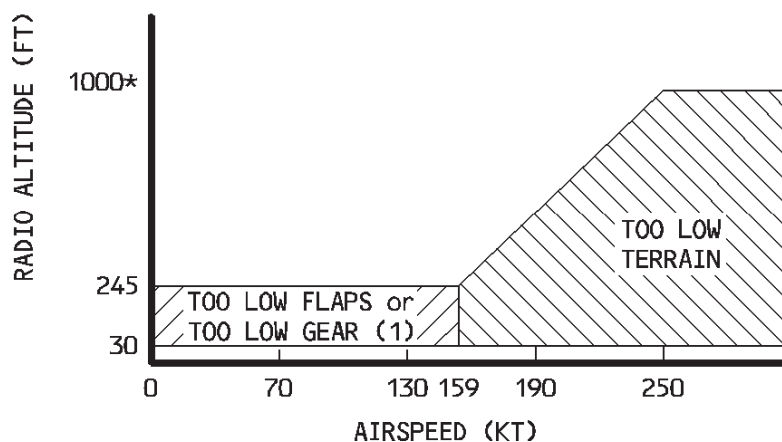
4A - Landing gear up and flaps not in landing configuration.



NFC5-01-3470-006-A001AA

R Two aural warnings may be triggered, depending on the area : "TOO LOW-GEAR" or "TOO LOW-TERRAIN". In addition, the GPWS light come on.

4B - Landing gear down, and flaps not in landing configuration or, Landing gear up, and flaps in landing configuration.



NFC5-01-3470-006-B001AA

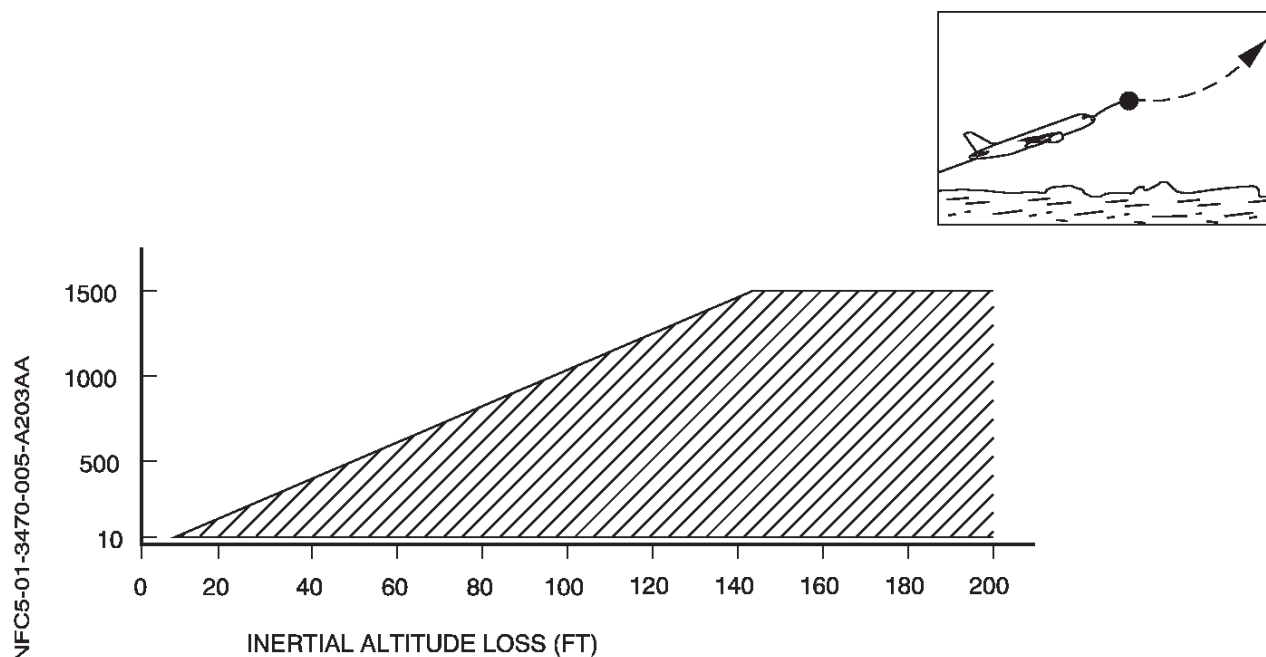
- \* : - OTHER MAXIMUMS ARE USED AT CERTAIN AIRPORTS TO MINIMIZE NUISANCE WARNINGS.  
 - THIS MAXIMUM IS ALSO REDUCED TO 800 FEET, WHEN AN OVERFLIGHT IS DETECTED  
 (1) "TOO LOW FLAPS" (If flaps not in landing configuration)  
 or  
 "TOO LOW GEAR" (If GEAR not extended)

R Three aural warnings may be generated, depending on the area and configuration : "TOO LOW-GEAR", "TOO LOW-FLAPS" or "TOO LOW-TERRAIN". In addition, the GPWS lights  
 R come on.

## 2B — Flaps in landing configuration

Lowering the flaps to the landing position automatically switches GPWS to Mode 2B. In this case lower boundary varies between 200 feet and 600 feet depending on altitude rate. In ILS approach (glide slope deviation  $< \pm 2$  dots) the lower boundary is fixed at 30 feet. When the aircraft enters the envelope, the alert is the same as for mode 2A. When gear and flaps are in the landing configuration, the aural message is "TERRAIN" only and is not followed by "PULL UP" if the aircraft remains in the envelope.

## MODE 3 : ALTITUDE LOSS AFTER TAKEOFF



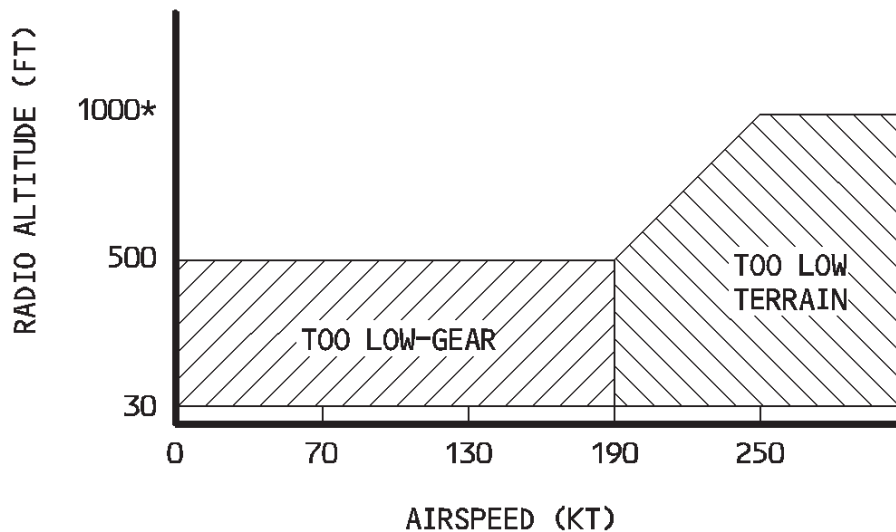
If the aircraft descends during the initial takeoff climb or during a go around, the GPWS lights come on and the aural alert "DON'T SINK" sounds repeatedly.

The lower cut-off limit is 10 feet radio altitude.

Mode 3 is desensitized according to the time accumulated after departure and the radio altitude.

## MODE 4 : UNSAFE TERRAIN CLEARANCE WHEN NOT IN LANDING CONFIGURATION

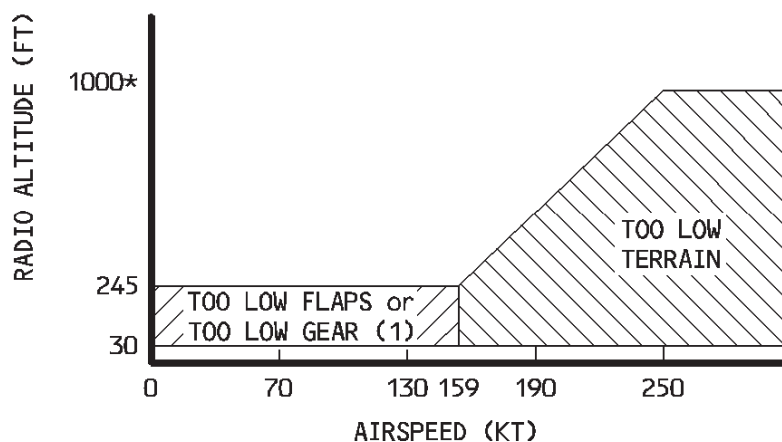
4A - Landing gear up and flaps not in landing configuration.



NFC5-01-3470-006-A001AA

R Two aural warnings may be triggered, depending on the area : "TOO LOW-GEAR" or "TOO LOW-TERRAIN". In addition, the GPWS light come on.

4B - Landing gear down, and flaps not in landing configuration or, Landing gear up, and flaps in landing configuration.



NFC5-01-3470-006-B001AA

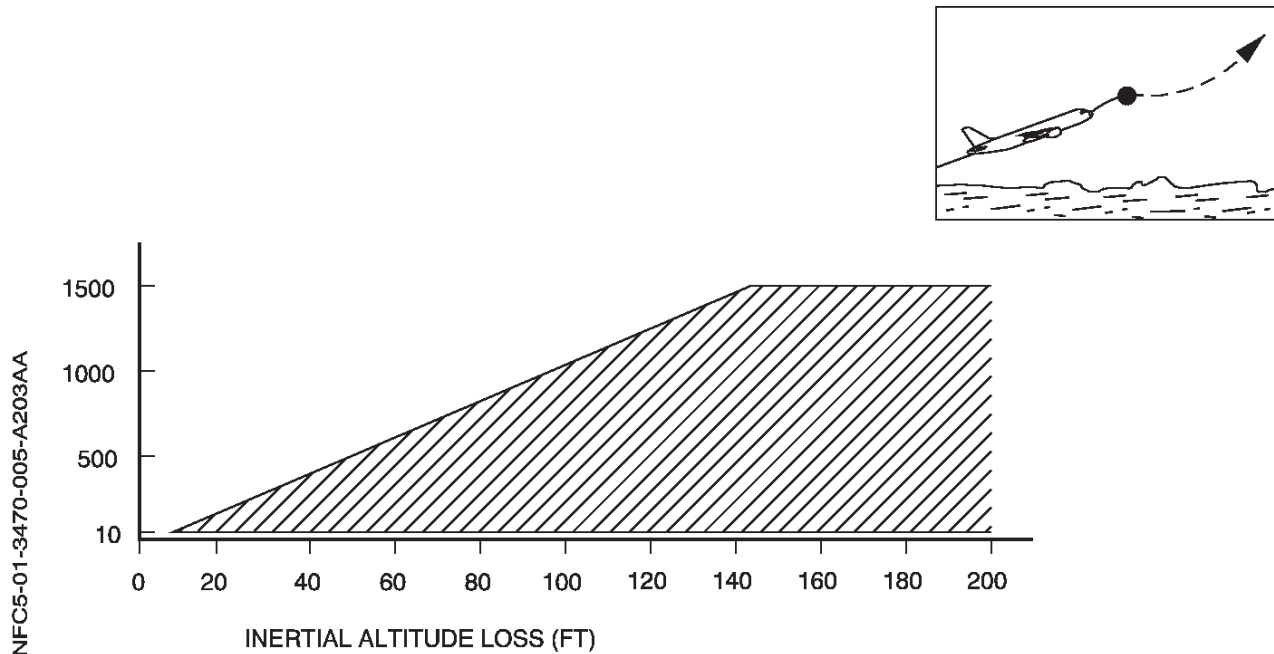
- \* : - OTHER MAXIMUMS ARE USED AT CERTAIN AIRPORTS TO MINIMIZE NUISANCE WARNINGS.  
 - THIS MAXIMUM IS ALSO REDUCED TO 800 FEET, WHEN AN OVERFLIGHT IS DETECTED  
 (1) "TOO LOW FLAPS" (If flaps not in landing configuration)  
 or  
 "TOO LOW GEAR" (If GEAR not extended)

R Three aural warnings may be generated, depending on the area and configuration : "TOO LOW-GEAR", "TOO LOW-FLAPS" or "TOO LOW-TERRAIN". In addition, the GPWS lights  
 R come on.

## 2B — Flaps in landing configuration

Lowering the flaps to the landing position automatically switches GPWS to Mode 2B. In this case lower boundary varies between 200 feet and 600 feet depending on altitude rate. In ILS approach (glide slope deviation  $< \pm 2$  dots) the lower boundary is fixed at 30 feet. When the aircraft enters the envelope, the alert is the same as for mode 2A. When gear and flaps are in the landing configuration, the aural message is "TERRAIN" only and is not followed by "PULL UP" if the aircraft remains in the envelope.

## MODE 3 : ALTITUDE LOSS AFTER TAKEOFF



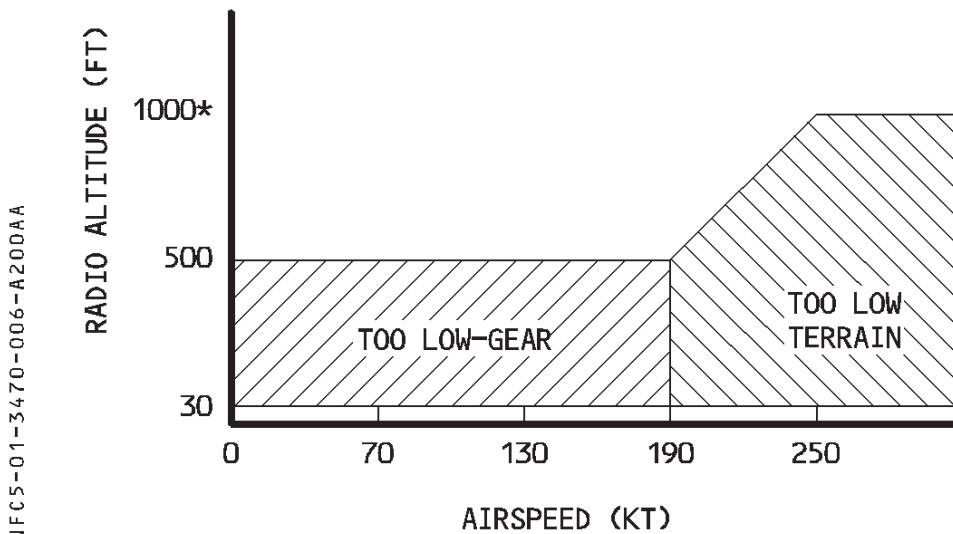
If the aircraft descends during the initial takeoff climb or during a go around, the GPWS lights come on and the aural alert "DON'T SINK" sounds repeatedly.

The lower cut-off limit is 10 feet radio altitude.

Mode 3 is desensitized according to the time accumulated after departure and the radio altitude.

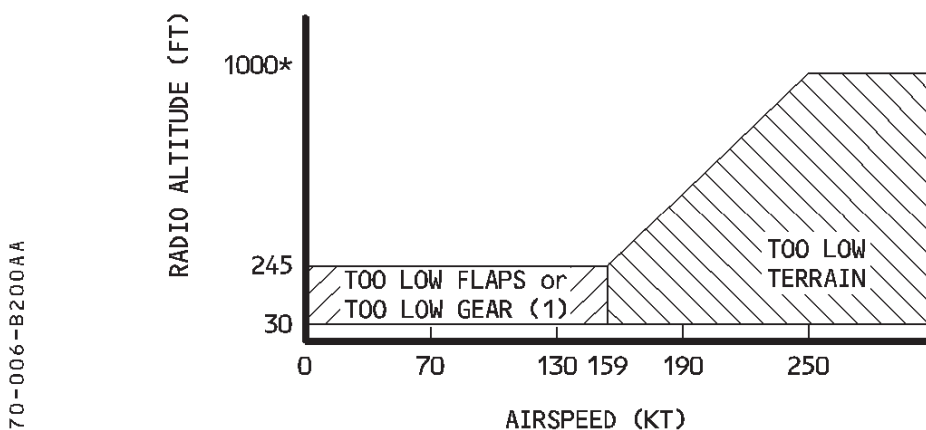
## MODE 4 : UNSAFE TERRAIN CLEARANCE WHEN NOT IN LANDING CONFIGURATION

4A - Landing gear up and flaps not in landing configuration.



R Two aural warnings may be triggered, depending on the area : "TOO LOW-GEAR" or "TOO LOW-TERRAIN". In addition, the GPWS lights come on.

4B - Landing gear down and flaps not in landing configuration or, Landing gear up, and flaps in landing configuration.



\* : OTHER MAXIMUMS ARE USED AT CERTAIN AIRPORTS TO MINIMIZE NUISANCE WARNING.

(1) "TOO LOW FLAPS" (If flaps not in landing configuration)

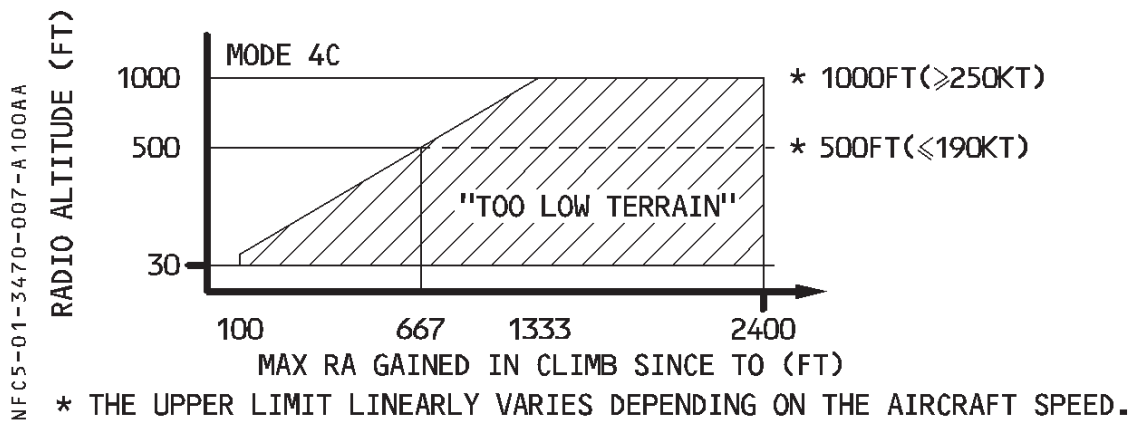
or

"TOO LOW GEAR" (If GEAR not extended)

R Three aural warnings may be generated, depending on the area and configuration : "TOO LOW-GEAR", "TOO LOW-FLAPS", or "TOO LOW-TERRAIN". In addition, the GPWS lights come on.

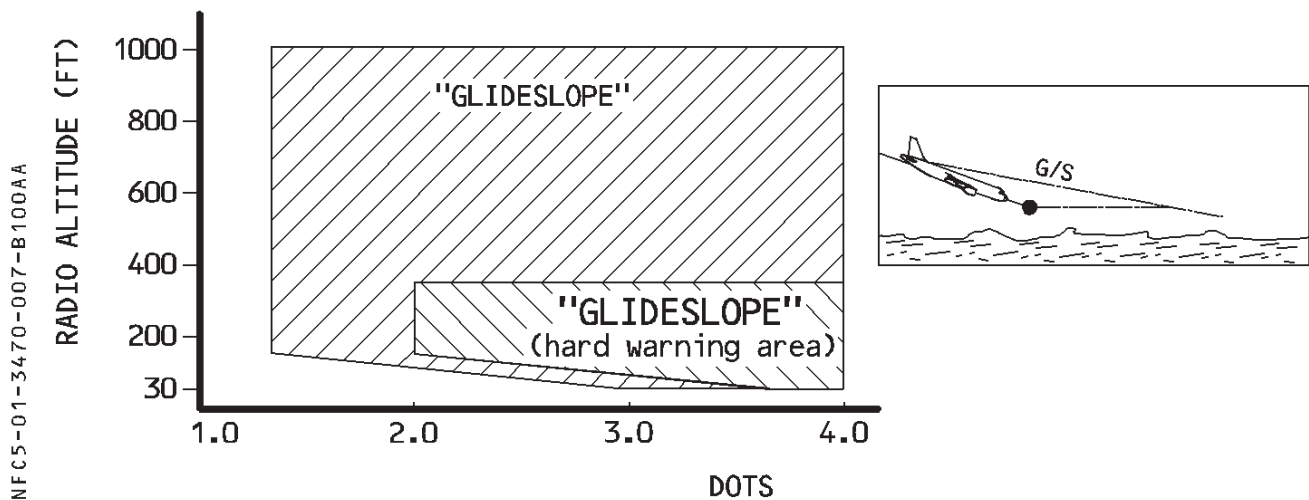
If the enhanced GPWS functions and the geometric altitude are of high integrity, the upper operational limit is reduced to 500 feet. If not, it is only reduced to 800 feet when an overflight is detected.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, which varies according to aircraft altitude, nearest runway altitude, distance to the nearest runway threshold, ground speed, and turn rate. When the boundary of these envelopes conflicts with the terrain, memorized in the database, the system generates the relevant alert :

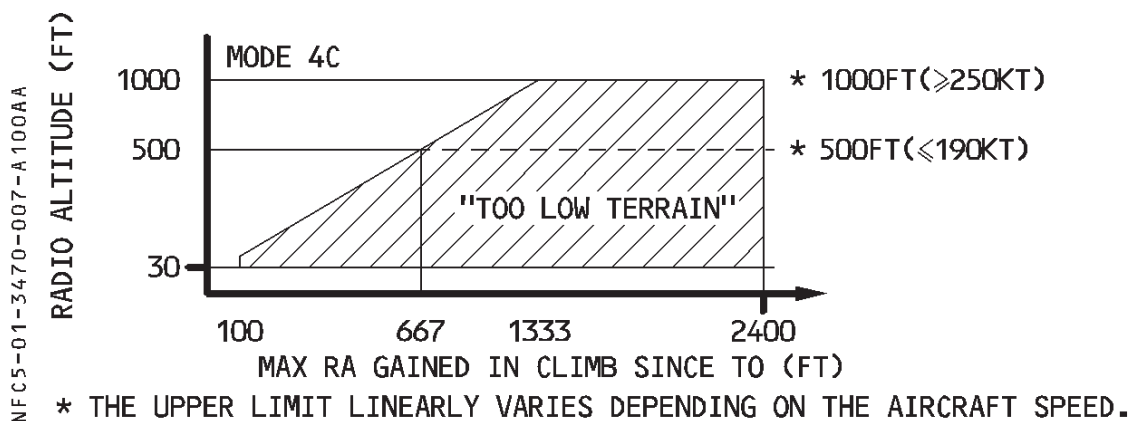
| Alert Level | Aural Warning             | ND (refer to 1.31.45)                                                                                                                             | Local Warning                                           |
|-------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Warning     | TERRAIN AHEAD,<br>PULL UP | <ul style="list-style-type: none"> <li>– Automatic terrain display*</li> <li>– Solid red areas</li> <li>– TERR AHEAD (red)</li> </ul>             | The pb light comes on, on each pilot's instrument panel |
| Caution     | TERRAIN AHEAD             | <ul style="list-style-type: none"> <li>– Automatic terrain display pop-up*</li> <li>– Solid yellow areas</li> <li>– TERR AHEAD (amber)</li> </ul> |                                                         |

- \* When the TERR ON ND pushbutton is selected ON, and ARC or ROSE mode is selected, the terrain is displayed on the ND. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat. (see 1.31.45, INDICATIONS ON ND). If an alert is generated (caution or warning) when TERR ON ND is not selected, the terrain will be automatically displayed and the ON light of the TERR ON ND pushbutton will come on.

Note : 1. When TERR ON ND is selected, the weather radar image is not displayed.  
 2. The relative height of the aircraft is computed using the Captain's baro setting. Thus, the Terrain Awareness Display (TAD) does not protect against baro setting errors.  
 3. The TAD and Terrain Clearance Floor (TCF) functions operate using the FMS 1 position. Thus, the system does not protect against FMS 1 position error.

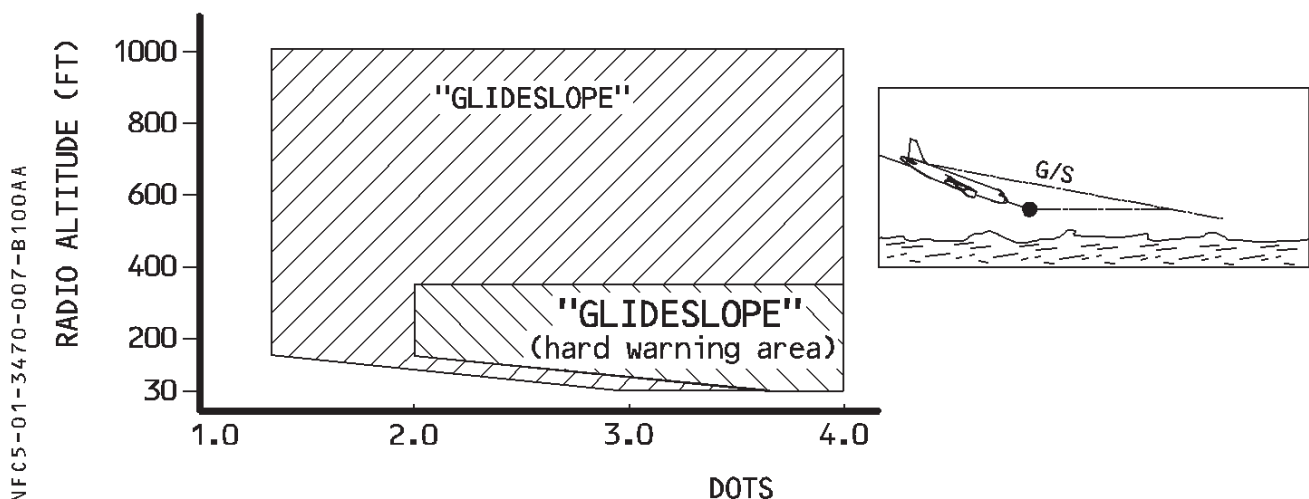
If the crew identifies that navigation accuracy is low, then they must select the enhanced modes off via the TERR pushbutton. The 5 GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights  
R come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, depending on the aircraft altitude, the nearest runway altitude, the distance to the nearest runway threshold, the ground speed, and the turn rate. When the boundary of these envelopes conflicts with the terrain, memorized in the database, the system generates the relevant alert :

R

| Alert Level | Aural Warning              | ND (refer to 1.31.45)                                                                                                              | Local Warning                                            |
|-------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Warning     | TERRAIN TERRAIN<br>PULL UP | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid red areas</li> <li>TERRAIN (red)</li> </ul>      | On each pilot's instrument panel, the pb light comes on. |
| Caution     | CAUTION TERRAIN            | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>TERRAIN (amber)</li> </ul> |                                                          |

- \* When the TERR ON ND pushbutton is set to ON, and ARC or ROSE mode is selected, the ND displays the terrain memorized in the database, depending on the aircraft's position. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat, (See 1.31.45 : INDICATIONS ON ND). If an alert is generated (caution or warning), and TERR ON ND is not selected, the terrain is automatically displayed, and the ON light of the TERR ON ND pushbutton comes on.

Note : 1. When TERR ON ND is selected, the weather radar image is not displayed.

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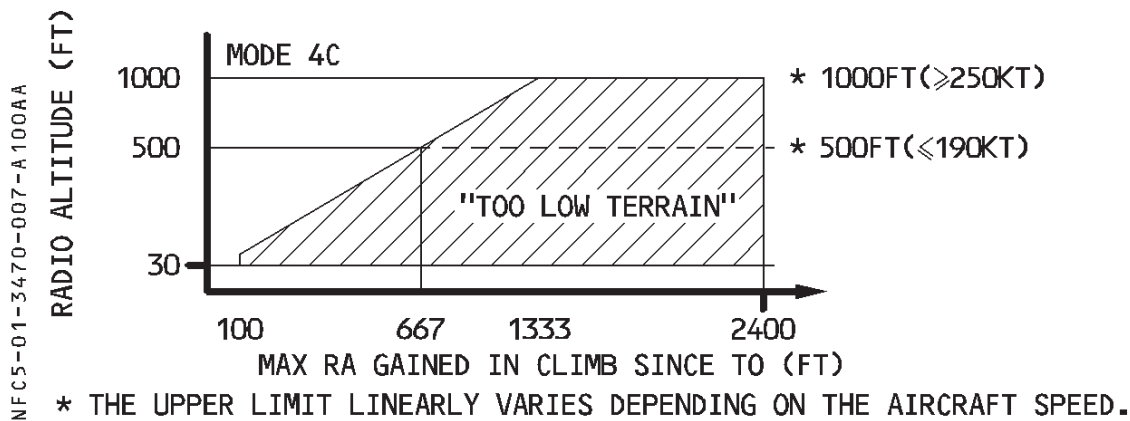
2. The Terrain Awareness Display (TAD) function computes the aircraft's relative altitude by using the Captain's baro setting information. Therefore, if the baro setting information is incorrect, the TAD function will also provide erroneous information and/or warnings

3. The TAD and Terrain Clearance Floor (TCF) functions use FMS 1 position information to perform their calculations. Therefore, in case of an FMS1 position error, the TAD and TCF will also provide erroneous information and/or warnings.

If the crew identifies that navigation accuracy is low, it must set the enhanced modes to off, via the TERR pushbutton. The 5 GPWS modes remain active.

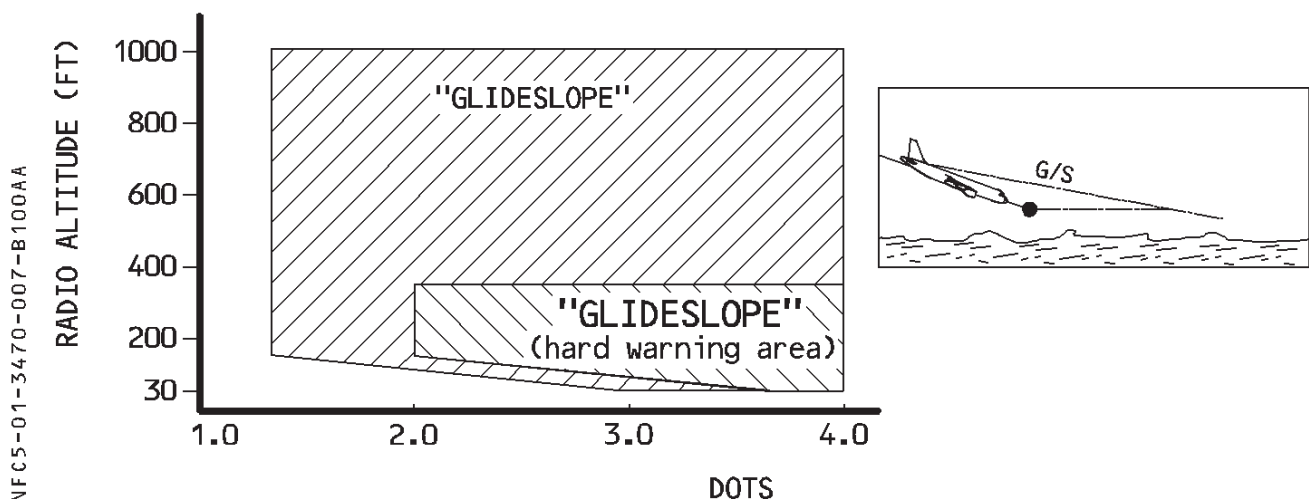
R

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, depending on the aircraft altitude, the nearest runway altitude, the distance to the nearest runway threshold, the ground speed, and the turn rate. When the boundary of these envelopes conflicts with the terrain memorized in the database, the system generates the relevant alert :

| Alert Level | Aural Warning             | ND (refer to 1.31.45)                                                                                                                      | Local Warning                                            |
|-------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Warning     | TERRAIN AHEAD,<br>PULL UP | <ul style="list-style-type: none"> <li>– Automatic terrain display*</li> <li>– Solid red areas</li> <li>– TERR AHEAD (red)</li> </ul>      | On each pilot's instrument panel, the pb light comes on. |
| Caution     | TERRAIN AHEAD             | <ul style="list-style-type: none"> <li>– Automatic terrain display*</li> <li>– Solid yellow areas</li> <li>– TERR AHEAD (amber)</li> </ul> |                                                          |

- \* When the TERR ON ND pushbutton is set to ON, and ARC or ROSE mode is selected, the ND displays the terrain memorized in the database, depending on the aircraft's position. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat , (See 1.31.45 : INDICATIONS ON ND). If an alert is generated (caution or warning), and TERR ON ND is not selected, the terrain and the obstacles are automatically displayed, and the ON light of the TERR ON ND pushbutton comes on.

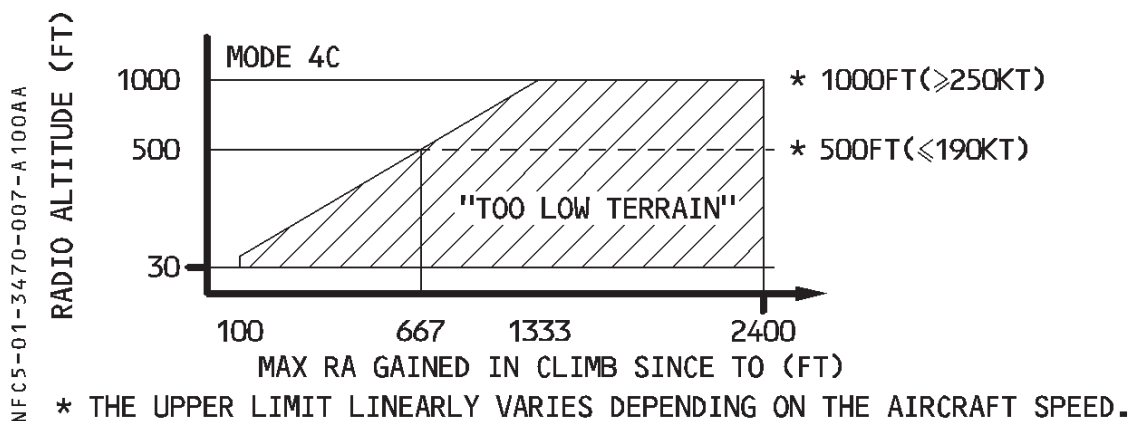
Note : 1. When TERR ON ND is selected, the weather radar image is not displayed.

2. The Geometric Altitude function can protect against certain baro setting errors, provided the components used to compute the Geometric Altitude are sufficiently valid and accurate.

3. The TAD and Terrain Clearance Floor (TCF) functions use FMS1 position information to perform their calculations. Therefore, in case of an FMS1 position error, the TAD and TCF will also provide erroneous information and/or warnings.

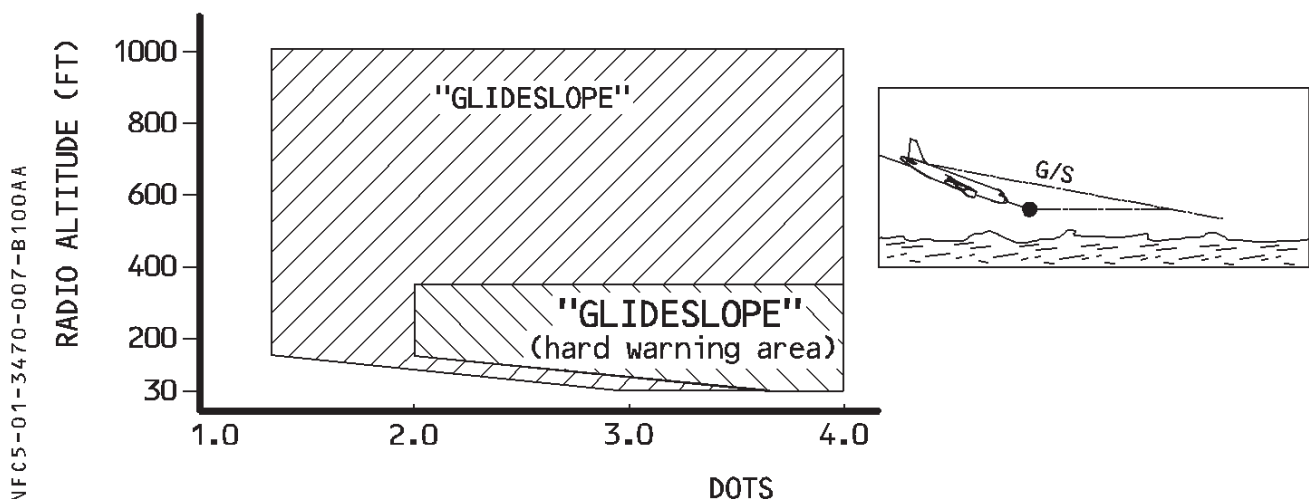
R If the crew identifies that navigation accuracy is low, it must set the enhanced modes to off, via the TERR pushbutton. The 5 GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights  
R come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, depending on the aircraft altitude, the nearest runway altitude, the distance to the nearest runway threshold, the ground speed, and the turn rate. When the boundary of these envelopes conflicts with the terrain, or with an obstacle memorized in the database, the system generates the relevant alert :

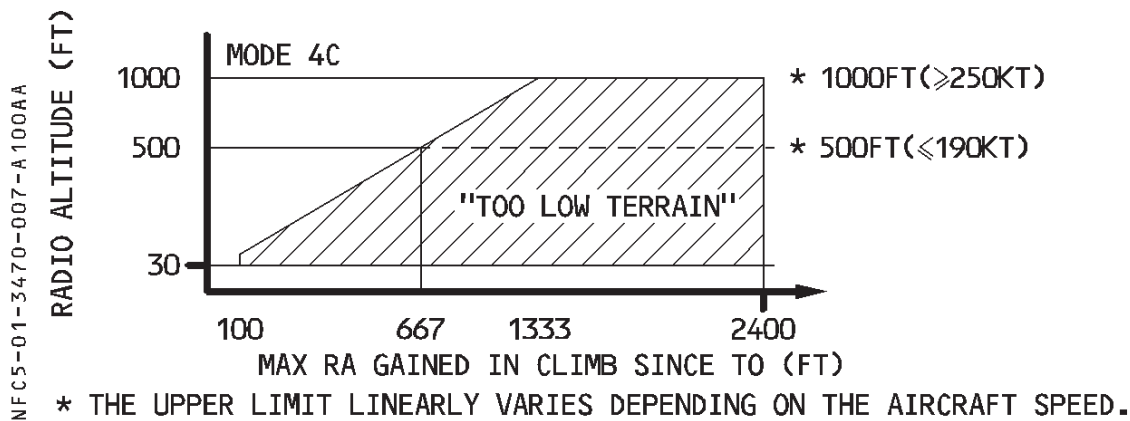
| Alert Level | Aural Warning                 | ND (refer to 1.31.45)                                                                                                              | Local Warning                                                    |
|-------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Warning     | TERRAIN TERRAIN,<br>PULL UP   | <ul style="list-style-type: none"> <li>Automatic terrain display*</li> <li>Solid red areas</li> <li>TERRAIN (red)</li> </ul>       | On each pilot's instrument panel, the pushbutton light comes on. |
|             | OBSTACLE OBSTACLE,<br>PULL UP | <ul style="list-style-type: none"> <li>Automatic terrain display*</li> <li>Solid red areas</li> <li>OBST (red)</li> </ul>          |                                                                  |
| Caution     | CAUTION TERRAIN               | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>TERRAIN (amber)</li> </ul> |                                                                  |
|             | CAUTION OBSTACLE              | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>OBST (amber)</li> </ul>    |                                                                  |

- \* When the TERR ON ND pushbutton is set to ON, and ARC or ROSE mode is selected, the ND displays the terrain and the obstacles memorized in the database, depending on the aircraft's position. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat (See 1.31.45 : INDICATIONS ON ND). If an alert is generated (caution or warning), and TERR ON ND is not selected, the terrain and the obstacles are automatically displayed, and the ON light of the TERR ON ND pushbutton comes on.

- Note :
1. When TERR ON ND is selected, the weather radar image is not displayed.
  2. The Terrain Awareness Display (TAD) function computes the aircraft's relative altitude by using the Captain's baro setting information. Therefore, if the baro setting information is incorrect, the TAD function will also provide erroneous information and/or warnings.
  3. The TAD and Terrain Clearance Floor (TCF) functions use FMS 1 position information to perform their calculations. Therefore, in case of an FMS 1 position error, the TAD and TCF will also provide erroneous information and/or warnings.

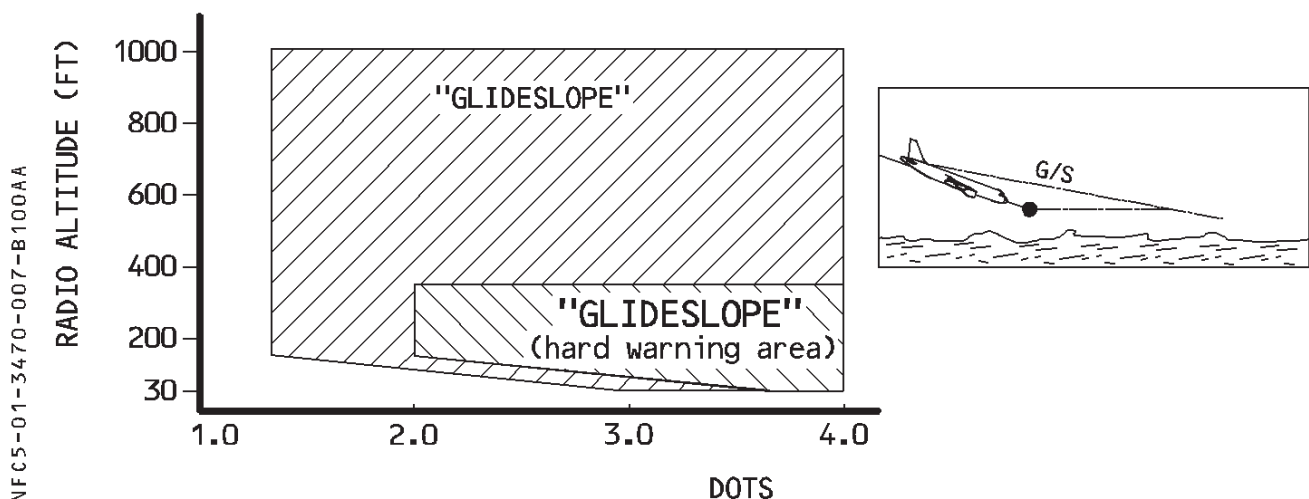
If the crew identifies that navigation accuracy is low, it must set the enhanced modes to off, via the TERR pushbutton. The 5 GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, depending on the aircraft altitude, the nearest runway altitude, the distance to the nearest runway threshold, the ground speed, and the turn rate. When the boundary of these envelopes conflicts with the terrain, or with an obstacle memorized in the database, the system generates the relevant alert :

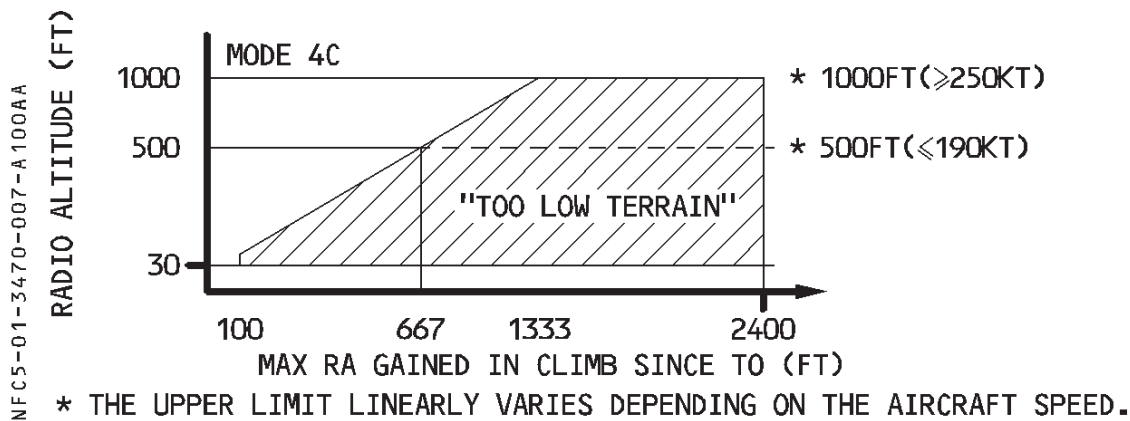
| Alert Level | Aural Warning                | ND (refer to 1.31.45)                                                                                                              | Local Warning                                                    |
|-------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Warning     | TERRAIN TERRAIN<br>PULL UP   | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid red areas</li> <li>TERRAIN (red)</li> </ul>      | On each pilot's instrument panel, the pushbutton light comes on. |
|             | OBSTACLE<br>OBSTACLE, PULLUP | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid red areas</li> <li>OBST (red)</li> </ul>         |                                                                  |
| Caution     | CAUTION TERRAIN              | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>TERRAIN (amber)</li> </ul> |                                                                  |
|             | CAUTION OBSTACLE             | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>OBST (amber)</li> </ul>    |                                                                  |

- \* When the TERR ON ND pushbutton is set to ON, and ARC or ROSE mode is selected, the ND displays the terrain and the obstacles memorized in the database depending on the aircraft's position. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat, (See 1.31.45 : INDICATIONS ON ND). If an alert is generated (caution or warning), and TERR ON ND is not selected, the terrain and the obstacles are automatically displayed, and the ON light of the TERR ON ND pushbutton comes on.

- Note :
1. When TERR ON ND is selected, the weather radar image is not displayed.
  2. The Geometric Altitude function can protect against certain baro setting errors, provided the components used to compute the Geometric Altitude are sufficiently valid and accurate.
  3. The TAD and Terrain Clearance Floor (TCF) functions use FMS1 position information to perform their calculations. Therefore, in case of an FMS1 position error, the TAD and TCF will also provide erroneous information and/or warnings.

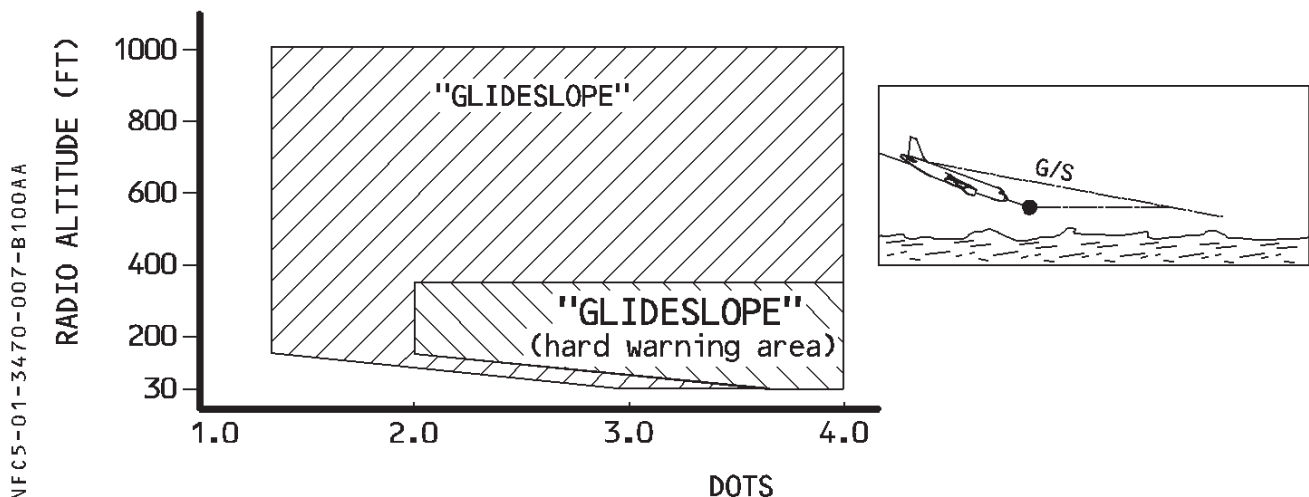
If the crew identifies that navigation accuracy is low, they must set the enhanced modes to off, via the TERR pushbutton. The 5 GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights  
R come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, depending on the aircraft altitude, the nearest runway altitude, the distance to the nearest runway threshold, the ground speed, and the turn rate. When the boundary of these envelopes conflicts with the terrain, or with an obstacle memorized in the database, the system generates the relevant alert :

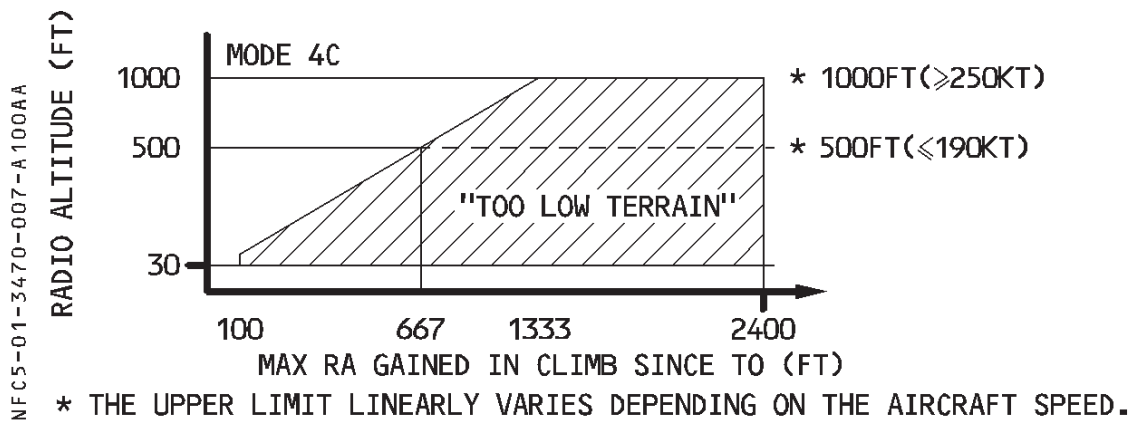
| Alert Level | Aural Warning              | ND (refer to 1.31.45)                                                                                                                 | Local Warning                                                    |
|-------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Warning     | TERRAIN AHEAD,<br>PULL UP  | <ul style="list-style-type: none"> <li>Automatic terrain display*</li> <li>Solid red areas</li> <li>TERR AHEAD (red)</li> </ul>       | On each pilot's instrument panel, the pushbutton light comes on. |
|             | OBSTACLE AHEAD,<br>PULL UP | <ul style="list-style-type: none"> <li>Automatic terrain display*</li> <li>Solid red areas</li> <li>OBST AHEAD (red)</li> </ul>       |                                                                  |
| Caution     | TERRAIN AHEAD              | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>TERR AHEAD (amber)</li> </ul> |                                                                  |
|             | OBSTACLE AHEAD             | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>OBST AHEAD (amber)</li> </ul> |                                                                  |

- \* When the TERR ON ND pushbutton is set to ON, and ARC or ROSE mode is selected, the ND displays the terrain and the obstacles memorized in the database, depending on the aircraft's position. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat (See 1.31.45 : INDICATIONS ON ND). If an alert is generated (caution or warning), and TERR ON ND is not selected, the terrain and the obstacles are automatically displayed, and the ON light of the TERR ON ND pushbutton comes on.

- Note :
1. When TERR ON ND is selected, the weather radar image is not displayed.
  2. The Geometric Altitude function can protect against certain baro setting errors, provided the components used to compute the Geometric Altitude are valid and accurate enough.
  3. The TAD and Terrain Clearance Floor (TCF) functions operate using the pure lateral GPS position and, the FMS 1 position as backup.

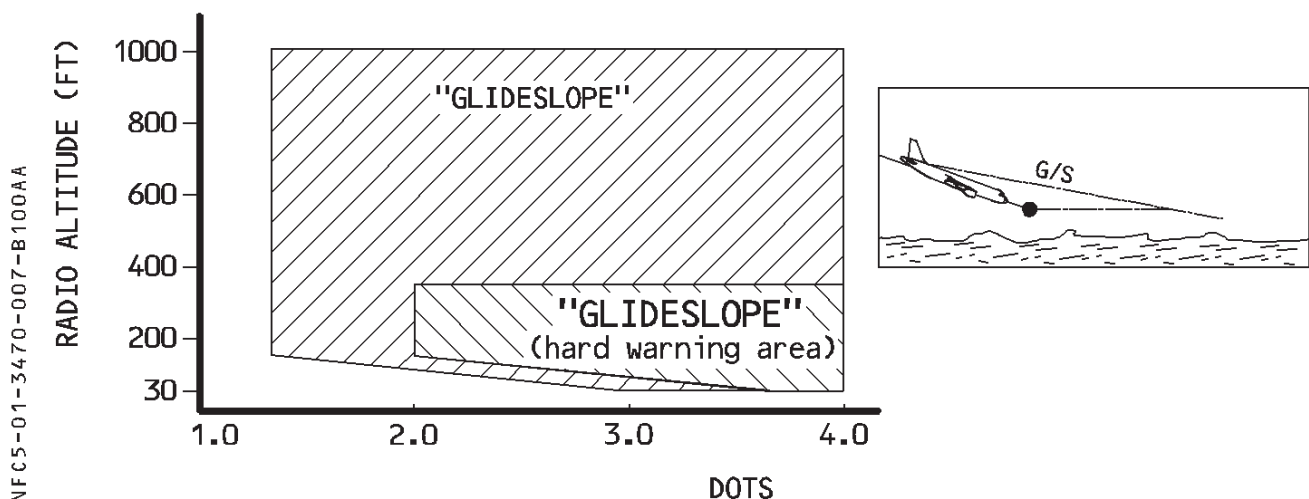
In case of low accuracy of the aircraft position computed by the EGPWS, the enhanced modes of the EGPWS are automatically deactivated. The five GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

R In both areas, the alert is a repeated "GLIDESLOPE" aural message, and the GPWS lights  
R come on.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, depending on the aircraft altitude, the nearest runway altitude, the distance to the nearest runway threshold, the ground speed, and the turn rate. When the boundary of these envelopes conflicts with the terrain, or with an obstacle memorized in the database, the system generates the relevant alert :

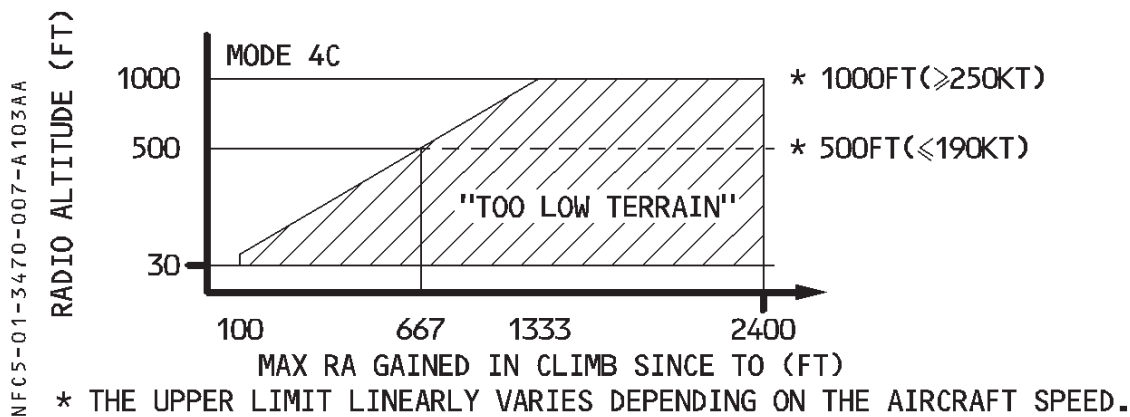
| Alert Level | Aural Warning                | ND (refer to 1.31.45)                                                                                                              | Local Warning                                                    |
|-------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Warning     | TERRAIN TERRAIN<br>PULL UP   | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid red areas</li> <li>TERRAIN (red)</li> </ul>      | On each pilot's instrument panel, the pushbutton light comes on. |
|             | OBSTACLE<br>OBSTACLE, PULLUP | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid red areas</li> <li>OBST (red)</li> </ul>         |                                                                  |
| Caution     | CAUTION TERRAIN              | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>TERRAIN (amber)</li> </ul> |                                                                  |
|             | CAUTION OBSTACLE             | <ul style="list-style-type: none"> <li>Automatic terrain display *</li> <li>Solid yellow areas</li> <li>OBST (amber)</li> </ul>    |                                                                  |

- \* When the TERR ON ND pushbutton is set to ON, and ARC or ROSE mode is selected, the ND displays the terrain and the obstacles memorized in the database depending on the aircraft's position. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat, (See 1.31.45 : INDICATIONS ON ND). If an alert is generated (caution or warning), and TERR ON ND is not selected, the terrain and the obstacles are automatically displayed, and the ON light of the TERR ON ND pushbutton comes on.

- Note :
1. When TERR ON ND is selected, the weather radar image is not displayed.
  2. The Geometric Altitude function can protect against certain baro setting errors, provided the components used to compute the Geometric Altitude are valid and accurate enough.
  3. The TAD and Terrain Clearance Floor (TCF) functions operate using the pure lateral GPS position and, the FMS 1 position as backup.

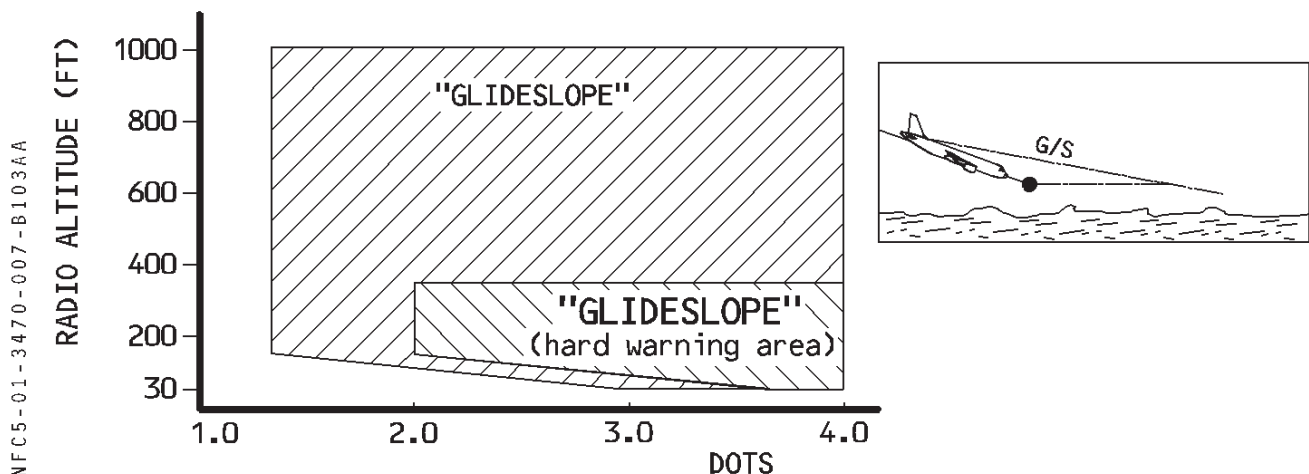
In case of low accuracy of the aircraft position computed by the EGPWS, the enhanced modes of the EGPWS are automatically deactivated. The five GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

In both areas, the alert is a repeated "GLIDESLOPE" aural message and lighting of both G/S lights.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, which varies according to aircraft altitude, nearest runway altitude, distance to the nearest runway threshold, ground speed, and turn rate. When the boundary of these envelopes conflicts with the terrain, memorized in the database, the system generates the relevant alert :

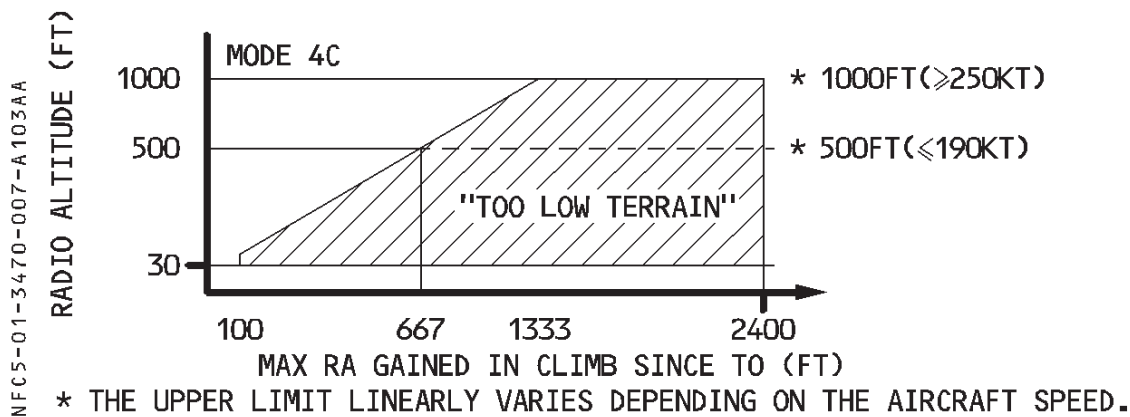
| Alert Level | Aural Warning             | ND (refer to 1.31.45)                                                                                                                             | Local Warning                                           |
|-------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Warning     | TERRAIN AHEAD,<br>PULL UP | <ul style="list-style-type: none"> <li>– Automatic terrain display*</li> <li>– Solid red areas</li> <li>– TERR AHEAD (red)</li> </ul>             | The pb light comes on, on each pilot's instrument panel |
| Caution     | TERRAIN AHEAD             | <ul style="list-style-type: none"> <li>– Automatic terrain display pop-up*</li> <li>– Solid yellow areas</li> <li>– TERR AHEAD (amber)</li> </ul> |                                                         |

- \* When the TERR ON ND pushbutton is selected ON, and ARC or ROSE mode is selected, the terrain is displayed on the ND. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat. (see 1.31.45, INDICATIONS ON ND). If an alert is generated (caution or warning) when TERR ON ND is not selected, the terrain will be automatically displayed and the ON light of the TERR ON ND pushbutton will come on.

Note : 1. When TERR ON ND is selected, the weather radar image is not displayed.  
 2. The relative height of the aircraft is computed using the Captain's baro setting. Thus, the Terrain Awareness Display (TAD) does not protect against baro setting errors.  
 3. The TAD and Terrain Clearance Floor (TCF) functions operate using the FMS 1 position. Thus, the system does not protect against FMS 1 position error.

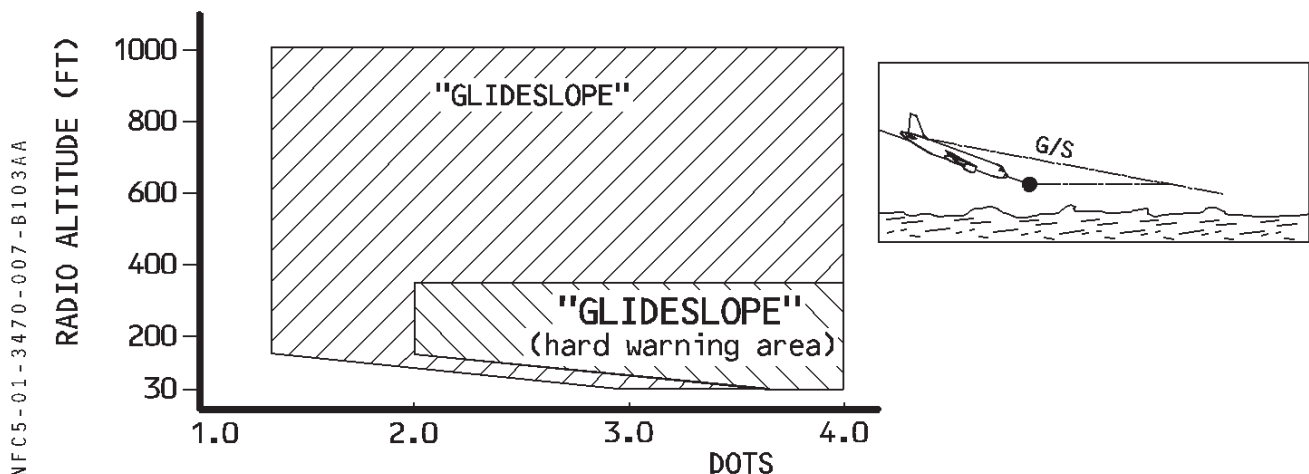
If the crew identifies that navigation accuracy is low, then they must select the enhanced modes off via the TERR pushbutton. The 5 GPWS modes remain active.

4C — Landing gear up, or flaps not in landing configuration.



If the aircraft starts an inadvertent controlled flight into the ground, during takeoff and climb, and penetrates the boundary, then the GPWS lights come on, and the "TOO LOW TERRAIN" aural alert sounds repeatedly.

### MODE 5 : DESCENT BELOW GLIDESLOPE



*Note : Normally, the glideslope alert is only triggered with the gear down. For a few airports, the gear down logic requirement is deleted and other upper limits are used to increase the warning envelope.*

In both areas, the alert is a repeated "GLIDESLOPE" aural message and lighting of both G/S lights.

The loudness of the aural message increases, when the aircraft enters the hard warning areas.

The mode is armed, when ILS 1 receives a valid signal.

Pressing the GPWS pushbutton cancels the warning. This is temporary ; the mode is automatically reactivated for a new envelope.

The upper cut-off limit is 1000 feet radio altitude.

The lower cut-off limit is 30 feet radio altitude.

## EGPWS FUNCTIONS

### TERRAIN AWARENESS AND DISPLAY

The Terrain Awareness and Display (TAD) function computes a caution and a warning envelope in front of the aircraft, which varies according to aircraft altitude, nearest runway altitude, distance to the nearest runway threshold, ground speed, and turn rate. When the boundary of these envelopes conflicts with the terrain memorized in the database, the system generates the relevant alert :

| Alert Level | Aural Warning             | ND (refer to 1.31.45)                                                                                                                       | Local Warning                                            |
|-------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Warning     | TERRAIN AHEAD,<br>PULL UP | <ul style="list-style-type: none"> <li>– Automatic terrain display *</li> <li>– Solid red areas</li> <li>– TERR AHEAD (red)</li> </ul>      | The pb light comes on, on each pilot's instrument panel. |
| Caution     | TERRAIN AHEAD             | <ul style="list-style-type: none"> <li>– Automatic terrain display *</li> <li>– Solid yellow areas</li> <li>– TERR AHEAD (Amber)</li> </ul> |                                                          |

- \* When the TERR ON ND pushbutton is selected ON, and ARC or ROSE mode is selected, the terrain is displayed on the ND. The terrain is displayed in various densities of green, yellow, red, or magenta, depending on the threat. (see 1.31.45, INDICATIONS ON ND). If an alert is generated (caution or warning) when TERR ON ND is not selected, the terrain will be automatically displayed and the ON light of the TERR ON ND pushbutton will come on.

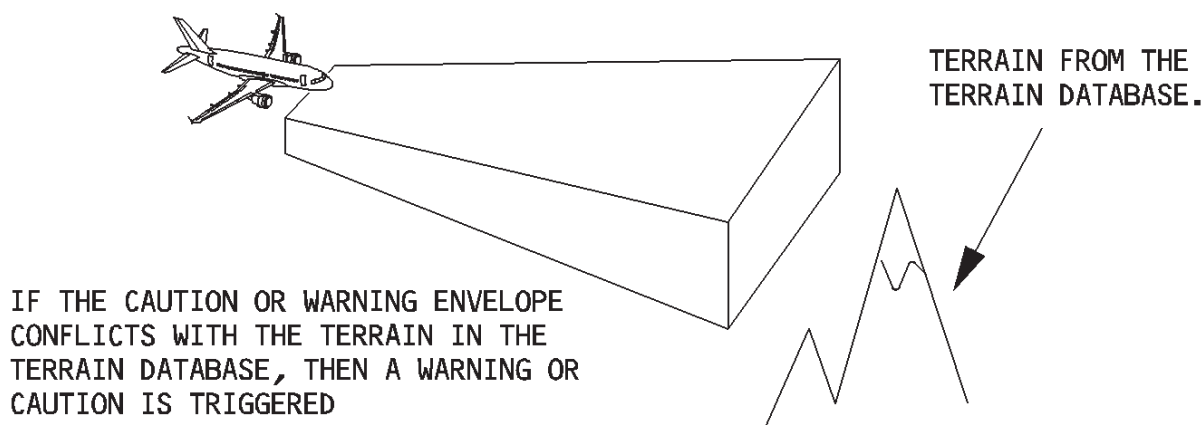
Note : 1. When TERR ON ND is selected, the weather radar image is not displayed.  
 2. The relative height of the aircraft is computed using the Captain's baro setting. Thus, the Terrain Awareness Display (TAD) does not protect against baro setting errors.  
 3. The TAD and Terrain Clearance Floor (TCF) functions operate using the FMS 1 position. Thus, the system does not protect against FMS 1 position error.

If the FMGS detects low navigation accuracy, then the enhanced modes of the EGPWS are automatically deactivated. The 5 GPWS modes remain active.

## TERRAIN CAUTION AND WARNING ENVELOPE

R

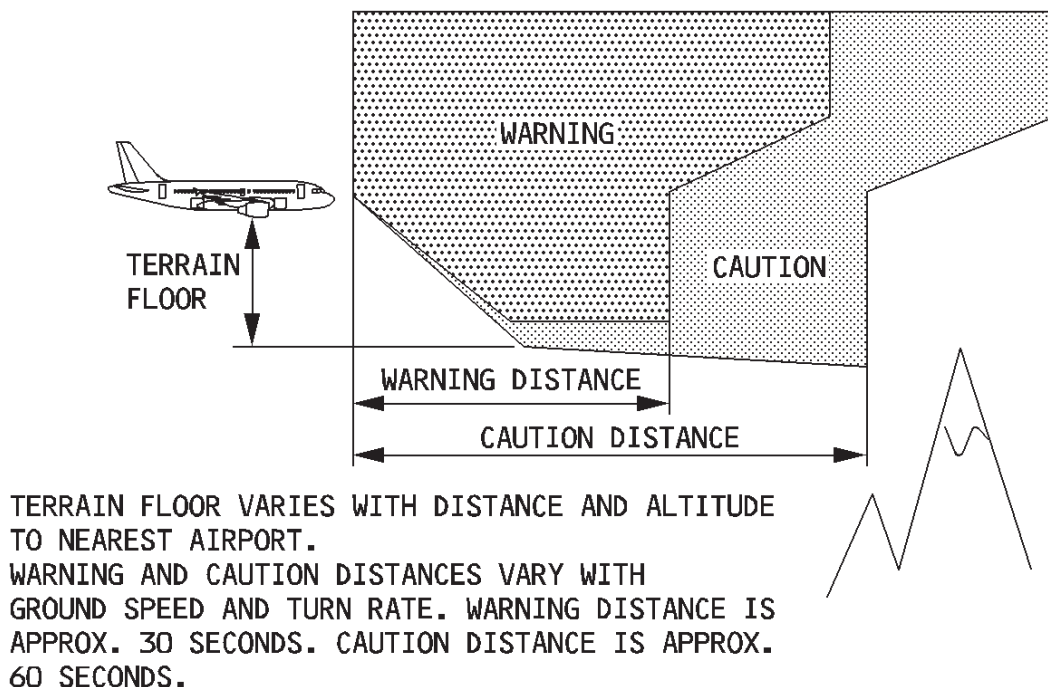
NFC5-01-3470-009-A100AA



## VERTICAL ENVELOPE

R

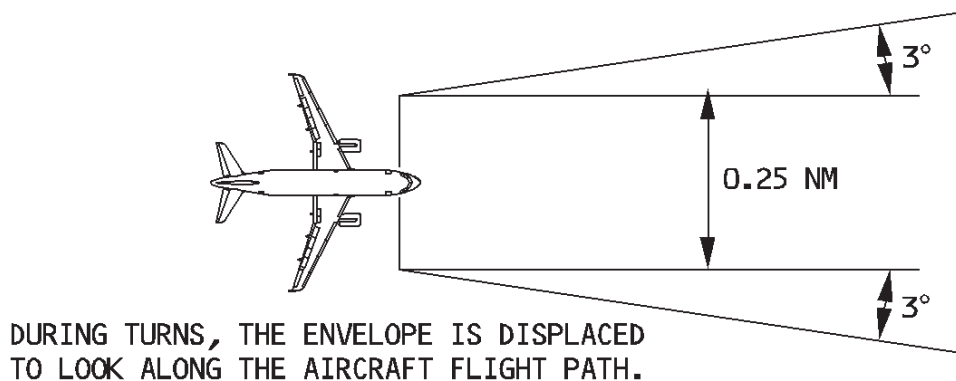
NFC5-01-3470-009-B100AA



## HORIZONTAL ENVELOPE

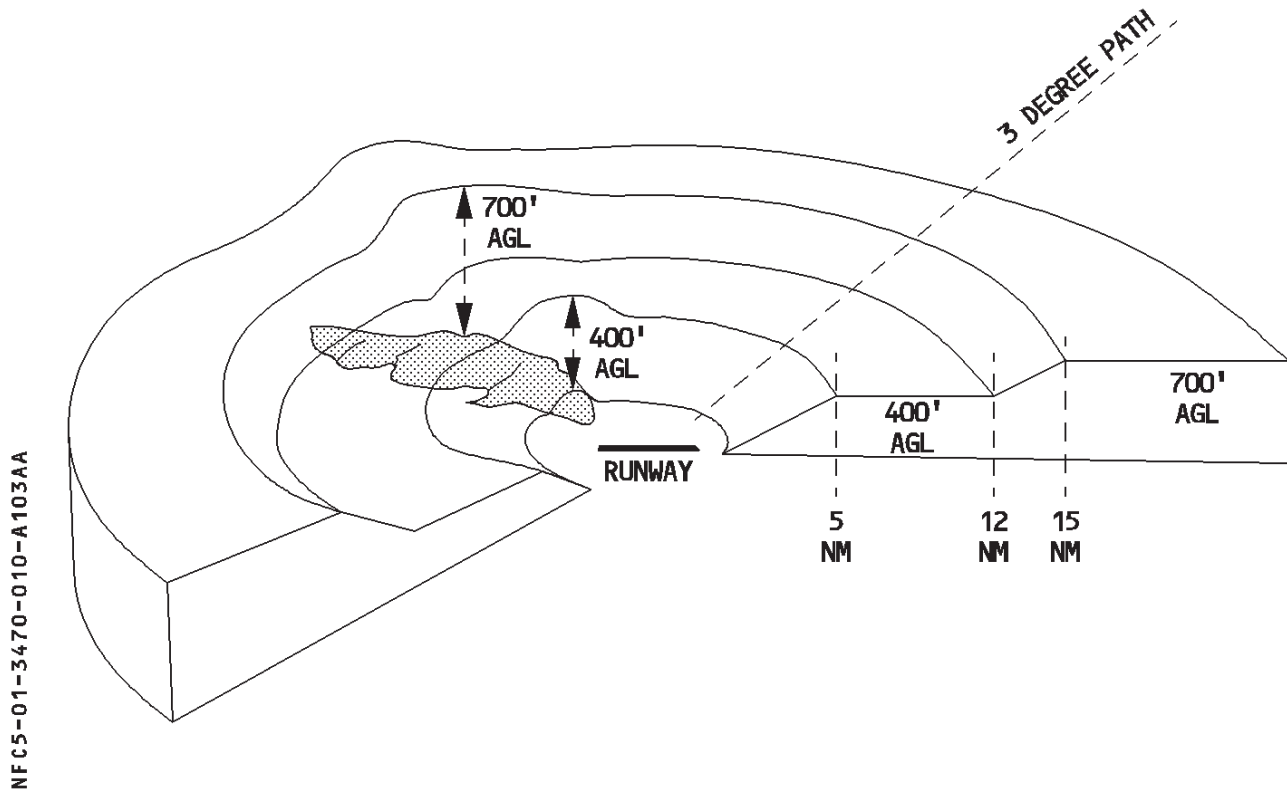
R

NFC5-01-3470-009-C100AA



## TERRAIN CLEARANCE FLOOR

A terrain clearance floor envelope is stored in the database for each runway for which terrain data exist. The Terrain Clearance Floor (TCF) function warns a premature descent below this floor, regardless of aircraft configuration.



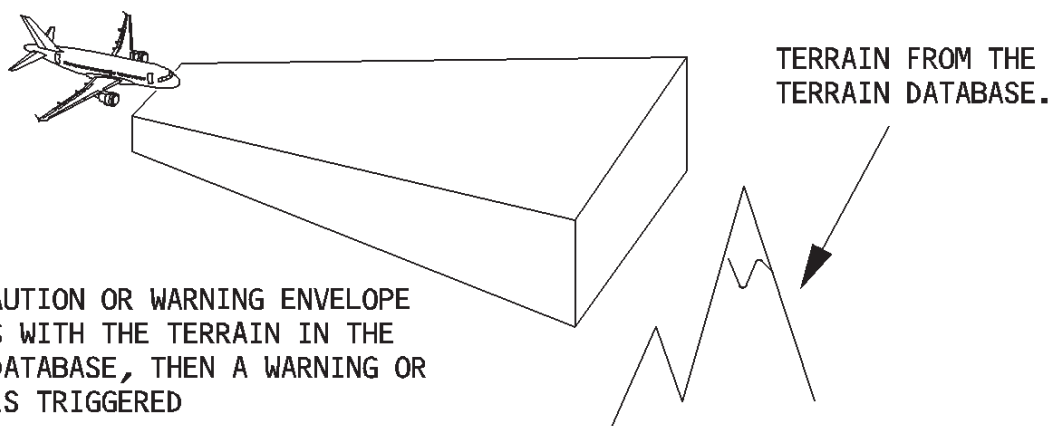
If the airplane descends below this floor, a TOO LOW TERRAIN aural warning is annunciated, and the pushbutton light comes on on glareshield.

## TERRAIN CAUTION AND WARNING ENVELOPE

R

NFC5-01-3470-009-A100AA

IF THE CAUTION OR WARNING ENVELOPE  
CONFLICTS WITH THE TERRAIN IN THE  
TERRAIN DATABASE, THEN A WARNING OR  
CAUTION IS TRIGGERED

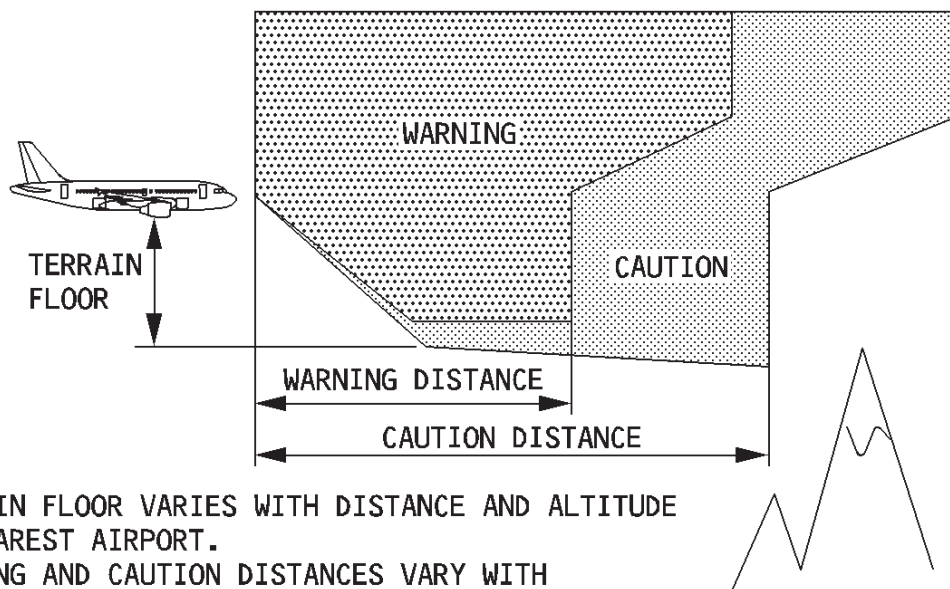


## VERTICAL ENVELOPE

R

NFC5-01-3470-009-B100AA

TERRAIN FLOOR VARIES WITH DISTANCE AND ALTITUDE  
TO NEAREST AIRPORT.  
WARNING AND CAUTION DISTANCES VARY WITH  
GROUND SPEED AND TURN RATE. WARNING DISTANCE IS  
APPROX. 30 SECONDS. CAUTION DISTANCE IS APPROX.  
60 SECONDS.

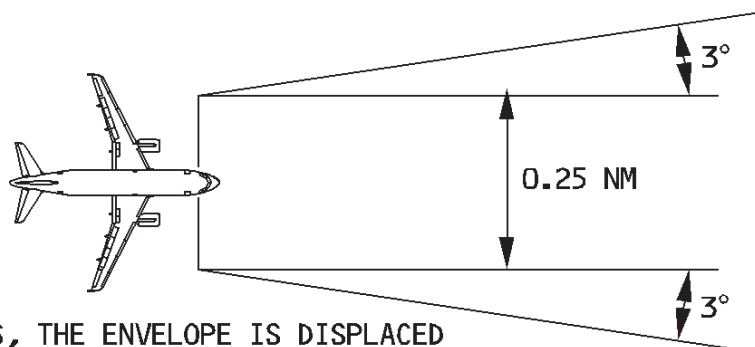


## HORIZONTAL ENVELOPE

R

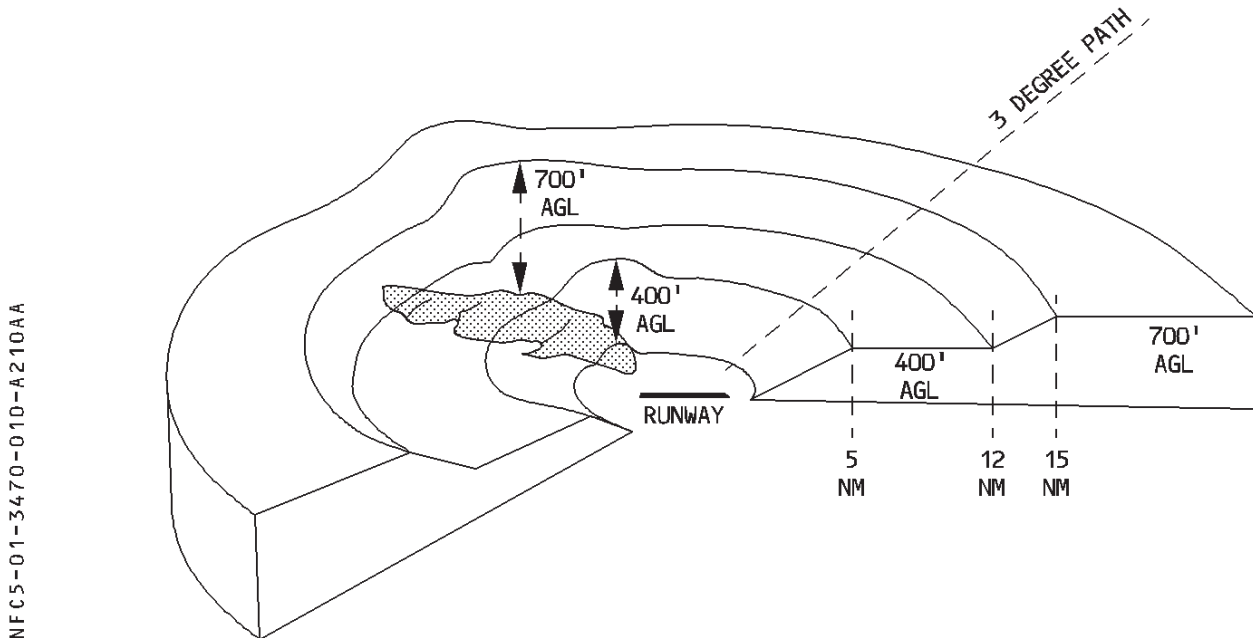
NFC5-01-3470-009-C100AA

DURING TURNS, THE ENVELOPE IS DISPLACED  
TO LOOK ALONG THE AIRCRAFT FLIGHT PATH.



## TERRAIN CLEARANCE FLOOR

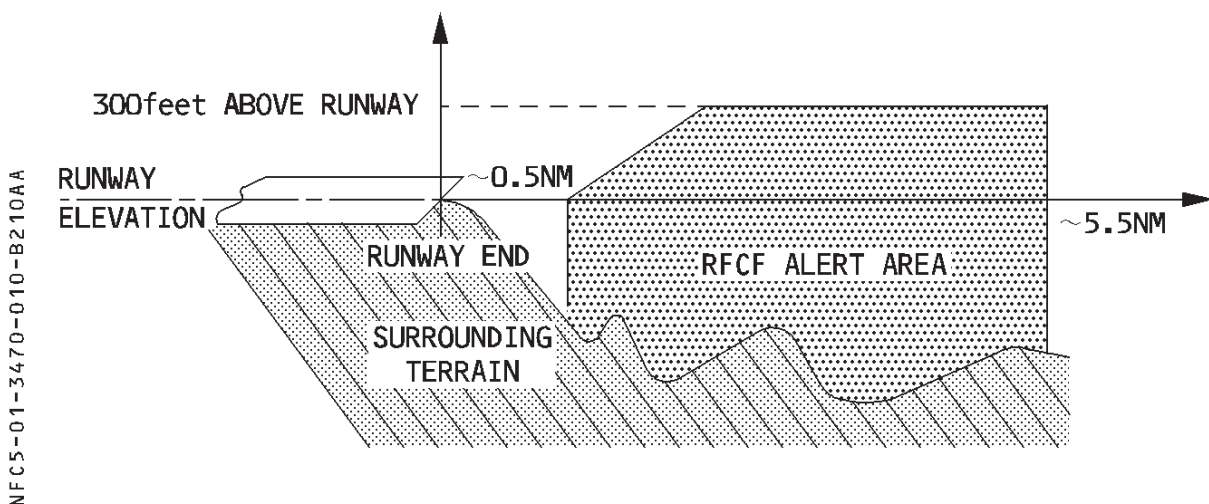
A terrain clearance floor envelope is stored in the database for each runway for which terrain data exist. The Terrain Clearance Floor (TCF) function warns of a premature descent below this floor, regardless of the aircraft's configuration.



If the airplane descends below this floor, a TOO LOW TERRAIN aural warning sounds and the pushbutton light comes on, on the glareshield.

## RUNWAY FIELD CLEARANCE FLOOR

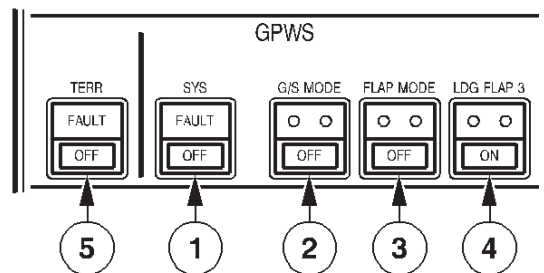
The Runway Field Clearance Floor (RFCF) provides an additional envelope protection, for runways that are significantly higher than the surrounding terrain. It is contained in a circle within the 5.5 NM of the runway threshold and it is based on the geometric altitude and the runway elevation.



## CONTROLS AND INDICATORS

### OVERHEAD PANEL

NFC5-01-3470-011-A 100AA



#### ① SYS pb

OFF : All basic GPWS alerts (Mode 1 to 5) are inhibited.

FAULT It : This amber light comes on, along with an ECAM caution, if the basic GPWS mode 1 to 5 malfunction.

*Note : If ILS 1 fails, only mode 5 is inhibited. Consequently, the FAULT light does not come on and the GPWS FAULT warning is not triggered.*

#### ② G / S MODE pb

OFF : Glideslope mode (mode 5) is inhibited.

#### ③ FLAP MODE pb

OFF : Flap mode ("TOO LOW FLAPS" mode 4) is inhibited.  
(To avoid nuisance warning in case of landing with flaps setting reduced).

#### ④ LDG FLAP 3 pb

ON : Flap mode is inhibited when FLAPS CONF 3 is selected (to avoid nuisance warning in case of landing in CONF 3).  
In this case, LDG MEMO displays FLAPS ... 3 instead of "CONF ... FULL".

#### ⑤ TERR pb

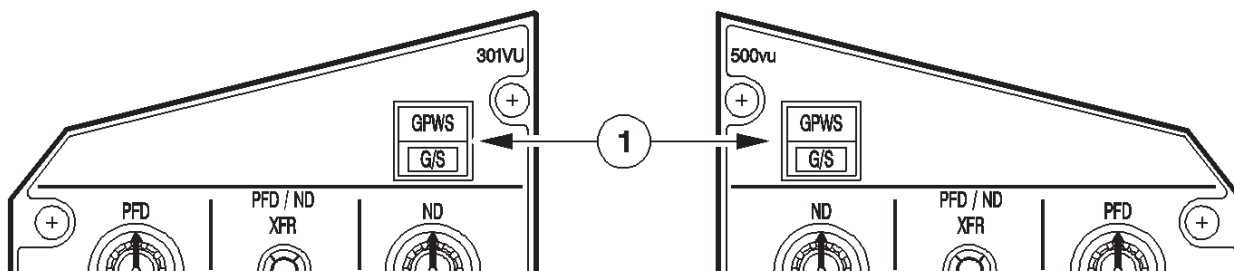
OFF : Inhibits the Terrain Awareness Display (TAD) and Terrain Clearance Floor (TCF) modes, and does not affect the basic GPWS mode 1 to 5. If OFF is selected the ECAM caution NAV GPWS TERR DET FAULT is displayed.

FAULT It : This amber light comes on, along with an ECAM caution, if the TAD or TCF mode fails. The terrain is not shown on the ND. The basic GPWS mode 1 to mode 5 are still operative if the SYS pushbutton OFF or FAULT lights are not illuminated.

R  
R  
R

## INSTRUMENT PANELS

NFC5-01-3470-012-A100AA



### ① GPWS – G / S pb

- R GPWS : This red light comes on when any mode from 1 to 4, or any TAD or TCF alert is activated. A specific voice alert accompanies it.
- G / S : This amber light comes on when mode 5 is activated. The aural "GLIDE SLOPE" warning accompanies it.

Note : If the flight crew presses this button briefly when a glide slope warning is on, the G/S light goes out and the "GLIDE SLOPE" aural warning (soft or loud) stops.

To test the GPWS, flight crew can push this button briefly :

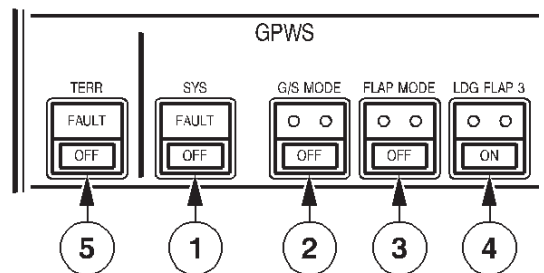
- In flight, above 2000 feet RA and below 8000 feet RA :
  - GPWS FAULT light comes on on the overhead panel
  - The soft "GLIDE SLOPE" aural warning sounds
  - The "PULL UP" aural warning sounds (once)
  - TERR FAULT light comes on
- R
  - the "TERRAIN AHEAD PULL UP" or "TERRAIN TERRAIN PULL UP" (⚠) aural warning sounds
  - the terrain self-test pattern is displayed on both NDs
  - The GPWS and G/S lights come on.
- On ground :
 

As above, plus pressing the switch either continually or during the "PULL UP" sequence, makes all aural warnings sound.

## CONTROLS AND INDICATORS

### OVERHEAD PANEL

NFC5-01-3470-011-A 100AA



#### ① SYS pb

OFF : All basic GPWS alerts (Mode 1 to 5) are inhibited.

FAULT It : This amber light comes on, along with an ECAM caution, if the basic GPWS mode 1 to 5 malfunction.

*Note : If ILS 1 fails, only mode 5 is inhibited. Consequently, the FAULT light does not come on and the GPWS FAULT warning is not triggered.*

#### ② G / S MODE pb

OFF : Glideslope mode (mode 5) is inhibited.

#### ③ FLAP MODE pb

OFF : Flap mode ("TOO LOW FLAPS" mode 4) is inhibited.  
(To avoid nuisance warning in case of landing with flaps setting reduced).

#### ④ LDG FLAP 3 pb

ON : Flap mode is inhibited when FLAPS CONF 3 is selected (to avoid nuisance warning in case of landing in CONF 3).  
In this case, LDG MEMO displays FLAPS ... 3 instead of "CONF ... FULL".

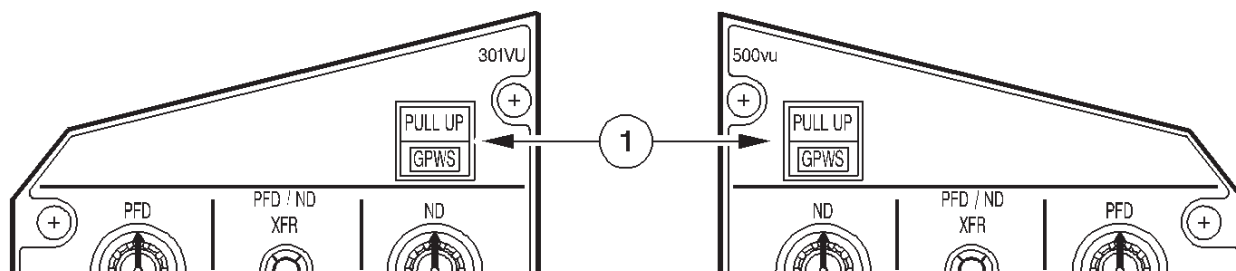
#### ⑤ TERR pb

OFF : Inhibits the Terrain Awareness Display (TAD) and Terrain Clearance Floor (TCF) modes, and does not affect the basic GPWS mode 1 to 5. If OFF is selected the ECAM caution NAV GPWS TERR DET FAULT is displayed.

FAULT It : This amber light comes on, along with an ECAM caution, if the TAD or TCF mode fails. The terrain is not shown on the ND. The basic GPWS mode 1 to mode 5 are still operative if the SYS pushbutton OFF or FAULT lights are not illuminated.

## INSTRUMENT PANELS

NFC5-01-3470-012-A200AA



### ① PULL UP – GPWS pb

- PULL UP** : Comes on red when the second boundary of Mode 1 is penetrated or when Mode 2, or any TAD or TCF alert is activated. It is accompanied by a corresponding voice alert.
- GPWS** : Comes on amber when any other mode is activated. It is accompanied by a corresponding voice warning.

Note : 1. If the flight crew briefly presses this button when one of these modes is activated, the GPWS light goes out and the voice warning stops.

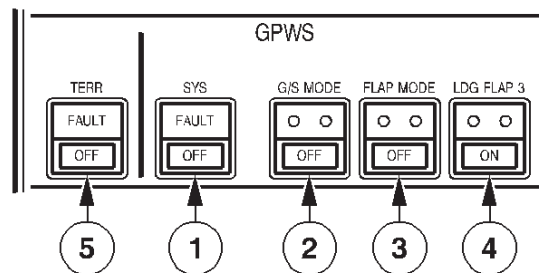
2. The GPWS can be tested by pressing this pushbutton. If the pushbutton is pressed briefly, some of the aural warnings sound and pushbutton captions, related to the GPWS, come on. If the pushbutton is pressed continuously, then all the aural warnings sound.

R  
R  
R  
R

## CONTROLS AND INDICATORS

### OVERHEAD PANEL

NFC5-01-3470-011-A 100AA



#### ① SYS pb

OFF : All basic GPWS alerts (Mode 1 to 5) are inhibited.

FAULT It : This amber light comes on, along with an ECAM caution, if the basic GPWS mode 1 to 5 malfunction.

*Note : If ILS 1 fails, only mode 5 is inhibited. Consequently, the FAULT light does not come on and the GPWS FAULT warning is not triggered.*

#### ② G / S MODE pb

OFF : Glideslope mode (mode 5) is inhibited.

#### ③ FLAP MODE pb

OFF : Flap mode ("TOO LOW FLAPS" mode 4) is inhibited.  
(To avoid nuisance warning in case of landing with flaps setting reduced).

#### ④ LDG FLAP 3 pb

ON : Flap mode is inhibited when FLAPS CONF 3 is selected (to avoid nuisance warning in case of landing in CONF 3).  
In this case, LDG MEMO displays FLAPS ... 3 instead of "CONF ... FULL".

#### ⑤ TERR pb

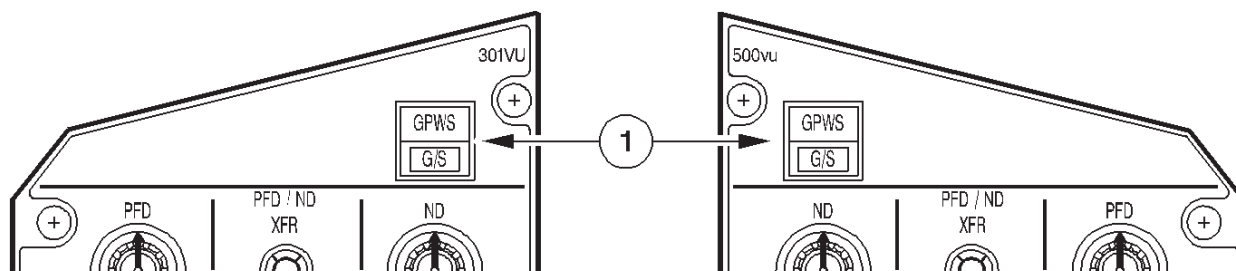
OFF : Inhibits the Terrain Awareness Display (TAD) and Terrain Clearance Floor (TCF) modes, and does not affect the basic GPWS mode 1 to 5. If OFF is selected the ECAM caution NAV GPWS TERR DET FAULT is displayed.

FAULT It : This amber light comes on, along with an ECAM caution, if the TAD or TCF mode fails. The terrain is not shown on the ND. The basic GPWS mode 1 to mode 5 are still operative if the SYS pushbutton OFF or FAULT lights are not illuminated.

R  
R  
R

## INSTRUMENT PANELS

NFC5-01-3470-012-A205AA



### ① GPWS – G / S pb

- R    GPWS    : This red light comes on when any mode from 1 to 4, or any TAD or TCF alert is activated. A specific voice alert accompanies it.
- G / S    : This amber light comes on when mode 5 is activated. The aural "GLIDE SLOPE" warning accompanies it.

Note : If the flight crew presses this button briefly when a glide slope warning is on, the G/S light goes out and the "GLIDE SLOPE" aural warning (soft or loud) stops.

To test the GPWS, flight crew can push this button briefly :

· In flight, above 2000 feet RA and below 5500 feet RA :

- GPWS FAULT light comes on on the overhead panel
- The soft "GLIDE SLOPE" aural warning sounds
- The "PULL UP" aural warning sounds (once)
- TERR FAULT light comes on

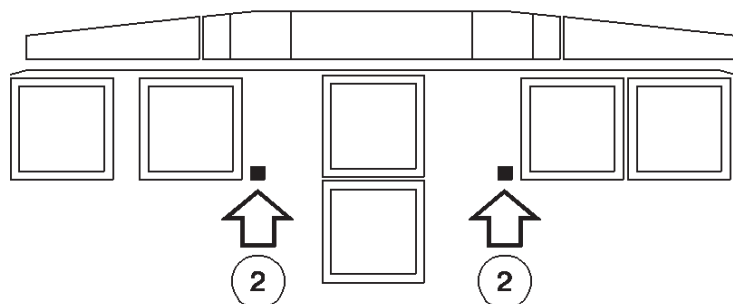
- R    – the "TERRAIN AHEAD PULL UP" or "TERRAIN TERRAIN PULL UP" (⚠) aural warning sounds
- The terrain self-test pattern is displayed on both NDs
  - The GPWS and G/S lights come on.

· On ground :

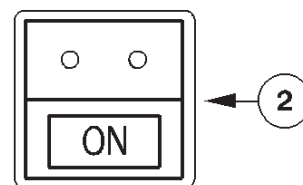
As above, plus pressing the switch either continually or during the "PULL UP" sequence, makes all aural warnings sound.

R

NFC5-01-3470-013-A100AA



TERR ON ND



## ② TERR ON ND pb sw

These switches are located on either side of the ECAM. Each switch controls the inside terrain display.

R  
R  
R  
R  
R

ON : The terrain is displayed on the ND if :

- TERR pushbutton is selected ON, and
- TERR FAULT light is not on, and
- the FMGS navigation accuracy is high.

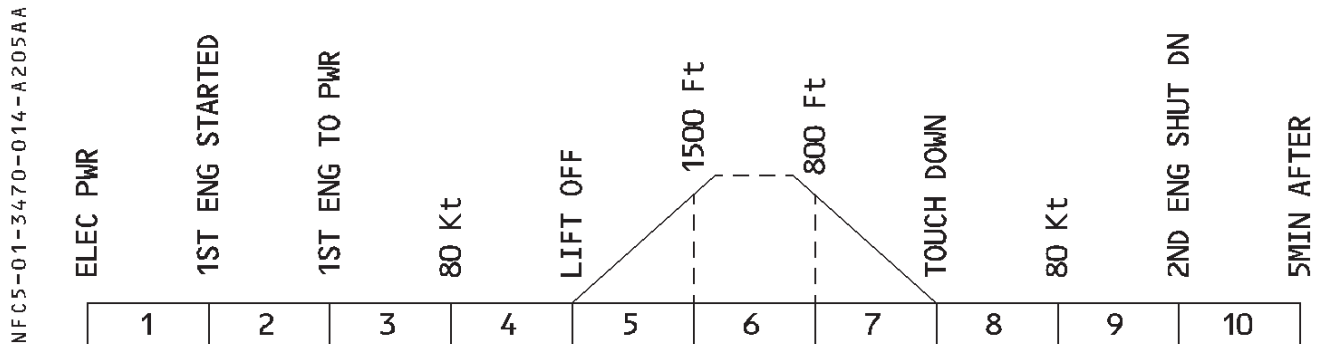
The ON light comes on.

OFF : The terrain data is not displayed on the ND.

Note : · If the Terrain Awareness Display (TAD) mode generates a caution or a warning while the TERR ON ND is not switched ON the terrain data is automatically displayed on the NDs (see EGPWS specific caution and warning due to TAD mode) and the ON light comes on, on the TERR ON ND pushbutton.

· To differentiate between the terrain and the weather display, the terrain display sweeps from the center outward to both sides of the ND.

## WARNINGS AND CAUTIONS



| E / WD : FAILURE TITLE<br>conditions                                                                                              | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING         | FLT<br>PHASE<br>INHIB   |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|--------------------------|-------------------------|
| GPWS FAULT                                                                                                                        | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | GPWS<br>SYS<br>FAULT It  | 1, 3, 4, 5,<br>8, 9, 10 |
| GPWS TERR DET FAULT<br>The enhanced terrain detection function is inoperative.<br>The basic GPWS mode 1 to 5 are still operative. |                  |                 |                      | GPWS<br>TERR<br>FAULT It | 1, 3, 4, 5<br>8,10      |

## MEMO DISPLAY

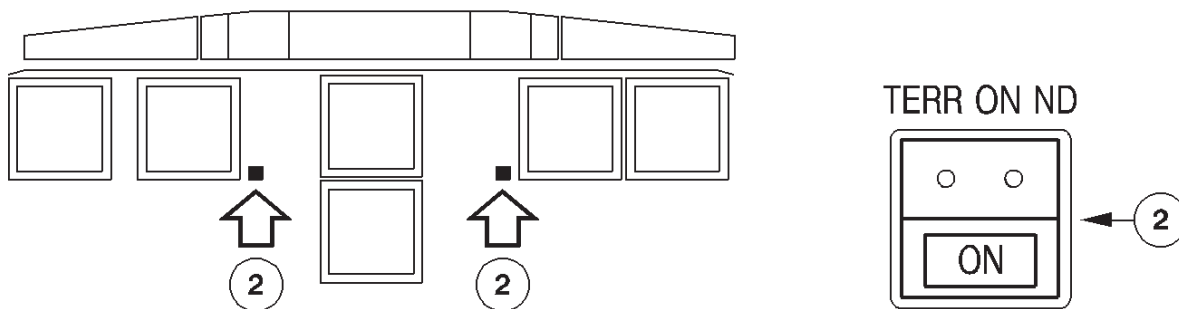
GPWS FLAP 3 is displayed in green when GPWS LDG FLAP 3 pushbutton switch is ON. GPWS FLAP MODE OFF is displayed in green when GPWS FLAP MODE pushbutton switch is OFF.

Airborne TERR STBY appears in green when the aircraft position accuracy (provided by the FMS) is not sufficient to allow the enhanced TCF and TAD modes to operate. These modes are not available until the TERR STBY memo disappears. If selected, the terrain data display on ND is automatically deselected when the TERR STBY memo is triggered."

TERR OFF is displayed when the TERR pushbutton is switched OFF :

- It appears in green in flight phase 2, before the Take Off Configuration test is launched, and in flight phase 6.
- It appears in amber in flight phase 2, after the Take Off Configuration test, and in flight phases 3, 4, 5, 7, 8 and 9.

NFC5-01-3470-013-A105AA



## ② TERR ON ND pb

These pushbuttons are located on either side of the ECAM. Each pushbutton controls the outside terrain display.

**ON** : The terrain is displayed on the ND, if the :

- TERR pushbutton is selected ON, and
- TERR FAULT light is not on.

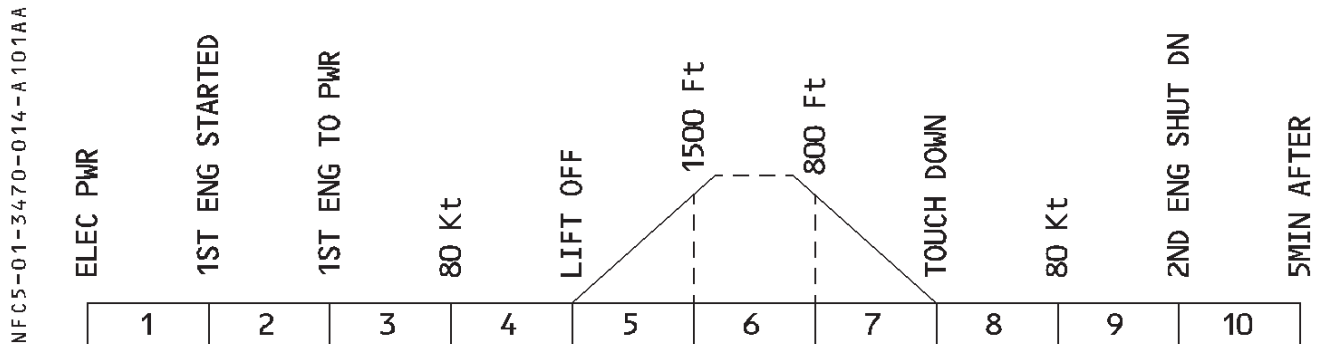
The ON light comes on.

**OFF** : The terrain data is not displayed on the ND.

Note : · If the Terrain Awareness Display (TAD) mode generates a caution, or a warning, while the TERR ON ND is not switched ON, the terrain is automatically displayed on the NDs (see EGPWS specific caution and warning due to TAD mode), and the ON light of the TERR ON ND pushbutton comes on.

· To differentiate between the terrain and the weather displays, the terrain display sweeps from the center outward to both sides of the ND.

## WARNINGS AND CAUTIONS



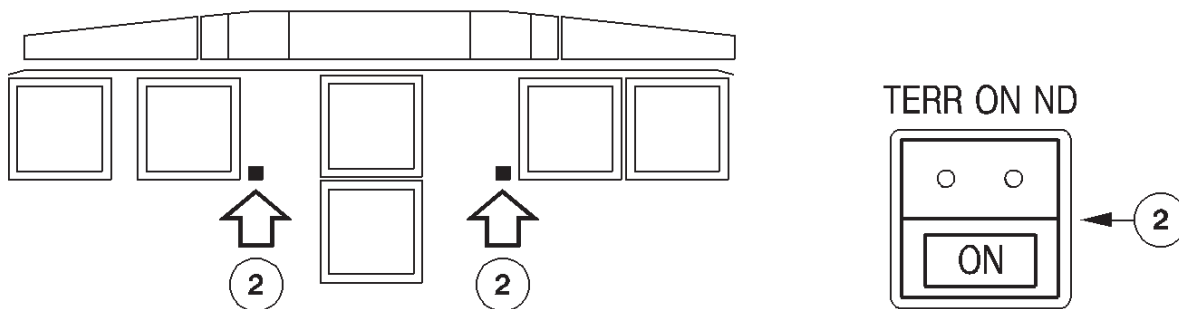
| E / WD : FAILURE TITLE<br>conditions                                                                                              | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING         | FLT<br>PHASE<br>INHIB   |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|--------------------------|-------------------------|
| GPWS FAULT                                                                                                                        | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | GPWS<br>SYS<br>FAULT It  | 1, 3, 4, 5,<br>8, 9, 10 |
| GPWS TERR DET FAULT<br>The enhanced terrain detection function is inoperative.<br>The basic GPWS mode 1 to 5 are still operative. |                  |                 |                      | GPWS<br>TERR<br>FAULT It | 1, 3, 4, 5<br>8,10      |

## MEMO DISPLAY

GPWS FLAP 3 is displayed in green when GPWS LDG FLAP 3 pushbutton switch is ON. GPWS FLAP MODE OFF is displayed in green when GPWS FLAP MODE pushbutton switch is OFF.

- R Airborne TERR STBY appears in green when the aircraft position accuracy (provided by the  
R FMS) is not sufficient to allow the enhanced TCF and TAD modes to operate. These modes  
R are not available until the TERR STBY memo disappears. If selected, the terrain data display  
R on ND is automatically deselected when the TERR STBY memo is triggered.”

NFC5-01-3470-013-A105AA



## ② TERR ON ND pb

These pushbuttons are located on either side of the ECAM. Each pushbutton controls the onside terrain display.

ON : The terrain is displayed on the ND, if the :

- TERR pushbutton is selected ON, and
- TERR FAULT light is not on.

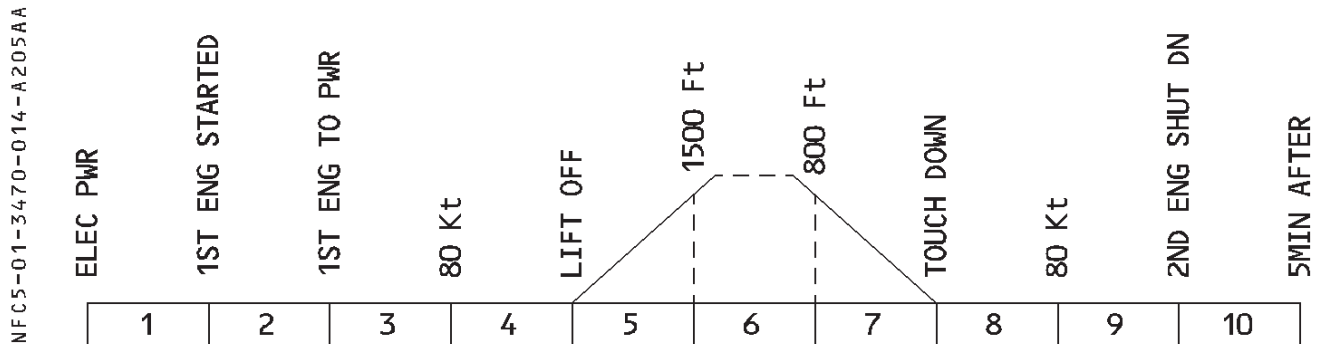
The ON light comes on.

OFF : The terrain data is not displayed on the ND.

Note : · If the Terrain Awareness Display (TAD) mode generates a caution, or a warning, while the TERR ON ND is not switched ON, the terrain is automatically displayed on the NDs (see EGPWS specific caution and warning due to TAD mode), and the ON light of the TERR ON ND pushbutton comes on.

- To differentiate between the terrain and the weather displays, the terrain display sweeps from the center outward to both sides of the ND.

## WARNINGS AND CAUTIONS



| E / WD : FAILURE TITLE<br>conditions                                                                                              | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING         | FLT<br>PHASE<br>INHIB   |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|--------------------------|-------------------------|
| GPWS FAULT                                                                                                                        | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | GPWS<br>SYS<br>FAULT It  | 1, 3, 4, 5,<br>8, 9, 10 |
| GPWS TERR DET FAULT<br>The enhanced terrain detection function is inoperative.<br>The basic GPWS mode 1 to 5 are still operative. |                  |                 |                      | GPWS<br>TERR<br>FAULT It | 1, 3, 4, 5<br>8,10      |

## MEMO DISPLAY

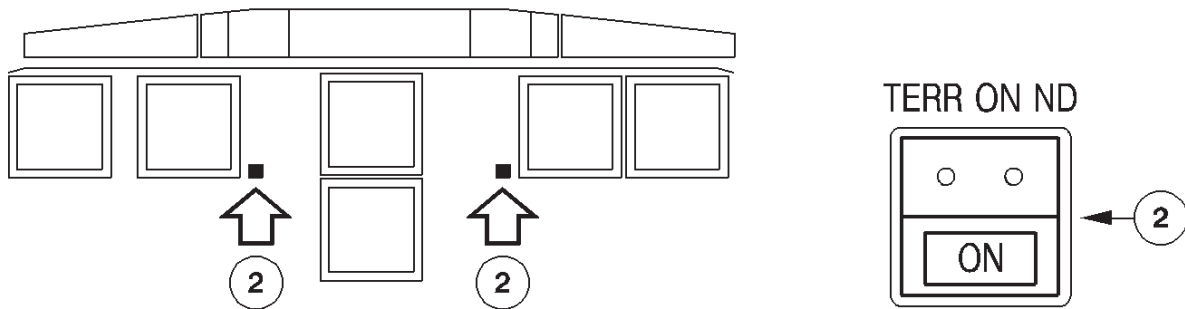
GPWS FLAP 3 is displayed in green when GPWS LDG FLAP 3 pushbutton switch is ON. GPWS FLAP MODE OFF is displayed in green when GPWS FLAP MODE pushbutton switch is OFF.

Airborne TERR STBY appears in green when the aircraft position accuracy (provided by the FMS) is not sufficient to allow the enhanced TCF and TAD modes to operate. These modes are not available until the TERR STBY memo disappears. If selected, the terrain data display on ND is automatically deselected when the TERR STBY memo is triggered."

TERR OFF is displayed when the TERR pushbutton is switched OFF :

- It appears in green in flight phase 2, before the Take Off Configuration test is launched, and in flight phase 6.
- It appears in amber in flight phase 2, after the Take Off Configuration test, and in flight phases 3, 4, 5, 7, 8 and 9.

NFC5-01-3470-013-A105AA



## ② TERR ON ND pb

These pushbuttons are located on either side of the ECAM. Each pushbutton controls the outside terrain display.

**ON** : The terrain is displayed on the ND, if the :

- TERR pushbutton is selected ON, and
- TERR FAULT light is not on.

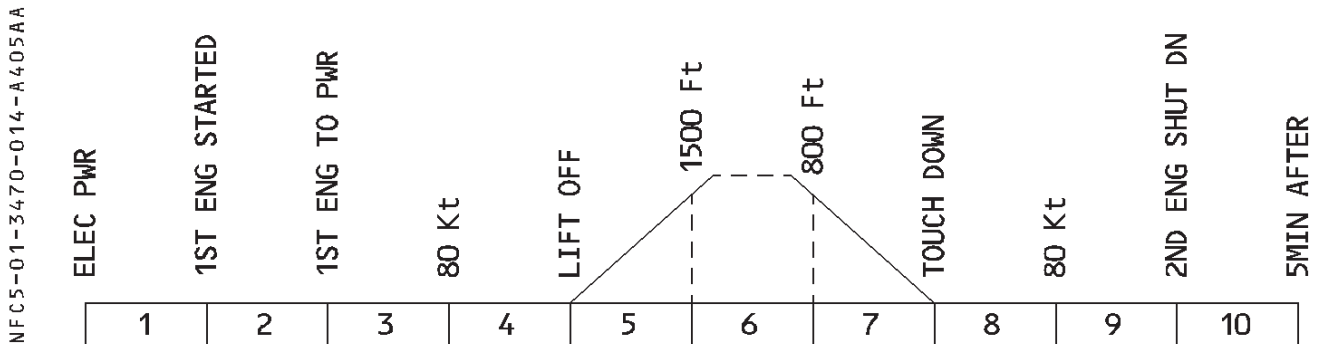
The ON light comes on.

**OFF** : The terrain data is not displayed on the ND.

Note : · If the Terrain Awareness Display (TAD) mode generates a caution, or a warning, while the TERR ON ND is not switched ON, the terrain is automatically displayed on the NDs (see EGPWS specific caution and warning due to TAD mode), and the ON light of the TERR ON ND pushbutton comes on.

· To differentiate between the terrain and the weather displays, the terrain display sweeps from the center outward to both sides of the ND.

## WARNINGS AND CAUTIONS



| E / WD : FAILURE TITLE<br>conditions                                                                                              | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING         | FLT<br>PHASE<br>INHIB   |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------------------|--------------------------|-------------------------|
| GPWS FAULT                                                                                                                        | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | GPWS<br>SYS<br>FAULT It  | 1, 3, 4, 5,<br>8, 9, 10 |
| GPWS TERR DET FAULT<br>The enhanced terrain detection function is inoperative.<br>The basic GPWS mode 1 to 5 are still operative. |                  |                 |                      | GPWS<br>TERR<br>FAULT It | 1, 3, 4, 5<br>8,10      |

## MEMO DISPLAY

GPWS FLAP 3 is displayed in green when GPWS LDG FLAP 3 pushbutton switch is ON. GPWS FLAP MODE OFF is displayed in green when GPWS FLAP MODE pushbutton switch is OFF.

Airborne TERR STBY appears in green when the aircraft position accuracy (computed by the EGPWS) is not sufficient to allow the enhanced TCF and TAD modes to operate. These modes are not available until the TERR STBY memo disappears. If selected, the terrain data display on ND is automatically deselected when the TERR STBY memo is triggered."

TERR OFF is displayed when the TERR pushbutton is switched OFF :

- It appears in green in flight phase 2, before the Take Off Configuration test is launched, and in flight phase 6.
- It appears in amber in flight phase 2, after the Take Off Configuration test, and in flight phases 3, 4, 5, 7, 8 and 9.

**DESCRIPTION****GENERAL**

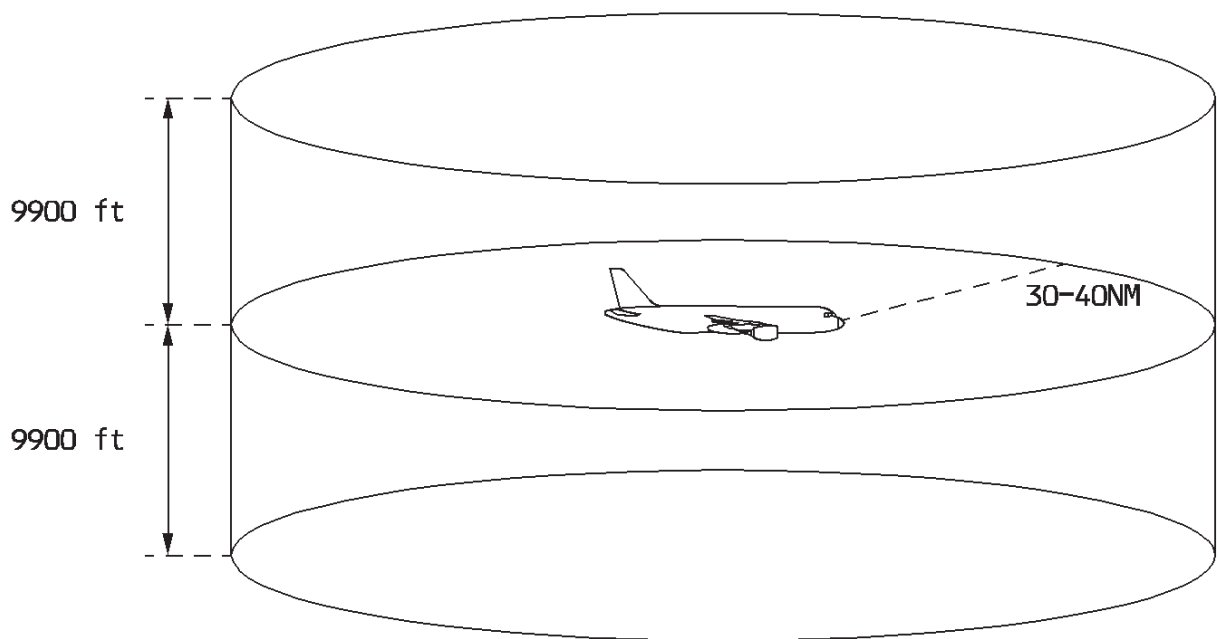
The TCAS (Traffic alert and Collision Avoidance System) :

- Detects any aircraft, equipped with transponders, flying in its vicinity ;
- Displays potential and predicted collision targets ;
- Issues vertical orders to avoid conflict.

The TCAS is normally independent of the ground-based air traffic control system.

The TCAS detection capability is limited to intruders flying within a maximum range of 30-40 NM (depending on aircraft configuration and external conditions), and within a maximum altitude range of 9900 feet (above and below the threatened aircraft).

R



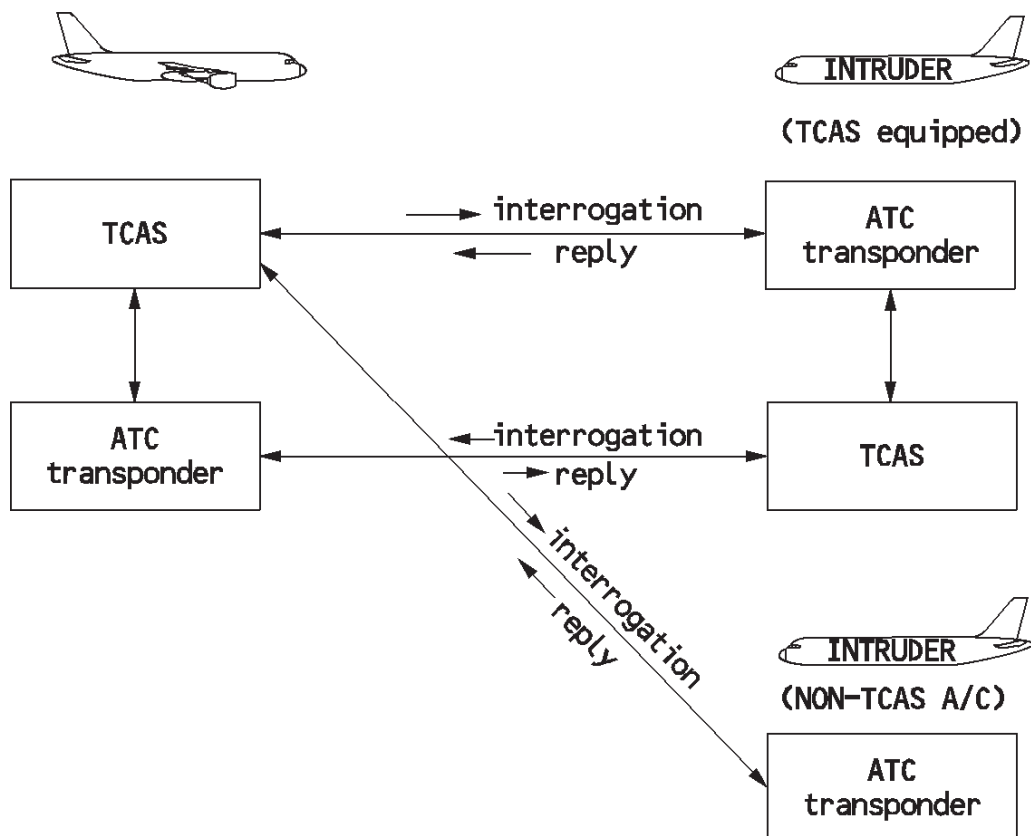
## MAIN COMPONENTS

The system includes :

- a single channel TCAS computer
- two TCAS antennas
- two mode S ATC transponders, one active the other in standby.

These transponders allow :

- interface between the ATC / TCAS control panel and the TCAS computer
  - communication between the aircraft and intruders equipped with a TCAS system
- an ATC /TCAS control panel.



NFC5-01-3480-002-A100AA

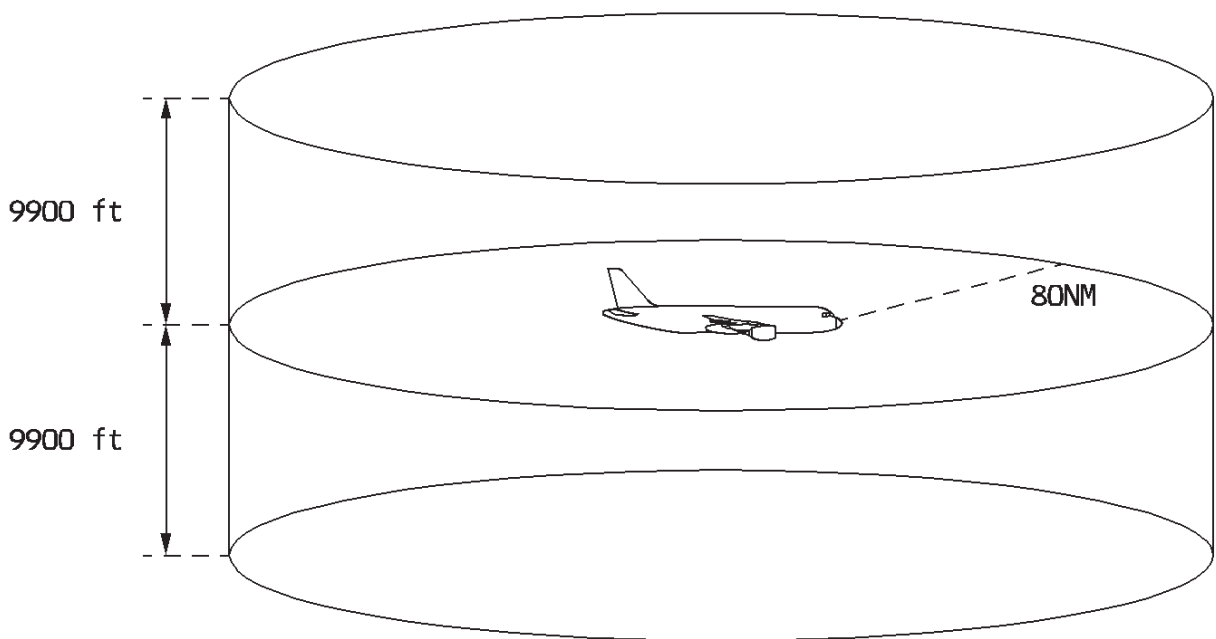
**DESCRIPTION****GENERAL**

The TCAS (Traffic alert and Collision Avoidance System) :

- Detects any aircraft, that is equipped with transponders and is flying in the vicinity
- Displays potential and predicted collision targets
- Issues vertical orders to avoid conflict.

TCAS is usually independent of ground-based air traffic control systems.

TCAS detection-capability is limited to the intruders that fly within a maximum range of approximately 80 NM (depending on aircraft configuration and on external conditions), and within a maximum altitude range of 9900 feet (above and below the threatened aircraft).



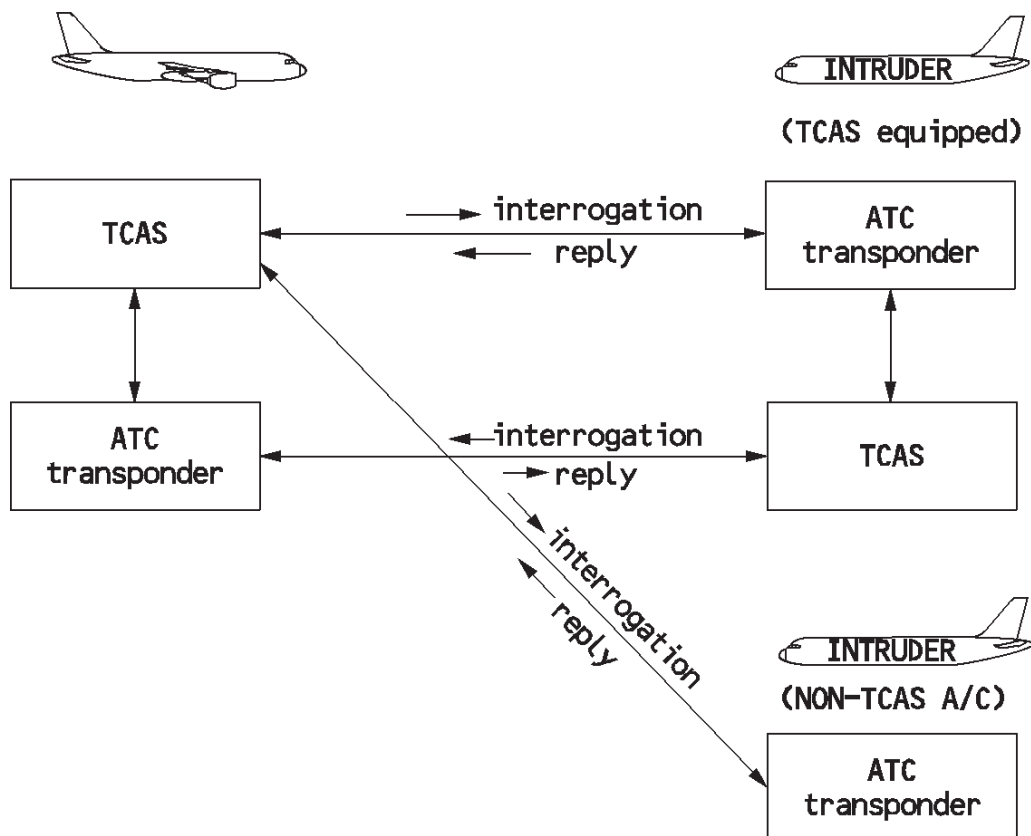
## MAIN COMPONENTS

The system includes :

- a single channel TCAS computer
- two TCAS antennas
- two mode S ATC transponders, one active the other in standby.

These transponders allow :

- interface between the ATC / TCAS control panel and the TCAS computer
  - communication between the aircraft and intruders equipped with a TCAS system
- an ATC /TCAS control panel.



NFC5-01-3480-002-A100AA

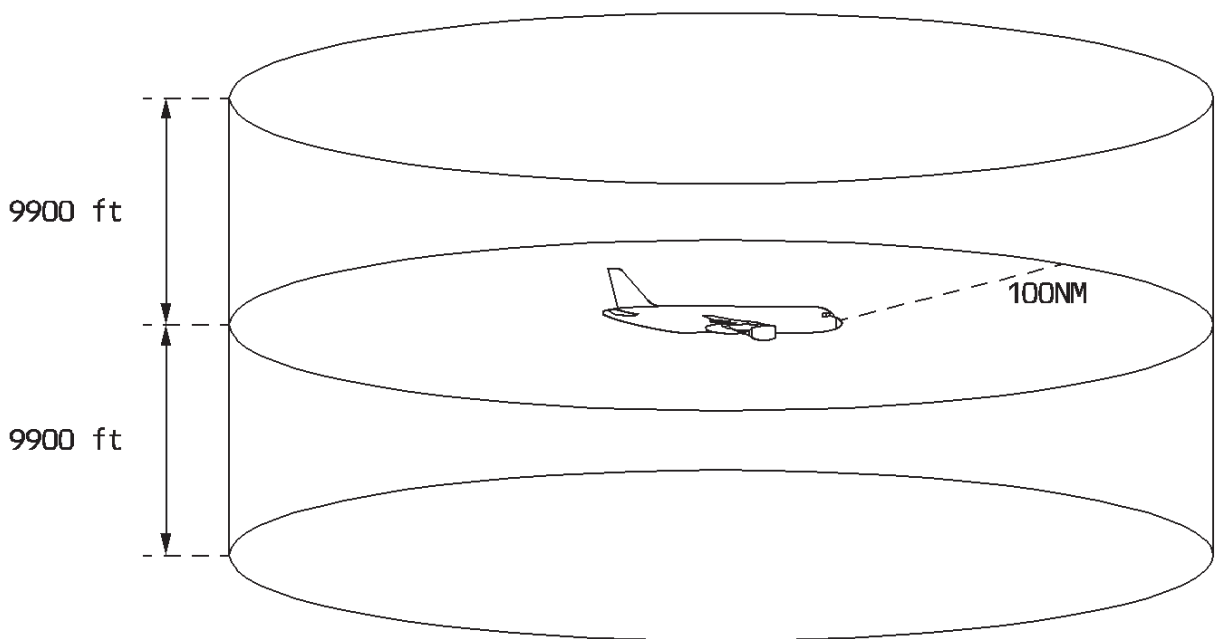
**DESCRIPTION****GENERAL**

The Traffic alert and Collision Avoidance System (TCAS) :

- Detects any aircraft, that is equipped with transponders and is flying in the vicinity
- Displays potential and predicted collision targets
- Issues vertical orders to avoid conflict.

The TCAS is normally independent of the ground-based Air Traffic Control system.

The TCAS's detection capability is limited to intruders flying within a maximum range of 100 NM (depending on aircraft configuration and external conditions), and within a maximum altitude range of 9900 feet above and below the threatened aircraft.



NFC5-01-3480-001-A210AA

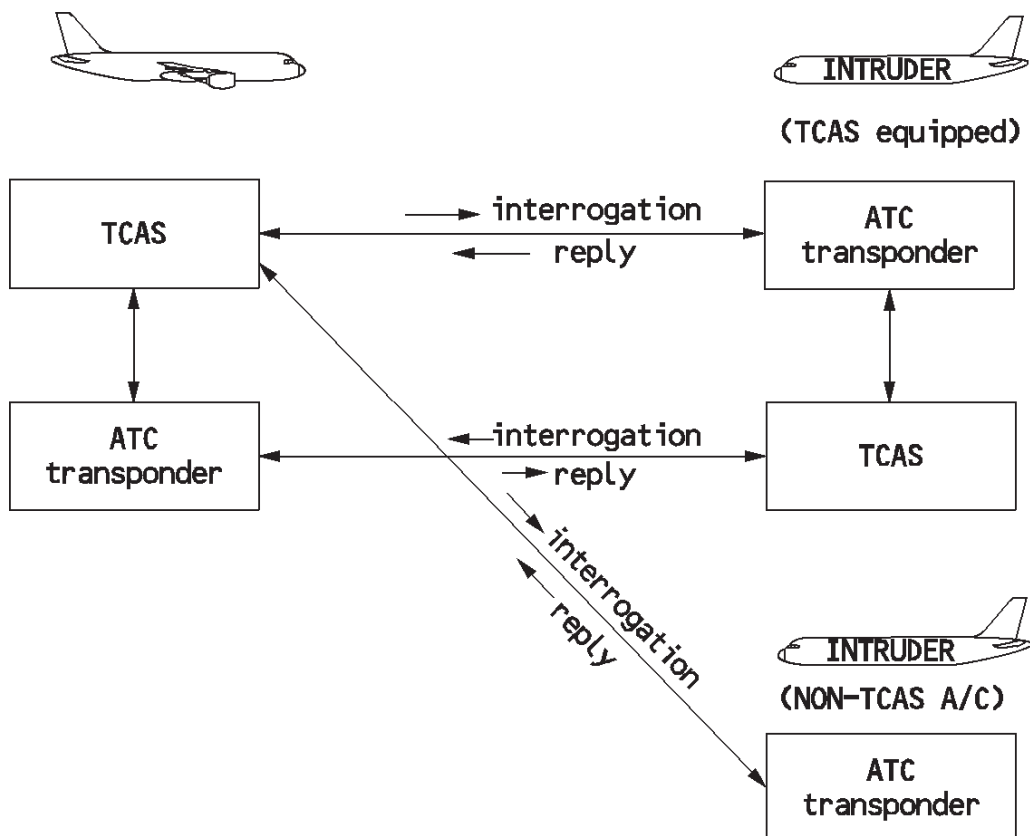
## MAIN COMPONENTS

The system includes :

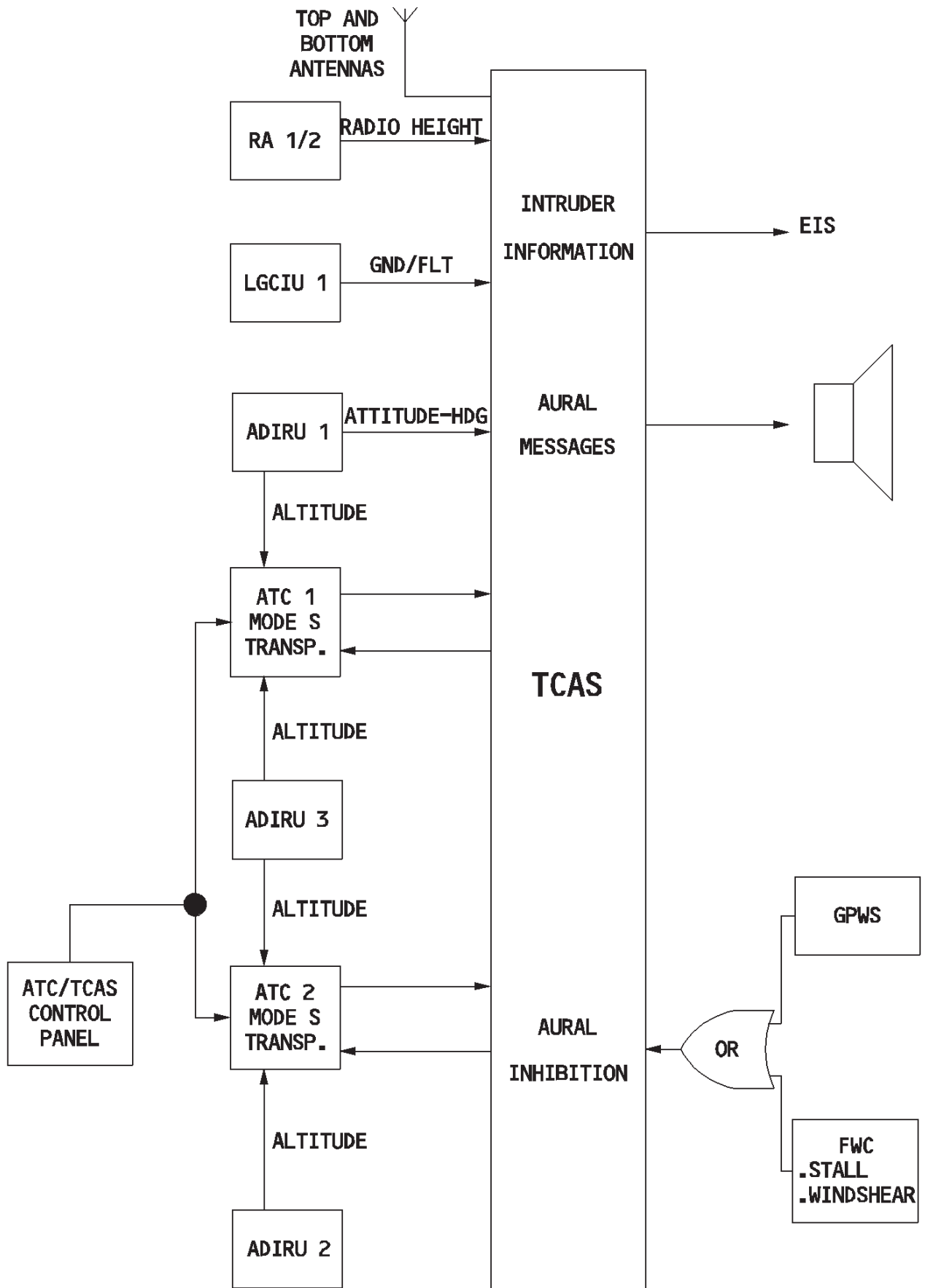
- a single channel TCAS computer
- two TCAS antennas
- two mode S ATC transponders, one active the other in standby.

These transponders allow :

- interface between the ATC / TCAS control panel and the TCAS computer
  - communication between the aircraft and intruders equipped with a TCAS system
- an ATC /TCAS control panel.



NFC5-01-3480-002-A100AA



NFC5-01-3480-003-A100AB

## **PRINCIPLE**

The TCAS interrogates transponder of intruders. From the transponder replies, the TCAS determines for each intruder :

- its relative bearing
- its range and closure rate
- its relative altitude if available (ATC mode C or S)

Then the TCAS computes the intruder trajectory, the Closest Point of Approach (CPA) and the estimated time (TAU) before reaching the CPA.

Each time the relative position of the intruder presents a collision threat, aural and visual advisories are triggered.

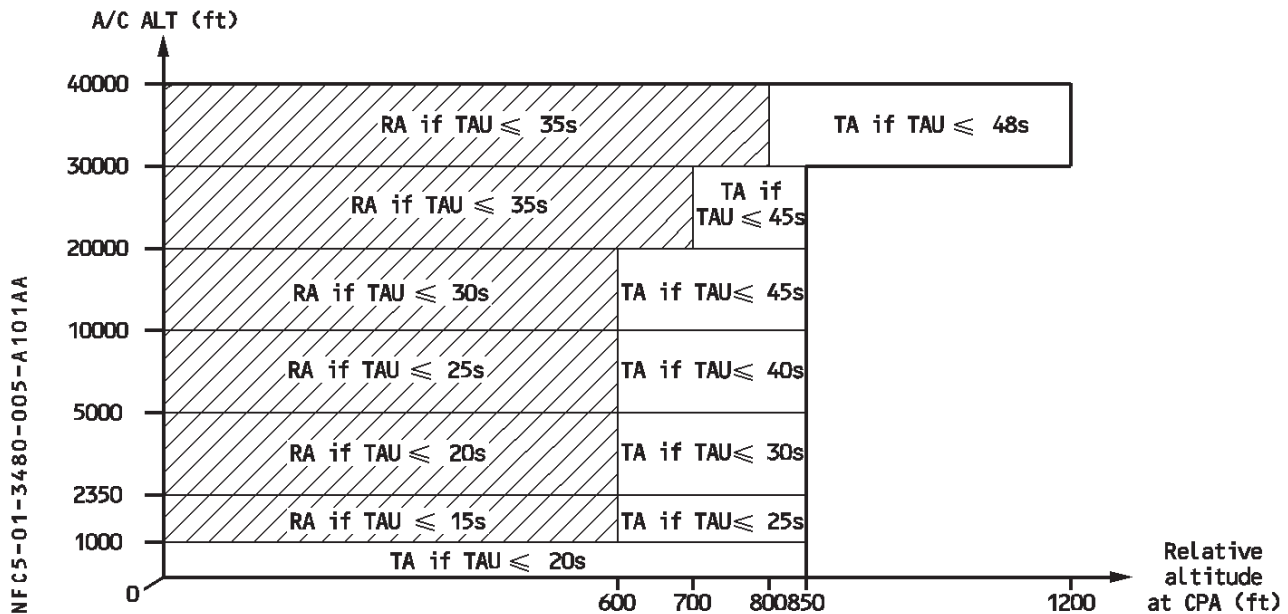
TCAS optimizes vertical orders to ensure a sufficient trajectory separation and a minimal vertical speed variation considering all intruders.

## **INTRUDER CLASSIFICATION**

The intruders are classified in four levels :

| LEVEL                    | INTRUDER POSITION                                                                                                                                                                                                        | DISPLAYED INFORMATION                                                                                                                                                                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proximate                | <ul style="list-style-type: none"> <li>– no collision threat</li> <li>– intruder in the vicinity of the a/c (closer than 6 NM laterally and <math>\pm 1200</math> ft vertically)</li> </ul>                              | <ul style="list-style-type: none"> <li>– ND : intruder position</li> </ul>                                                                                                                                                                                                                                 |
| Traffic Advisory (TA)    | <ul style="list-style-type: none"> <li>– potential collision threat</li> <li>– TAU is about 40 seconds</li> </ul>                                                                                                        | <ul style="list-style-type: none"> <li>– ND : intruder position</li> <li>– Aural messages</li> </ul>                                                                                                                                                                                                       |
| Resolution Advisory (RA) | <ul style="list-style-type: none"> <li>– real collision threat</li> <li>– TAU is about 25 seconds</li> </ul>                                                                                                             | <ul style="list-style-type: none"> <li>– ND : intruder position</li> <li>– Aural messages</li> <li>– PFD : vertical orders               <ul style="list-style-type: none"> <li>· Maintain actual V/S (Preventive Advisory)</li> <li>or</li> <li>· Modify V/S (Corrective Advisory)</li> </ul> </li> </ul> |
| Other intruders (<4)     | <ul style="list-style-type: none"> <li>– no collision threat</li> <li>– any non proximate, TA, RA within the surveillance envelope (lateral range : closer than 30 NM. vertical range : Refer to 1.34.80 p 7)</li> </ul> | <ul style="list-style-type: none"> <li>– ND : intruder position</li> </ul>                                                                                                                                                                                                                                 |

## TA / RA THRESHOLDS

**FOR INFO**


## TCAS MODES

The TCAS has 2 modes of operation :

- TA/RA : This mode allows the display of all intruders.
- TA : Can be selected by :
  - the crew, on the ATC/TCAS panel, in case of aircraft degraded performance (engine failure, landing gear extended) or when operating near closely spaced parallel runways, or
  - automatically, if TA/RA is previously selected, and :
    - The windshear (⚠) message is triggered
    - The stall message is triggered
    - GPWS messages are triggered
    - Aircraft is below 1000 ft AGL.

Consequently :

- all RA are inhibited and converted into TA
- TA threshold is set to  $TAU \leq 20$  seconds independent of aircraft altitude.
- no vertical speed advisories are indicated on the PFDs
- TA ONLY is displayed on the NDs

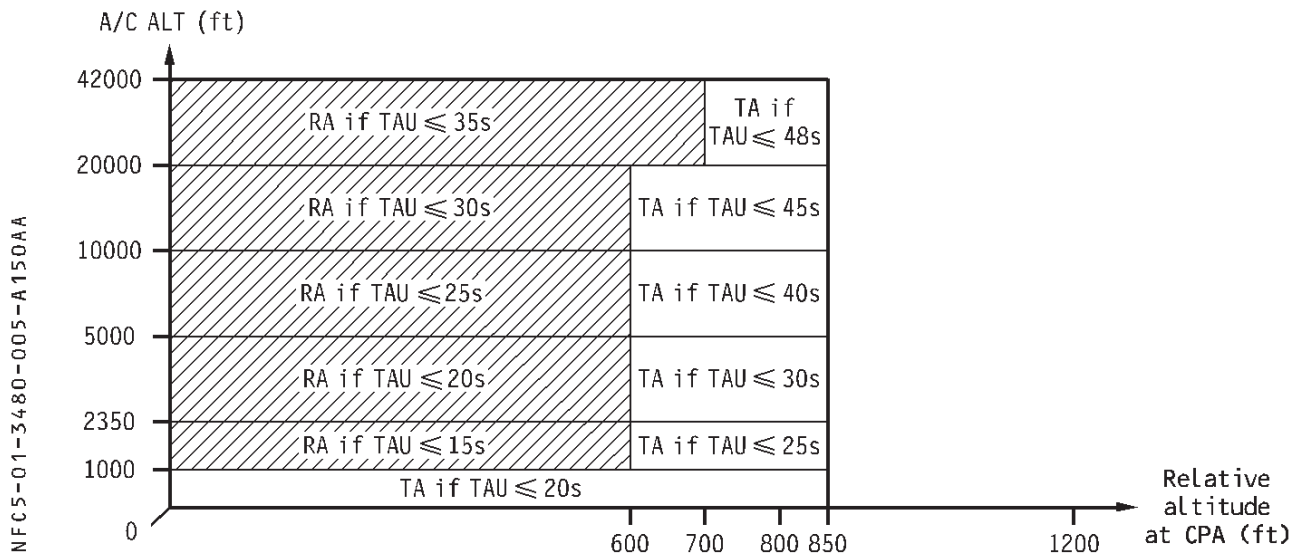
If windshear, stall or GPWS messages are triggered, or aircraft below 500 ft AGL, all the TCAS aural messages are suppressed.

## **ADVISORY INHIBITION**

Some advisories are inhibited depending on the aircraft altitude :

- all intruders flying below 380 feet AGL when the own aircraft altitude is below 1700 feet AGL.
- all TA aural messages and all RAs below 1100 feet in climb and 900 feet in descent. In this case, the RAs are converted into TAs.
- “Descend” type advisory below 1200 feet AGL at takeoff or 1000 feet AGL in approach.
- “Increase Descent” RA below 1450 feet.

## TA / RA THRESHOLDS

**FOR INFO**


## TCAS MODES

TCAS has 2 modes of operation :

TA/RA : This mode allows the display of all the intruders.

TA : Can be selected by :

- the crew, on the ATC/TCAS panel, in case of aircraft degraded performance (engine failure, landing gear extended) or when operating near closely spaced runways, or
- automatically, if TA/RA is previously selected, and :
  - The windshear (⚠) message is triggered
  - The stall message is triggered
  - GPWS messages are triggered
  - Aircraft is below 1000 ft AGL.

Consequently :

- All RAs are inhibited and converted into TAs
- the TA threshold is set to  $TAU \leq 20$  seconds, irrespective of the aircraft's altitude.
- No vertical speed advisories are indicated on the PFDs
- "TA ONLY" is displayed on the NDs

If windshear, stall or GPWS messages are triggered, or aircraft below 500 ft AGL, all the TCAS aural messages are suppressed.

## **ADVISORY INHIBITION**

Some advisories are inhibited depending on the aircraft altitude :

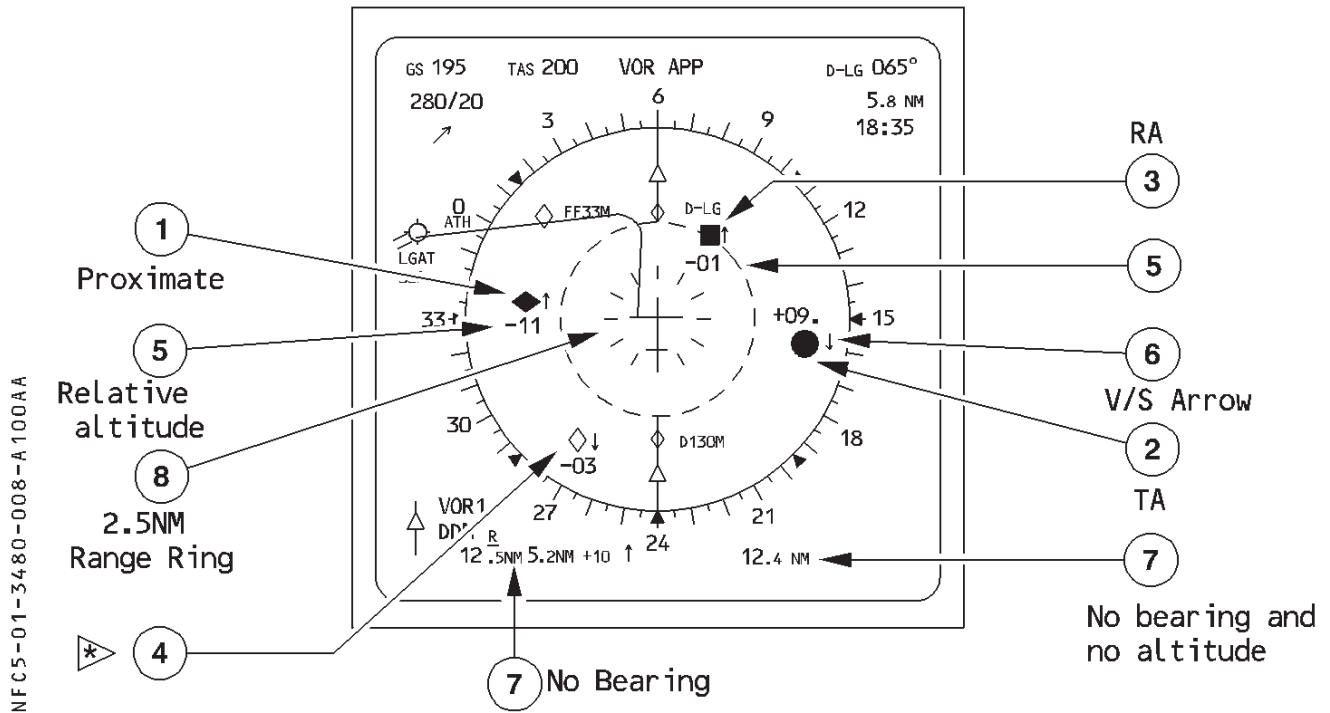
- All intruders flying below 380 feet AGL when the own aircraft altitude is below 1700 feet AGL.
  - All RA below 1100 feet in climb and 900 feet in descent. In this case, the RAs are converted into TAs.
  - “Descend” type advisory below 1200 feet AGL at takeoff or 1000 feet AGL in approach.
  - “Increase Descent” RA below 1450 feet.
- R    – All TA aural messages below 600 feet AGL in climb and below 400 feet AGL in descent.



## ND INDICATIONS

The traffic is displayed in all ROSE modes and ARC mode when 10, 20 or 40 NM range is selected.

Only the 8 most threatening intruders are displayed.



### ① Proximate intruder

Indicated by a white filled diamond.

### ② TA intruder

Indicated by an amber circle.

Associated with the TRAFFIC-TRAFFIC aural message.

### ③ RA intruder

Indicated by a red square.

Associated with vertical orders displayed on the PFD and aural messages.

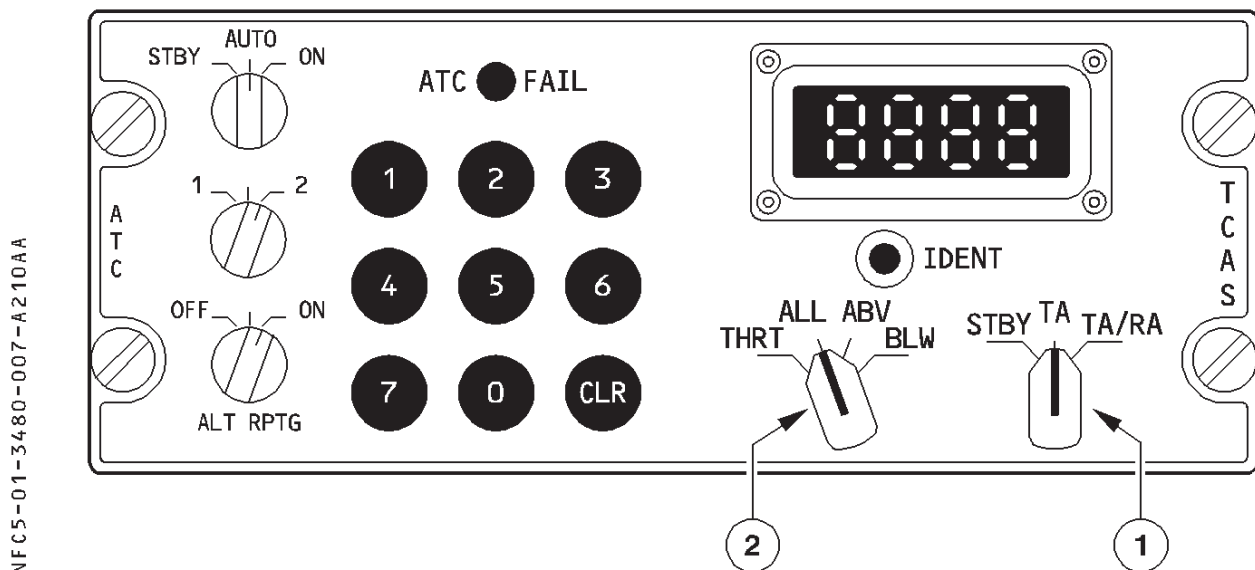
### ④ Other intruders

Indicated by a white empty diamond.

*Note : If the range of an intruder is not available, the intruder is not displayed.  
An intruder may be partially displayed when its range is out of scale.*

## CONTROLS AND INDICATORS

### ATC/TCAS PANEL



#### ① Mode sel

TA/RA : Normal position.

The RAs, TAs and proximate intruders are displayed, if the ALT RPTG switch is ON and the transponder is not on STBY.

TA : The TCAS does not generate any vertical orders. This mode should be used in case of aircraft degraded performance (engine failure, landing gear extended...), or on parallel runways.

All RAs are converted into TAs. TAs, proximate and other intruders are displayed, if the ALT RPTG switch is ON and the transponder is not on STBY.

The TA ONLY white memo is displayed on the NDs.

STBY : The TCAS is on standby.

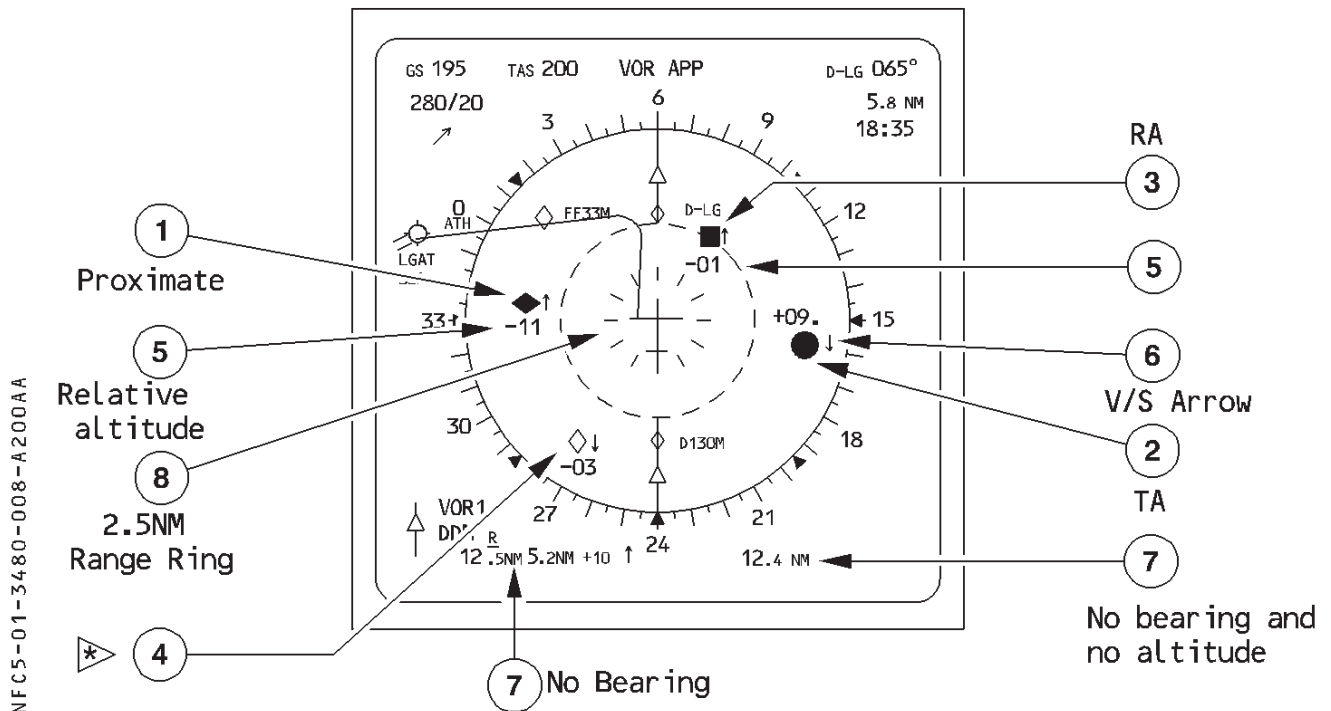
#### ② TRAFFIC sel

- R THRT : Proximate and other intruders are displayed only if a TA or RA is present,  
 R and they are within 2700 feet above and 2700 feet below the aircraft.  
 R ALL : Proximate and other intruders are displayed even if no TA or RA is present  
 R (full time function). The altitude range is – 2700 feet to + 2700 feet.  
 R ABV : The same as ALL, except that the other intruders are displayed if within  
 R 9900 feet above the aircraft and 2700 feet below.  
 R BLW : The same as ALL, except that the other intruders are displayed if within  
 R 9900 feet below the aircraft and 2700 feet above.

## ND INDICATIONS

The traffic is displayed in all ROSE modes and ARC mode. Only the 8 most threatening intruders are displayed.

R



### ① Proximate intruder

Indicated by a white filled diamond.

### ② TA intruder

Indicated by an amber circle.

Associated with the TRAFFIC-TRAFFIC aural message.

### ③ RA intruder

Indicated by a red square.

Associated with vertical orders displayed on the PFD and aural messages.

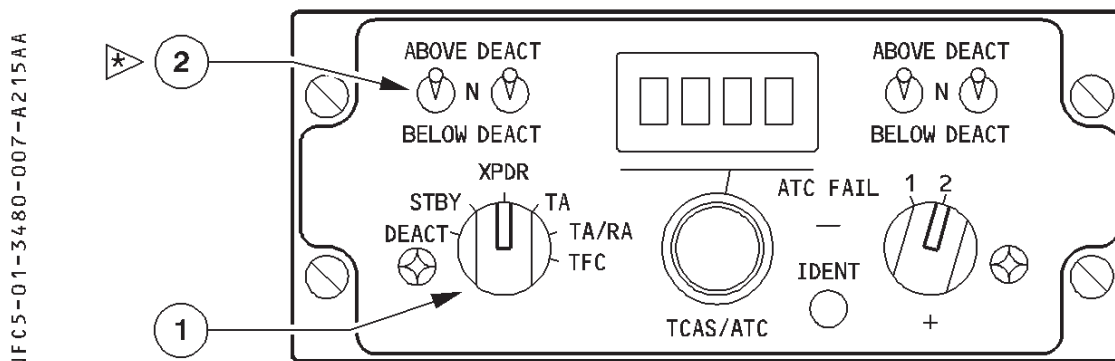
### ④ Other intruders

Indicated by a white empty diamond.

Note : If the range of an intruder is not available, the intruder is not displayed.  
An intruder may be partially displayed when its range is out of scale.

## CONTROLS AND INDICATORS

### ATC/TCAS PANEL



#### ① Mode selector

TFC  : TAs and RAs are displayed. The proximate and other intruders are displayed, even if no TA or RA is present at the same time (full-time function).

TA/RA : Normal position.

Proximate and other intruders are displayed, only if a TA or RA is present at the same time.

TA : The TCAS does not generate any vertical orders. This mode should be used, in case of degraded aircraft performance (engine failure, landing gear extended, or approach on parallel runways).

All RAs are converted into TAs. Proximate and other intruders are displayed, only if a TA is already displayed.

The "TA ONLY" white memo is displayed on the NDs.

XPDR : – The TCAS is on standby.

– On ground: The selected ATC Transponder only operates in the selective aircraft interrogation mode of Mode S.

– In flight : The selected ATC Transponder operates.

STBY : The TCAS and the two Transponders are on standby.

#### ② Altitude range switch

Sets the altitude range, within which other intruders will be displayed on the NDs, irrespective of the selected mode (TA, TA/RA or TFC).

ABOVE : The altitude range is – 2700 feet to + 7000 feet

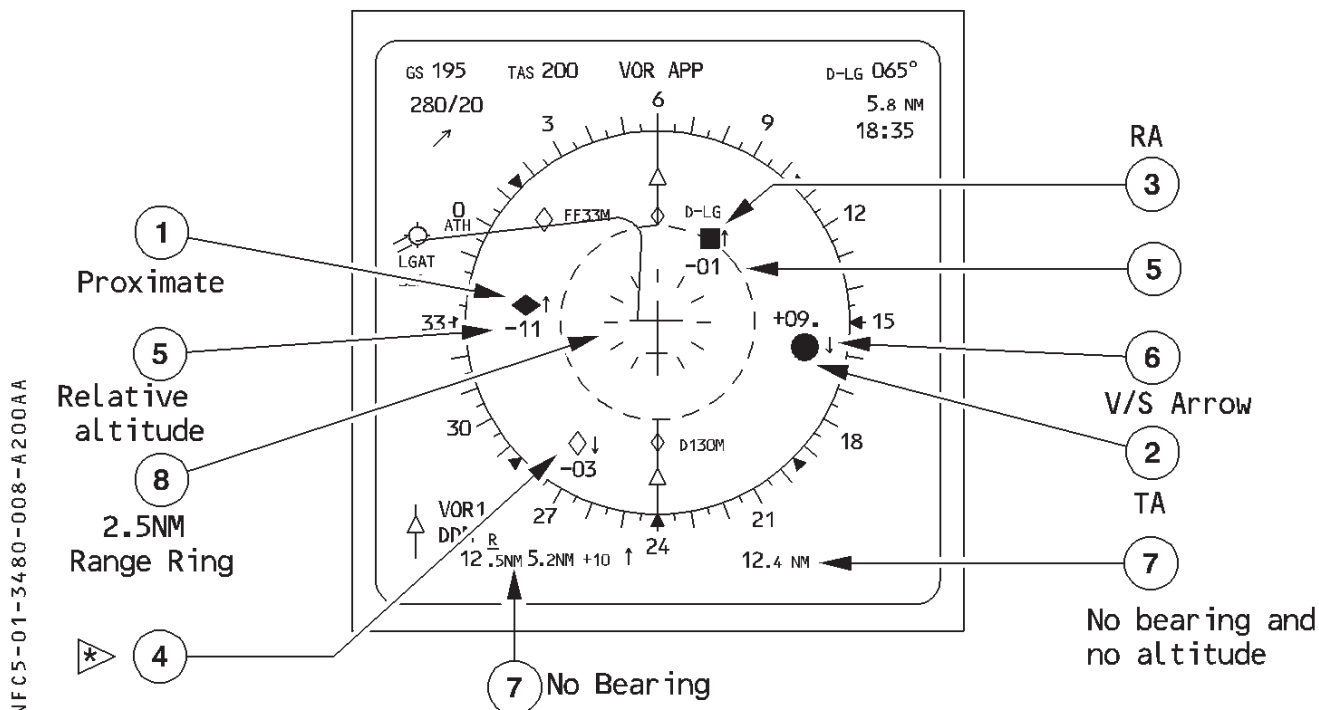
N : The altitude range is – 2700 feet to + 2700 feet

BELOW : The altitude range is – 7000 feet to + 2700 feet

## ND INDICATIONS

The traffic is displayed in all ROSE modes and ARC mode. Only the 8 most threatening intruders are displayed.

R



### ① Proximate intruder

Indicated by a white filled diamond.

### ② TA intruder

Indicated by an amber circle.  
Associated with the TRAFFIC-TRAFFIC aural message.

### ③ RA intruder

Indicated by a red square.  
Associated with vertical orders displayed on the PFD and aural messages.

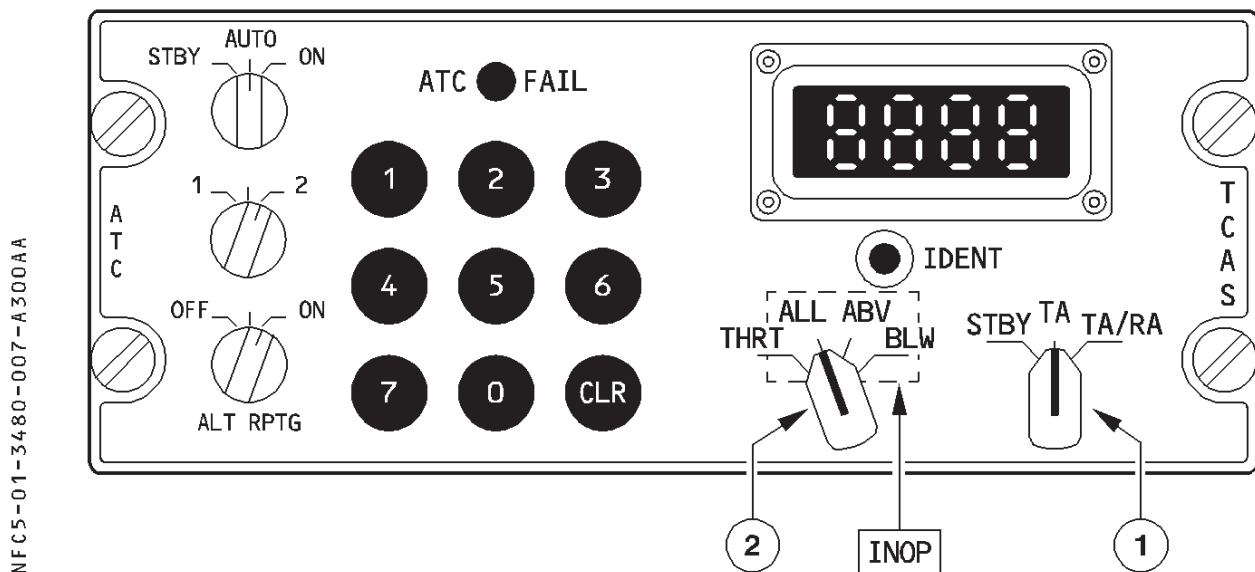
### ④ Other intruders

Indicated by a white empty diamond.

Note : If the range of an intruder is not available, the intruder is not displayed.  
An intruder may be partially displayed when its range is out of scale.

## CONTROLS AND INDICATORS

### ATC/TCAS PANEL



#### ① Mode sel

TA/RA : Normal position.

The RAs, TAs and proximate intruders are displayed if the ALT RPTG switch is ON and the transponder is not on STBY.

TA : The TCAS does not generate any vertical orders. This mode should be used in case of aircraft degraded performance (engine failure, landing gear extended...) or on parallel runways.

All RAs are converted into TAs. TAs, proximate and other intruders are displayed if the ALT RPTG switch is ON and the transponder is not on STBY. The TA ONLY white memo is displayed on the NDs.

STBY : The TCAS is in standby.

#### ② TRAFFIC sel

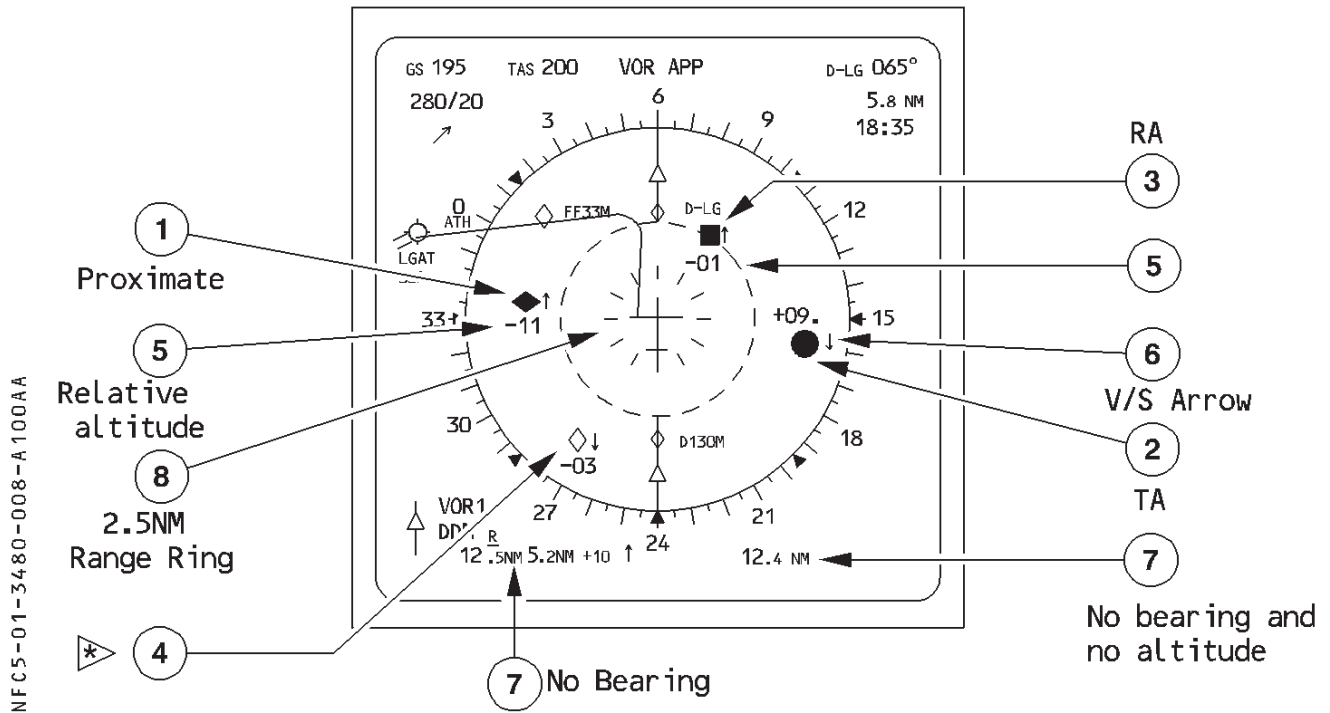
THRT : The other and proximate intruders are displayed only if a TA or a RA is present at the same time. The altitude range is – 2700 feet to + 2700 feet.

R

## ND INDICATIONS

The traffic is displayed in all ROSE modes and ARC mode when 10, 20 or 40 NM range is selected.

Only the 8 most threatening intruders are displayed.



### ① Proximate intruder

Indicated by a white filled diamond.

### ② TA intruder

Indicated by an amber circle.

Associated with the TRAFFIC-TRAFFIC aural message.

### ③ RA intruder

Indicated by a red square.

Associated with vertical orders displayed on the PFD and aural messages.

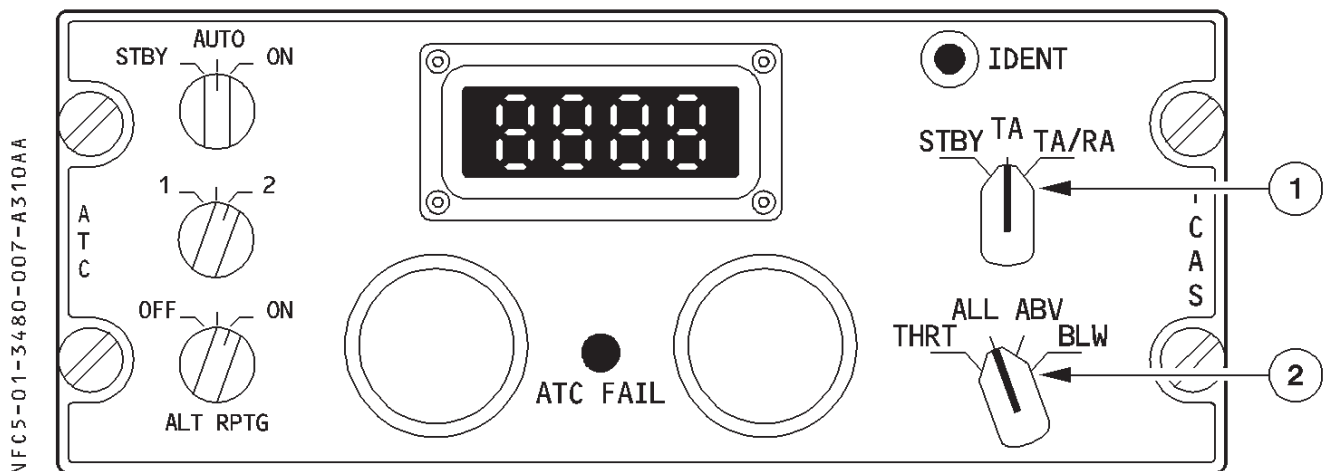
### ④ Other intruders

Indicated by a white empty diamond.

*Note : If the range of an intruder is not available, the intruder is not displayed.  
An intruder may be partially displayed when its range is out of scale.*

## CONTROLS AND INDICATORS

### ATC/TCAS PANEL



#### ① TCAS Mode sel

TA/RA : Normal position.

The RAs, TAs and proximate intruders are displayed if the ALT RPTG switch is ON and the transponder is not on STBY.

TA : The TCAS does not generate any vertical orders. This mode should be used in case of aircraft degraded performance (engine failure, landing gear extended...) or on parallel runways.

All RAs are converted into TAs. TAs, proximate and other intruders are displayed if the ALT RPTG switch is ON and the transponder is not on STBY. The TA ONLY white memo is displayed on the NDs.

STBY : The TCAS is in standby.

#### ② TRAFFIC sel

THRT : The other and proximate intruders are displayed only if a TA or a RA is present at the same time.

ALL : The other and proximate intruders are displayed even if no TA or RA is present at the same time (full time function).

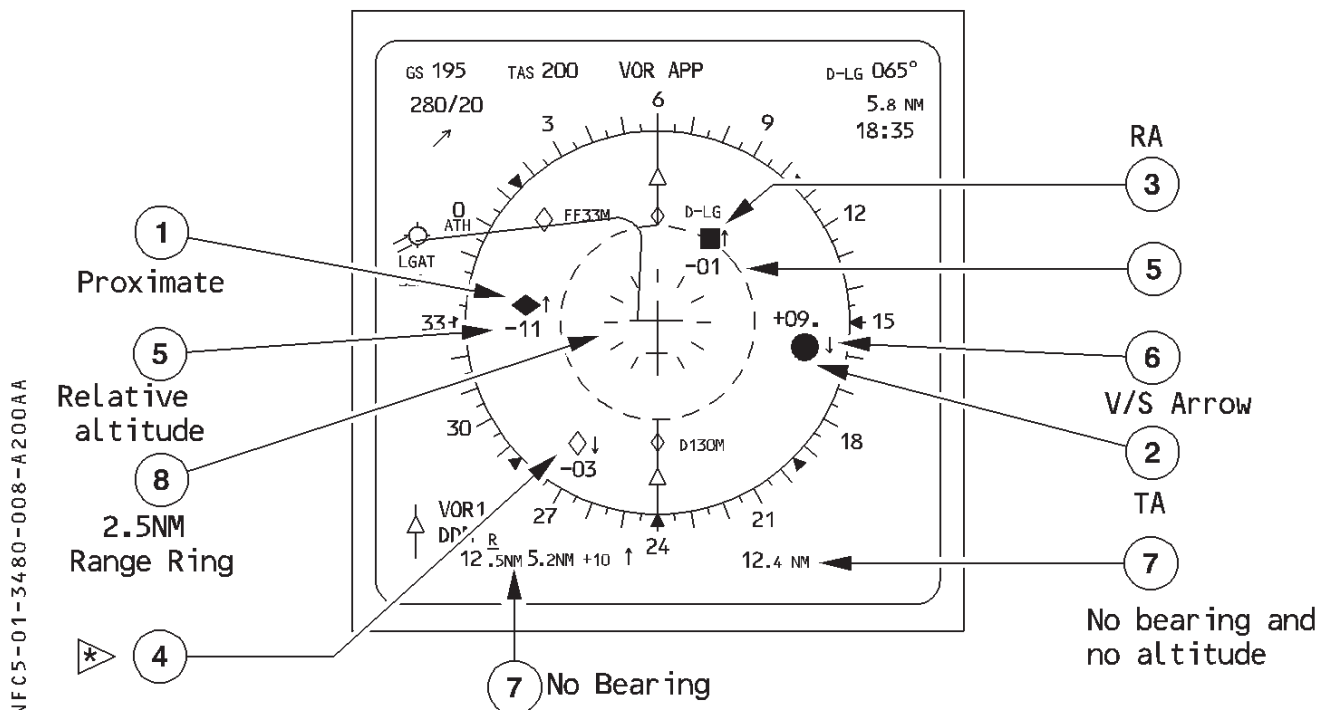
ABV : The other intruders are displayed within 9900 feet above the aircraft and 2700 ft below.

R BLW : The other intruders are displayed within 9900 feet below the aircraft and 2700 feet above.

## ND INDICATIONS

The traffic is displayed in all ROSE modes and ARC mode. Only the 8 most threatening intruders are displayed.

R



### ① Proximate intruder

Indicated by a white filled diamond.

### ② TA intruder

Indicated by an amber circle.  
Associated with the TRAFFIC-TRAFFIC aural message.

### ③ RA intruder

Indicated by a red square.  
Associated with vertical orders displayed on the PFD and aural messages.

### ④ Other intruders

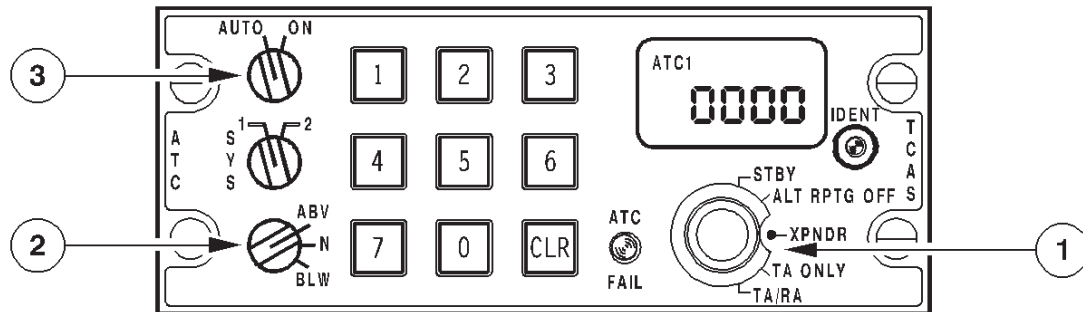
Indicated by a white empty diamond.

Note : If the range of an intruder is not available, the intruder is not displayed.  
An intruder may be partially displayed when its range is out of scale.

## CONTROLS AND INDICATORS

### ATC/TCAS PANEL

NFC5-01-3480-007-A315AA



#### ① Mode selector

- TA/RA : Normal position.  
TAs, RAs, proximate and other intruders are displayed.
- TA ONLY : This mode should be used, in case of degraded aircraft performance (engine failure, landing gear extended, or approach on parallel runways). All RAs are converted into TAs. TAs, proximate and other intruders are displayed.
- STBY : The TCAS and ATC are on standby.
- XPNDR : – The TCAS is on standby.  
– On ground: The selected ATC Transponder only operates in the selective aircraft interrogation mode of Mode S.  
– In flight : The selected ATC Transponder operates.

#### ② TRAFFIC selector

- ABV : The altitude range is set to + 7000 feet above the aircraft, and – 2700 feet below the aircraft.
- N : The altitude range is set to – 2700 feet below the aircraft, and + 2700 feet above the aircraft.
- BLW : The altitude range is set to – 7000 feet below the aircraft, and + 2700 feet above the aircraft.

#### ③ AUTO/ON selector or THRT/ALL selector

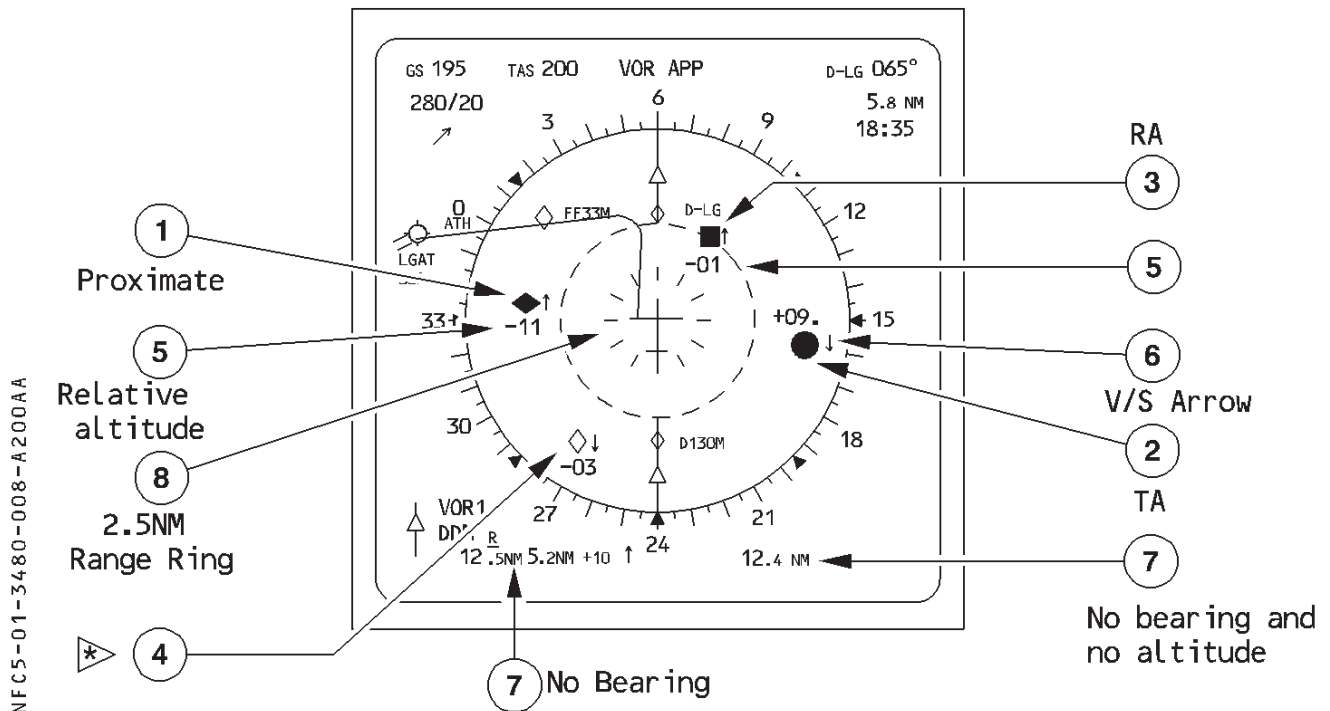
- ON (or ALL) : All intruders are displayed.
- AUTO (or THRT) : Proximate and other intruders are only displayed, if a TA or RA is already presented.

*Note : Some TCAS control panels are equipped with a THRT/ALL selector, instead of the AUTO/ON selector. The associated functions remain unchanged.*

## ND INDICATIONS

The traffic is displayed in all ROSE modes and ARC mode. Only the 8 most threatening intruders are displayed.

R



⑤ Relative altitude

Indicated in hundred of feet above or below the symbol depending on the intruder position.

⑥ Vertical speed arrow

Displayed only if the intruder V/S > 500 feet/minute

Relative altitude and vertical speed arrow are displayed in the same color as the associated intruder symbol.

*Note : If the altitude of an intruder is not available, neither altitude nor vertical speed indications are displayed.*

⑦ No bearing intruder

If the bearing of TA or RA intruder is not available the following data is presented in digital form at the bottom of the ND :

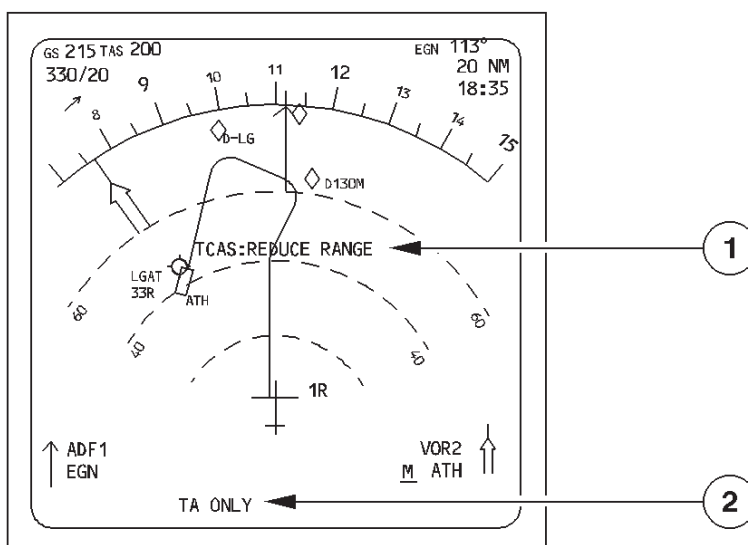
- range
  - relative altitude and vertical speed arrow if available.
- Displayed amber or red according to threat level.

⑧ Range ring

A 2.5 NM white range ring is displayed when a 10 or 20 NM range is selected.

## TCAS MESSAGES

NFC5-01-3480-010-A100AA



### ① Mode and Range Messages

The following messages can be displayed to draw pilot's attention :

**TCAS : REDUCE RANGE** : Displayed when a TA or RA is detected and ND range above 40 NM

**TCAS : CHANGE MODE** : Displayed when a TA or RA is detected and ND mode is PLAN.

R It is displayed amber or red depending on the advisory level (TA or RA).

### ② TCAS Operation Messages

R **TCAS** : Displayed in red, in the case of TCAS internal failure.

**TA ONLY** : Displayed white when the TA mode is selected automatically, or manually by the flight crew.

⑤ Relative altitude

Indicated in hundred of feet above or below the symbol depending on the intruder position.

⑥ Vertical speed arrow

Displayed only if the intruder V/S > 500 feet/minute

Relative altitude and vertical speed arrow are displayed in the same color as the associated intruder symbol.

*Note : If the altitude of an intruder is not available, neither altitude nor vertical speed indications are displayed.*

⑦ No bearing intruder

If the bearing of TA or RA intruder is not available the following data is presented in digital form at the bottom of the ND :

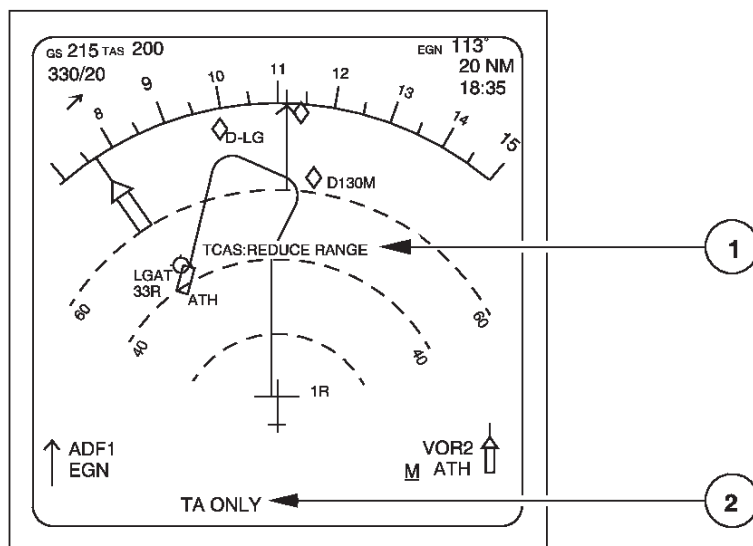
- range
  - relative altitude and vertical speed arrow if available.
- Displayed amber or red according to threat level.

⑧ Range ring

A 2.5 NM white range ring is displayed when a 10 or 20 NM range is selected.

## TCAS MESSAGES

NFC5-01-3480-010-A211AA



### ① Mode and Range Messages

The following messages can be displayed to draw pilot's attention :

TCAS : REDUCE RANGE : Displayed when a TA or RA is detected and ND range above 40 NM

TCAS : CHANGE MODE : Displayed when a TA or RA is detected and ND mode is PLAN.

R It is displayed amber or red depending on the advisory level (TA or RA).

### ② TCAS Operation Messages

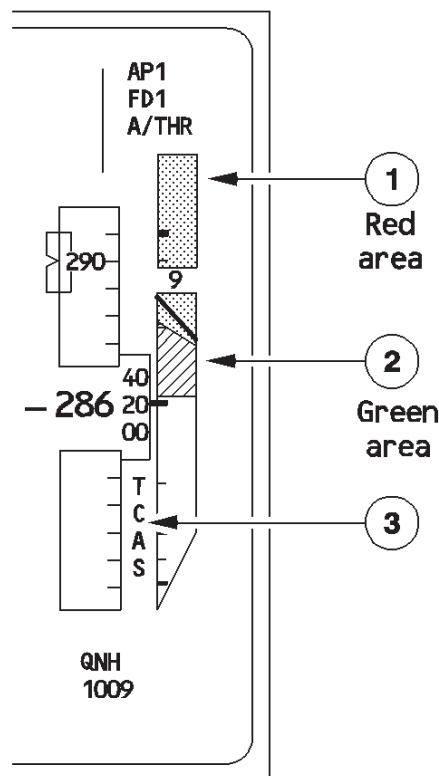
R TCAS : Displayed in amber in the case of TCAS internal failure.

TA ONLY : Displayed white when the TA mode is selected automatically, or manually by the flight crew.

## PFD INDICATIONS

In case of RA detection, the PFD presents vertical orders on the vertical speed scale. The vertical speed scale background is normally grey, but may be partially replaced by green and/or red areas.

NFC5-01-3480-011-A100AA



### ① Red area

Indicates the vertical speed range, where the risk of conflict is high.

### ② Green area

Indicates the recommended vertical speed range.

Note : – The aircraft can also fly in the grey vertical speed range without the risk of conflict (preventive RA).  
 – The color of the vertical speed needle and the digits corresponds to the appropriate area.

### ③ TCAS message

Appears in red when TCAS cannot deliver RA data, or in case of TCAS internal failure.

## **AURAL MESSAGES**

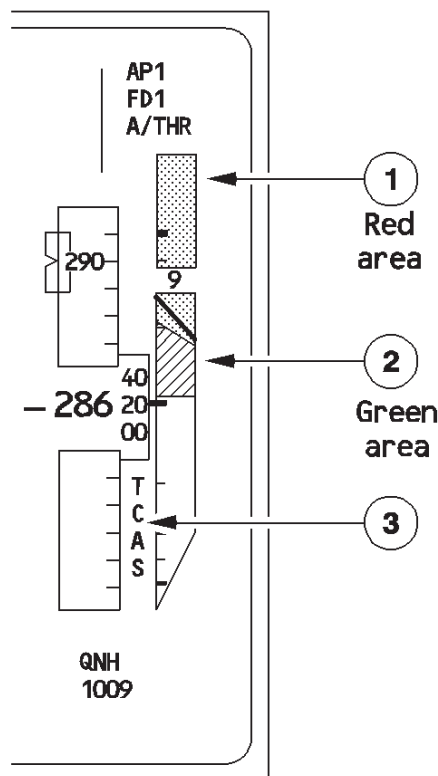
TA/RA detection is associated with the following messages :

|                                  |                                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------|
| TRAFFIC TRAFFIC                  | : only in case of TA detection.                                                                          |
| CLIMB CLIMB CLIMB                | : climb at the vertical speed indicated by the green area on PFD.                                        |
| CLIMB CROSSING CLIMB (twice)     | : same as above. Indicates that you will cross through the intruder altitude.                            |
| INCREASE CLIMB (twice)           | : triggered after CLIMB message if vertical speed not sufficient to achieve safe vertical separation.    |
| REDUCE CLIMB (twice)             | : reduce vertical speed to that indicated on PFD green area.                                             |
| DESCEND DESCEND DESCEND          | : descend at the vertical speed indicated by the green area on PFD.                                      |
| DESCEND CROSSING DESCEND (twice) | : same as above. Indicates that you will cross through the intruder altitude.                            |
| INCREASE DESCEND (twice)         | : triggered after DESCEND message if vertical speed not sufficient to achieve safe vertical separation.  |
| REDUCE DESCEND (twice)           | : reduce vertical speed to that indicated on the PFD green area.                                         |
| CLIMB CLIMB NOW (twice)          | : triggered after DESCEND message if the intruder trajectory has changed.                                |
| DESCEND DESCEND NOW (twice)      | : triggered after CLIMB message if the intruder trajectory has changed.                                  |
| MONITOR VERTICAL SPEED (twice)   | : ensure that vertical speed remains outside the red area.<br>Triggered only when leaving a previous RA. |
| CLEAR OF CONFLICT                | : range is increasing and separation is adequate.<br>Return to assigned clearance.                       |

## PFD INDICATIONS

In case of RA detection, the PFD presents vertical orders on the vertical speed scale. The vertical speed scale background is normally grey, but may be partially replaced by green and/or red areas.

NFC5-01-3480-011-A100AA



### ① Red area

Indicates the vertical speed range, where the risk of conflict is high.

### ② Green area

Indicates the recommended vertical speed range.

Note : – The aircraft can also fly in the grey vertical speed range without the risk of conflict (preventive RA).  
 – The color of the vertical speed needle and the digits corresponds to the appropriate area.

### ③ TCAS message

Appears in red when TCAS cannot deliver RA data, or in case of TCAS internal failure.

## AURAL MESSAGES

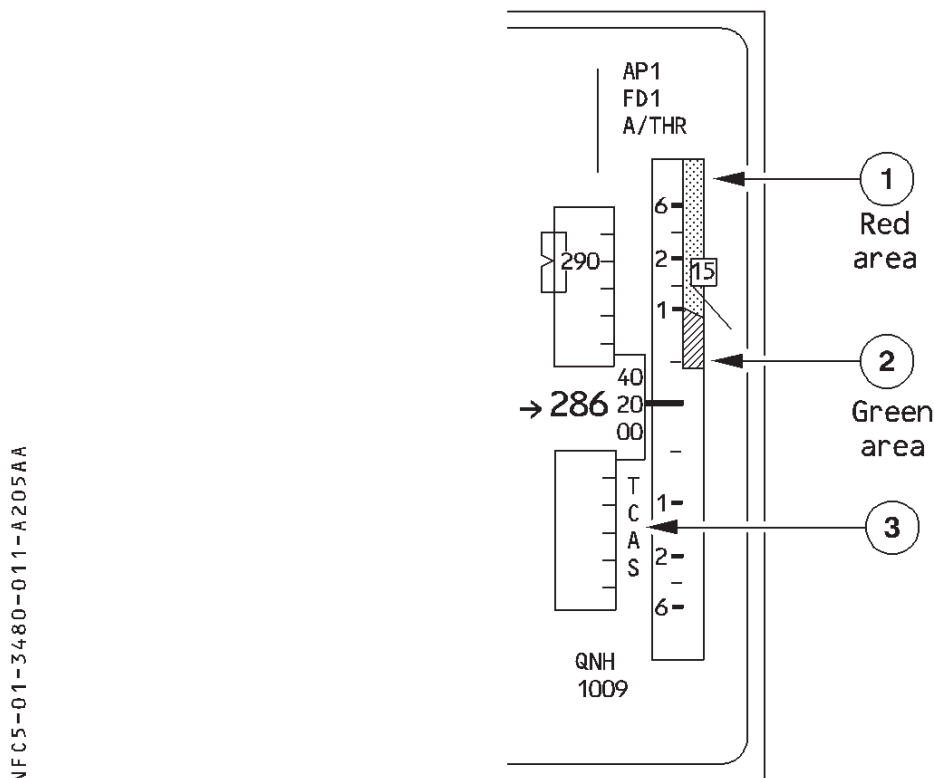
TA/RA detection is associated with the following messages :

|                                              |                                                                                                                                                      |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| "TRAFFIC TRAFFIC"                            | : Only in case of TA detection.                                                                                                                      |
| "CLIMB CLIMB"                                | : Climb at the vertical speed indicated by the green area on the PFD.                                                                                |
| "CLIMB, CROSSING CLIMB" (twice)              | : Same as above. Indicates that you will cross through the intruder altitude.                                                                        |
| "INCREASE CLIMB" (twice)                     | : Triggered after the CLIMB message, if vertical speed is insufficient to achieve safe vertical separation.                                          |
| "DESCEND DESCEND"                            | : Descend at the vertical speed indicated by the green area on the PFD.                                                                              |
| "DESCEND, CROSSING DESCEND" (twice)          | : Same as above. Indicates that you will cross through the intruder altitude.                                                                        |
| "INCREASE DESCEND" (twice)                   | : Triggered after the DESCEND message, if the vertical speed is insufficient to achieve safe vertical separation.                                    |
| "ADJUST VERTICAL SPEED, ADJUST"              | : Adjust the vertical speed to that indicated on the green area of the PFD, reducing climb vertical speed or descent vertical speed, as appropriate. |
| "CLIMB CLIMB NOW" (twice)                    | : Triggered after the DESCEND message, if the intruder trajectory has changed.                                                                       |
| "DESCEND DESCEND NOW" (twice)                | : Triggered after the CLIMB message, if the intruder trajectory has changed.                                                                         |
| "MONITOR VERTICAL SPEED"                     | : Ensure that the vertical speed remains outside the red area. Triggered only once, in case of preventive RA.                                        |
| "MAINTAIN VERTICAL SPEED, MAINTAIN"          | : Maintain the vertical speed indicated on the green area of the PFD.                                                                                |
| "MAINTAIN VERTICAL SPEED, CROSSING MAINTAIN" | : Maintain the vertical speed indicated on the green area of the PFD. Indicates that you will cross through the intruder altitude.                   |
| "CLEAR OF CONFLICT"                          | : The range increases and separation is adequate. Return to assigned clearance.                                                                      |

R

## PFD INDICATIONS

In case of RA detection, the vertical speed scale becomes rectangular and the PFD presents vertical orders on the vertical speed scale. The vertical speed scale background is normally grey, but may be partially replaced by green and/or red areas.



### ① Red area

Indicates the vertical speed range, when there is a high risk of conflict.

### ② Green area

Indicates the recommended vertical speed range.

Note : — The aircraft can also fly in the grey vertical speed range, without the risk of conflict (preventive RA).  
 — The color of the vertical speed needle and the digits corresponds to the appropriate area.

### ③ TCAS message

Appears in amber, when the TCAS cannot deliver RA data, or in case of an internal TCAS failure.

## AURAL MESSAGES

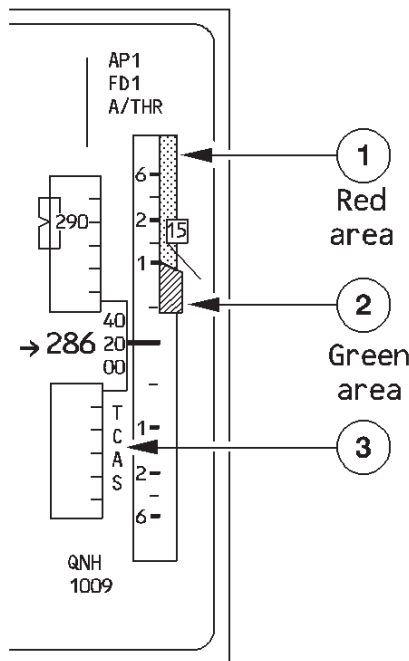
TA/RA detection is associated with the following messages :

|                                              |                                                                                                                                                      |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| "TRAFFIC TRAFFIC"                            | : Only in case of TA detection.                                                                                                                      |
| "CLIMB CLIMB"                                | : Climb at the vertical speed indicated by the green area on the PFD.                                                                                |
| "CLIMB, CROSSING CLIMB" (twice)              | : Same as above. Indicates that you will cross through the intruder altitude.                                                                        |
| "INCREASE CLIMB" (twice)                     | : Triggered after the CLIMB message, if vertical speed is insufficient to achieve safe vertical separation.                                          |
| "DESCEND DESCEND"                            | : Descend at the vertical speed indicated by the green area on the PFD.                                                                              |
| "DESCEND, CROSSING DESCEND" (twice)          | : Same as above. Indicates that you will cross through the intruder altitude.                                                                        |
| "INCREASE DESCEND" (twice)                   | : Triggered after the DESCEND message, if the vertical speed is insufficient to achieve safe vertical separation.                                    |
| "ADJUST VERTICAL SPEED, ADJUST"              | : Adjust the vertical speed to that indicated on the green area of the PFD, reducing climb vertical speed or descent vertical speed, as appropriate. |
| "CLIMB CLIMB NOW" (twice)                    | : Triggered after the DESCEND message, if the intruder trajectory has changed.                                                                       |
| "DESCEND DESCEND NOW" (twice)                | : Triggered after the CLIMB message, if the intruder trajectory has changed.                                                                         |
| "MONITOR VERTICAL SPEED"                     | : Ensure that the vertical speed remains outside the red area. Triggered only once, in case of preventive RA.                                        |
| "MAINTAIN VERTICAL SPEED, MAINTAIN"          | : Maintain the vertical speed indicated on the green area of the PFD.                                                                                |
| "MAINTAIN VERTICAL SPEED, CROSSING MAINTAIN" | : Maintain the vertical speed indicated on the green area of the PFD. Indicates that you will cross through the intruder altitude.                   |
| "CLEAR OF CONFLICT"                          | : The range increases and separation is adequate. Return to assigned clearance.                                                                      |

R

## PFD INDICATIONS

In case of RA detection, the vertical speed scale becomes rectangular and the PFD presents vertical orders on the vertical speed scale. The vertical speed scale background is normally grey, but may be partially replaced by green and/or red areas.



NFC5-01-3480-011-A210AA

### ① Red area

Indicates the vertical speed range, when there is a high risk of conflict.

### ② Green area

Indicates the recommended vertical speed range. It is wider than the red area.

Note : – The aircraft can also fly in the grey vertical speed range, without the risk of conflict (preventive RA).  
 – The color of the digits corresponds to the appropriate area.  
 – In case of RA detection, the vertical speed needle that is normally green, becomes white.

### ③ TCAS message

Appears in amber, when the TCAS cannot deliver RA data, or in case of an internal TCAS failure.

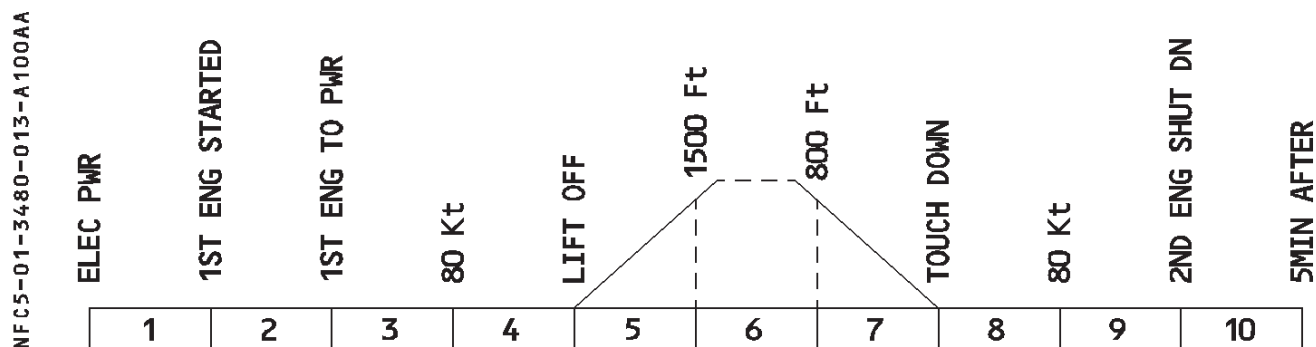
## AURAL MESSAGES

TA/RA detection is associated with the following messages :

|                                              |                                                                                                                                                      |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| "TRAFFIC TRAFFIC"                            | : Only in case of TA detection.                                                                                                                      |
| "CLIMB CLIMB"                                | : Climb at the vertical speed indicated by the green area on the PFD.                                                                                |
| "CLIMB, CROSSING CLIMB" (twice)              | : Same as above. Indicates that you will cross through the intruder altitude.                                                                        |
| "INCREASE CLIMB" (twice)                     | : Triggered after the CLIMB message, if vertical speed is insufficient to achieve safe vertical separation.                                          |
| "DESCEND DESCEND"                            | : Descend at the vertical speed indicated by the green area on the PFD.                                                                              |
| "DESCEND, CROSSING DESCEND" (twice)          | : Same as above. Indicates that you will cross through the intruder altitude.                                                                        |
| "INCREASE DESCEND" (twice)                   | : Triggered after the DESCEND message, if the vertical speed is insufficient to achieve safe vertical separation.                                    |
| "ADJUST VERTICAL SPEED, ADJUST"              | : Adjust the vertical speed to that indicated on the green area of the PFD, reducing climb vertical speed or descent vertical speed, as appropriate. |
| "CLIMB CLIMB NOW" (twice)                    | : Triggered after the DESCEND message, if the intruder trajectory has changed.                                                                       |
| "DESCEND DESCEND NOW" (twice)                | : Triggered after the CLIMB message, if the intruder trajectory has changed.                                                                         |
| "MONITOR VERTICAL SPEED"                     | : Ensure that the vertical speed remains outside the red area. Triggered only once, in case of preventive RA.                                        |
| "MAINTAIN VERTICAL SPEED, MAINTAIN"          | : Maintain the vertical speed indicated on the green area of the PFD.                                                                                |
| "MAINTAIN VERTICAL SPEED, CROSSING MAINTAIN" | : Maintain the vertical speed indicated on the green area of the PFD. Indicates that you will cross through the intruder altitude.                   |
| "CLEAR OF CONFLICT"                          | : The range increases and separation is adequate. Return to assigned clearance.                                                                      |

R

## WARNINGS AND CAUTIONS



| E / WD: FAILURE TITLE<br>conditions         | AURAL<br>WARNING | MASTER<br>LIGHT | SD<br>PAGE<br>CALLED | LOCAL<br>WARNING         | FLT<br>PHASE<br>INHIB |
|---------------------------------------------|------------------|-----------------|----------------------|--------------------------|-----------------------|
| TCAS FAULT in case of TCAS internal failure | SINGLE<br>CHIME  | MASTER<br>CAUT  | NIL                  | Flag on<br>PFD and<br>ND | 3, 4, 5,<br>7         |

## MEMO DISPLAY

TCAS STBY is displayed green when :

- ATC STBY or TCAS STBY is selected by the crew, or
- ALT RPTG is selected at off, or
- both ATC or both RA are failed.

## GENERAL

The Para Visual Indicator (PVI) is a device which helps ground roll out for takeoff in poor visibility conditions. It may also be used for landing roll.

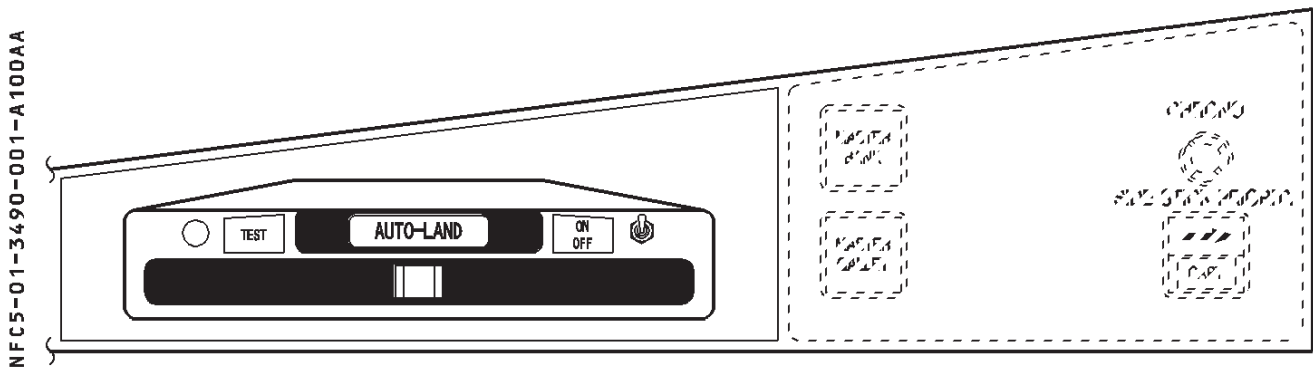
PVI is located on the glareshield in front of the left pilot and displays with a para visual image an information for aircraft guidance during ground roll.

It is normally signalled from DMC 1 (or DMC 3 when the EIS DMC selector is at CAPT 3 position), the order computation being done by FMGC 1 (or FMGC 2 if FMGC 1 fails).

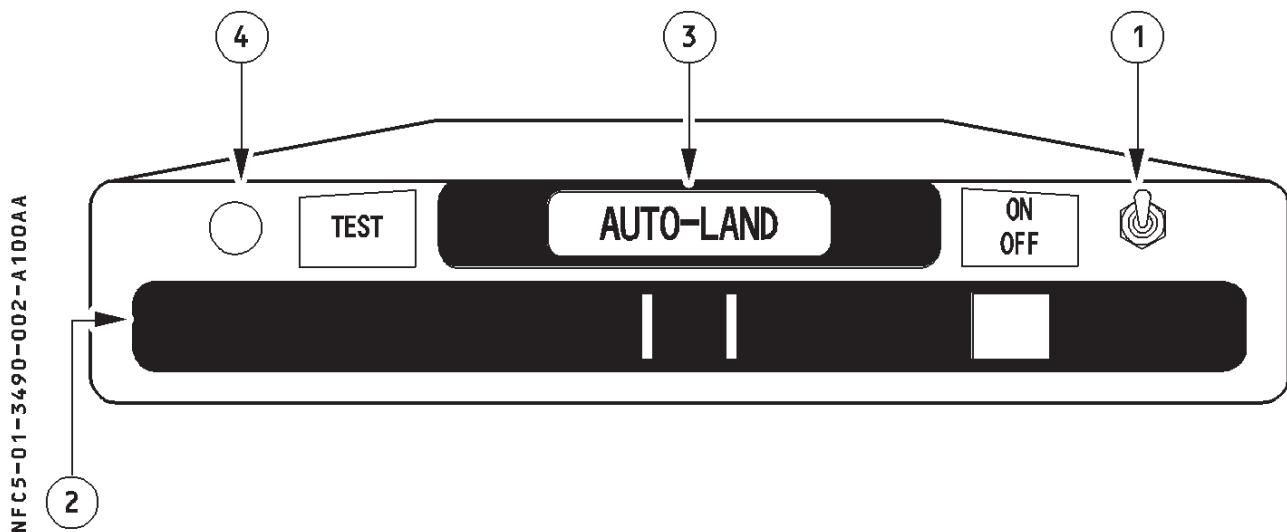
Two fixed white bars and a moving white square are displayed on a black background. The white square provides yaw orders similar to the FD yaw bar. It provides informations about direction and amount of pedal input to recover smoothly and to maintain the runway axis.

- Displacement of the white square is towards the right (left) for a correction by the pilot with the right (left) rudder pedal.
- The moving white square is centered when no correction is needed.

The AUTO-LAND legend is integrated in the indicator and flashes in black on a red background according to a FWC logic.



## CONTROLS AND INDICATORS



### ① ON/OFF sw

ON : The PVI is electrically supplied

### ② PVI symbols

Provided PVI switch is at ON :

- PVI yaw orders are displayed when
  - RUNWAY or ROLL OUT mode is active, and
  - both AP are disengaged, and
  - at least one FD is engaged.
- PVI is in standby when at least one of the above condition, is not fulfilled. In this case two bands are displayed at each end of the display.

*Note : When the ON/OFF switch is at OFF or in case of PVI failure no symbol is displayed. In case of external failure the PVI window is illuminated white.*

### ③ AUTO-LAND legend

Refer to FMGS chapter

### ④ TEST pb

When pressed :

- AUTO-LAND legend is illuminated.
- Display of the white square for 1 second on the display centre then, displacement towards the far left for 1.5 second, then, displacement towards the far right for 3 seconds, then, moving back to the centre for 1.5 second.

**BUS EQUIPMENT LIST**

R

|                 |                | NORM |    | EMER ELEC |        |                         |
|-----------------|----------------|------|----|-----------|--------|-------------------------|
|                 |                | AC   | DC | AC ESS    | DC ESS | HOT                     |
| ADIRU           | ADIRU 1        |      |    | X         |        | HOT 2 *                 |
|                 | AOA RESOLVER 1 |      |    | X         |        |                         |
|                 | ADIRU 2        | AC2  |    |           |        | HOT 2 **<br>during 5 mn |
|                 | AOA RESOLVER 2 | AC2  |    |           |        |                         |
|                 | ADIRU 3        | AC1  |    |           |        | HOT 1 ***               |
|                 | AOA RESOLVER 3 | AC1  |    |           |        |                         |
| STD BY INST     | HORIZON        |      |    |           | X      |                         |
|                 | ALTIMETER      |      |    |           | SHED   |                         |
|                 | COMPASS        |      |    |           | X      |                         |
| NAVAIDS         | VOR 1          |      |    | X         |        |                         |
|                 | VOR 2          | AC2  |    |           |        |                         |
|                 | ILS 1          |      |    | X         |        |                         |
|                 | ILS 2          | AC2  |    |           |        |                         |
|                 | ADF 1          |      |    | SHED      |        |                         |
|                 | ADF 2 ◀        | AC2  |    |           |        |                         |
|                 | DDRFI          |      |    | X         |        |                         |
|                 | DME 1          |      |    | SHED      |        |                         |
|                 | DME 2          | AC2  |    |           |        |                         |
| RADIO ALTIMETER | RA 1           | AC1  |    |           |        |                         |
|                 | RA 2           | AC2  |    |           |        |                         |
| ATC             | ATC 1          |      |    | SHED      |        |                         |
|                 | ATC 2          | AC2  |    |           |        |                         |
| GPWS            |                | AC1  |    |           |        |                         |
| WEATHER RADAR   | WX 1           | AC1  |    |           |        |                         |
|                 | WX 2           | AC2  |    |           |        |                         |
| TCAS ◀          |                | AC1  |    |           |        |                         |

\* Backup supply.

\*\* Backup supply for 5 minutes.

\*\*\* Backup supply, when ATT HDG is in the CAPT 3 position.  
Backup supply for 5 minutes, when the ATT HDG is in the NORM or F/O 3 position.

**BUS EQUIPMENT LIST**

R

|                 |                | NORM |    | EMER ELEC |        |                         |
|-----------------|----------------|------|----|-----------|--------|-------------------------|
|                 |                | AC   | DC | AC ESS    | DC ESS | HOT                     |
| ADIRU           | ADIRU 1        |      |    | X         |        | HOT 2 *                 |
|                 | AOA RESOLVER 1 |      |    | X         |        |                         |
|                 | ADIRU 2        | AC2  |    |           |        | HOT 2 **<br>during 5 mn |
|                 | AOA RESOLVER 2 | AC2  |    |           |        |                         |
|                 | ADIRU 3        | AC1  |    |           |        | HOT 1 ***               |
|                 | AOA RESOLVER 3 | AC1  |    |           |        |                         |
| STD BY INST     | HORIZON        |      |    |           | X      |                         |
|                 | ALTIMETER      |      |    |           | SHED   |                         |
|                 | COMPASS        |      |    |           | X      |                         |
| NAVAIDS         | VOR 1          |      |    | X         |        |                         |
|                 | VOR 2          | AC2  |    |           |        |                         |
|                 | MMR 1          |      |    | X         |        |                         |
|                 | MMR 2          | AC2  |    |           |        |                         |
|                 | ADF 1          |      |    | SHED      |        |                         |
|                 | ADF 2 ◀        | AC2  |    |           |        |                         |
|                 | DDRMI          |      |    | X         |        |                         |
|                 | DME 1          |      |    | SHED      |        |                         |
|                 | DME 2          | AC2  |    |           |        |                         |
| RADIO ALTIMETER | RA 1           | AC1  |    |           |        |                         |
|                 | RA 2           | AC2  |    |           |        |                         |
| ATC             | ATC 1          |      |    | SHED      |        |                         |
|                 | ATC 2          | AC2  |    |           |        |                         |
| GPWS            |                | AC1  |    |           |        |                         |
| WEATHER RADAR   | WX 1           | AC1  |    |           |        |                         |
|                 | WX 2           | AC2  |    |           |        |                         |
| TCAS ◀          |                | AC1  |    |           |        |                         |

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\*\*\* Backup supply, when the ATT HDG is in the CAPT 3 position.  
Backup supply for 5 minutes, when the ATT HDG is in the NORM or F/03 position.

**BUS EQUIPMENT LIST**

|                 |                | NORM |    | EMER ELEC |        |                         |
|-----------------|----------------|------|----|-----------|--------|-------------------------|
|                 |                | AC   | DC | AC ESS    | DC ESS | HOT                     |
| ADIRU           | ADIRU 1        |      |    | X         |        | HOT 2 *                 |
|                 | AOA RESOLVER 1 |      |    | X         |        |                         |
|                 | ADIRU 2        | AC2  |    |           |        | HOT 2 **<br>during 5 mn |
|                 | AOA RESOLVER 2 | AC2  |    |           |        |                         |
|                 | ADIRU 3        | AC1  |    |           |        | HOT 1 ***               |
|                 | AOA RESOLVER 3 | AC1  |    |           |        |                         |
| STD BY INST     | ISIS           |      |    |           | X      | HOT 1 *                 |
|                 | COMPASS        |      |    |           | X      |                         |
| NAVAIDS         | VOR 1          |      |    | X         |        |                         |
|                 | VOR 2          | AC2  |    |           |        |                         |
|                 | MMR 1          |      |    | X         |        |                         |
|                 | MMR 2          | AC2  |    |           |        |                         |
|                 | ADF 1          |      |    | SHED      |        |                         |
|                 | ADF 2 ◀        | AC2  |    |           |        |                         |
|                 | DDRMI          |      |    | X         |        |                         |
|                 | DME 1          |      |    | SHED      |        |                         |
|                 | DME 2          | AC2  |    |           |        |                         |
| RADIO ALTIMETER | RA 1           | AC1  |    |           |        |                         |
|                 | RA 2           | AC2  |    |           |        |                         |
| ATC             | ATC 1          |      |    | SHED      |        |                         |
|                 | ATC 2          | AC2  |    |           |        |                         |
| GPWS            |                | AC1  |    |           |        |                         |
| WEATHER RADAR   | WX 1           | AC1  |    |           |        |                         |
|                 | WX 2           | AC2  |    |           |        |                         |
| TCAS ◀          |                | AC1  |    |           |        |                         |

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**BUS EQUIPMENT LIST**

|                 |                | NORM |    | EMER ELEC |        |                         |
|-----------------|----------------|------|----|-----------|--------|-------------------------|
|                 |                | AC   | DC | AC ESS    | DC ESS | HOT                     |
| ADIRU           | ADIRU 1        |      |    | X         |        | HOT 2 *                 |
|                 | AOA RESOLVER 1 |      |    | X         |        |                         |
|                 | ADIRU 2        | AC2  |    |           |        | HOT 2 **<br>during 5 mn |
|                 | AOA RESOLVER 2 | AC2  |    |           |        |                         |
|                 | ADIRU 3        | AC1  |    |           |        | HOT 1 ***               |
|                 | AOA RESOLVER 3 | AC1  |    |           |        |                         |
| STD BY INST     | HORIZON        |      |    |           | X      | HOT 1 *                 |
|                 | ALTIMETER      |      |    |           | SHED   |                         |
|                 | COMPASS        |      |    |           | X      |                         |
| NAVAIDS         | VOR 1          |      |    | X         |        |                         |
|                 | VOR 2          | AC2  |    |           |        |                         |
|                 | MMR 1          |      |    | X         |        |                         |
|                 | MMR 2          | AC2  |    |           |        |                         |
|                 | ADF 1          |      |    | SHED      |        |                         |
|                 | ADF 2 ◀        | AC2  |    |           |        |                         |
|                 | DDRMI          |      |    | X         |        |                         |
|                 | DME 1          |      |    | SHED      |        |                         |
|                 | DME 2          | AC2  |    |           |        |                         |
| RADIO ALTIMETER | RA 1           | AC1  |    |           |        |                         |
|                 | RA 2           | AC2  |    |           |        |                         |
| ATC             | ATC 1          |      |    | SHED      |        |                         |
|                 | ATC 2          | AC2  |    |           |        |                         |
| GPWS            |                | AC1  |    |           |        |                         |
| WEATHER RADAR   | WX 1           | AC1  |    |           |        |                         |
|                 | WX 2           | AC2  |    |           |        |                         |
| TCAS ◀          |                | AC1  |    |           |        |                         |

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